

GETINGE MAQUET FLOW-I 4.2 USER MANUAL



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User's Manual

FLOW-i 4.2
Anesthesia System

MAQUET
GETINGE GROUP



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1.1 Indications for use

The indication for FLOW-i Anesthesia System is administering inhalation anesthesia while controlling the entire ventilation of patients with no ability to breathe, as well as in supporting patients with a limited ability to breathe. The system is intended for use on neonatal to adult patient populations. The system is intended for use in hospital environments, except MRI environment, by healthcare professionals trained in inhalation anesthesia administration.

1.2 Intended use

The system is intended for use in administering anesthesia while controlling the entire ventilation of patients with no ability to breathe, as well as in supporting patients with a limited ability to breathe.

The system is intended for use by healthcare professionals, trained in the administration of anesthesia.

The system is intended for use on neonate to adult patient populations.

The system is intended for use in hospital environments, except the MRI environment.

When not in operation, the system is designed for in-hospital transport.

1.3 Using this manual

This manual is divided into the following chapters:

- Chapters 1 and 2: Important information before use, along with a basic introduction to the system and User's Manual.
- Chapter 3: Overview of the functional parts of the system.
- Chapters 4 through 11: Understanding and working with the system.
- Chapter 12: Routine cleaning and maintenance procedures.
- Chapter 13: The technical specification of the system.
- Chapter 14: EMC Declaration
- Chapter 15: Terminology and definitions
- Chapter 16: Certificates
- Chapter 17: Index
- Chapter 18: Log sheets

References made to optional accessories within the FLOW-i User's Manual are accompanied by the text 'option' in brackets.

| [Introduction](#) 1

Important information is highlighted with Warning or Caution, where:

WARNING! Indicates critical information about a potential serious outcome to the patient or the user.

CAUTION: Indicates instructions that must be followed in order to ensure the proper operation of the equipment.

2 Important Table of contents

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2.1 General information

- The following applies throughout this User's Manual:
 - 'FLOW-i', 'anesthesia system' and 'system' represent FLOW-i anesthesia system version 4.2.
- This manual applies to the FLOW-i anesthesia system models C20, C30 and C40, and optional equipment that can be fitted onto these.
- Only authorized personnel who are well trained in its use should operate the anesthesia system. It must be operated according to the instructions in this User's Manual.
- Gas volumes, flows and leakages associated with the breathing system are stated in the technical specifications and adhere to BTPS reference conditions. (Body temperature, ambient pressure, Saturated).
- All gas concentration readings are normally referenced to dry gas conditions, ambient room temperature and atmospheric pressure (ATPD).
- The condition for measured inlet gas pressures and flows is STPD (Standard Temperature and Pressure Dry); 20° C, standard pressure at 101.3 kPa and 0 % relative humidity (dry).
- All data on pressures are given in either cmH₂O or bar, where 1 cmH₂O = 1 hPa = 1 mbar
1 bar = 15 psi = 1 atm = 1 kgf/cm² (kp/cm²)
- The anesthesia system is not made with natural latex.
- Applied parts, i.e. equipment making physical contact with the patient, comprise gases (including agents) and the patient mask.

If the mains power supply is interrupted, the internal battery will provide temporary power to the system (approx. 90 minutes when fully charged).
- The fresh gas/gas supply outlets are not affected by switch to battery power.
- Malfunction of the central gas supply can potentially cause one or several of the devices connected to the system to stop their operation simultaneously.
- When the system is in use, a backup gas supply shall always be available.
- If the central gas supply is interrupted, the backup gas cylinders O₂/N₂O or O₂/Air (option) will provide gas to the system.
- In case of automatic ventilation failure, switch to manual ventilation. In case of manual ventilation failure, switch to emergency ventilation.
- In case of a total power (mains power and battery) or other electronic failures, the built-in emergency ventilation system can be used.
- The system maintains its performance when tilted up to two degrees.
- In case of a complete system failure, immediate access to alternative means of ventilation, e.g. a manually powered resuscitator, must be ensured to avoid possible patient injury.
- MAQUET takes full responsibility for compliance of the CE mark requirements for the MAQUET CO₂ cartridge produced by Molecular Products Ltd (MPL).

MAQUET takes full responsibility for the supply of the instructions for use in accordance with the legislation relevant for the intended use for this product.

2.2

WARNING! To avoid the risk of electric shock, this equipment must only be connected to a supply mains with protective earth.

Connection

- A full System checkout procedure must be performed at least once a day.
- The system must never be left unattended while

CAUTION: Federal law restricts this device to sale by or on the order of a physician.

connected to a patient.

- Electronic accessories and auxiliary equipment other than the vaporizers must not be connected or disconnected during operation or when the system is plugged into a mains power outlet. Such connection or disconnection may interfere with the functioning of the system.
- Supplied gases shall meet the requirements for medical grade gases according to applicable standards.
- The backup gas supply should only be turned ON (valves open) when the backup gas supply is in use, or during System checkout.

2.3 Operation

- The system shall always be used in combination with other vital signs monitoring devices and/or professional human judgements of patient condition.

- To protect the patient from high airway pressure, an upper pressure limit must always be set. See Chapter 10 for details.
- To protect the patient, an alarm limit must

always be set for low expired minute volume. See Chapter 10 for details.

- Only anesthetic agents recommended by MAQUET are suitable for use.
- Anesthetic agent bottles without keying may not be used with the system, nor is it allowed to tamper with the keying of anesthetic agent bottles.
- Ensure that the operating room is properly ventilated.
- The system is not intended for use during interhospital transportation.
- The system is not intended for use in an MR environment.
- Antistatic or electrically conductive breathing tubes should not be used. If such breathing tubes are used in combination with high frequency electric surgery

equipment, burns may occur.

- To allow for mains power disconnection, make sure that the power cable connected

to the mains power supply remains visible and fully accessible during patient treatment and not obstructed in any way by ME

(Medical Electrical) equipment.

- If the integrity of the protective earth conductor or the protective earthing system

- in the installation is in doubt, unplug the mains power cable and use battery power.

The equipotentiality terminal is designed for the connection of a potential equalization conductor according to DIN 42 801 and IEC 60601-1. The function of the equipotentiality terminal is to equalize potentials between the system and other medical electrical devices that can be touched simultaneously. The equipotentiality terminal must not be used

- for a protective earth connection. The anesthesia system has been designed and tested to comply with requirements specified in electromagnetic compatibility standard IEC 60601-1-2. It is the responsibility of the user to take necessary measures to ensure that the EMC environment in which the workstation will be used is compatible with the requirements of IEC 60601-1-2 and that the installation is carried out according to the EMC information. See Chapter 16 for details on EMC environments. If limits are exceeded, the accuracy and safety of the system may be impaired. Proactive measures include, but are not limited to, avoiding the use of portable and mobile radio-emitting devices, such as cellular phones and high frequency apparatus, in the proximity of the system. Full performance is reached after a 15 minute warm-up. In case of an emergency, the system can be used immediately. FLOW-i shall be connected to a centre-tapped single phase supply circuit when connected to a 240 Vac (L1-L2-N-GND) supply in the United States. The breathing system can handle negative pressures down to -200 cmH₂O, but is not designed to withstand pressures below that. The operator must not touch the patient and any of the following parts simultaneously:

- Accessible contacts of connectors
- Contacts of fuse holders that are accessible during replacement of the fuse.
- Contacts of lamp holders that are accessible after removal of the lamp.
- Parts inside access covers that can be opened without the use of a tool, e.g. patient cassette lid connector.

WARNINGS!

- In case of system failure, the lack of immediate access to appropriate alternative means of ventilation can result in patient injury.
- Gas inlets and outlets shall not be covered or in any other way be obstructed.
- FLOW-i is not designed to be resistant to direct exposure to high ionizing radiation. Such exposure may result in memory erasure and/or interruption of ventilation.

2.4 Installation and service

- Installation, service and maintenance of the system must be performed by personnel trained and authorized by MAQUET.
- Instructions for installation, service and maintenance, i.e. a service manual, is available for personnel trained and authorized by MAQUET.
- Only original spare parts from MAQUET must be used in the system.

2.5 Accessories and auxiliary equipment

- External equipment intended for connection to signal input, signal output or other connectors shall comply with relevant IEC standards (e.g. IEC 60950 for IT equipment and the IEC 60601 series for medical electrical equipment). In addition, all such combinations – systems – shall comply with the standard IEC 60601-1 'Safety requirements for medical electrical systems'. Equipment not complying with IEC 60601-1 shall be kept outside the patient environment, as defined in the standard.
- Any person who connects external equipment to signal input, signal output or other connectors has formed a system and is therefore responsible for ensuring that the whole system complies with the requirements of IEC 60601-1. If in doubt, contact a qualified medical technician or your local representative.
- For optional equipment and accessories, refer to the user documentation supplied by the manufacturer.
- Use of an anesthesia gas scavenging system is compulsory and must comply with ISO 7396-2 (wall connection) and ISO 80601-2-13 (tubings).
- Values measured at the signal outputs of the anesthesia system, which have been processed in auxiliary equipment, must not be used as a substitute for therapeutic or diagnostic decisions. Such decisions can be made only by staff with medical expertise, and according to established and accepted practice.

If there should be any deviation between information shown on the system and that shown by auxiliary equipment, the parameters shown on the system shall be considered the primary source of information.

- When electrical equipment is connected to the auxiliary power outlet or communications

interface, a medical electrical system (ME system) is effectively created, potentially reducing the level of safety. This could result in previously unidentified risks to patient, users or third parties. It is the responsibility of the user to ensure that the connected equipment is compatible with the requirements of IEC 60601-1.

- Connecting auxiliary equipment to the auxiliary power outlet can potentially increase leakage currents to values above the allowable limits.
- External monitors or similar devices connected to the VGA port of the system, must be powered via a medical grade isolation transformer. No other use is allowed.
- The responsible organization should identify, analyze, evaluate and control these risks.
- Subsequent changes to the medical electrical system could introduce new risks and require additional analysis.
- Changes to the medical electrical system include configuration changes, connection of additional items, disconnection of items, update or upgrade of connected equipment.

- The use of O₂ and Air gas outlets may, depending on central gas supply pressure and ventilation settings, affect ventilation performance.
- Reprocessing parts labelled 'Single-use' will degrade biocompatibility and cleanliness.

WARNINGS!

- The use of other accessories, transducers and cables other than those specified by MAQUET may result in increased emissions or decreased immunity (EMC) of the system.
- No other electrical equipment other than those described in this user's manual may be placed on, or in the immediate vicinity of the system.
- The patient monitor power connector must only be used for patient monitors or equipment mounted on the top shelf.

CAUTIONS:

- Only MAQUET recommended accessories, supplies and auxiliary equipment must be connected to or used in conjunction with the system. Use of other unvalidated accessories, supplies and auxiliary equipment may impair the performance and safety of the system.
- To prevent the system from tilting, follow the restrictions for patient monitors and auxiliary equipment specified in Chapter 15, page 268. Maximum torque on the side rails by the table is 20 Nm.
- Equipment placed on shafts or rails must not restrict opening of the Emergency Ventilation cover.

3 System overview Table of contents

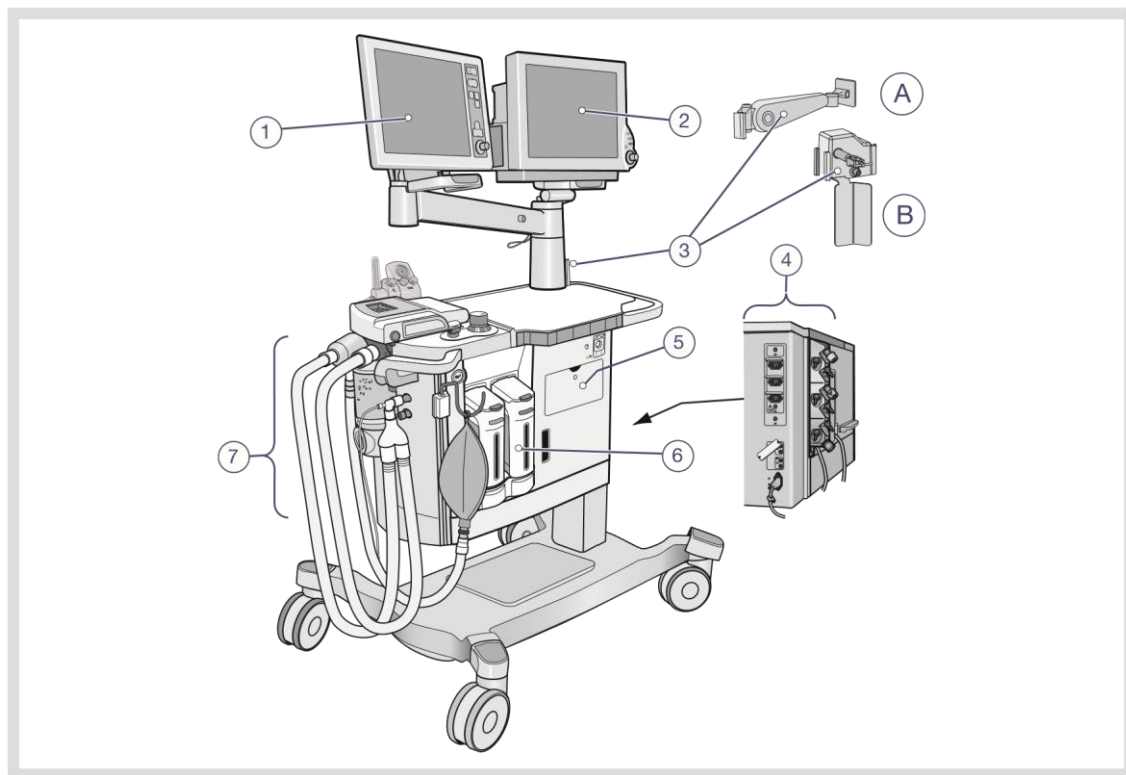
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The anesthesia system is designed to enable the operator to work with the basic parts of the system in the most suitable way for each procedure. As the system is mounted on wheels, and the control panel is mounted on a rotatable arm, it can be easily moved into an ergonomically-suitable position.

There are a number of different system models available. For more information on the different models, see page 40.

3.1 System parts

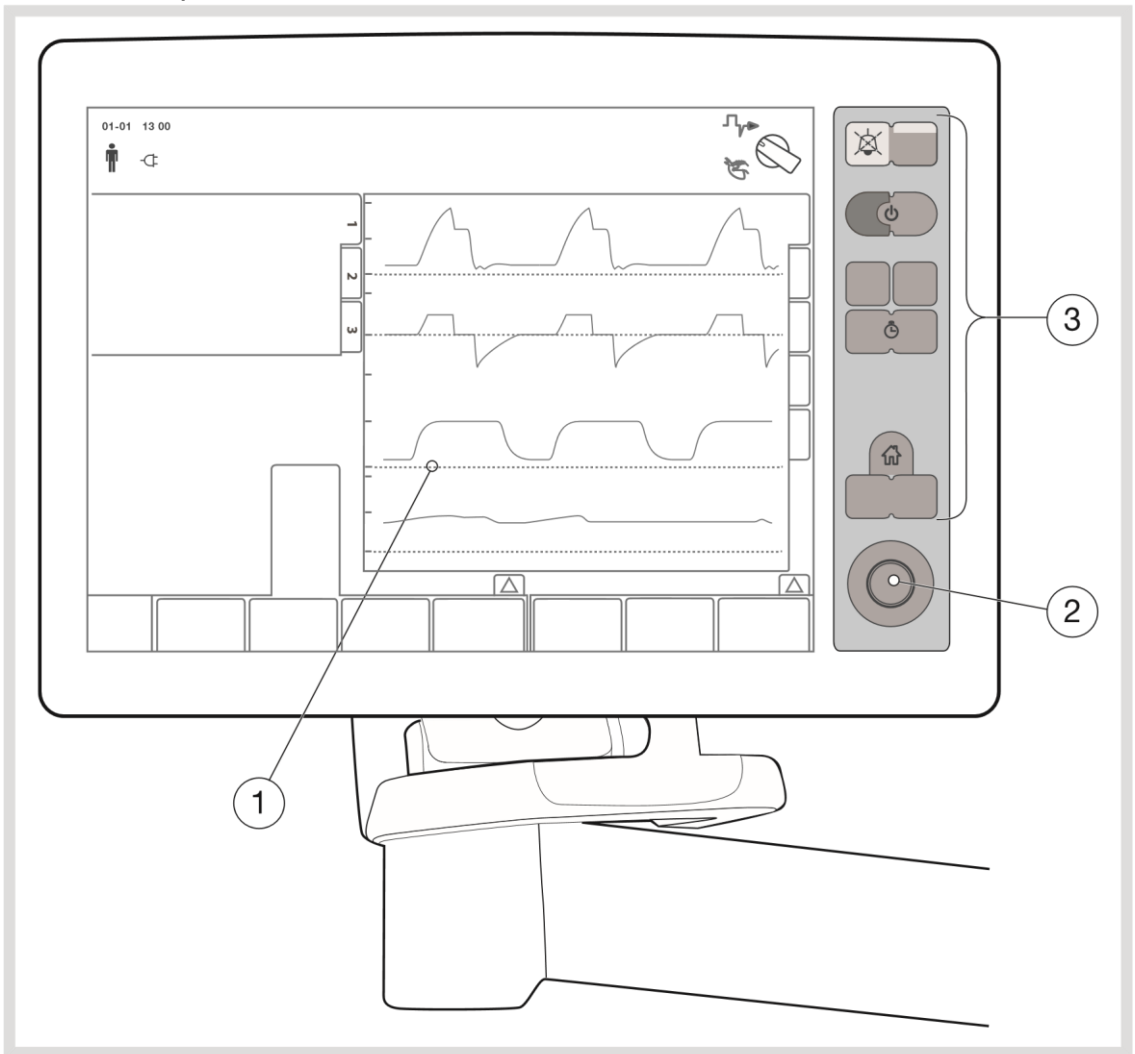
The system comprises the following basic parts:



- | | |
|---|---------------------------------|
| 1. Control panel | 5. Emergency ventilation system |
| 2. Patient monitor (option) | 6. Vaporizer unit |
| 3. Additional arm 'A' (option) and/or 1
Gas backup holder 'B' (option) | 7. Breathing system |
| 4. External connections | |

1. C30 can only be equipped with the additional arm or the extra backup gas holder.

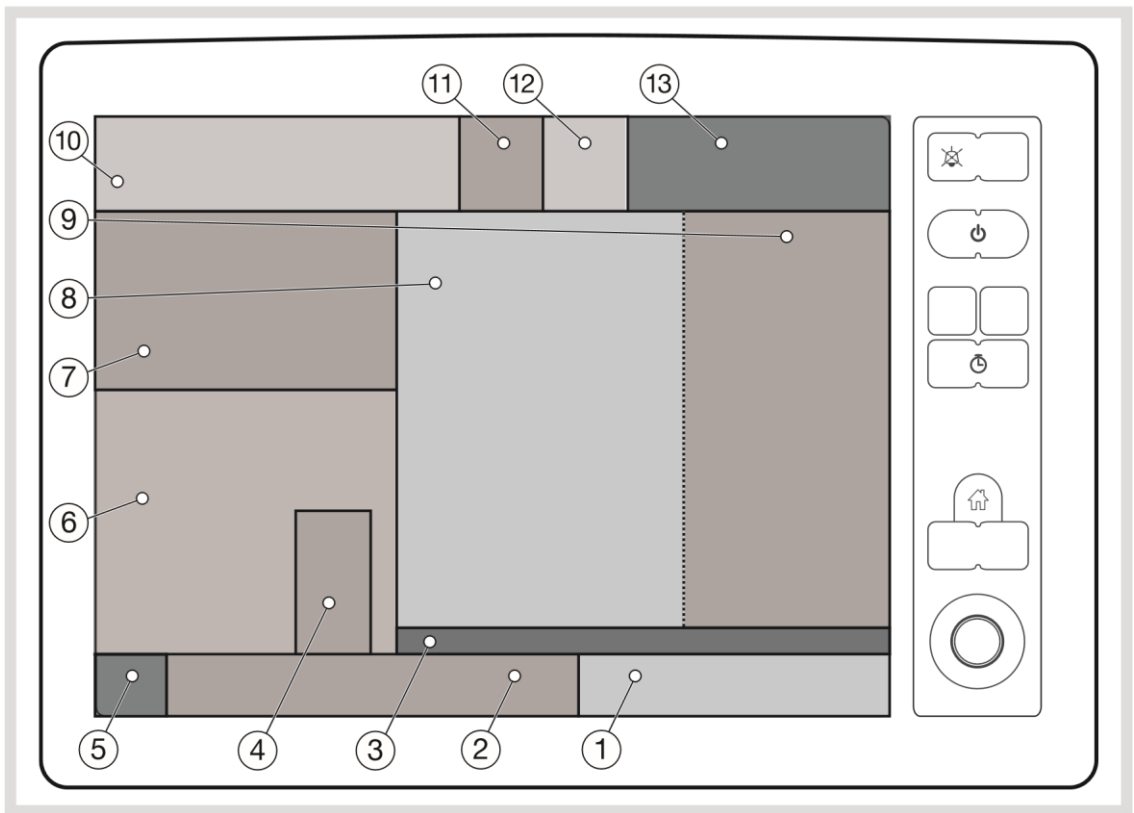
3.2 Control panel



The control panel includes:

1. Screen with active touch pads
2. Rotary knob
3. Membrane buttons

3.2.1 Areas of the screen



The screen is divided up into a number of different areas:

- | | |
|---|-----------------------------------|
| 1. Ventilation direct access settings | 9. Tab area: |
| 2. Gas direct access settings | - Short trends |
| 3. Activate additional settings/gas settings - Loops window | - Volume Reflector Indicator |
| 4. Fresh gas mix rotameter - Gas supply pressure | |
| 5. APL valve value | |
| 6. Gas measurement area | 10. Current alarm and status area |
| 7. Ventilation measurement area | 11. Alarm functions |
| 8. Waveform area | 12. Timer |
| | 13. Mode indicator |

The touch screen and rotary knob allow the Messages are displayed in the Alarm message operator to control the primary functions of area or System message area. The following

the anesthesia system, where the patient's condition is monitored through measured values and waveform displays.

Measured values and waveforms are displayed on the screen in the following color groups:

Color for measured values and waveforms	
Pressure	Yellow
Flow	Green
Volume	Blue
CO ₂	Light gray
O ₂	White
N ₂ O	Blue
MAC	Gray
Isoflurane	Purple
Desflurane	Blue
Sevoflurane	Yellow

The colors for O₂ and N₂O in the above table may vary due to country-specific standards.

3.2.2 Navigating the screen

There are several ways of navigating the screen and setting values.

Using the touch screen

1. Press the required touch pad. The touch pad becomes active, which is indicated by a blue highlight.
2. Turn the rotary knob to the required value.
3. Press the touch pad to confirm setting.

An activated touch pad is only active for 20 seconds. The system will prompt the user to enter a value when 10 seconds have passed. If no value is entered and confirmed within the following 10 seconds, the touch pad setting returns to its previous setting and is deactivated.

Using the rotary knob

1. Turn the Rotary knob to move between the touch pads on the screen. The selected touch pad is indicated by a **blue frame**.

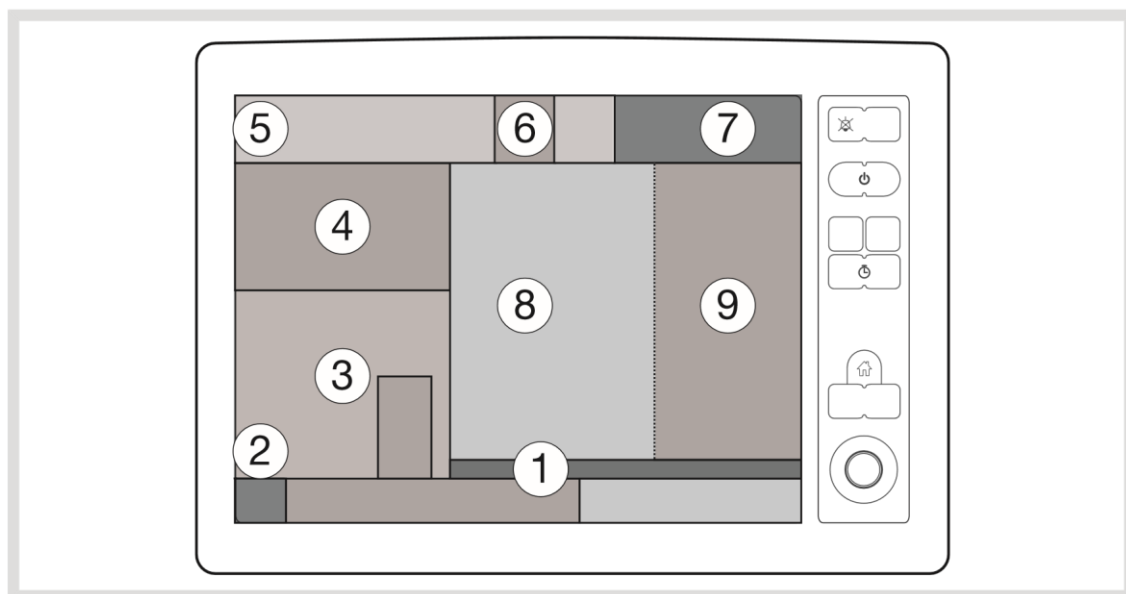
color scheme is adopted:

Color for alarms and system messages	
High priority alarms	Black text on red background
Medium priority alarms	Black text on yellow background
Low priority alarms	Black text on blue background
Technical alarms, i.e. alarms with the prefix TExXX, where XXX is an integer.	Black text on red, yellow or blue background, dependent on priority (high/medium/low)
System messages	White text on black background

A detailed description of alarms and patient safety is found in Chapter 10.

2. On required touch pad, press the rotary knob to activate the touch pad. This will highlight the touch pad in blue.
3. Turn the Rotary knob to the required value.
4. Press the Rotary knob to confirm setting.

Active screen

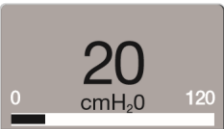


Pressing any of the areas labelled with a number in the illustration, displays a window according to the table below.

Screen area		Dialog/window produced
1.	Gas settings/ Ventilation settings	Multiple windows, see Chapter 5
2.	MAC valuePatient category, see Chapter 5, page 75.	
3.	Gas measurement area	Alarm profile window, see Chapter 10, page 187. 4.
	Ventilation measurement area	Alarm profile window, see Chapter 10, page 187.
5.	Patient category symbol	Patient category, see Chapter 5, page 75.
6.	Alarm functions	Alarm profile window, see Chapter 10, page 187.
7.	Mode indicator area	Ventilation mode selection, see Chapter 5, page 87.
8.	Waveform area	Waveforms and scales, see Chapter 9, page 163.
9.	Tab area (displayed window depends on selected tab) -	- Waveforms and scales, see Chapter 9, page 163. Screen layout, see Chapter 9, page 163.

3.2.3 Touch pad settings

A range bar is located at the bottom of all ventilation and gas touch pads, where a numeric value can be specified by the user. If the entered value deviates too far from the norm given the current ventilation mode and other parameter settings, the bar will change color according to the table below.

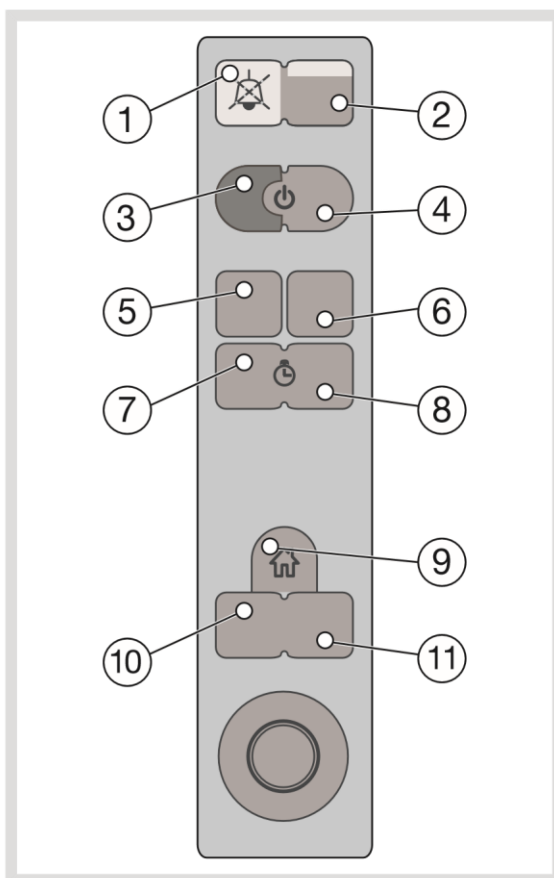
Touch pad setting	Description
	A black range bar indicates normal parameter setting.
	A yellow range bar indicates that the parameter setting is high (or low, depending on the parameter).
	A red range bar indicates that the parameter setting is very high (or very low, depending on the parameter).

The change in color of the range bar is accompanied by a system message that remains displayed on the screen until the trigger condition is relieved.

3.2.4 Membrane buttons

2. Alarm profile
3. Start case
4. End case
5. Save screen
6. Trends
7. Start/Stop timer

8. Reset timer



1. Audio pause
9. Home
10. Screen layout
11. Menu

See Chapter 9 for more information.

3.3 Patient monitor (option)

The system can be connected to a selection of different patient monitors. For full details, contact your local MAQUET representative.

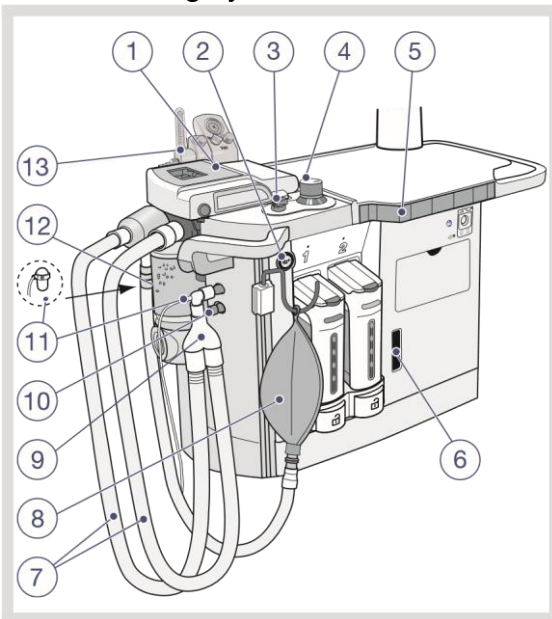
The patient monitor must comply with IEC 60601-1 ed. 3.

During mains power failure, the backup battery in the anesthesia system will not power the patient monitor.

3.3.1 Panel interchangeability

If desired, the placement of the control panel and the patient monitor can be switched so that the patient monitor is placed on the display arm. This procedure shall only be performed by a service technician trained and authorized by MAQUET. Contact your local MAQUET supplier for more information.

3.4 Breathing system



The breathing system comprises the following:

1. Patient cassette
2. O₂ flush
3. MAN/AUTO ventilation switch
4. APL valve
5. Volume reflector
6. AGS (Anesthetic Gas Scavenging) flow indicator
7. Patient tubing
8. Manual breathing bag with tubing
9. Y-piece
10. AFGO - Additional Fresh Gas Outlet (option)
11. Water trap and sampling line
12. CO₂ absorber
13. Auxiliary O₂ and suction module (option)

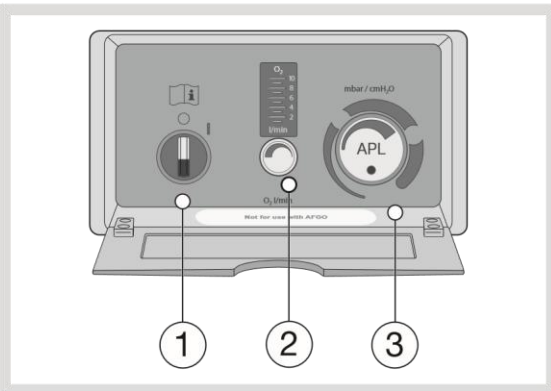
MAQUET recommends that a bacterial/viral filter is always connected to the expiratory connection on the patient cassette. This will minimize the risk of cross-contamination.

3.6 Emergency ventilation system



In case of a total power (i.e. mains power and battery) or system failure, this system will allow the patient to be manually ventilated. The emergency ventilation system comprises:

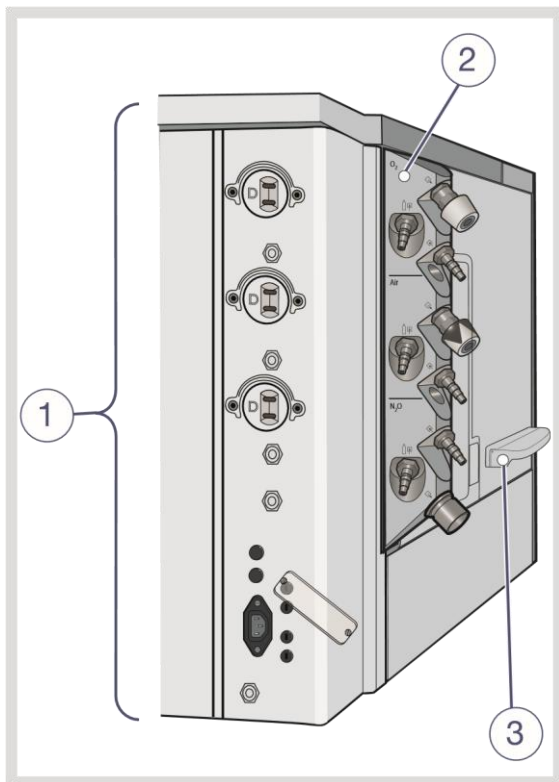
1. Activation switch
2. O₂ gas supply and flowmeter, graded up to 10 l/min.
3. Mechanical APL



WARNING! If the emergency ventilation

system is activated while the anesthesia system is in operation, the anesthesia system will be shutdown.

3.7 External connections

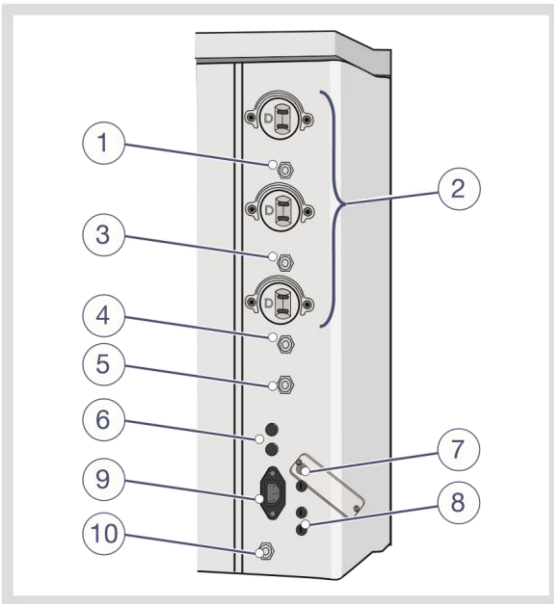


The following external connections exist on the system:

1. Power supply and fuses
2. Gas connections
3. Input/Output ports

3.7.1 Power supply and fuses

The patient monitor outlet and the auxiliary power outlets do not have a battery backup.



1. Auxiliary power outlets fuse 2A (option)
2. Auxiliary power outlets (option)
3. Auxiliary power outlets fuse 1A (option)
4. Auxiliary power outlets fuse 1A (option)
5. Patient monitor fuse (option)
6. Isolation transformer fuse (option)
7. Mains power inlet fuses
8. Lift fuses (C30 model only)
9. Mains power inlet
10. Equipotential terminal (earth)

Auxiliary power supply outlet connections vary depending on country specific standards.

In addition to the individual fuses, the maximum current delivered through the system is regulated by a shared fuse.

In case of mains power failure, a battery backup will power the system for a limited time.

WARNING!

Restrictions apply to the use of auxiliary power outlets:

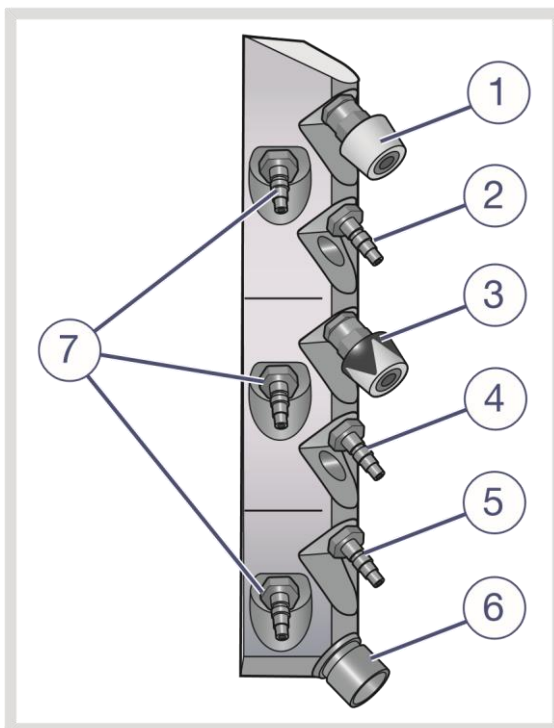
- Outlets must not be used to power life support equipment unless the life support equipment itself is equipped with battery backup.
- A multiple socket extension cord must not be connected to any of the outlets.

For more details on the power supply and status, see Chapter 10, page 203.

CAUTION: The auxiliary power outlets shall only be used for supplying power to equipment intended to form part of the medical electric system.

3.7.2 Gas connections

2. O₂ inlet
3. Air outlet (option)
4. Air inlet
5. N₂O inlet
6. AGS outlet (country-specific)



1. O₂ outlet (option) connections)
7. Gas cylinder inlets for O₂, Air and N₂O (option)

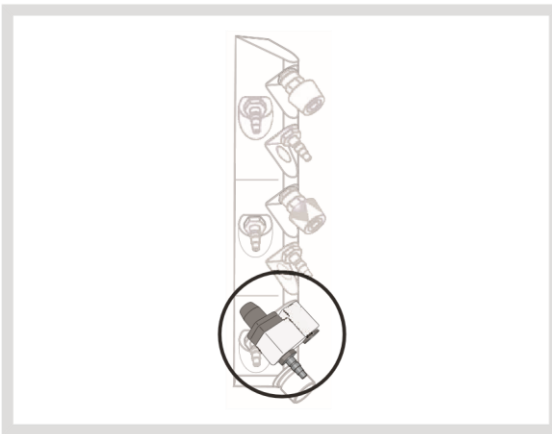
If equipment with high gas consumption are connected to the gas outlets, the central gas supply pressure should be above 3 bar (300 kPa, 44 PSI). Make sure that the central gas supply is sufficient for the extra equipment.

The intended use for the Auxiliary O₂ outlet is to provide oxygen for patient therapy.

The intended use for the Auxiliary AIR outlet is to provide drive gas for small ejector driven suction devices.

Maximum allowed air flow from each of the O₂ and Air outlets is 60 l/min.

The central N₂O gas supply (wall supply) is connected to the system via a pressure regulator attached to the N₂O gas inlet.



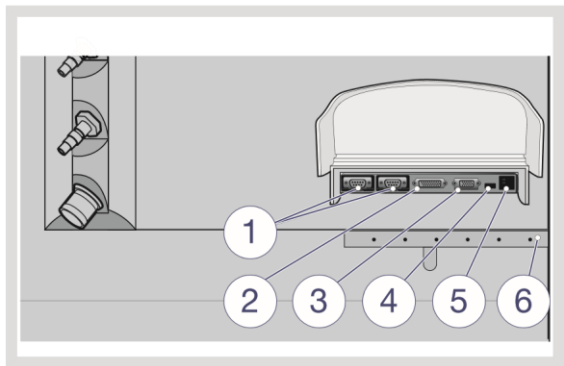
Clinical situations that require high gas consumption might affect the supply gas pressure.

If the gas supply is low, i.e. close to 2.5 bar (250 kPa, 36 PSI), the system's ability to deliver gas according to all possible settings might be compromised by additional gas hoses and connections.

The network connection (LAN) port is for service use, and should only be used by personnel trained and authorized by MAQUET.

External cables connected to the I/O ports must be secured using cable restrainers where available.

3.7.3 Input/Output ports



1. RS232 serial data communication ports
2. Control panel connection
3. VGA connector
4. USB port
5. Network connection
6. Cable restrainer

The intended use of the USB port is only for USB memory flash drives. Restricted items include, but are not limited to, external hard drives, USB hubs and any equipment using the USB port primarily as a power source.

CAUTION: The operator must not touch any of the input/output ports, e.g. RS232, VGA connector and USB port, and the patient simultaneously.

3.8 Explanation of symbols

3.8.1 Labels

The  White drawing on a blue background. Consulting accompanying documentation is a mandatory action.

Symbol	Description
--------	-------------

Symbol	Description
	Worn-out batteries must be recycled or disposed of properly in accordance with appropriate standards.

following symbols are shown on the industrial and environmental system:



Special waste to be disposed of in accordance with appropriate industrial and environmental standards.

Pb



Caution must be taken when moving the system up or down a slope. Refer to Transport conditions, page 38.



CE label. The device complies with the requirements of the European Council Directive



CSA label—Indicates compliance with Canadian and US standards



Manufacturer



Manufacturing date



UDI Label - Unique Device Identification. See technical specifications, page 290.



Class 1 equipment, Type B. The device classification according to IEC 60601-1.



Federal law restricts this device to sale by or on the order of a physician.



IP classification: IPX1, drip proof



Indicates instructions that must be followed in order to ensure the proper operation of the equipment.



Use by date



Black cross over an orange background. Broken CO₂ absorber canisters may cause skin irritation.



Use no oil. Applicable to parts marked with this symbol.



Serial number



Batch code



Article number



Do not use if packaging is damaged

93/42/EEC (i.e. the Medical Device Directive).

Black border, black exclamation mark over a yellow background.

Symbol	Description
	Indicates critical information about a potentially serious outcome to the patient or the user.



Red circle with a single red line over a black drawing.
Pushing prohibited



Equipotentiality terminal.

Anesthesia Gas Scavenging

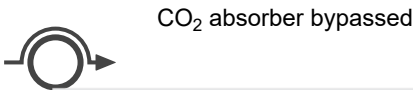
AGS



Medical device maximum with three drawers). C30 and see technical specifications,

Symbol	Description
	CO ₂ absorber connected and locked in position

weight (C20 C40 are lighter, page 266.



CO₂ absorber bypassed

Home touch pad symbol, see



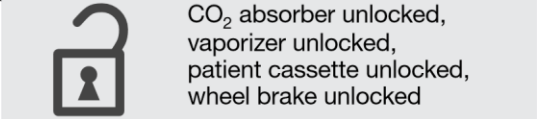
Red circle with a single red line over a black drawing. Stepping prohibited



Red circle with a single red line over a black drawing. Sitting prohibited



Lift button, 10% lift dutycycle, for more information see Chapter 7, page 146.



CO₂ absorber unlocked, vaporizer unlocked, patient cassette unlocked, wheel brake unlocked



Wheel brake locked



Labelled part may be disinfected using a steam autoclave



Use centre of floating ball as reference when reading from the Auxiliary O₂ flow scale



Manual (MAN) ventilation



Inspiratory connection to the patient cassette



Automatic (AUTO) ventilation



Expiratory connection to the patient cassette



Manual breathing bag connection

Vac

O₂ flow meter/suction equipment



Y-piece connection for System checkout

O₂ +

O₂ flush

Mains power On Off button



Chapter 9, page 162

Description
Timer touch pad symbol Chapter 9, page 16

Symbol	Description
	Data communication input/output ports

Emergency ventilation

Fuse

Emergency ventilation

Not for use with AFGO. Emergency O ₂ flow is always delivered through the patient cassette.
--

N ₂ O inlet pressure range: 250-650 kPa/2.5-6.5 bar/36-94 PSI

AFGO Max 120 mbar/cmH ₂ O

Reading lamp

Cylinder gas inlet

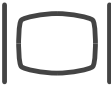
Backup gas system pressure increase/decrease
--

RS232 serial port

Gas supply inlet

Gas supply outlet

Network connection



Control panel connection

VGA connection

USB connection

3.8.2 Screen-displayed symbols

The following symbols are shown on the screen:

Symbol	Description
+2	Overlay loops

Symbol	Description
	Standby mode
	Active alarm
	Multiple active alarms
	Audio pause, pre-mute alarm, mute active alarm or remove resolved alarms.
	Audio off
	Alarm off
	Power indicator - indicates AC power connected
	Infant
	Adult
	Ventilation mode indicator
	Manual ventilation selected
	Automatic ventilation selected
	Reference loop

	Numerical trends selected
	Graphical trends selected
	Activates a cursor in the trends display window, and allows for use of the rotary knob to scroll through values.
	Checkbox - can be selected to mark an option or to choose a feature for display on the screen
	Gas analyzer has no information to display
	Audio pause—silence or confirm an alarm
	Power indicator - indicates battery operation, along with estimated time remaining
	Timer activated
	Date and time

3.9 Ergonomical positioning



- The C20 and C30 systems are on wheels and can be rotated 360°.
- The C40 system is mounted onto a ceiling pendant and can be moved and rotated as allowed by the pendant solution.
- The control panel can be tilted up and down 15° and rotated 220° (without moving the display arm).
- The display arm can be rotated 170°.
- The height of the C30 working surface can be set anywhere between approx. 30 in. and 40 in.

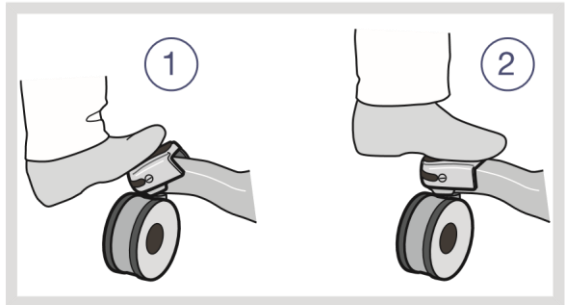
CAUTION: Be careful when moving the display arm to avoid damaging equipment placed on the writing table or top shelf.

3.9.2 Working position

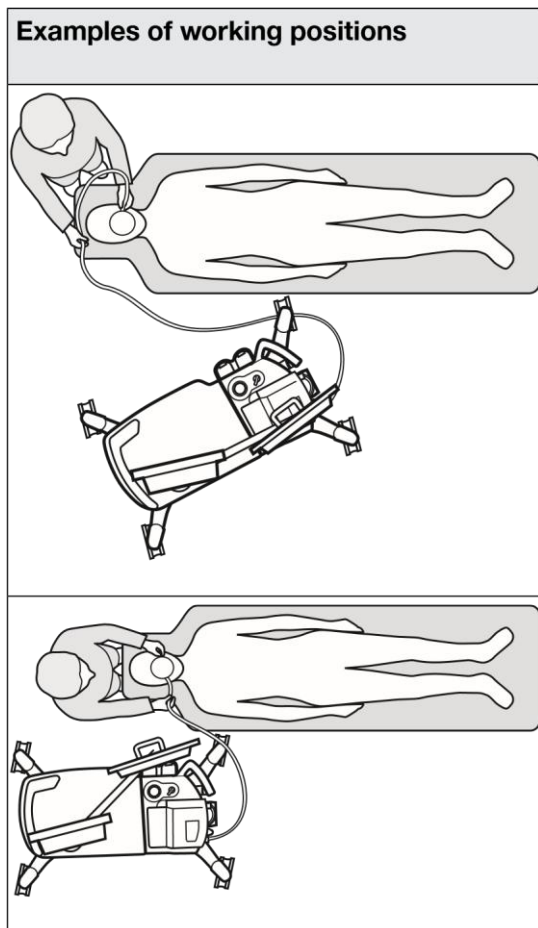
The flexibility of the system allows the operator many alternative working positions:

3.9.1 Using the brake

Once a suitable position has been found, the wheels should be locked into position.



1. Push the brake down to lock the wheel.
2. Push the brake up to unlock the wheel.



3.10 Storage and transportation

3.10.1 Before transport

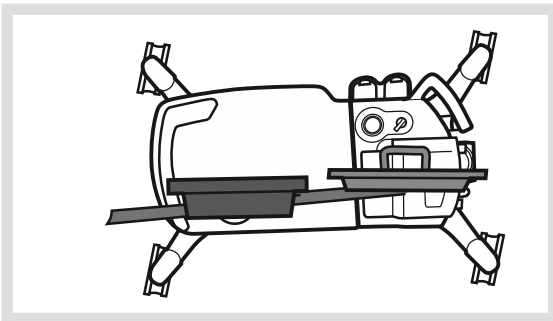
- If the system is to be moved to another room or transported a longer distance, ensure mounted accessories are securely attached to the system.

- The C40 system is intended for transport inside the operation room only when undocked from the ceiling pendant. Transportation to other locations requires the use of a suitable cart or trolley according to hospital routines.
- Be careful when moving the system down a slope.

CAUTION: Make sure extra equipment and accessories, e.g. support arm and additional table, are folded close to the system during transport to minimize the risk of tipping. The display arm should be positioned as shown in the illustration.

3.10.2 During transport

- Move the system using the handles on the main unit and not those on the control panel or patient monitor. This will reduce the risk of tipping and/or system damage.
- If the optional extra table is installed, make sure it is folded into a vertical position.



3.10.3 Storage

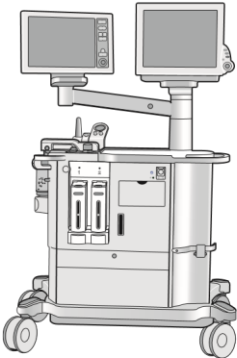
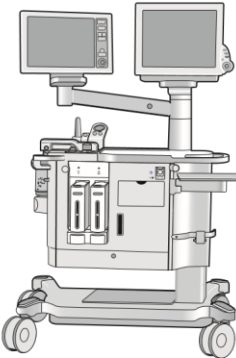
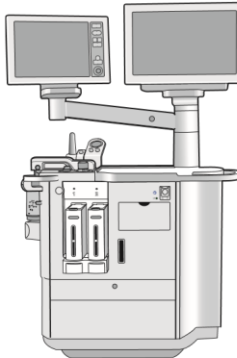
- During storage keep the system connected to the mains power supply so that the batteries maintain a full charge.
- If the system is disconnected from a power source, ensure the batteries are fully charged before storage to avoid deterioration of battery performance.
- When the system is disconnected from a mains power supply, a fully charged battery can be stored in the FLOW-i system for up to six weeks at temperatures between +5°C (+40°F) and +40°C (+105°F). At temperatures between +50°C (+125°F) and 60°C (+140°F) storage time is one week. If these limits are exceeded, battery performance can no longer be guaranteed.
- MAQUET recommends that the vaporizers are emptied before long term storage (>30 days).
- For fire safety purposes, the system requires functional batteries for system Startup.
- Ensure the system is not exposed to temperatures below -25°C (-13°F) or above +60°C (140°F).
- For information on CO₂ absorber storage, see Chapter 7, page 128.
- Ensure the system is not exposed to a relative humidity above 95%.

3.11 System models

The anesthesia system is available in different models:

- C20
- C30
- C40

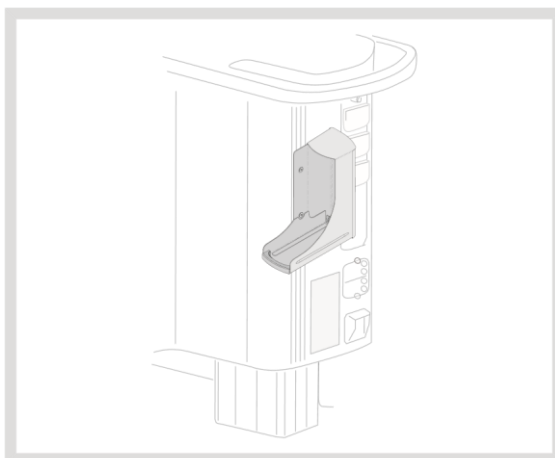
The following table shows the standard equipment for these different models (main components):

C20	C30	C40
		
• Working surface/writing table • Reading lamp • Two drawers, one lockable • Battery for approx. 90 minutes support (fully charged) • Vertical shafts for optional horizontal rails • Four wheels with individual locking brake	• Height adjustable • Working surface/writing table • Reading lamp • One lockable drawer • Battery for approx. 90 minutes support (fully charged) • Vertical shafts for optional horizontal rails • Four wheels with individual locking brake	• Ceiling pendant • Working surface/writing table • Reading lamp • One lockable drawer • Battery for approx. 90 minutes support (fully charged) • Vertical shafts for optional horizontal rails

3.12 Optional equipment

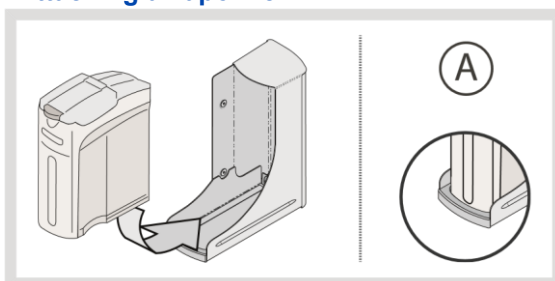
3.12.1 Vaporizer holder

Not all listed optional equipment may be available in your country, contact your local MAQUET representative for more information. The vaporizer holder provides easy access to available an additional vaporizer during surgical procedures.



The vaporizer holder may only be installed or moved by a service technician trained and authorized by Maquet.

Attaching a vaporizer



Slide the vaporizer into the slot until the front locks into place (A).

To remove the vaporizer, lift up the front and gently pull outwards.

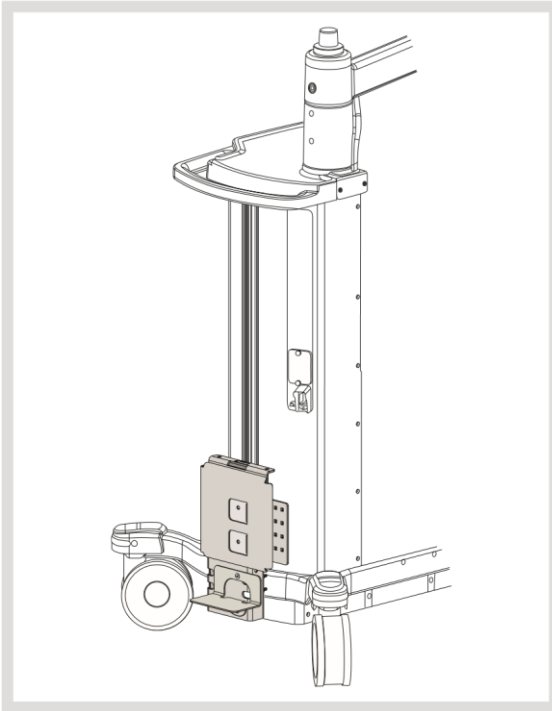
3.12.2 Universal bracket for C20

The universal bracket for C20 is intended for and secure use with the FLOW-i C20 model only. It bag and provides additional space for mounting

3.12.3 Manual breathing bag support arm

The support arm provides a static support for the manual breathing bag associated tubings.

accessory equipment.



The universal bracket is permanently attached to the core unit, and shall only be removed by a service technician trained and authorized by MAQUET.

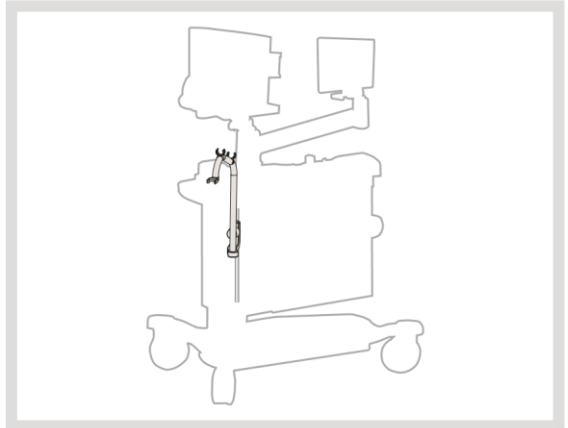
Maximum dimensions of mounted equipment is approximately 430x340x180 mm (WxHxD).

Always ensure mounted equipment is properly secured before starting a patient case.

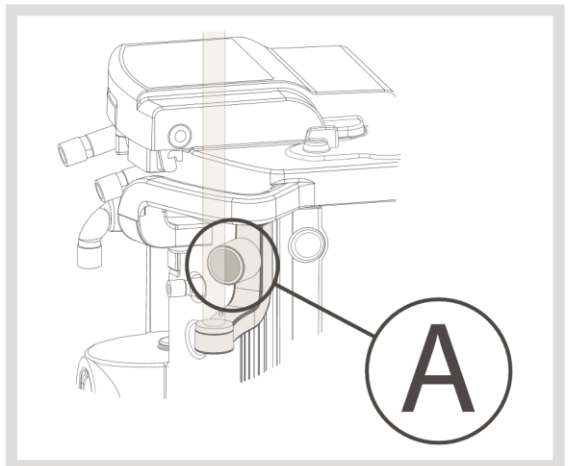
3.12.4 EVAC Restrictor

The EVAC restrictor acts to reduce the pressure difference between high vacuum

The support arm shall only be used to secure the manual breathing bag and associated tubings.



Connection



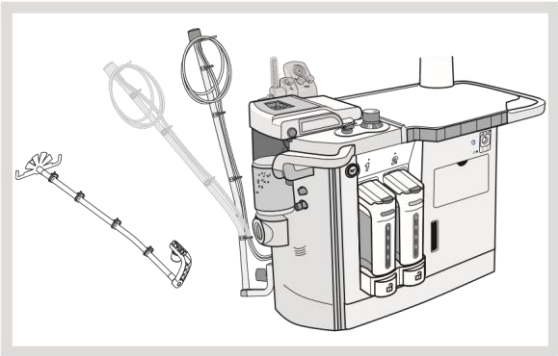
The support arm is fastened to the vertical railing on the core unit by turning the fastening knob (A).

evacuation systems and the FLOW-i AGS outlet. This ensures proper gas evacuation without adverse side effects.

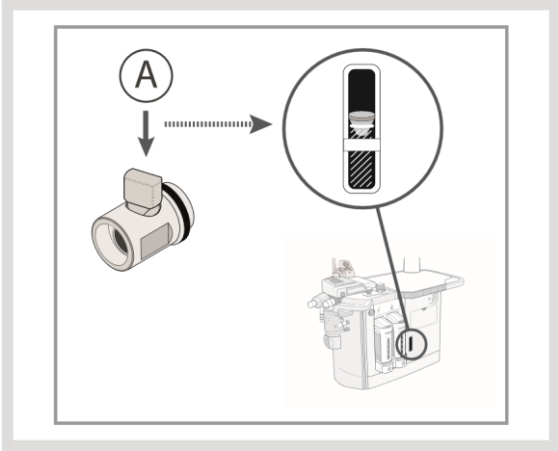
The EVAC restrictor may only be installed by a service technician trained and authorized by Maquet.

AGS hose assemblies connected to the system shall comply with ISO 80601-2-13.

If equipped with alternate connector systems, the connectors shall comply with DISS: CGA-V5:2008 (WAGD), SS 875 24 30:2004 (UTS).



Restrictor unit



The appropriate evacuation flow is achieved by turning the adjusting knob (A), so that the evacuation floater in the AGS flow meter hovers above the dashed area. This ensures sufficient evacuation flow.

3.12.5 Cable support arm

To organize and manage cables connecting the patient to the patient monitor, a cable support arm can be attached to any of the system's four vertical shafts.

The arm can be bent and angled to provide the most suitable cable arrangement. Ensure that the emergency ventilation hatch is not obscured by the cable support arm.

The cable support arm is to be used for electrical cables only. It shall not be used for patient tubings or any other equipment.

The maximum load is 1.1 lbs.

CAUTION

- The weight of the cables should not exceed 1.1 kg.
- The cable support arm should not be used for patient tubing or any other equipment.
-
-
-
- Ensure the cable support arm is properly secured to the vertical shaft.
- Be careful not to obstruct the emergency ventilation hatch.

3.12.6 Top shelf

The top shelf is attached to the display column. Various types of equipment, e.g.

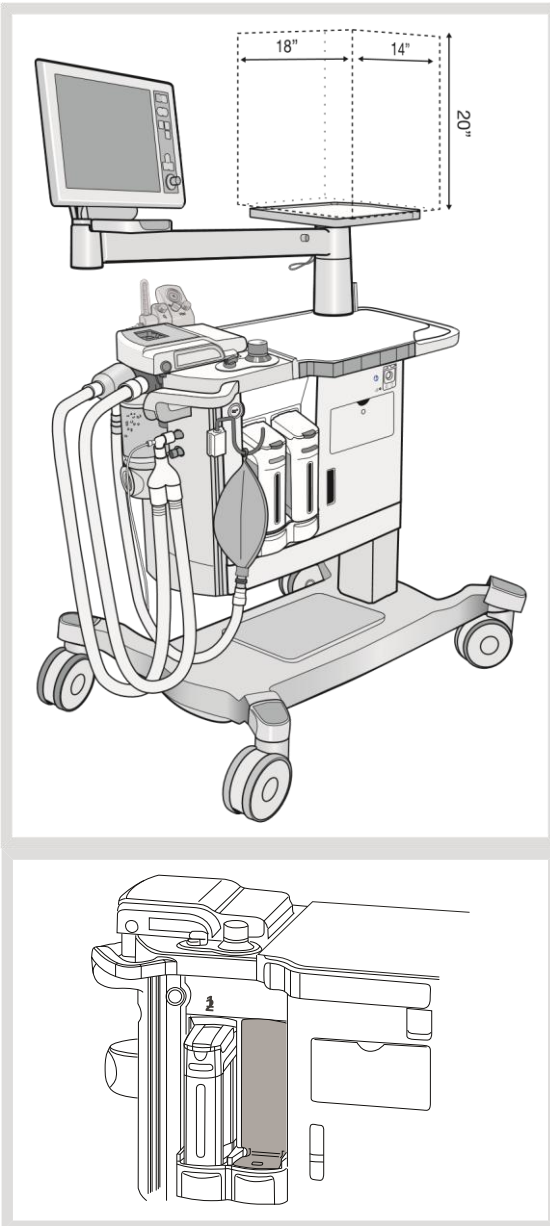
parameter box and patient monitor, can be used in combination with the shelf.

The top shelf, along with any associated equipment, shall be assembled by a hospital technician.

3.12.7 Vaporizer slot cover

3.12.8 Backup gas system

The vaporizer slot cover is intended to protect the backup gas rack and backup gas cylinder. The gas and electrical connections inside the holder are described in Chapter 10, page 204. The vaporizer slot when no vaporizer is connected.

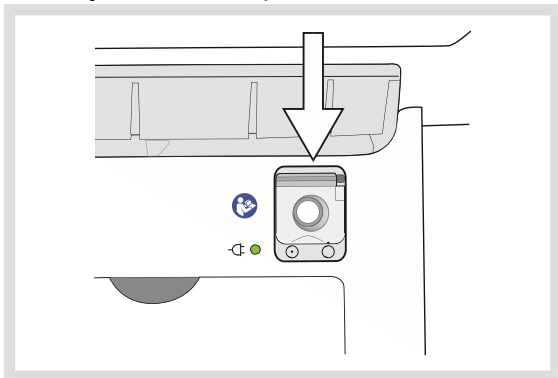


Slide the cover at a slight angle into the vacant vaporizer slot until it locks into place..

4 Startup and system checkout Table of contents

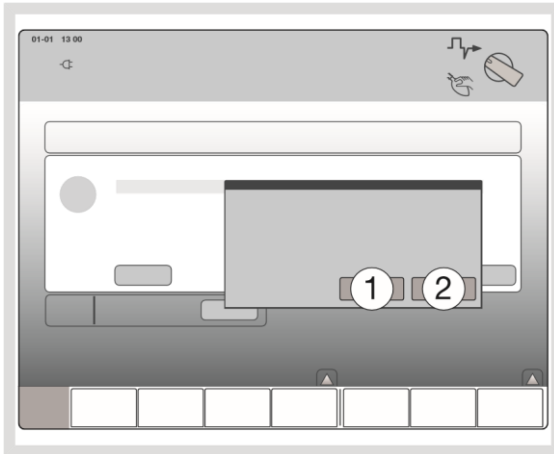
4.1	System startup		48
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4.5	Understanding results		61
4.6	Standby mode		62
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4.1 System startup



- Start the system by means of the Power button found directly above the Emergency ventilation system.

4.2 System checkout



If the system needs to be used immediately, i.e. during an emergency, values and results from the last successful System checkout remain in effect.

Press the Start touch pad in the System checkout window to begin the procedure.

A prompt to start the System checkout procedure is automatically displayed at system startup:

1. Bypass the System checkout
2. Start the System checkout

The System checkout can also be initiated via the Menu membrane button during Standby, see Chapter 9, page 166.

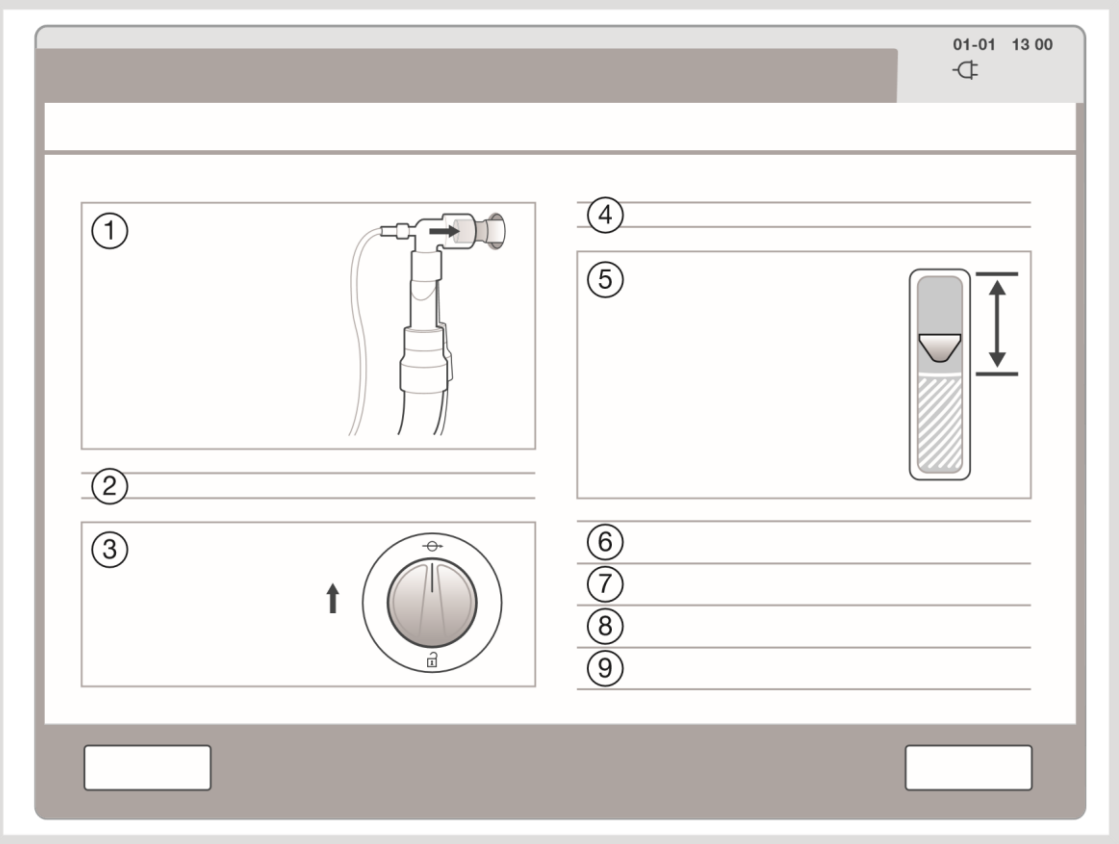
The procedure includes the following main areas:

- Preparations
- Checks requiring user interaction
- Automatic checks

To ensure correct system functionality, optimal performance and patient safety, the System checkout procedure must be performed as follows:

- Once a day, or before connecting the first patient within a running 24 hour period.
- After replacing the patient cassette.
- After the system has been transported.

4.2.1 Preparations



The first part of the System checkout procedure ensures that the system is correctly on page 52. prepared for use.

Check the components listed on the screen and press the 'Continue' touch pad located in the bottom right of the screen when finished.

If an external Patient Gas Analyzer is used (not the one in FLOW-i), the sampling has to be switched off in order not to remove gas from the system. The Leakage check may otherwise fail.

An in-depth description of the different steps is presented in the table

If the system has backup cylinders installed, an extra check point appears on the screen. Make sure these cylinders are open for the duration of the System checkout.

If only the O₂ backup gas cylinder is used, the Air and/or N₂O backup gas cylinder pressure check can be disabled in Service and Settings – Startup configuration, see Chapter 9, page 178.

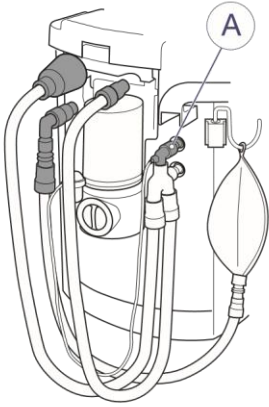
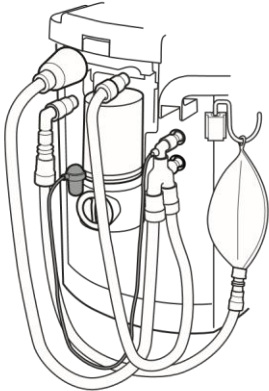
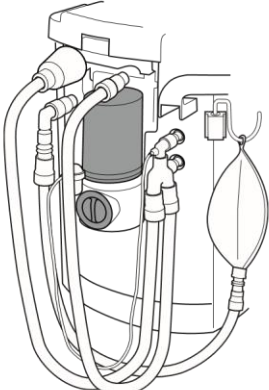
If any of the connected vaporizers contain <5% of the total agent volume, the automatic vaporizer test will not be initiated for that vaporizer. A prompt window appears. Pressing 'bypass test' will exclude that vaporizer from the check. Pressing 'redo test' will repeat the test.

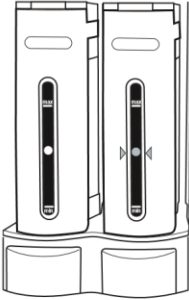
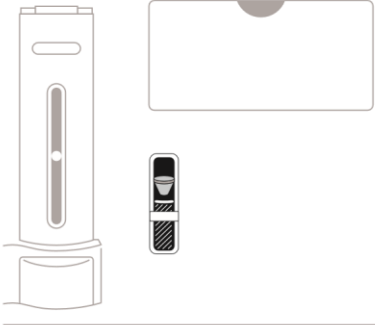
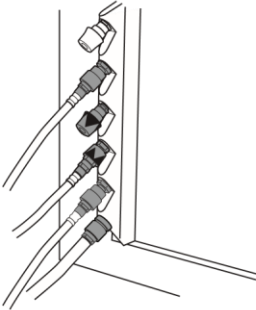
4.2.2 Gas sampling leakage

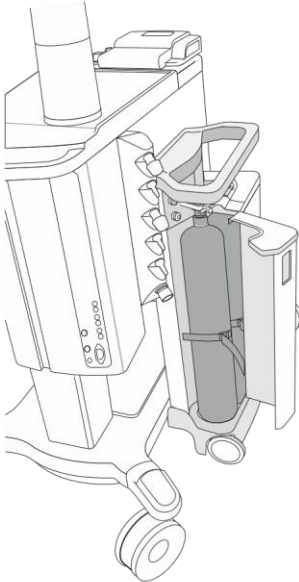
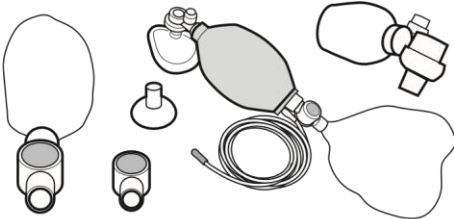
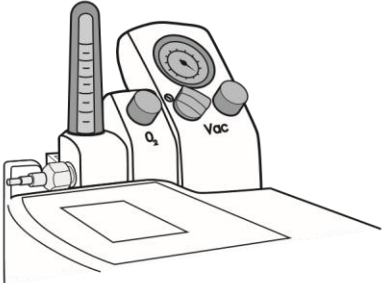
The 'sampling leakage alarm' responds to amounts of leakage sufficient to interfere with the gas sampling precision.

If the alarm is activated, perform any or all of the following actions before contacting Maquet service:

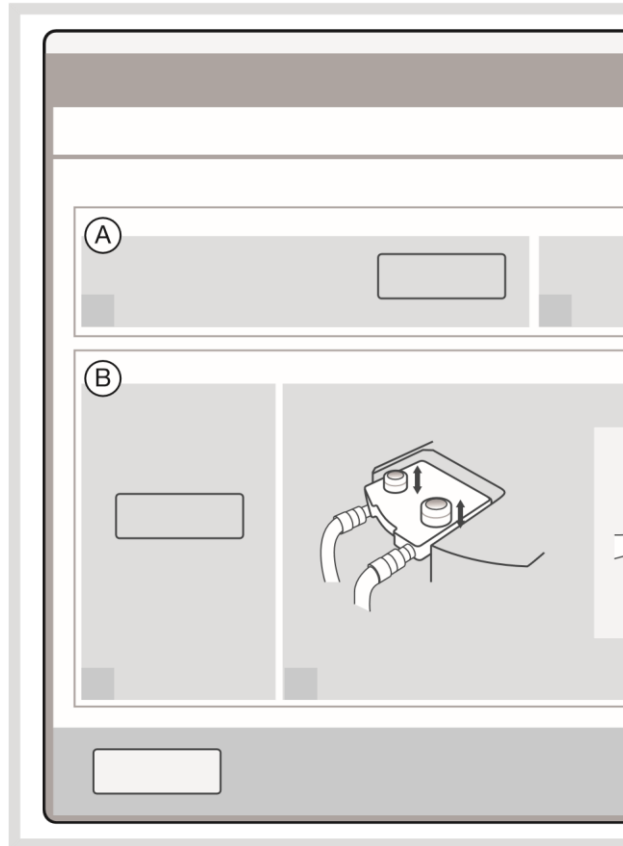
- Check the sampling line connections
- Check the patient tubing connections
- Replace the patient cassette

Preparatory steps	Additional description
Step 1 Check that the breathing circuit is correctly mounted and connected to the test plug (A).	 A line drawing of a medical device, likely a ventilator or breathing circuit component. It features a cylindrical main body with various ports and tubes connected. A label 'A' in a circle points to a specific plug or connector on the right side of the device.
Step 2 Check the water trap and sampling line. Discard/empty any water present in the water trap.	 A line drawing of the same medical device as in Step 1, but from a slightly different angle or with different components highlighted. It shows a complex arrangement of tubes and connectors, including a water trap and a sampling line as mentioned in the text.
Step 3 Check that the absorber is correctly mounted and unsaturated. Also check that the switch is in the 'locked' position.	 A line drawing of the medical device, showing the absorber and a switch. The absorber is a cylindrical component, and the switch is a lever or button on the side. The diagram illustrates the correct mounting and connection of these components.

Preparatory steps	Additional description
Step 4 Check that the vaporizer(s) contains sufficient agent.	 A line drawing of two vertical vaporizers standing side-by-side on a base. Each vaporizer has a vertical scale with a dot indicating the agent level. The right vaporizer also has a small arrow pointing to the right.
Step 5 Check that the Anesthesia Gas Scavenging (AGS) flow indicator is above the dashed area. CAUTION: Ensure that the AGS is correctly connected. Running the system with a poor AGS connection may result in anesthetic agent being emitted into the operation environment.	 A line drawing showing a vertical AGS flow indicator on the left and a rectangular dashed area on the right. Below the dashed area is a small vertical scale with a shaded section at the bottom.
Step 6 Check that the central gas supplies are connected.	 A line drawing showing several gas supply lines with connectors plugged into a panel on the side of a device.

Preparatory steps	Additional description
Step 7 (only if backup cylinders are installed) Check that the Backup cylinders are opened. Close the cylinders after completing System checkout. The system checkout will not initiate if the system is not connected to the central gas supply. WARNING! If the backup gas supply pressure is higher than the central gas supply pressure, the backup cylinder gas will be used during the System checkout. Make a note of the backup cylinder pressure presented on the Control Panel screen during the System checkout.	 A line drawing of a medical device, likely a ventilator, showing a backup gas cylinder installed on the side. The cylinder is connected to the main unit via a hose and a valve. The device has a control panel with various buttons and a screen.
Step 8 Check that the manual resuscitator is readily available and is working correctly.	 A line drawing of a manual resuscitator, which is a device used to provide manual ventilation to a patient. It consists of a main body with a valve, a mask, and a tube. The diagram shows the main body, the mask, and the tube separately.
Step 9 Check for adequate suction pressure in the suction unit and that the auxiliary O₂ flow is functioning properly, refer to Chapter 7, page 129, for instructions.	 A line drawing of a medical device, likely a suction unit and an oxygen flowmeter. The device has a control panel with a gauge, a valve, and a tube. The gauge is labeled 'O₂' and 'Vac'. The tube is connected to the main unit.

4.2.3 Checks requiring user interaction



The second part of the System checkout procedure requires the user to perform a few actions before proceeding to the next step:

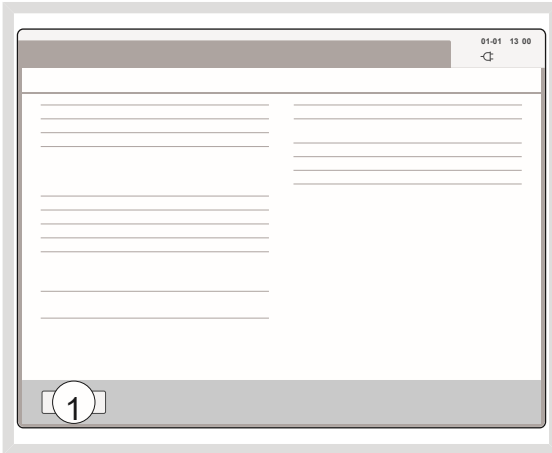
A. Manually test the O₂ flush:

- Press the 'Start check' touch pad.
- Fully depress the O₂ flush button for approx. 3 seconds. If the test is successful, 'Passed' appears on the panel screen. Continue to the next step (B). Otherwise repeat the test.

B. Inspect the function of the inspiratory and expiratory unidirectional valves (some systems may be fitted with a patient cassette lid that needs to be opened to see the valves):

- If necessary, open the patient cassette lid.
- Press 'Start check' and ensure that the unidirectional valves are moving up and down.
- Confirm by pressing 'Yes' on the panel screen prompt.
- If necessary, close the patient cassette lid and press 'Continue' or simply press 'Continue' to move on to the next step.

4.2.4 Automatic checks



The third part of the System checkout procedure contains a number of tests that the system automatically performs. These are as follows:

- Internal tests
- Barometer
- Gas supply pressure
- Pressure transducers
- Safety valve
- Vaporizer inlet/outlet valve
- Flow transducer
- AUTO ventilation leakage
- MAN ventilation leakage
- Gas analyzer
- Battery
- Vaporizer 1
- Vaporizer 2
- Technical alarms

Components listed on the screen are individually tested. 'Passed' (in green text) or 'Failed' (in red text) is displayed after each test depending on the outcome. 'Running' indicates the test currently being performed.

If the gas analyzer test fails, the vaporizer tests will not be performed. A dialog box will appear to inform the user of this.

The automatic check procedure ends when all tests are performed. This procedure can at any time be bypassed by pressing the Bypass touch pad located in the lower left area of the panel screen (1).

The current leakage for manual and automatic ventilation is displayed separately in ml/min. A maximum leakage of up to 150 ml/min is allowed for each of the ventilation modes. The leakage tests are performed using a pressure of 50 cmH₂O and 30 cmH₂O for AUTO and MAN ventilation mode respectively.

Some automatic tests can be repeated after the automatic test sequence has finished; a prompt to check the relevant part and redo check appears on the screen.

If only the O₂ backup gas cylinder is used, the Air and/or N₂O backup gas cylinder pressure check can be disabled in Service and Settings – Startup configuration, see Chapter 9, page 178.

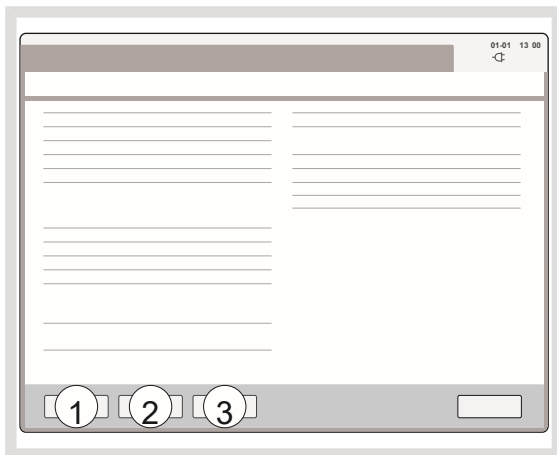
Circuit compliance compensation is calculated during the leakage tests and presented with the test results.

If N₂O central gas supply pressure is connected while the N₂O function is deactivated, the system checkout will fail.

WARNING! If the System checkout is unsuccessful, the system must not be connected to the patient until the malfunction is corrected. If the malfunction cannot be corrected by the operator, the

system must be turned Off, taken out of operation and serviced by personnel trained and authorized by MAQUET.

4.2.5 Finalization



Once all checks are completed, a summary is displayed on the screen. Press the Standby touch pad located at the bottom right of the screen to enter Standby mode.

Checks can be repeated by pressing any of the touch pad buttons located in the bottom left area of the panel screen:

1. Redo System checkout
2. Redo leakage check
3. Redo vaporizer check

These checks can also be initiated in Standby mode via the Menu membrane button.

The results of each test in the System checkout procedure can also be viewed by pressing the 'Results' touch pad in the Standby mode main screen, or by accessing the System checkout sub-menu via the Menu membrane button.

CAUTION: Close the backup cylinders after completing the System checkout to avoid unintentional use or leakage of the backup gas.

4.2.6 Bypassing System checkout

In emergencies, the System checkout can be bypassed at any stage of the procedure. This, however, is not recommended.

With an emergency startup situation, Manual ventilation and fresh gas dosage are ready for use within 15 seconds. Monitored parameters have full accuracy after a maximum of 15 minutes. However, pressure, flow and volume displays will be 90-95% accurate after approx. two minutes.

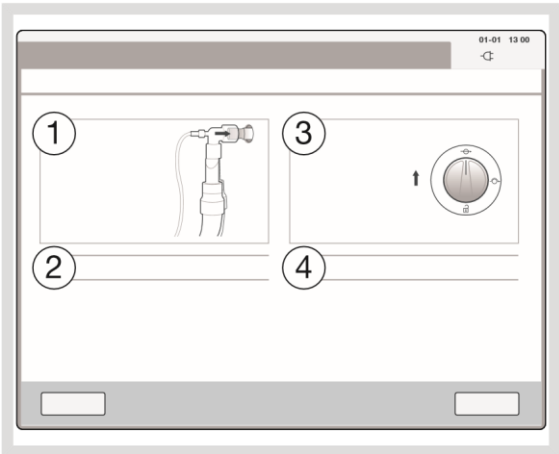
If the system is still warm, full accuracy is obtained much more quickly.

If the System checkout is bypassed, the default system value (without patient tubing) for circuit compliance compensation will be used.

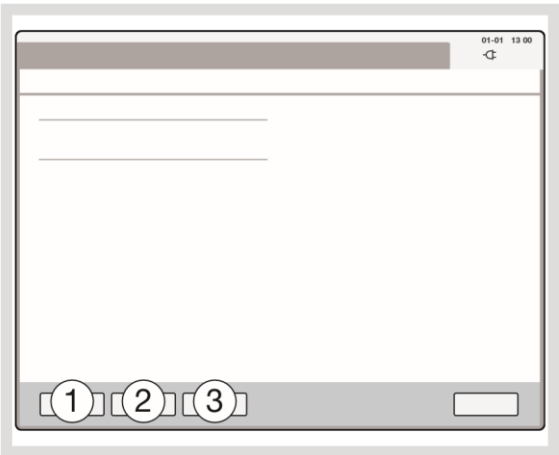
4.3 Leakage check

Performs a Leakage check of the manual and automatic breathing circuits. It shall be

The automatic check sequence includes automatic and manual breathing circuit leakage checks.



performed after each change of tubings, breathing bags or filters.



A summary of the results is displayed after the automatic check. Press the Standby touch pad to enter Standby mode or press any of the touch pads marked 1-3 to repeat any of the checks.

The circuit compliance compensation is re-calculated during the Leakage check and the log is updated with the new measured values for leakage.

Vaporizer check preparatory phase contains six steps, equivalent to steps 1 to 6 performed during the Full system check, see section 4.2.1.

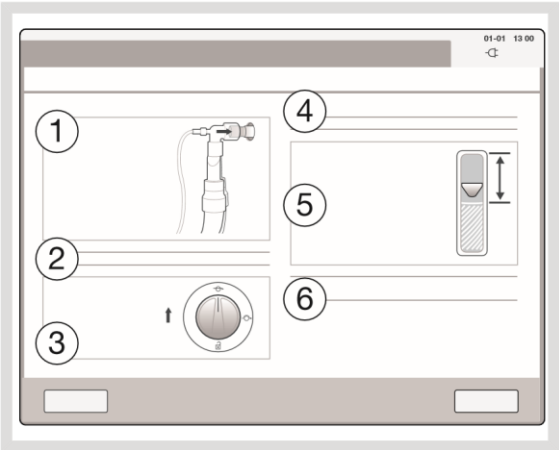
Leakage check preparatory phase contains four steps, equivalent to steps 1, 2, 3 and 6 performed during the Full system check, see section 4.2.1.

If any of the connected vaporizers contain <5% of the total agent volume, the automatic vaporizer test will not be initiated for that vaporizer. A prompt window appears. Pressing 'bypass test' will exclude that vaporizer from the check. Pressing 'redo test' will repeat the test.

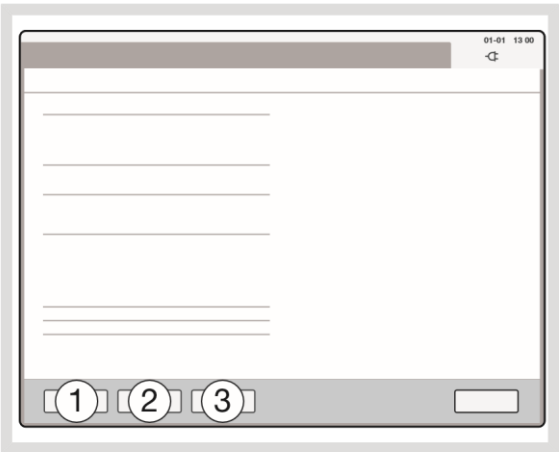
4.4 Vaporizer check

Performs a leakage check of the manual and automatic breathing circuits and a check of vaporizer functionality. It shall be performed after a vaporizer has been connected to the system.

The first part is identical to the Leakage check and is made to ensure that no agents leak out into the operating room during Run mode.



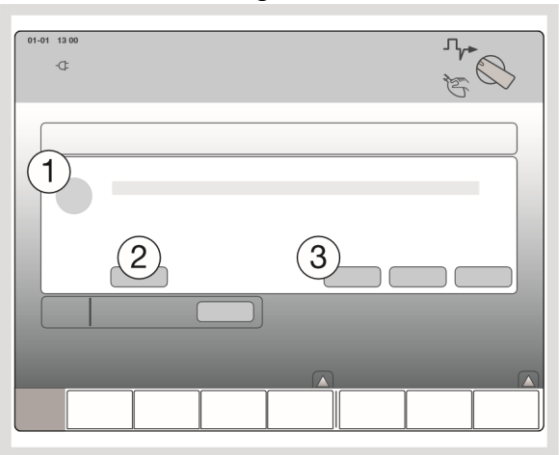
When the vaporizer is connected and active, the Agent concentration touch pad displays the name of the agent, and the agent concentration is by default set to OFF.



The automatic check sequence includes vaporizer checks in addition to the automatic and manual breathing circuit leakage checks.

A summary of the Leakage check and the Vaporizer check results is displayed after the automatic check. Press the Standby touch pad to enter Standby mode or press any of the touch pads marked 1-3 to repeat any of the checks.

4.5 Understanding results



In Standby mode, the central area of the screen displays the result of performed

system checks, along with date, time and outcome. The symbol (1) can be either green, yellow or red, depending on the current status of system checkouts. A detailed view is obtained by pressing 'View results' (2). A full system checkout, leakage check or vaporizer check is initiated by pressing the corresponding touch pad (3).

Symbol color codes correspond to the following:

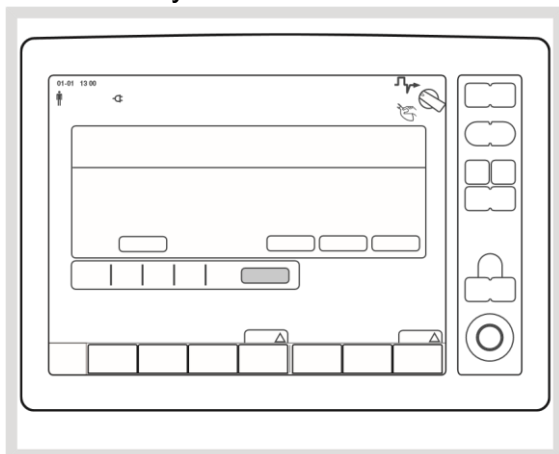
Patient cases should not be started if only

Green symbol	Yellow symbol	Red symbol
Passed. A patient case can be started.	More than 24 hours since a successful full system checkout was performed. Perform a System checkout. Not performed. A patient case can be started, but this is not recommended. Perform a System checkout.	Failed. One or several checks failed during the System checkout/Leakage check/Vaporizer check. Make appropriate adjustments and redo check.

Leakage or Vaporizer checks have been performed since system startup or during the last 24 hours period.

Press the 'Results' touch pad to display a window describing the system checkout results in more detail.

4.6 Standby mode



If the System checkout was bypassed, or if the System checkout failed, a corresponding system message will be displayed in the message area.

When a patient case is started using AUTO ventilation or a switch is made during Run mode to AUTO ventilation, the preset gas and ventilation settings are activated.

The Standby screen provides a summary of the System status, the current gas/ventilation settings and the selected ventilation mode. If a vaporizer is connected and selected, the name of the agent is presented on the screen.

Listed below are examples of functions that can be performed in Standby mode either by pressing the active touch pads, or by accessing the menu using the Menu button.

- The gas and ventilation settings can be preset before starting a patient case.
- Define patient settings.
- Any or all system checks can be started; System checkout, Leakage test and Vaporizer test.
- The Service and settings menu can be accessed.
- A patient case can be started.
- Alarm limits can be preset before starting a patient case.
- Test results of the latest performed checks can be reviewed by pressing the results touch pad.

4.7 Start case

Once a successful system checkout has been performed, the system is ready to start a patient case. Customize the case by defining patient category, ventilation mode, ventilation mode settings, and alarm limits. Press the 'start case' membrane button to start the case. See Chapter 9, page 157, for more information.

Patient category selection is described in Chapter 5, page 75.

Ventilation mode selection and parameter settings are described in Chapter 5.

Alarm limit settings are described in Chapter 9, page 187.

WARNING! Ensure that all alarm limits and ventilator settings are appropriately set before connecting the system to the patient.

4.7.1 Compliance compensation

Part of the volume of each inspiration will not reach the patient because of gas compression in the anesthesia system and

expansion of the tubing. All components in the patient circuit affect such losses.

When compliance compensation is calculated, the delivered and measured volume and flow values are automatically compensated for these losses. The leakage test must be passed in order for the system to calculate correct compliance compensation. If the leakage test is not passed, the default system value (without patient tubing) for circuit compliance compensation will be used.

The compliance compensation calculations are made based on the current set-up of the system, and are subject to changes to the breathing system volume.

These calculations become obsolete when, for example, patient tubings are changed (e.g. from adult to infant), or if additional equipment is installed which increases the breathing system volume.

Update the system by performing a new leakage check to ensure correct delivery of gas flow.

4.8 Emergency ventilation If the emergency ventilation is activated, either during a patient case or when testing the

4.8.1 Manual check of Emergency ventilation system

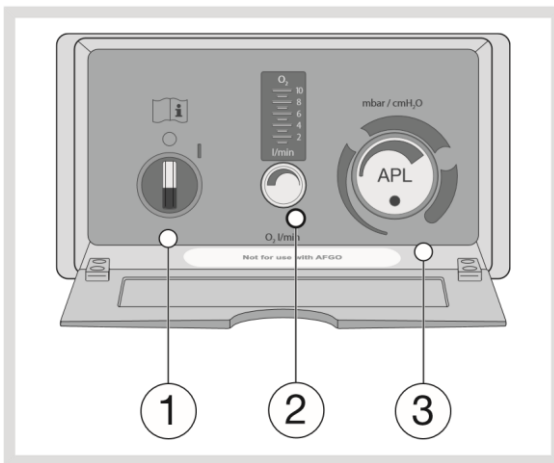
system, the emergency ventilation APL valve must be set to minimum before resuming normal ventilation (AUTO or MAN) or starting

To test its operation, perform the following: a new patient case.

1. Activate the Emergency ventilation system using the activation switch for emergency ventilation.
2. Set the O₂ flow to 10 l/min - the current flow is indicated on the flowmeter when the center of the ball is aligned with the printed scale.
3. Check at various APL settings that the test lung can be ventilated using the Manual breathing bag.

CAUTIONS:

- The emergency flow is always delivered via the patient cassette outlet and thus the patient tubings must always be connected to patient cassette outlets when emergency ventilation is used.
- Staff working with the system should practise regularly using the emergency ventilation system.



5 System functionality Table of contents

5.1	FLOW-i Anesthesia system	66
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5.1 FLOW-i Anesthesia system

The FLOW-i is a software controlled circle system for anesthesia. It features several components distinguishing it from other anesthesia systems; gas modules regulating fresh gas flow, the volume reflector and the injector based vaporizer.

The volume reflector replaces the 'bag in bottle' used in many traditional systems. It has a fixed volume of 1.2 liters and no moving parts. Gas flow is regulated by the reflector gas module. This facilitates precise control of gas flow and pressure during automatic ventilation.

The Vaporizer works with an electronic based injector technology capable of operating at low tidal volumes. This allows for fresh gas to be led directly through the vaporizer chamber and be mixed with the anesthetic agent. Changes made in agent concentrations will therefore be instantaneous. The vaporizer technology

also enables high agent concentrations at high fresh gas flows, i.e. up to 20 l/min.

5.1.1 Tidal volume delivery

The tidal volume delivered to the patient comes from two sources: the fresh gas flow and exhaled gas in the volume reflector pushed back towards the patient by the reflector gas module. Refer to page 72 for a schematic view.

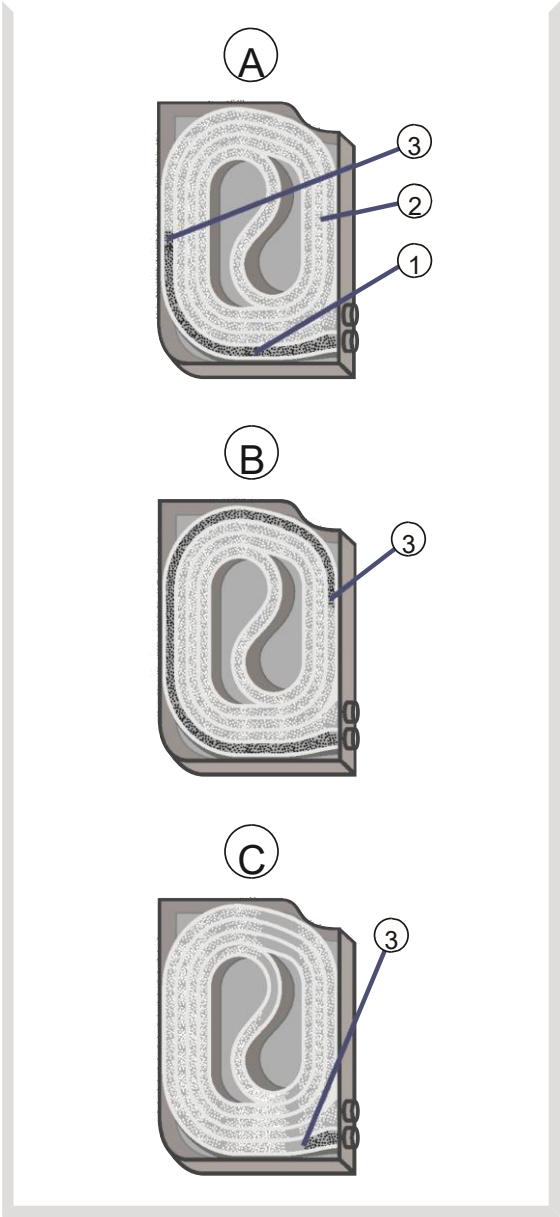
When the part of the tidal volume delivered from the volume reflector exceeds one litre, gas from the reflector gas module may pass through the volume reflector and add oxygen to the breathing system. This will trigger the 'Leakage' alarm.

In these situations, the oxygen concentration will increase and the agent concentration will decrease. However, ventilation will always be maintained.

The situation can be corrected by increasing the fresh gas flow.

Carefully monitor displayed gas measurements when using low fresh gas flow settings and high tidal volumes.

5.1.2 Low flow anesthesia



The use of oxygen as the drive gas in the volume reflector diminishes the risk for hypoxia should a leakage occur during low, or minimal flow anesthesia. However, should a leakage occur (the source of which is either known or unknown), then the volume reflector always ensures ventilation (it will never collapse like a bellows).

A leakage moves the borderline between exhaled gas and oxygen towards the breathing circuit, see illustration below. During

5.1.3 Safety functions in FLOW-i to avoid potential hypoxia

To prevent the oxygen concentration in the breathing circuit from becoming too low and potentially causing hypoxia, the system has two features; O₂Guard and the Safety Flush.

a leakage the volume reflector adds oxygen to the breathing circuit. The leakage is obvious to the anesthetist as there is a simultaneous slow increase of oxygen concentration and a slow decrease of agent concentration at the y-piece. A leakage alarm will also be activated.

The illustration shows the relative positioning of oxygen to exhaled gas during three different situations; Inhalation (A), Exhalation (B) and Leakage (C).

1. Exhaled gas
2. Oxygen from the reflector module
3. Oxygen/exhaled gas border

O₂Guard

During a patient case, exhaled gas from the volume reflector and fresh gas from the system are mixed in the breathing circuit. This results in an oxygen concentration in the breathing circuit that is lower than the set oxygen concentration.

During low flow anesthesia the oxygen concentration in the total inspiratory flow may

decrease over time. This may result in a situation whereby the inspiratory oxygen concentration becomes so low that the patient experiences hypoxia.

If the system detects an inspiratory oxygen concentration below a certain level (see image below), it will initiate several actions to mitigate the situation. The gas mix is set to Air/O₂, the fresh gas flow and oxygen concentration are both adjusted. A dialog window appears informing of the alterations. These changes remain in effect until new settings are made.

The trigger level is always lower than the relevant alarm, 'FiO₂: low, thus avoiding interference.

Safety flush

If the inspiratory oxygen concentration falls even lower, possibly due to gas delivery malfunction, then the system performs a safety flush replacing the entire breathing circuit volume with gas containing a high oxygen concentration. In addition, the fresh gas flow and O₂ concentration are increased to predefined values. These actions are accompanied by a dialog window informing of the changes.

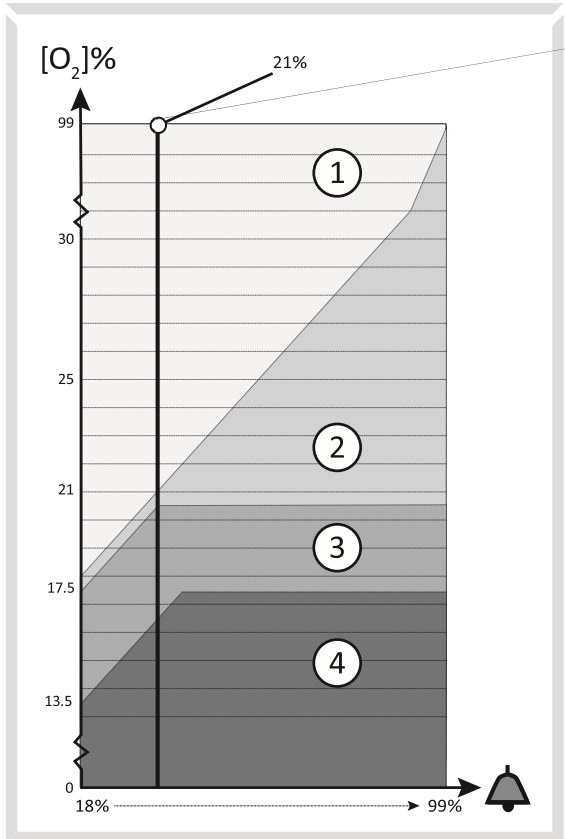
Inspiratory oxygen concentration and alarm limits

The illustration opposite shows how the settings for the 'FiO₂: Low' alarm affect the activation thresholds for O₂Guard and the Safety Flush. At lower alarm settings, the respective thresholds for O₂Guard and Safety Flush decrease accordingly to avoid interference between the functions. At higher alarm settings, these thresholds remain at 20.5% and 17.5% for O₂Guard and the Safety Flush respectively.

Each shaded area corresponds to a system action that is dependant on the current inspiratory oxygen concentration (Y-axis), and current alarm limit settings (X-axis). A vertical line can be drawn from the X-axis (21% in the illustration), representing the current alarm level setting. In this case, as long as the inspiratory oxygen level remains above 21%, no O₂-alarm is triggered.

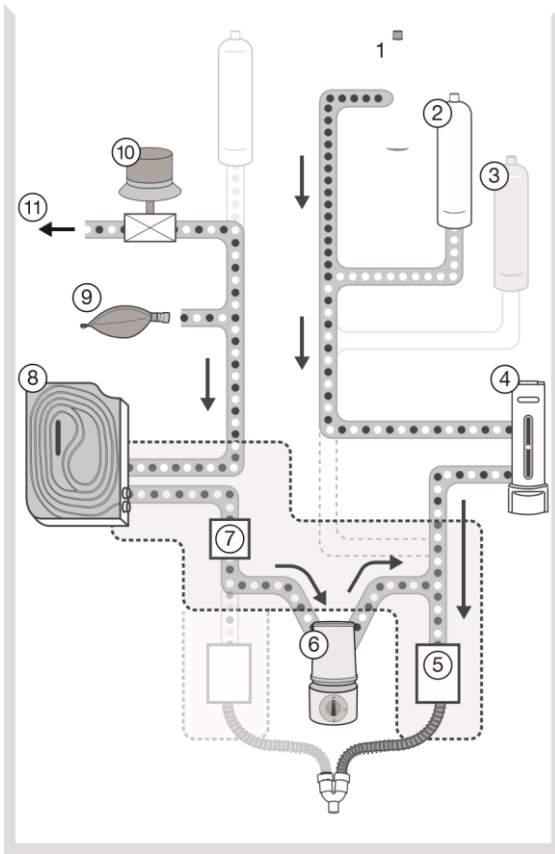
However, if at an alarm setting of 21%, the inspiratory oxygen falls below 21%, the 'FiO₂: Low'-alarm is triggered. If the inspiratory oxygen decreases even further, then O₂Guard is activated at 20.5%, and the Safety Flush at 16.5% respectively.

The O₂Guard and Safety Flush are only activated after the trigger requirements have been met for more than 20 consecutive seconds.



Inspiratory oxygen concentration is listed on the Y-axis, and inspiratory oxygen alarm level on the X-axis.

1. Normal operation (no alarm)
2. Activation of the low inspiratory oxygen alarm
3. Activation of O₂Guard
4. Activation of the Safety Flush



5.1.4 Manual ventilation

The fresh gas flow is constant during both inspiration and expiration when manual breathing mode is selected.

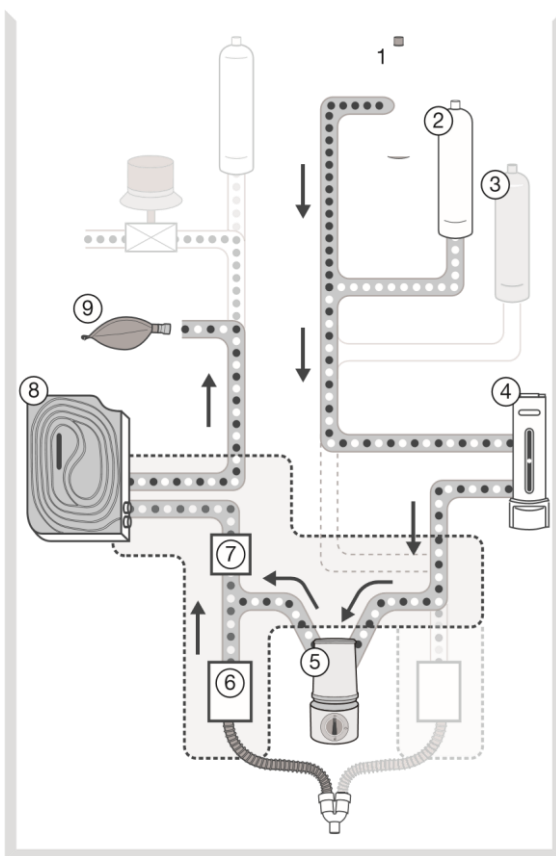
Inspiration

Expired gas is, together with new fresh gas from the gas modules, re-administered to the patient by compression of the manual breathing bag. This resulting pressure build-up also serves to evacuate gas through the APL valve when the set APL pressure is reached.

Direction of gas flow is indicated by the arrows.

1. Air gas module
2. O₂ gas module

3. N₂O gas module
4. Vaporizer
5. One-way valve
6. CO₂ absorber
7. Flow transducer
8. Volume reflector
9. Manual breathing bag
10. APL valve
11. Gas evacuation



Expiration

Exhaled gas enters the volume reflector. Fresh gas is added to the circle system from the gas modules.

Due to the counter pressure in the inspiratory channel caused by flow from the patient, fresh gas delivered during the expiration phase will fill the CO₂ absorber. It will be returned to the inspiratory channel during the next inspiration.

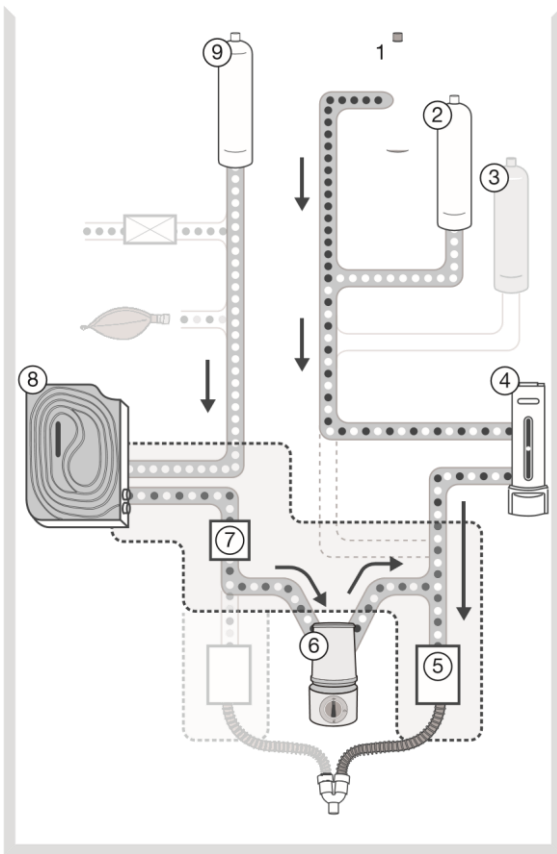
Direction of gas flow is indicated by the arrows.

1. Air gas module
2. O₂ gas module
3. N₂O gas module
4. Vaporizer
5. CO₂ absorber
6. One-way valve
7. Flow transducer
8. Volume reflector
9. Manual breathing bag

A larger proportion of the fresh gas is added during the inspiration phase, also contributing to minimizing agent consumption. The set fresh gas flow is the mean of fresh gas flow delivered during a full breathing cycle.

5.1.5 Automatic ventilation

The fresh gas delivery is designed to be cost effective with regards to agent consumption. The fresh gas flow is thus limited to the set minute volume, i.e. the delivered fresh gas can never exceed the set minute volume.



- 2. O₂ gas module
- 3. N₂O gas module
- 4. Vaporizer
- 5. One-way valve
- 6. CO₂ absorber
- 7. Flow transducer
- 8. Volume reflector
- 9. Reflector gas module

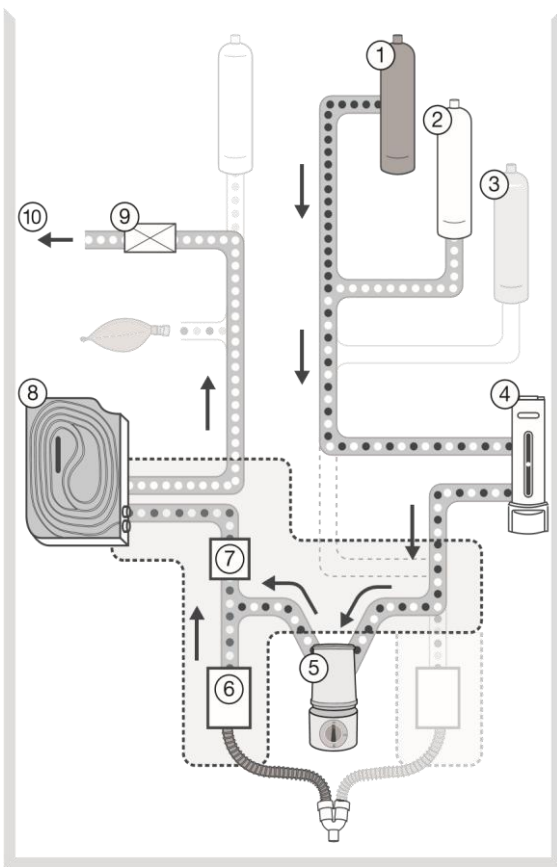
Inspiration

In automatic (controlled) ventilation, reflector gas is used to mechanically push the exhaled gas mixture in the volume reflector, back into the circle system. The mixture passes through the CO₂ absorber, washing the gas free of CO₂, before re-administration to the patient.

The gas mixture from the volume reflector, along with new fresh gas, form the patient's tidal volume. This ensures precise delivery of the tidal volume irrespective of changes in the fresh gas flow.

Direction of gas flow is indicated by the arrows.

- 1. Air gas module



Expiration

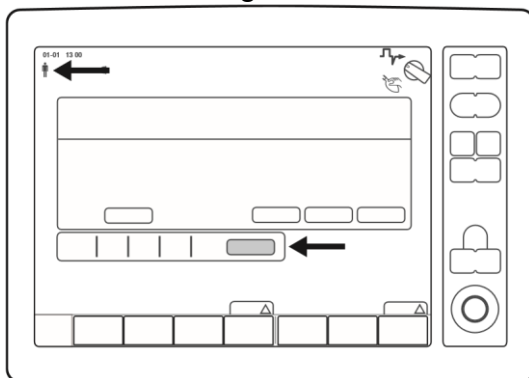
Exhaled gas enters the volume reflector. Reflector gas is pushed back and evacuated through the PEEP valve.

Due to the counter pressure in the inspiratory channel caused by flow from the patient, fresh gas delivered during the expiration phase will fill the CO₂ absorber. It will be returned to the inspiratory channel during the next inspiration.

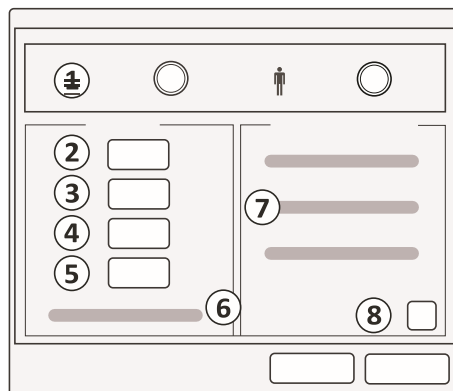
Direction of gas flow is indicated by the arrows.

1. Air gas module
2. O₂ gas module
3. N₂O gas module
4. Vaporizer
5. CO₂ absorber
6. One-way valve
7. Flow transducer
8. Volume reflector
9. PEEP valve
10. Gas evacuation

5.2 Patient settings



Press 'Patient settings' in standby, or press the patient category symbol in the upper left corner to open the patient settings menu.



1. *Category*
2. *Age*
3. *Weight*
4. *Height*
5. *Gender*
6. *Predicted body weight*
7. *Recommended values*
8. *Use recommended values*

Press each data field and use the rotary knob to set and confirm the patient data value.

The patient data (1-5) is used to determine allowable ventilatory setting ranges, relative MAC_{age} and recommended ventilatory values.

To estimate PBW, all fields need to be filled in. The PBW, together with the patient category, is then used to define the recommended values for ventilation (7) displayed on the right.

The 'Use recommended values' touch pad (8) may be used to apply these recommended values.

5.2.1 Patient category

The selected patient category is either 'Infant' or 'Adult'. This determines the allowable setting ranges for ventilatory parameters and patient data.

Recommended ventilatory parameters outside the allowed patient category range are always highlighted in yellow.

If the patient category is changed in standby, all entered patient data except age is removed.

Patient age has the default setting. Recommended ventilation parameters are also removed.

If the patient category is changed during running mode, entered patient data is generally conserved but patient details outside the allowed patient category range are automatically changed to the closest allowable value.

If patient data is changed while the system is in AUTO running mode, it will not be possible to select and use the resulting 'recommended values'.

Changing the patient category can also be done by using options available under the Menu membrane button, see Chapter 9, page 165.

Changing the patient category affects the following:

- Alarm ranges and defaults
- Circuit compliance compensation
- Parameter setting ranges
- Step values, i.e. differences in increments when increasing/decreasing parameter values
- Automatic changes in parameter settings when the current value is outside the patient category parameter range

If there is no circuit compliance compensation data available after the patient category has been changed, the default value will be used until a full check, or leakage check, has been performed.

CAUTION: Alarm limits and ventilation settings may change after having selected a different patient category. Re-check the set alarm limits after the patient category has been changed.

5.2.2 Patient Age

This field is never blank. If no information is entered, the default value will be used.

Select 'Age' in the patient settings menu and use the rotary knob to adjust the value.

5.2.6 Predicted body weight

If all patient details have been filled in, PBW will be displayed. In some cases, it will be the same as the actual body weight, while in others the PBW may be either higher or lower than actual body weight.

Category	Range	Default
Infant	1 - 15 years	1
Adult	5 - 120 years	40

Patient category age default values can be changed in Service and Settings, see Chapter 9 page 178.

Patient age is used to estimate the relative MAC for the current patient, see page 145.

5.2.3 Weight

Used for calculating PBW and recommended ventilatory values according to the description on page 74.

5.2.4 Height

Used for calculating PBW and recommended ventilatory values according to the description on page 74.

5.2.5 Gender

Used for calculating PBW and recommended ventilatory values according to the description on page 74.

The PBW estimate is based on various formulae depending on patient gender, height and weight, as shown in the tables.

Gender - male	Height < 130 cm	Height > 130 cm
Weight < 30 kg	Actual body weight used for PBW	PBW based on a combination of actual body weight and the formula used by Devine et al, 1974.
Weight > 30 kg	PBW based on a limited version of the formula used by Devine et al, 1974.	PBW estimate based on the formula used by Devine et al, 1974.

Gender - female	Height < 135 cm	Height > 135 cm
Weight < 30 kg	Actual body weight used for PBW	PBW based on a combination of actual body weight and the formula used by Devine et al, 1974.
Weight > 30 kg	PBW based on a limited version of the formula used by Devine et al, 1974.	PBW estimate based on the formula used by Devine et al, 1974.

The PBW value displayed here forms the basis for all values for ventilatory parameters recommended in this window.

The estimated PBW may be displayed alongside the patient category symbol in the top left corner depending on the setting in the Screen layout window. See section Screen layout on page 163.

When PBW is shown, the figure is used as the basis of an estimated parameter, set VT/PBW, displayed among estimated parameters in the ventilation mode window and in the additional ventilation settings for the following ventilation modes:

- Volume Control (VC)
- Pressure Regulated Volume Control (PRVC)

- SIMV (VC) + PS

It also forms the basis of a measured value, expired VT/PBW, found on the selected page of Measured values when the system is running. See section Ventilation measurements on page 104.

This measured value is also trended, see section Page information on page 161.

5.2.7 Recommended values

The values recommended here for Tidal volume, Resp. rate and Minute volume are based on the patient category selected and the patient data entered.

Values that are out of range for the patient category are highlighted in yellow.

5.2.8 Use recommended values

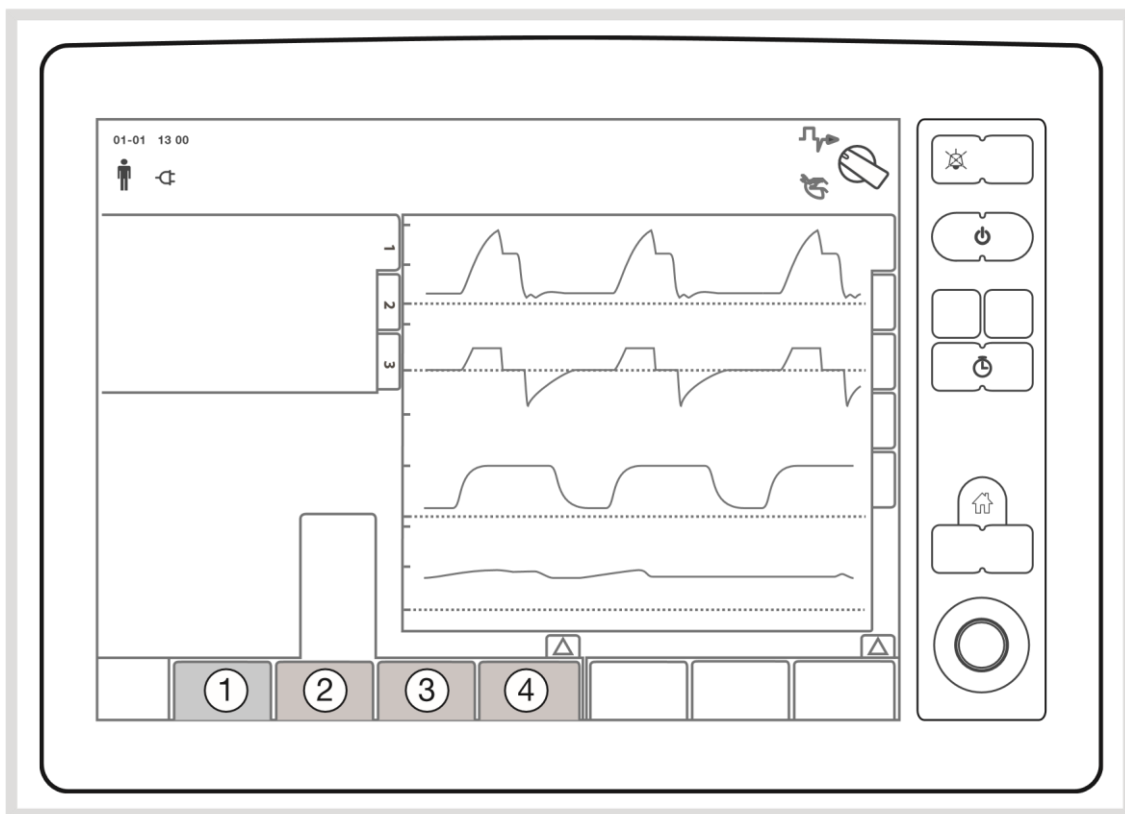
When the patient data fields have been completed, the suggested ventilatory parameters are presented.

The 'Use recommended values' touch pad may then be either automatically checked or empty depending on settings made in Service & Settings/Startup configuration/Ventilation & gas. See section Startup configuration on page 174.

If the recommended ventilatory values are satisfactory, either press 'Accept' if the touch pad is pre-checked, or check it and then press 'Accept'. The direct settings at the bottom right of the control panel are then updated accordingly.

This can be done when the system is in standby. When it is running, it can only be done if using manual ventilation (MAN) but NOT when using automatic ventilation (AUTO). See section MAN/AUTO ventilation switch on page 84. This restriction also applies if the Volume setting in Service & Settings/Startup configuration/Ventilation & gas is set to Minute Volume. See section Startup configuration on page 174.

5.3 Fresh gas flow settings

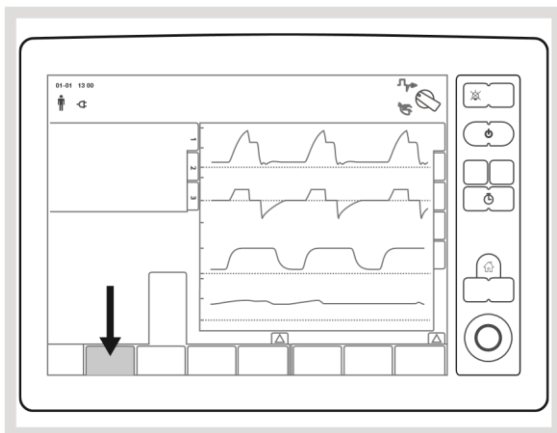


The fresh gas supply is set using the following touch buttons:

1. Gas mix
2. Fresh gas flow

3. O₂ concentration
4. Agent concentration

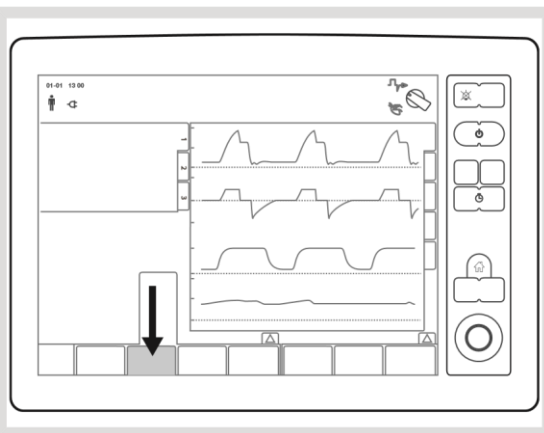
5.3.1 Selection of gas mixture



There are two settings:

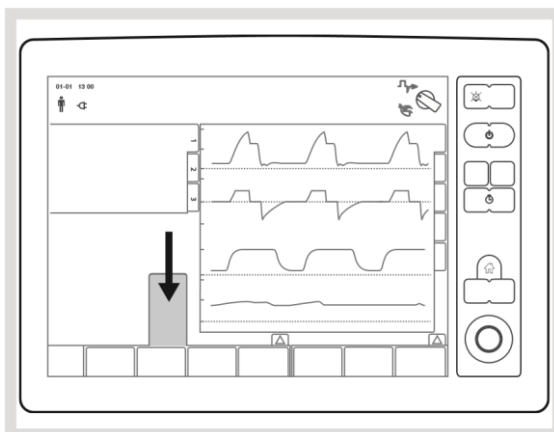
- O₂/Air (oxygen/air)

5.3.2 Setting fresh gas flow



Total fresh gas flow in liters/minute.

The minimum O₂ flow is 200 ml/min. The electronic gas mixer will automatically adjust

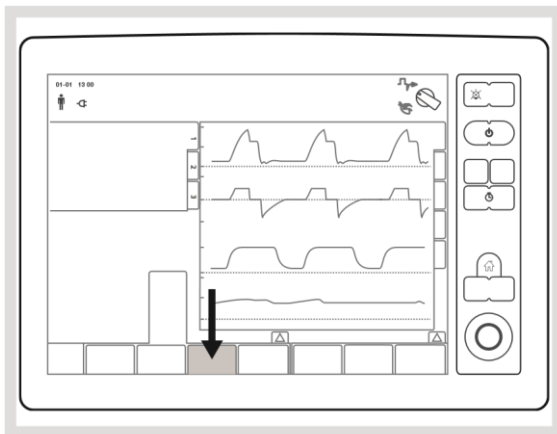


Additionally, the fresh gas flow can be shown using the rotameter, configurable via the Screen layout membrane button. It is a graphic representation of the individual flow rates for each gas.

the fresh gas flow settings to match this if necessary.

- O₂/N₂O (oxygen/nitrous oxide).

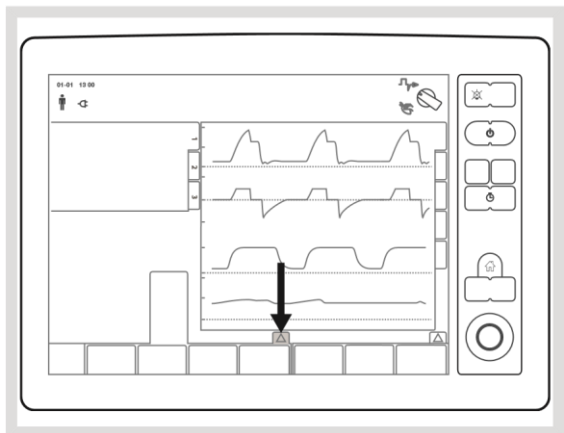
5.3.3 Setting O₂ concentration



Percentage oxygen set in the fresh gas mixture. The range of the fresh gas concentration is 28-100% for O₂/N₂O and 21-100% for O₂/Air.

The O₂ flow can never be less than 200 ml/min for any patient category. The current O₂ concentration in the gas mix is automatically adjusted to match this requirement if the fresh gas flow is decreased sufficiently. In addition, the O₂ concentration touch pad is highlighted to inform of this adjustment, and a system message is displayed for as long as the limitation is active: 'O₂ lower limit reached'. Confirming the new fresh gas flow will also confirm the new O₂ concentration.

5.3.4 Setting anesthesia agent



concentration

Sets the anesthesia agent concentration for the selected vaporizer.

The name of the selected vaporizer agent is shown on the touch pad.

The agent concentration can have four states:

State	Description
No agent	No connected vaporizer in the selected slot
OFF	Vaporizer ready to be disconnected
0 %	Pressurized and active vaporizer
0.3 - 5 % (ISO) 0.3 - 8 % (SEV) 1.0 - 18 % (DES)	Selected vaporizer/ agent concentration range

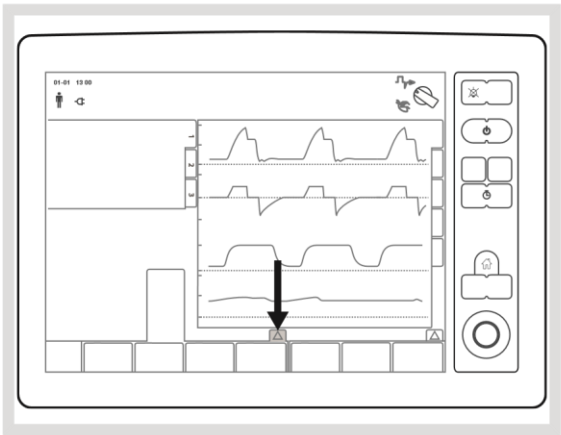
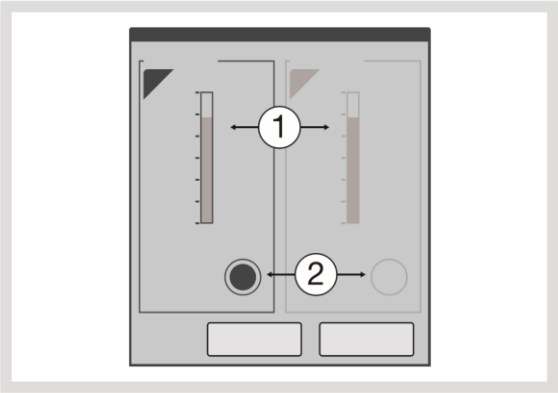
Agent concentration values on the screen are grayed out when the vaporizer lid is open.

5.3.5 Selecting active vaporizer

Selecting an active vaporizer is necessary if:

- The vaporizer is connected into the non-selected slot.
- There is a need to change the active vaporizer when there are two vaporizers connected to the system.

1. Press the vaporizer touch button on the screen.



2. The Select vaporizer window appears. It displays the connected vaporizer's current liquid level (1) and vaporizer type (2).

3. To select a vaporizer, press the required vaporizer type and confirm by pressing Accept. This activates the vaporizer. The agent name is shown on the Vaporizer setting touch pad. affecting parameter setting limits and step values.

If the new vaporizer contains another agent, the suggested agent concentration displayed on the control panel is based on the equivalent MAC value of the previous agent.

When a second vaporizer is activated, the other vaporizer is automatically de-activated; the agent concentration is set to OFF, the vaporizer is depressurized, flushed and unlocked.

It is possible to manually set the agent concentration to OFF, and the vaporizer to standby, using the Agent concentration touch pad, see page 141 for details.

5.4 Ventilation settings

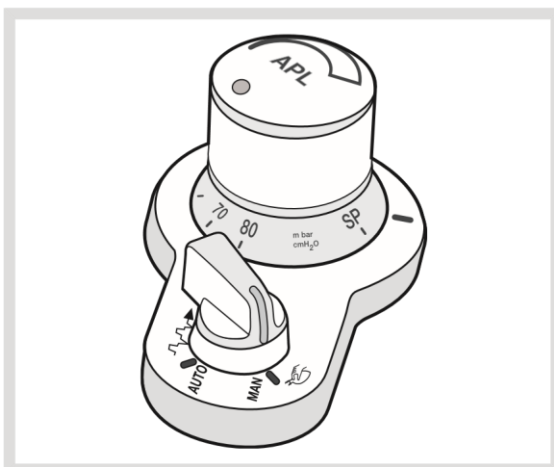
The system can be run using manual (MAN) or automatic (AUTO) ventilation. Ventilation modes available for automatic ventilation include the following:

- Pressure Control (PC)
- Volume Control (VC)
- Pressure Support (PS)

Manual ventilation modes include manual ventilation using the circle system and manual ventilation using an external breathing system connected to the AFGO outlet (option).

The system also allows for the patient category (Infant or Adult) to be specified,

5.4.1 MAN/AUTO ventilation switch



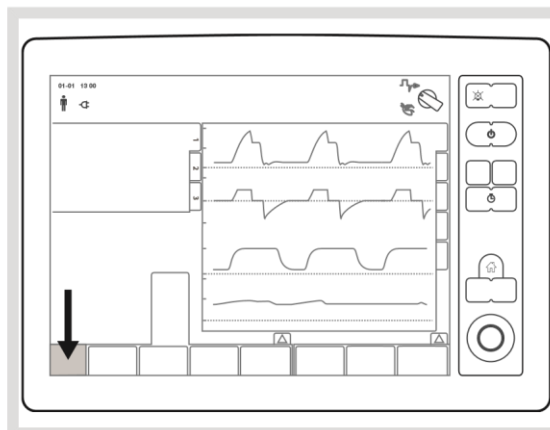
Sets the status of the ventilator:

- Manual ventilation (MAN)
- Automatic ventilation (AUTO)

Using manual ventilation

Manual ventilation is performed using the manual breathing bag to administer breaths, and the APL valve to regulate the pressure limit. For details on the APL valve, see page 122.

The set value of the APL valve is displayed at the bottom-left of the screen:



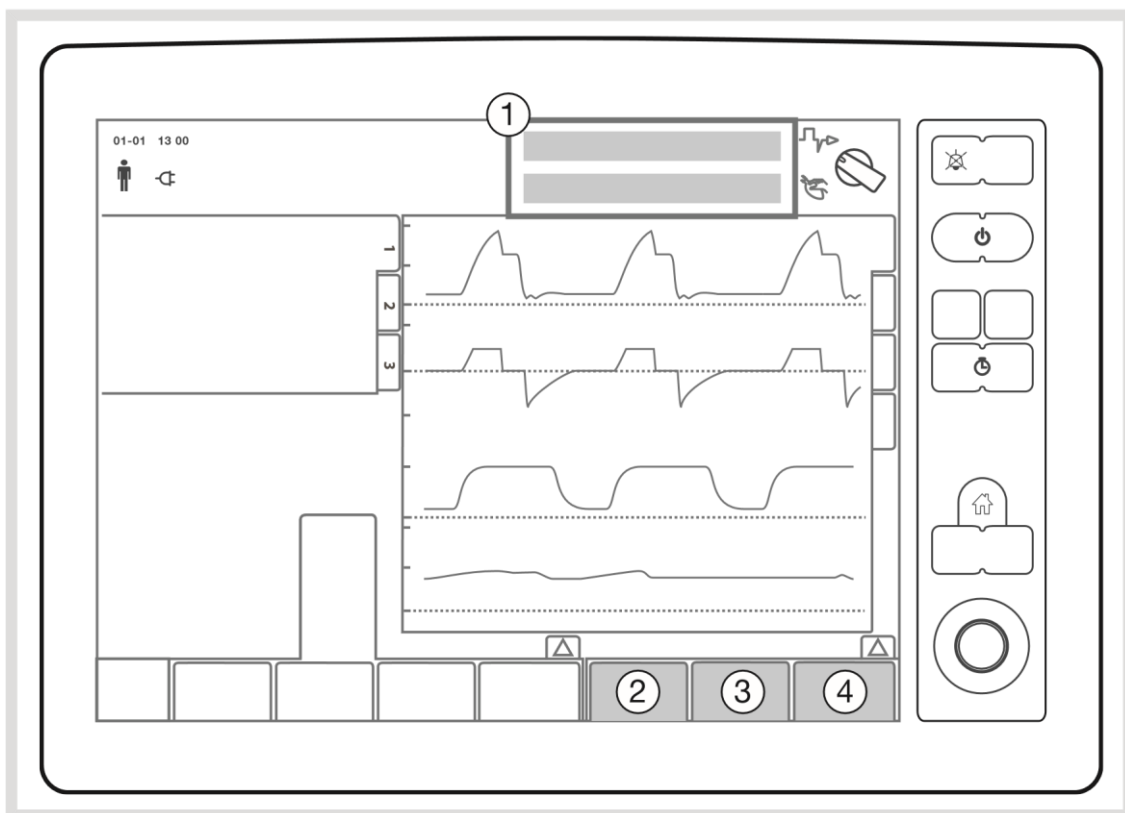
Using automatic ventilation

Automatic ventilation is performed using the built-in ventilator, where the ventilation mode initially selected will depend on the system's startup configuration, see Chapter 9.

For a detailed description of each automatic ventilation mode, see Chapter 11.

The required ventilation settings can be pre-set before switching to automatic ventilation. The ability to pre-set these values is also true when switching back to manual ventilation.

5.4.2 Parameter settings



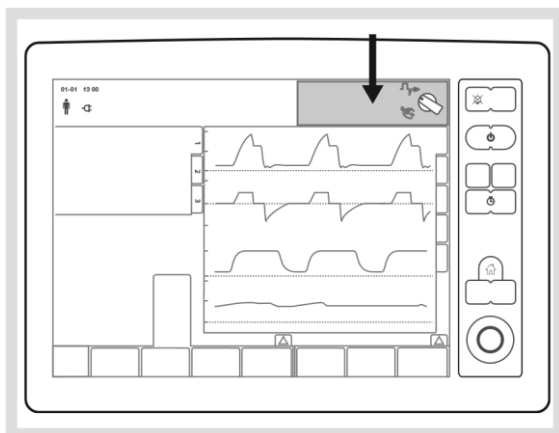
The ventilation parameter settings are controlled using the following touch pads on the screen:

1. Ventilation mode selection
2. PEEP
3. Respiratory rate
4. PC above PEEP/Tidal volume

PC above PEEP will be replaced with Tidal/Minute volume in the Volume Control ventilation mode.

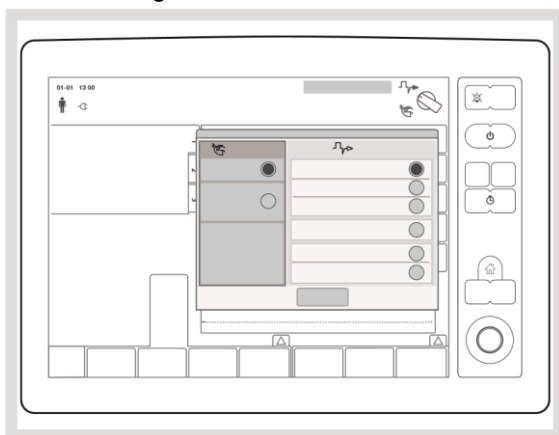
Due to interdependencies between parameter settings, certain setting combinations are not possible, particularly settings relating to breathing cycle phases (T_i , T_{insp} rise, T_{pause} and RR) in combination with Tidal volume (VC) and Pressure over PEEP (PC, PS). These parameters cannot be set so that the subsequent gas flow exceeds the system's performance, e.g. high tidal volumes with a short T_i .

5.4.3 Selection of automatic ventilation mode



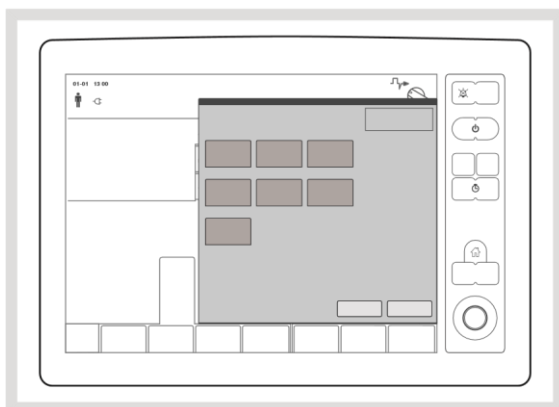
If AFGO is installed on the system then manual and AFGO ventilation appear as selectable options to the left of this window. These options can be preset in preparation for switching to manual breathing mode using the MAN/AUTO switch, see Chapter 7, page 151.

Pressing the ventilation mode text (e.g. Volume control) on the control panel, displays the following window:



Available ventilation modes appear and can be selected. Manual ventilation modes are grayed out.

When selecting the required ventilation mode, are always the set ventilation mode parameter window the control appears:

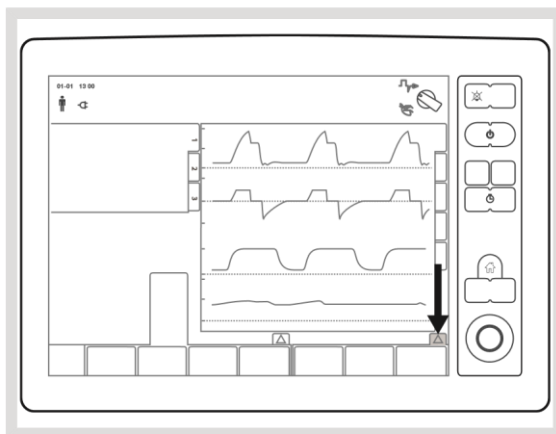


Adjust the ventilation settings as desired and press 'Accept'. This will activate (or preset if current mode is AFGO or MAN) the chosen automatic ventilation mode.

The following pages describe each ventilation mode parameter window individually.

Direct access ventilation settings accessible at the bottom right of panel.

Additional settings can be viewed anytime by pressing the triangular 'Additional settings' touchpad located above the direct settings area. The additional settings window always corresponds to the current ventilation mode.

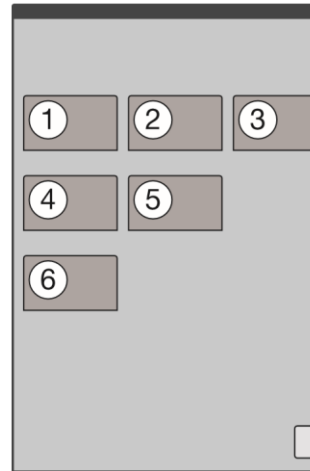


Refer to page 95 for a description of direct access settings.

Refer to page 99 for a description of additional settings.

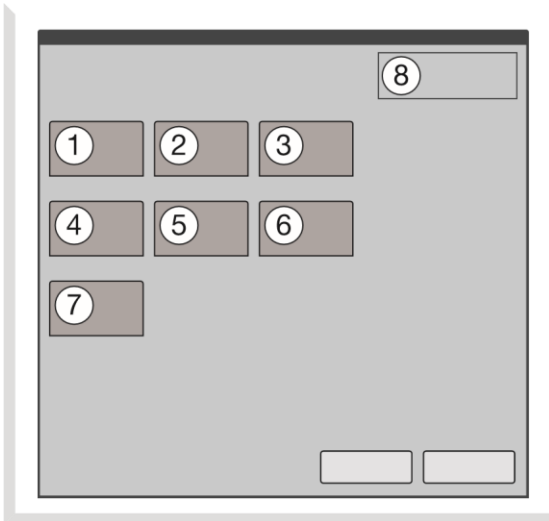
Ventilation mode parameters

Pressure Control

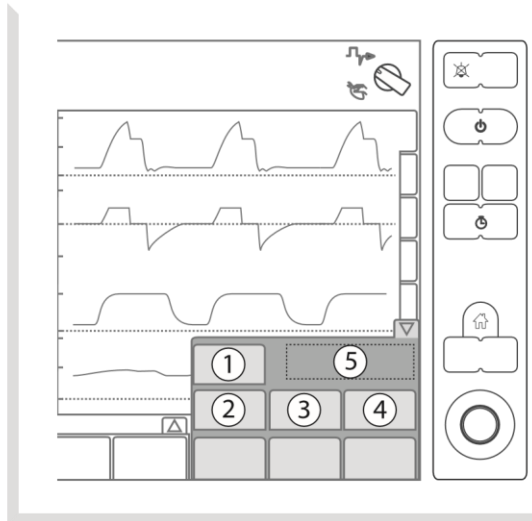


1. PEEP
2. Respiratory Rate
3. PC above PEEP
4. I:E or Ti (inspiration time)
5. T insp. rise
6. Trigger
7. Estimated parameters
1. I:E or Ti (inspiration time)
2. T insp. rise
3. Trigger
4. Estimated parameters

Ventilation mode parameter window Volume Control

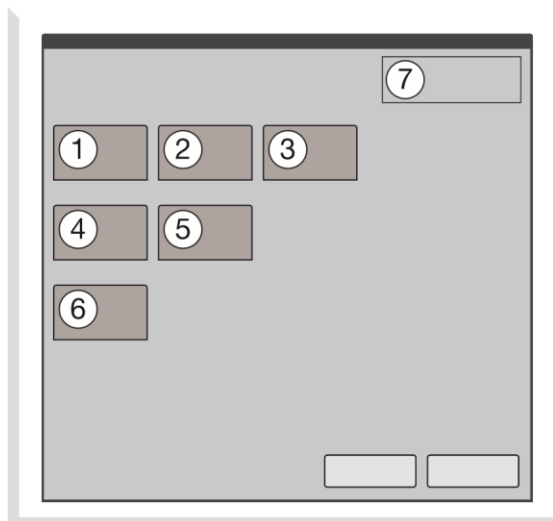


Additional settings window

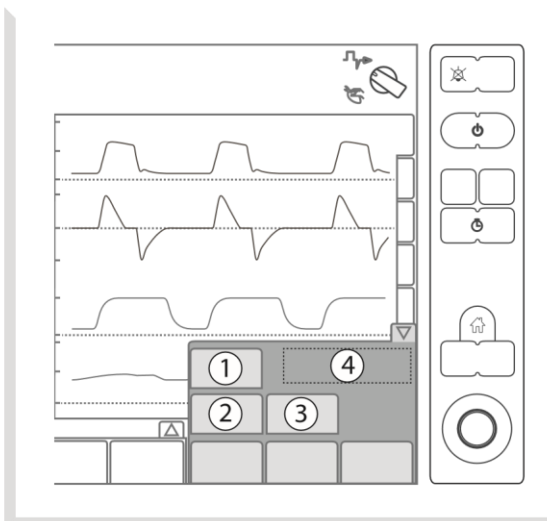


1. PEEP
 2. Respiratory Rate
 3. Tidal/Minute volume
 4. I:E or Ti (inspiration time)
 5. T pause
 6. T insp. rise
 7. Trigger
 8. Estimated parameters
1. Trigger
 2. I:E or Ti (inspiration time)
 3. T pause
 4. T insp. rise
 5. Estimated parameters

Ventilation mode parameter window PRVC



Additional settings window

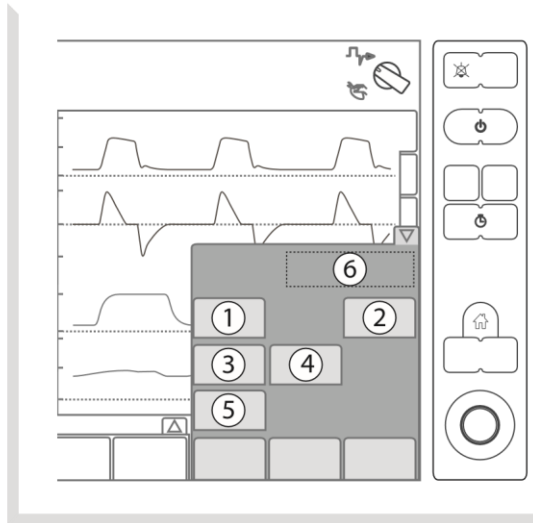
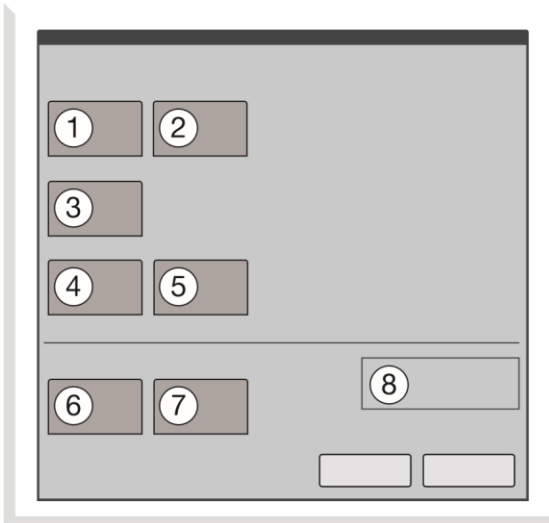


1. PEEP
 2. Respiratory Rate
 3. Tidal/Minute volume
 4. I:E or Ti (inspiration time)
 5. T insp. rise
 6. Trigger
 7. Estimated parameters
-
1. Trigger
 2. I:E or Ti (inspiration time)
 3. T insp. rise
 4. Estimated parameters

Ventilation mode parameter window

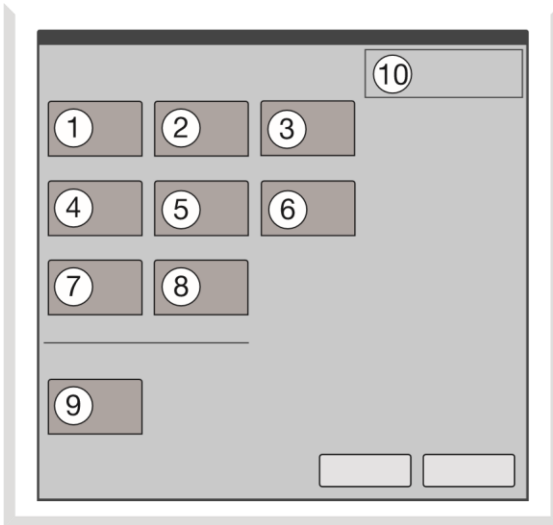
Pressure Support

Additional settings window

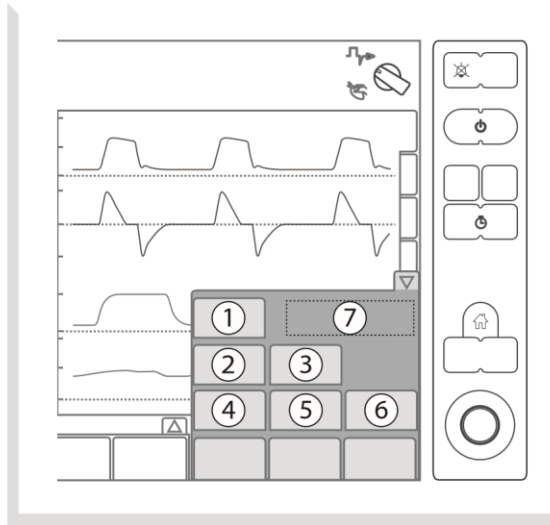


1. PEEP
2. PS above PEEP
3. T insp. rise
4. Trigger
5. Insp. cycle off
6. Estimated parameters
1. PS above PEEP
2. PC above PEEP
3. Trigger
4. Insp. cycle off
5. T insp. rise
6. Estimated parameters

Ventilation mode parameter window **SIMV (PC) + PS**

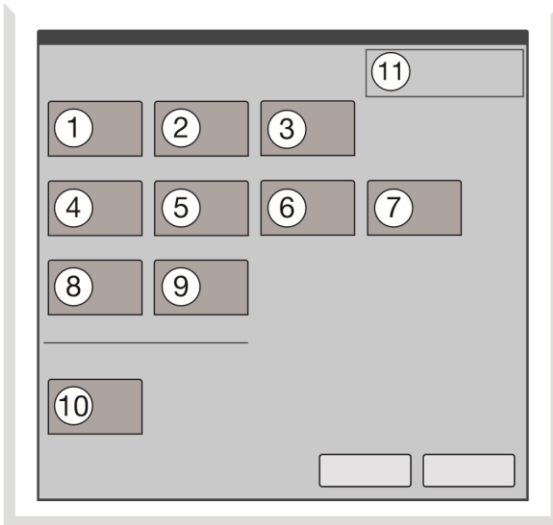


Additional settings window



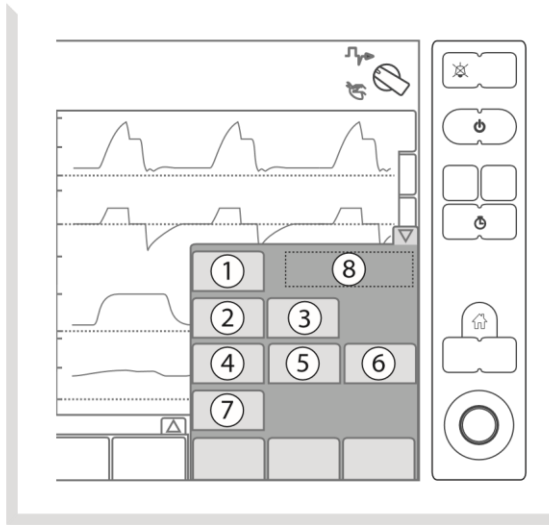
1. PEEP
 2. SIMV rate
 3. PC above PEEP
 4. I:E or Ti (inspiration time)
 5. T insp. rise
 6. Breath cycle time
 7. Trigger
 8. Insp. cycle off
 9. PS above PEEP
 10. Estimated parameters
-
1. PS above PEEP
 2. Trigger
 3. Insp. cycle off
 4. I:E or Ti (inspiration time)
 5. T insp. rise
 6. Breath cycle time
 7. Estimated parameters

Ventilation mode parameter window **SIMV (VC) + PS**



1. PEEP
2. SIMV rate
3. Tidal/Minute volume
4. I:E or Ti (inspiration time)
5. T pause
6. T insp. rise
7. Breath cycle time
8. Trigger
9. Insp. cycle off
10. PS above PEEP
11. Estimated parameters

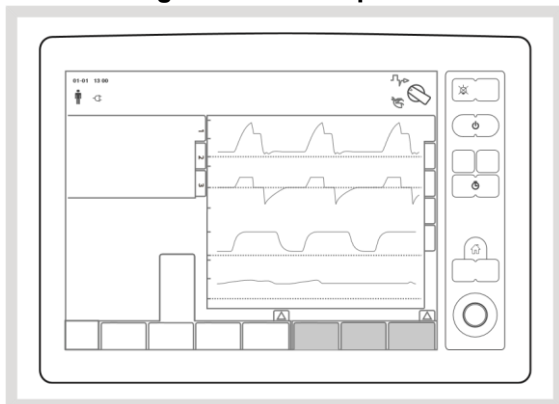
Additional settings window



7. Breath cycle time
8. Estimated parameters

1. PS above PEEP
2. Trigger
3. Insp. cycle off
4. I:E or Ti (inspiration time)
5. T pause
6. T insp. rise

5.4.4 Setting direct access parameters

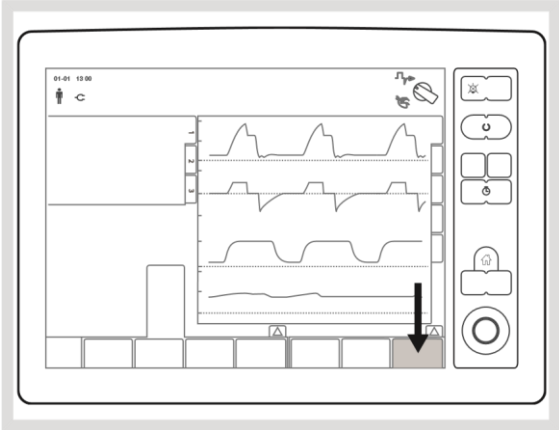


The following direct access parameters can be set for the selected ventilation mode:

Ventilation mode	Tidal Volume	PC above PEEP	PS above PEEP	Respiratory Rate	Backup RR	SIMV rate	PEEP
PC		X		X			X
VC	X			X			X
PRVC	X			X			X
PS		X ²	X ²		X		X
SIMV (PC) + PS		X				X	X
SIMV (VC) + PS	X					X	X

2. In PS backup ventilation, the direct access parameter toggles between 'PS above PEEP' and 'PC above PEEP' depending on if pressure support or backup ventilation is the current active mode.

Tidal/Minute volume



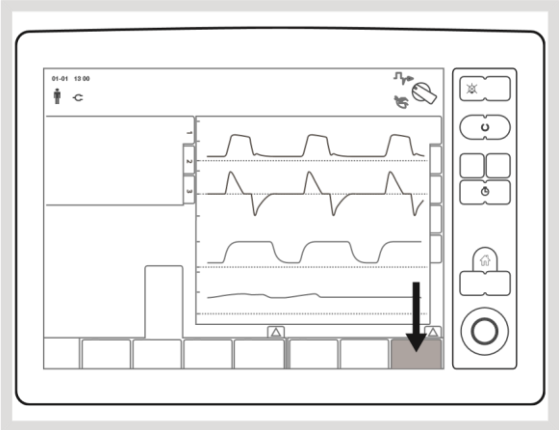
Sets the tidal/minute volume.

Different setting intervals apply for adult and reached, the system generates a pressure infant patient categories, see table below. plateau until the start of the expiration phase.

Patient Category	Tidal Volume	Minute Volume
Adult	100 - 2000 ml	0.5 - 60 l/min
Infant	20 - 350 ml	0.3 - 20 l/min

Either tidal (MTi) or minute (MV) volume can be used and is selected as part of the system configuration, see Chapter 9.

Pressure control above PEEP



Sets the pressure level above PEEP. When the set inspiration pressure above PEEP is reached, the system generates a pressure infant plateau until the start of the expiration phase.

Different setting intervals apply for adult and infant patient categories, see table below.

Patient category	PC above PEEP
Adult	0 - 120 cmH ₂ O

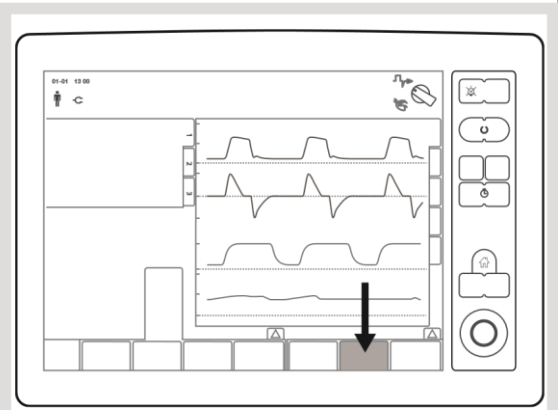
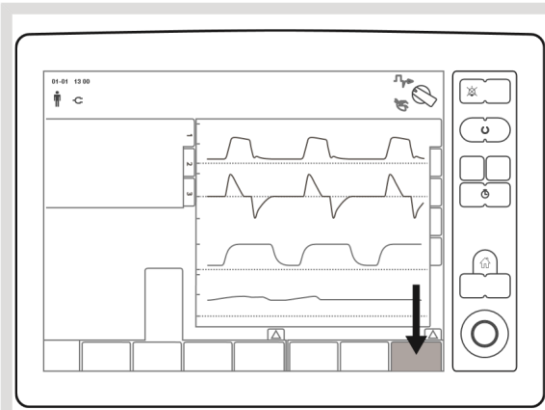
Infant 0 - 80 cmH₂O

Pressure support above PEEP

PS above PEEP is the set inspiratory pressure support level for triggered

Sets the frequency of administered breaths.

The system allows a breathing frequency within the range of 4 to 100

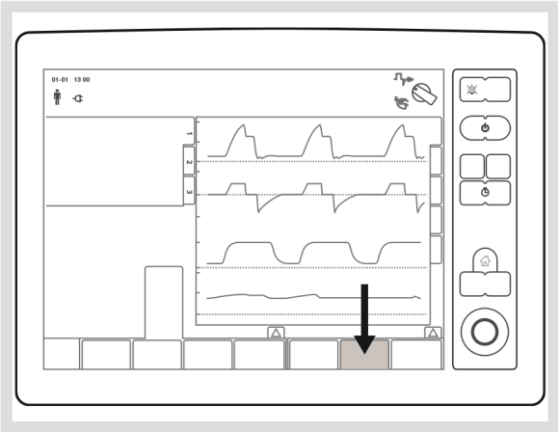


breaths in pressure support. Same restrictions apply as for 'PC above PEEP'.

breaths/minute.

Backup respiratory rate

Respiratory rate

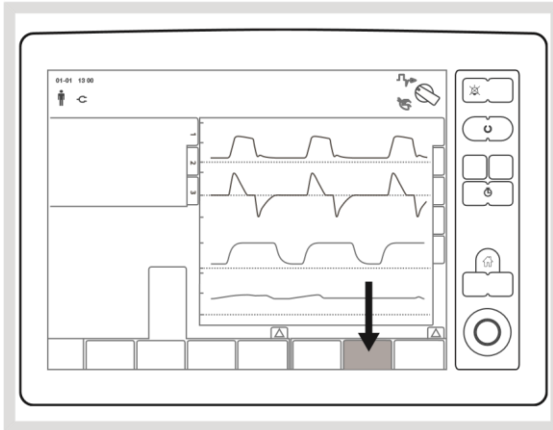


Sets the frequency of pressure controlled administered breaths, i.e. backup rate, should the patient not initiate a breathing attempt. The time-out period, after which a breath is administered, is 60 seconds divided by 'backup respiratory rate'.

The system allows a backup breathing frequency within the range of 2 to 60 breaths/minute and OFF.

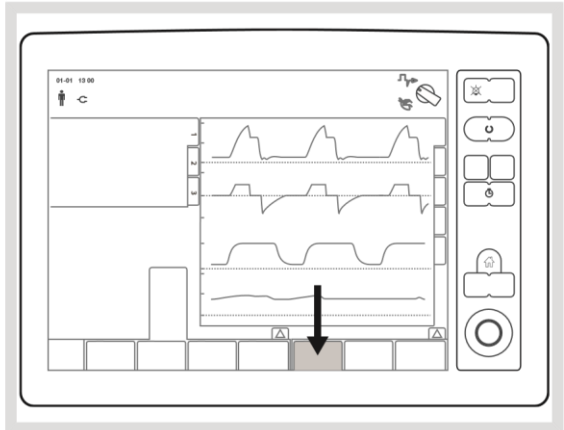
Setting the backup respiratory rate to OFF will deactivate this parameter when running PS backup ventilation.

SIMV rate



Rate of controlled mandatory breaths (b/min).
Active only in SIMV modes.

PEEP



Sets the Positive End Expiratory Pressure.

The system allows PEEP settings within the range 0 to 50 cmH₂O.

5.4.5 Setting additional parameters

The following additional parameters can be set for the selected ventilation mode:

Ventilation mode	Trigger	I:E/Ti	T pause	T insp. rise	PC above PEEP	PS above PEEP	Insp. cycle off	Breath cycle time
PC	X	X		X				
VC	X	X	X	X				
PRVC	X	X		X				
PS	X			X	X		X	
SIMV (PC) + PS	X	X		X		X	X	X
SIMV (VC) + PS	X	X	X	X		X	X	X

Setting inspiration time

The following parameters can be set to control inspiration time:

- I:E/Ti
- T pause
- T insp. rise

I:E/Ti

Sets the ratio between the inspiratory and expiratory time for each breath.

Either I:E or Ti can be used and is selected as part of the system configuration. See Chapter 9, page 178, for more information.

If I:E is set to 1:4, the fresh gas flow is limited to the highest allowed flow for this setting, and may result in the delivered fresh gas flow being less than the set fresh gas flow.

T pause

Sets the inspiratory pause time as a percentage of the breath cycle time. During the pause, the inspiratory flow is closed.

The range is 0 to 30%.

T insp. rise

Sets the time to full inspiratory flow or pressure at the start of each breath, as a percentage of the breath cycle time.

The range is 0 to 20%.

Insp. cycle off

Inspiratory Cycle-off is the point at which inspiration changes to expiration in

spontaneous and supported modes of ventilation. A decrease of the inspiratory flow to a preset level causes the system to switch to expiration. This preset level is measured as a percentage of the maximum flow during inspiration.

The range is 1 - 70%.

Breath cycle time

This is the length of the breath i.e. the total cycle time of the mandatory breath in SIMV (inspiration, pause plus expiration). This is set in seconds within the range:

- Infants: 0.5 -15 seconds in half second steps.
- Adults: 1-15 seconds in one second steps.

See Chapter 11, page 227, for more background on breath cycle time.

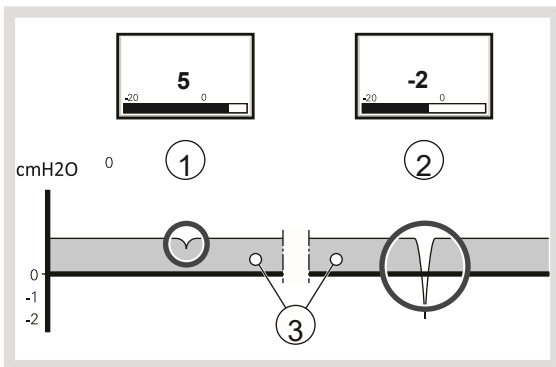
Setting trigger

The following parameter can be set to control the trigger.

through the PEEP valve (1 on the scale), to 10% of this flow (10 on the scale).

Trigg. Flow/Trigg. Pressure

Flow triggering or Pressure triggering sets the level of patient effort (circled area in the picture below) required to trigger inspiration. Flow triggering can be set in the range 1 to 10 (green and red area on the bar). This roughly corresponds to the effort that the patient has to make to trig a new breath. The sensitivity ranges from 100% of the flow



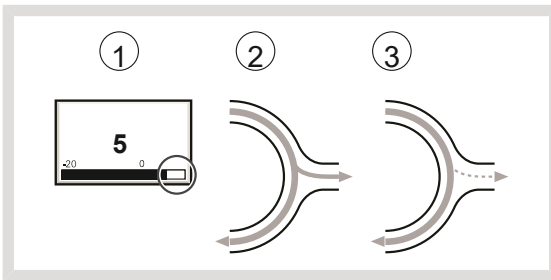
1. Flow triggering
2. Pressure triggering
3. Flow through PEEP valve

Trigger sensitivity can be set in pressure triggering (Trigg. Pressure) or flow triggering (Trigg. Flow). Flow triggering enables the patient to breathe with less effort.

The sensitivity is set as high as possible without self-triggering. This ensures that triggering is patient initiated and avoids auto-cycling by the system.

If an external gas analyzer is connected to the system, it might be necessary to adjust the trigger sensitivity.

Weak patient effort



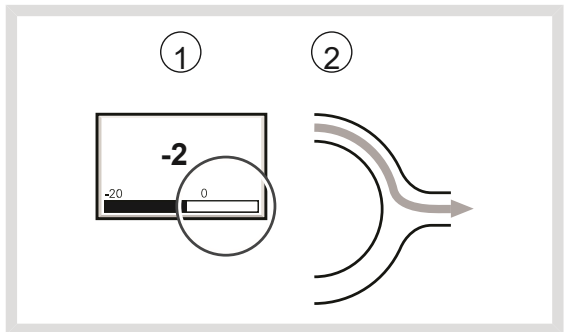
1. At a Trigger sensitivity level above zero (0), the system senses deviations in the flow through the PEEP valve caused by inspiratory efforts of the patient. The more to the right on the scale, the more sensitive is the trigger function.
2. Weak inspiratory effort.
3. Very weak inspiratory effort.

WARNING! If the trigger sensitivity is set too high, a self triggering (auto-triggering) condition may be reached (red area on the bar). This condition can also be reached if there is leakage in the breathing system, e.g. if an uncuffed endotracheal tube is used. Triggering will then be initiated by the system and not by the patient. This should always be avoided by decreasing the trigger sensitivity.

CAUTION: If System checkout has been bypassed, the trigger sensitivity may need to be adjusted to avoid self triggering.

At a Trigger sensitivity level below zero (0), the system senses negative pressures created by the patient. Required preset negative pressure to initiate a breath is shown numerically. The more to the left on the scale, the more effort is required to trigger.

Stronger patient effort

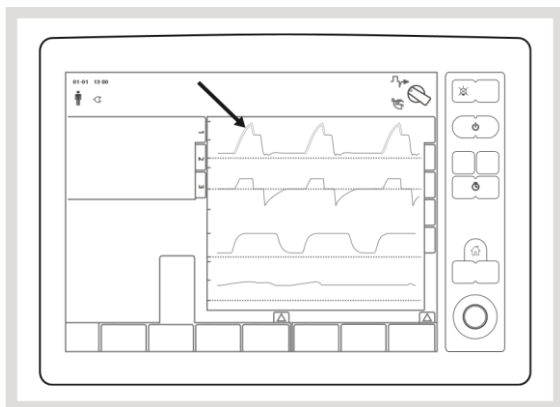


Stronger patient effort.

2

Monitoring patient effort

Administered breaths triggered by patient effort are highlighted in the waveform display area:

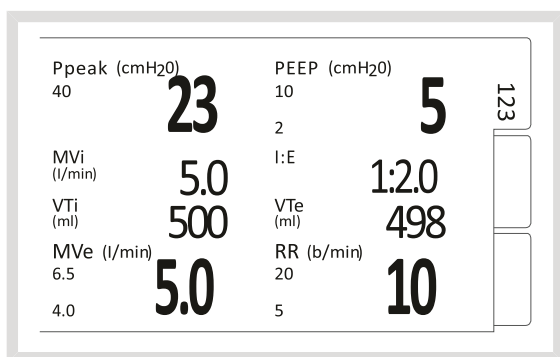


A purple color is superimposed on the inspiratory phase of either the pressure curve (pressure triggering) or the flow curve (flow triggering). In addition, "Triggering" appears in the upper left part of the waveform display area.

5.5 Measured values

The following ventilation values can be **5.5.1**

Ventilation measurements displayed in two pages of information:



Flow

Volume

- RR (mandatory)

MVe (mandatory)

- I:E

- **Ti**

- T_i/T_{tot}

- RRsp

- MVe sp

- VT/PBW

Pressure

datory)

- MVi

- VTi

- VTe

- Vee

Other

- P_{peak}
- PEEP
- P_{mean}
- P_{plat}
- PEEP_{tot}
- C_{dyn}
- SBI

Page 1

Four values can be displayed in a large font, and four values in a smaller font.

Large font values have the following extra information displayed on the panel screen:

- Associated upper and lower alarm limits
- An alarm OFF symbol if the alarm limit has been set to OFF in the alarms profile window.

Page 2

12 values can be displayed in a small font.

The ventilation values to be displayed are specified via Menu/Service and settings.

The user need not select all page values, i.e. value areas can be left blank. The amount of information presented can thus be defined by the user.

Page 3

If the inspiratory/expiratory hold feature is activated from the menu, a third metrics tab appears, displaying the following estimates values:

- PEEP_{tot}
- C static
- E (elastance)
- PEEP
- T_c
- P_{plat}

5.5.2 Gas measurements

	Fi	Et.
CO ₂ (%)	6.5	5.3
0.5	0.1	4.0
ISO (%)	2.4	2.0
2.4	2.0	2.0
⊗	⊗	
O ₂ (%)	30	35
⊗	40	30
30		
N ₂ O (%)	58	58
MAC	1.2	
Age: 40		

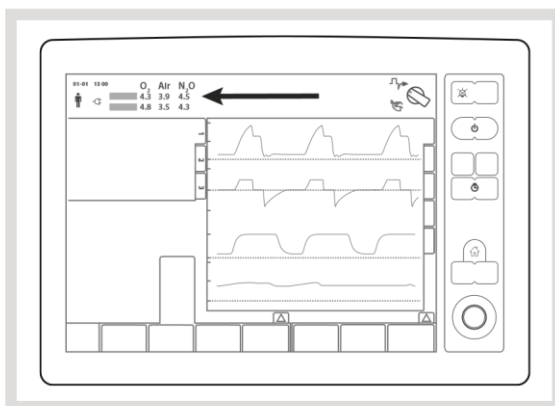
The following values are monitored with the patient gas analyzer:

- Fi/Et O₂
- Fi/Et CO₂
- Fi/Et N₂O
- Fi/Et Agent
- MAC

A reference measurement is made once every four hours or if there is a shift in temperature greater than 1°C. The message 'Gas analyzer zeroing' will be displayed. During this calibration period, the color of the gas measurement values will turn gray, and the last measured values will remain visible on the screen. This reference measurement takes approximately five seconds.

The same behaviour is observed in the case of occlusion in the gas sampling line.

5.5.3 Gas supply pressure



The current gas supply pressure (mains supply and backup gas supply) can be displayed in the upper left area of the control panel. See 'Startup configuration', Chapter 9, page 174. If N₂O is not activated in the startup configuration, 'N₂O' and associated pressure values are not shown.

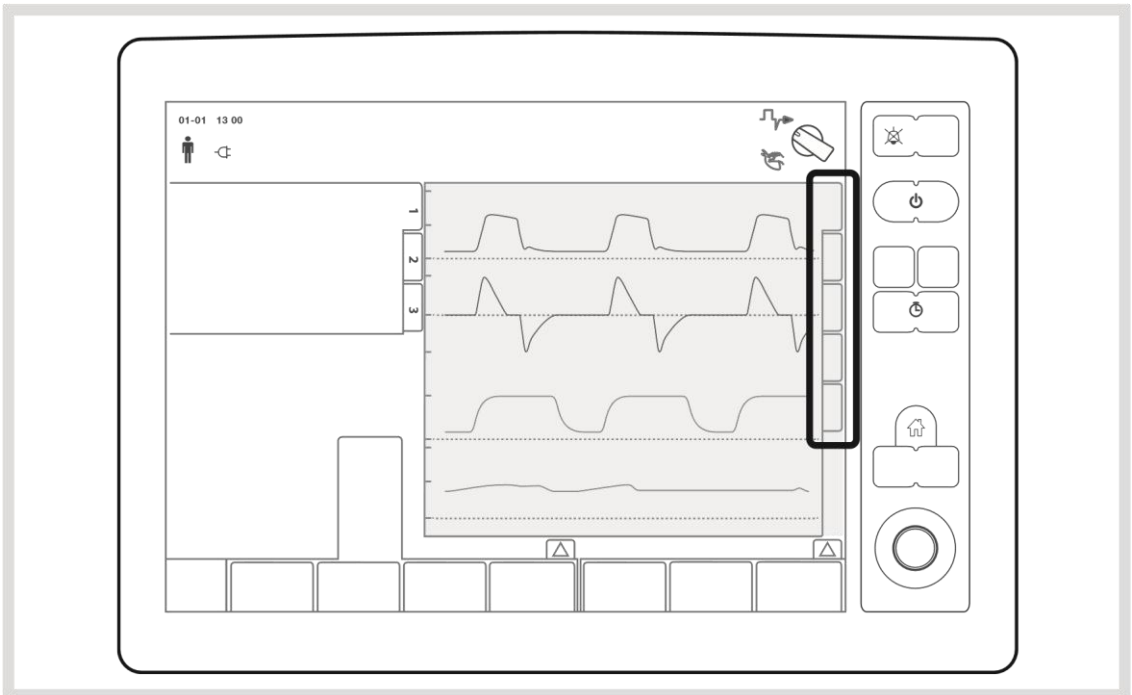
If no backup gas is connected to the system, the backup gas supply pressure values are replaced with asterisks; '****'.

Current gas pressure can also be monitored separately by activating the 'Gas pressure tab', see page 112.

The waveform area is customizable, allowing

- Startup Configuration. Settings affect the number and type of tabs shown at the start

5.6 Waveform area



for different arrangements of displayed information.

In addition to the real-time waveforms, any one of the following can be shown simultaneously by selecting the corresponding tab located to the right in the waveform area:

1. Waveforms
2. Loops
3. Short trends
4. Volume reflector indicator
5. Gas pressure

There are two methods of activating and displaying any one of these tabs:

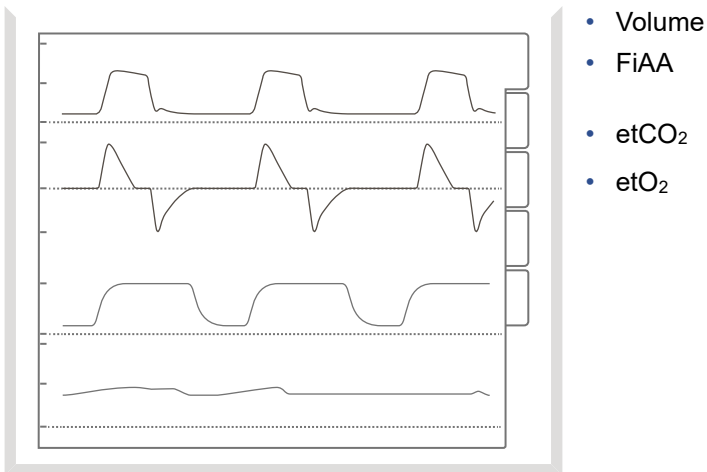
of every patient case.

- Screen layout window. These settings affect the current patient case. Any changes made during a patient case revert to Startup configuration when the patient case is ended.

Startup configuration settings are described in Chapter 9, page 174.

The Screen layout, accessed via the Screen layout membrane button, is described in Chapter 9, page 163.

Waveform tab



The waveform tab is by default always selected at the start of a patient case.

The order and number of displayed waveforms can be modified at any time using the screen layout membrane button. A maximum of six waveforms can be displayed.

Any modified settings revert to default when the patient case is ended. Default set-up is selected during system installation.

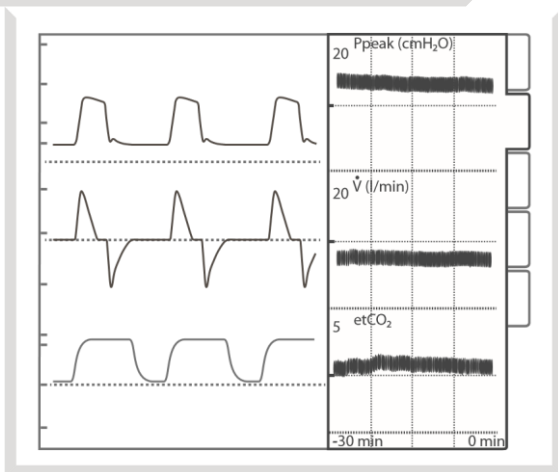
The Pressure waveform is mandatory and always displayed at the top of the waveform display area.

Displayed gas waveforms have a four second delay, and are thus not synchronized with the pressure, flow or volume waveforms.

The following information can be displayed as waveforms:

- Pressure
- Flow

Short trends tab



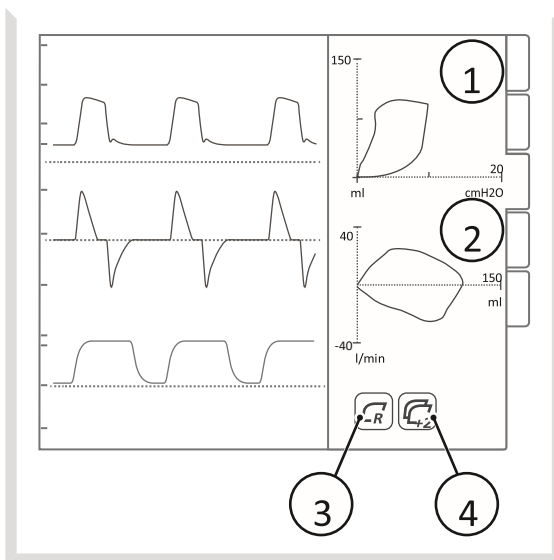
The short trend tab depicts waveform data going back 10, 30 or 60 minutes. The type and number of these shortened waveforms match the screen layout of the displayed real time waveforms.

The short trends are updated every 60 seconds.

If agent concentration is tracked using the short trend function, the type of anesthetic agent (ISO, DES, SEV) is distinguished using agent specific color.

'Short trends' and associated settings are configured in the screen layout window.

Loops



The loops tab gives visual access to real-time relationships between administered volume/pressure (1), and administered flow/volume (2) respectively. These metrics combined yield a characteristic 'loop' that can be used to discern certain properties of the patient's lungs.

Loops are updated with each patient breath. A reference loop can be stored and shown together with the current loop by pressing the reference loop touch pad (3). The reference loop remains visible in the loops window until the touch pad is pressed again. The sampling time point of the reference loop appears above the reference loop touch pad.

By pressing the overlay loops touch pad (4), the two latest loops are displayed together with the current loop. Pressing both touch pads (3+4) will display the reference loop and the two latest loops together with the real-time loop.

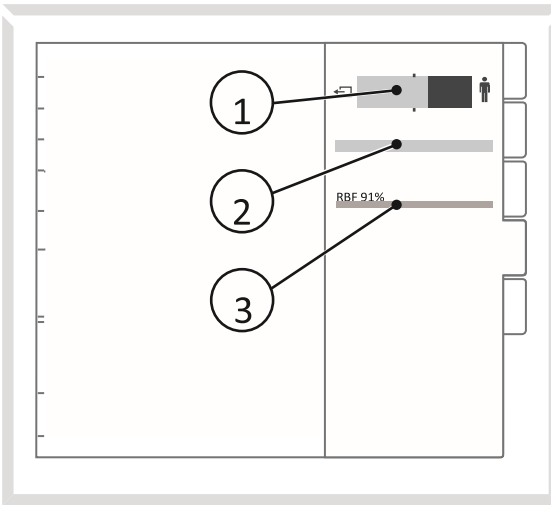
The reference/overlay loops appear with a less prominent white color, distinguishing them from the current loop.

If the option 'Go to Standby' is selected in the end case window, the reference loop is stored and displayed again if the patient case is resumed.

Selecting 'Go to Standby and return to startup configuration' will erase the reference loop information.

This function is not available in AFGO.

Volume reflector



The volume reflector tab shows real time data relevant to the volume reflector and level patient re-breathing:

1. VRI (Volume Reflector Indicator)
2. VR Balance (Volume Reflector Balance)
3. RBF (Rebreathing Fraction)

This function is not available in MAN mode. When switching from MAN to AUTO, or between AUTO modes, volume reflector data becomes available after 3-4 complete breathing cycles.

VRI (Volume Reflector Indicator)

The VRI is a graphic representation of the ratio of reflector gas (white) to exhaled gas

(gray) in the volume reflector. The diffusion point, i.e. border between white and gray, moves back and forth with each breathing cycle.

The color used to distinguish reflector gas from exhaled gas may vary depending on country specific standards.

The left half of the VRI represents gas that is being continuously evacuated via the AGS (Anesthetic Gas Scavenging) system. The majority of this gas volume is usually the reflector module drive gas (O_2).

At the end of expiration, the diffusion point should be left of the rectangle centre, indicating that patient gas is also being evacuated.

If no patient gas is being evacuated, i.e. the diffusion point never reaches half way during expiration, reflector gas (O_2) will eventually enter the breathing system, creating a 'wash-out' situation. Most likely, this is caused by a leak in the breathing circuit.

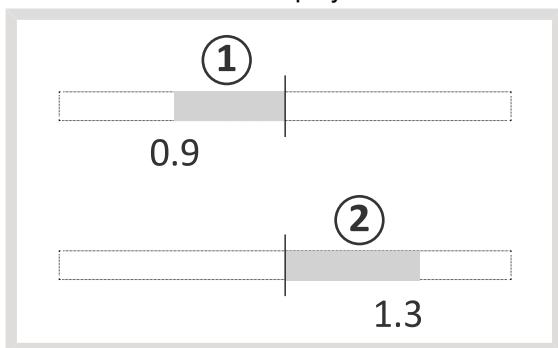
The VRI is complemented with the volume reflector balance metric; VR Balance. This numerical value describes the net flow of gas in the volume reflector, see next section.

VR Balance (Volume Reflector Balance)

This metric describes the net flow of gas from the volume reflector into the AGS system. It is measured during the last complete breath cycle.

A colored bar indicates the direction- and amount of gas that flows through the volume reflector.

The size of the bar is proportional to the flow, a numerical value is displayed under the bar.



1. Exhaled gas flows toward the AGS.
2. Reflector gas (O_2) flows toward the breathing circuit.

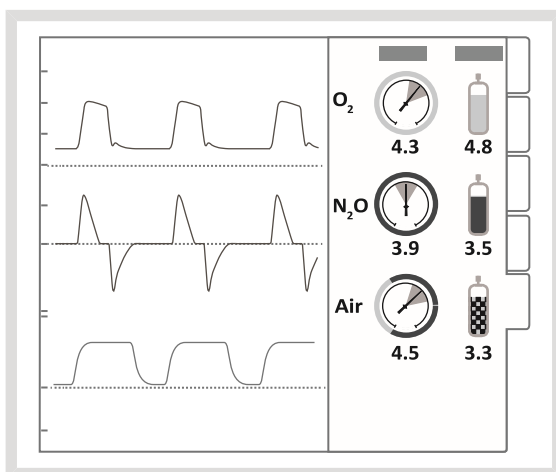
RBF (Rebreathing Fraction)

This metric describes how much of the exhaled gas is being re-used during each successive breath (during inspiration). A value higher than 50% means that exhaled gas comprises more than half of the inspiratory volume.

No single parameter alone controls the RBF; in general a low FGF (Fresh Gas Flow) causes a high re-breathing fraction.

Careful monitoring of the inspiratory O_2 is advised when the RBF is high. Oxygen concentration in the breathing circuit only gets replenished via the FGF. Set O_2 concentration and FGF must be combined in a way so that the patient's oxygen demand is met.

The amount of fresh gas added to the breathing circuit when minimal flow anesthesia is used is very low due to high rebreathing fractions. The system will allow re-breathing fractions up to 95%. **Gas pressure**



The Gas pressure tab offers a clear visual representation of the mains gas supply and the backup gas supply.

Amount of gas remaining in connected backup cylinders is approximated and displayed in the right section.

To get the actual cylinder pressure, press the 'Menu' membrane button and access the 'system info' sub-category.

5.7 Inspiratory and expiratory hold

Using the inspiratory and expiratory hold function enables certain lung characteristics e.g. static compliance and total PEEP to be estimated.

This is achieved by closing the inspiratory and expiratory valves immediately at the end of an inspiration or an expiration. The system then proceeds to estimate the specified metrics for as long as the valves are closed.

The valves remain closed the whole time the touch pad is pressed down. Ventilation resumes as before when pressure is removed.

Measurements start at the end of expiration/inspiration irrespective of the time when the touch pad was pressed down. If 'Insp. hold' is pressed down in-between breaths, the actual function will not initiate until the flow of gas into the lungs stops. If the respiratory rate is low, this intermediate time could be several seconds.

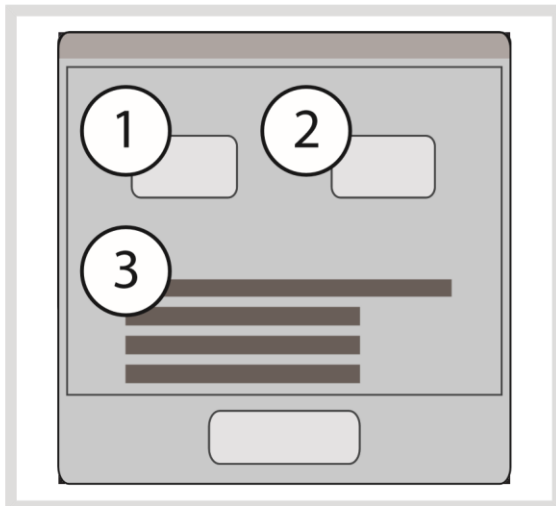
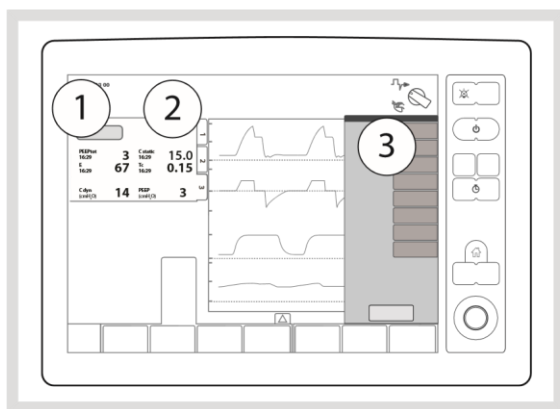
To get accurate estimates of metrics, it is usually sufficient to keep the touch pad pressed down for the equivalent of a complete breathing cycle (a few cycles if the respiratory rate is high).

The time of the last performed measurement is displayed below each estimated parameter.

All alarms will be muted for 60 seconds when either insp/exp hold is used.

The 'Insp Exp hold' dialog can be accessed The 'Insp. Exp. Hold' dialog has two active by either of the following: touch pads:

1. Press the 'Calculate static measurements' touch pad, located on page three in the ventilation measurements area.
2. Press anywhere on page three in the ventilation measurements area.
3. Press the 'Menu' membrane button and select the 'Insp hold / exp hold' from the menu.

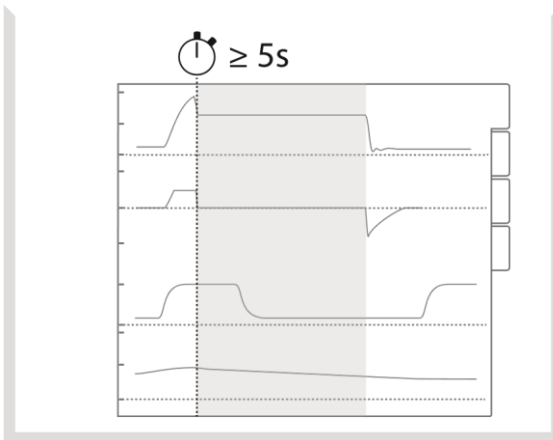


1. Insp. hold
2. Exp. hold
3. Suggested holding time to calculate static measurements.

Press the appropriate touch pad to initiate the desired function. The function is active as long as the touch pad remains pressed down.

Insp./exp hold can also be initiated using the rotary knob. Select the desired function and press down the rotary knob. The function is active as long as the rotary knob remains pressed down.

Inspiratory hold



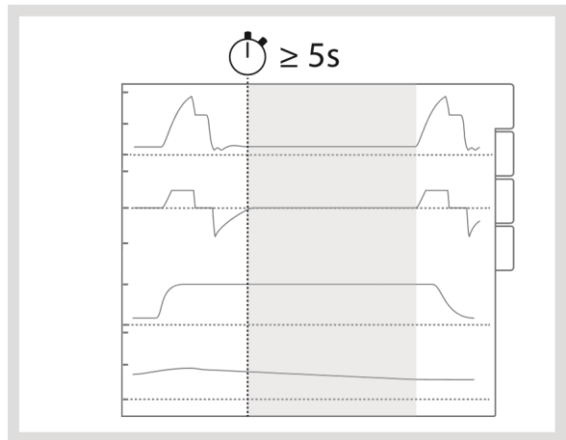
Typical appearance of waveforms during inspiration hold (volume control). Note the lack of flow when the feature is active (gray).

Inspiratory hold provides measurements of the following metrics:

- Static compliance (Cstat)
- Elastance (E)
- Time Constant (Tc)
- "PEEP" and "Pplat" (patient airway pause pressure) are periodically updated during inspiratory hold
- Expiratory resistance (Re)

The inspiratory hold function times out after 40 seconds, after which ventilation resumes.

Expiratory hold



Typical appearance of waveforms during expiration hold (volume control). Note the lack of flow when the feature is active (gray).

Expiratory hold provides measurements of the following metrics:

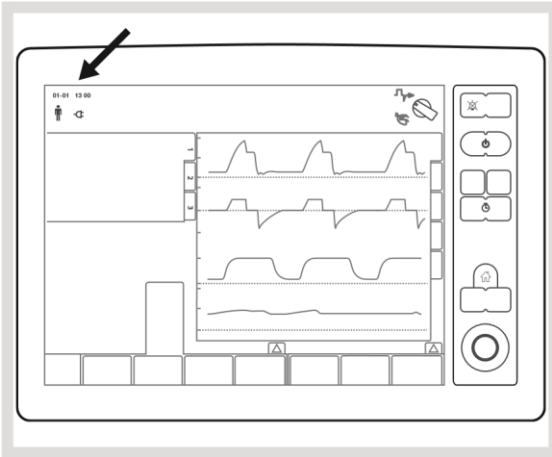
- Static compliance (Cstat)
- Elastance (E)
- Total PEEP (PEEP_{tot})
- "PEEP" is periodically updated during expiratory hold

The expiratory hold function times out after 60 seconds, after which ventilation resumes.

5.8 Miscellaneous

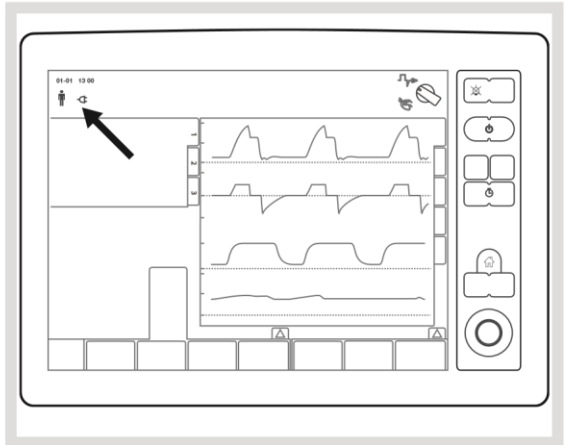
5.8.1 Date and time

The displayed date and time at the top of the touch screen is set during system installation.



The date and time and display format can be

5.8.2 Power and battery status

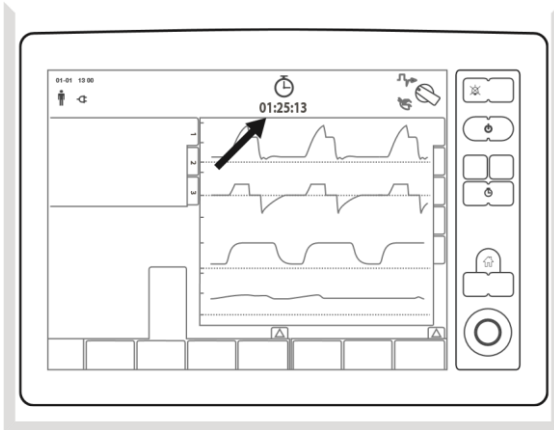


The power and battery status is shown at the top-left on the control panel.

When the system is using battery power, then the estimated time remaining for the battery is also displayed.

changed in Service and Settings, available under Menu. For a detailed description of the power supply, refer to Chapter 10 Alarms and patient safety.

5.8.3 Timer

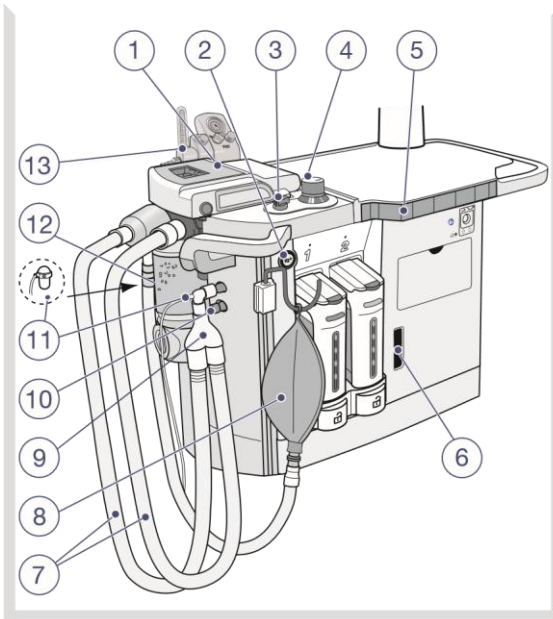


See Chapter 9 'Membrane buttons' on how to operate the timer functions.

6 Breathing system Table of contents

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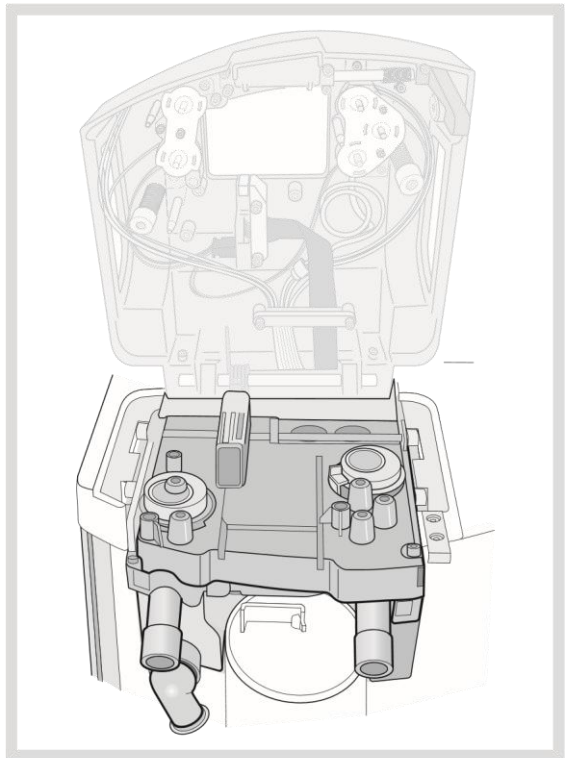
6.1 Breathing system



The breathing system comprises the following:

1. Patient cassette
2. O₂ flush
1. 1.
2. 2.
3. MAN/AUTO ventilation switch
4. APL valve
5. Volume reflector
6. AGS (Anesthetic Gas Scavenging) flow indicator
7. Patient tubing
8. Manual breathing bag with tubing
9. Y-piece
10. AFGO - Additional Fresh Gas Outlet (option)
11. Water trap and sampling line
12. CO₂ absorber

13. Auxiliary O₂ and suction module



(option)

MAQUET recommends that a bacterial/viral filter is always connected to the expiratory connection on the patient cassette. This will minimize the risk of cross-contamination.

6.2 Patient cassette

The patient cassette is the center of gas flow in the system.

The cassette is connected to the following:

- Fresh gas inlet
- Volume reflector
- Manual breathing bag

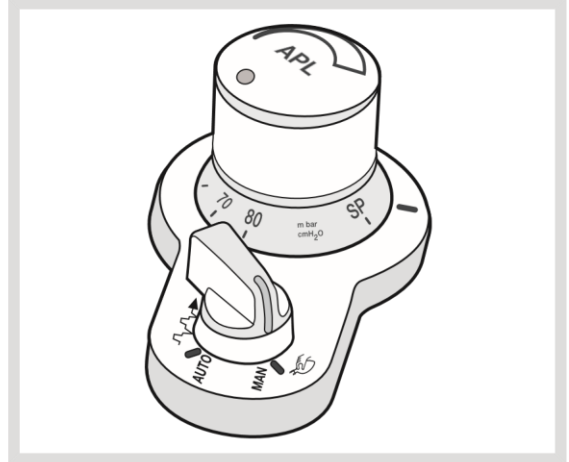
- CO₂ absorber
- Pressure transducers

6.3 O₂ flush



- Gas evacuation
- Patient tubing
- Gas sampling line

6.4 MAN/AUTO ventilation switch



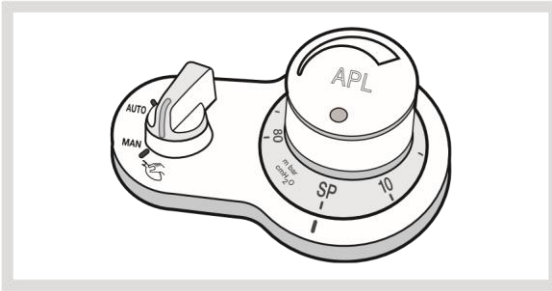
See page 84 for a detailed description.

The O₂ flush is used to manually supply the breathing system with an additional gas flow from the O₂ gas supply. For details refer to section Fresh gas flow on page 273. This gas supply will not go through the vaporizer. Press and hold in. The flow will continue as long as the button is depressed.

If the O₂ flush is used during AUTO mode ventilation, the excess pressure is released via the expiratory valve. Set PEEP is maintained throughout the flush procedure.

6.5 APL valve

Adjustable Pressure Limit valve.

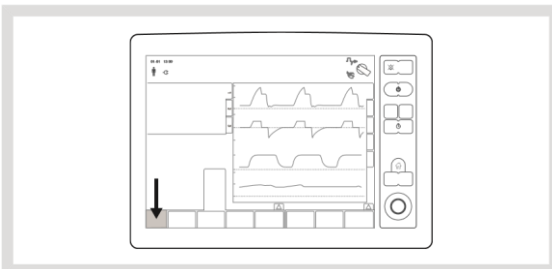


The APL valve is mechanically regulated by means of the APL knob, i.e it is not regulated using any of the control panel touch pads. It is only active during manual ventilation and can be set between spontaneous breathing (= fully open) and 80 cmH₂O.

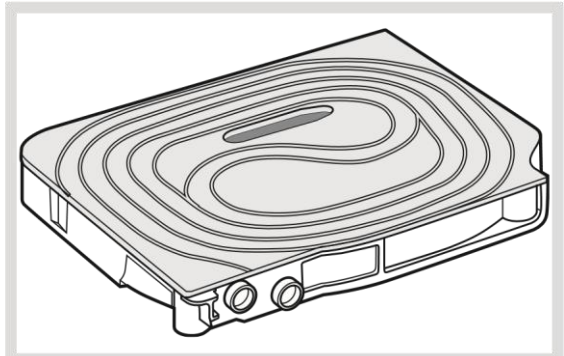
Turning the valve clockwise increases the pressure.

At values above 30 cmH₂O, an increased tactile resistance can be felt and a slight 'click' heard at every 5 cmH₂O interval.

The set value of the APL valve (it is not an active touch pad) is displayed at the bottom-left of the screen:



6.6 Volume reflector



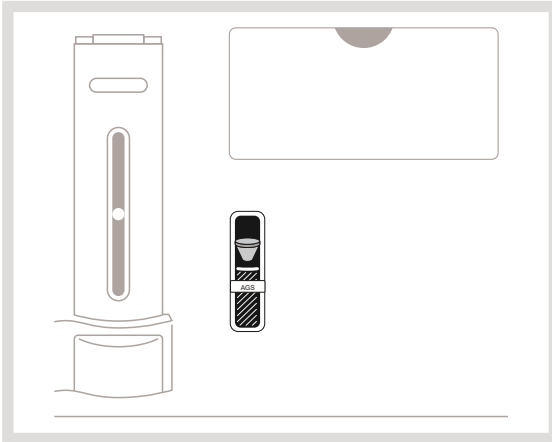
The re-breathing function in FLOW-i is supported by the volume reflector. It functions as a reservoir of exhaled gases, allowing the partial re-administration of these to the patient via the circle system.

The main difference compared to traditional anesthesia systems is that the volume reflector acts as a rigid reservoir with no moving parts. This allows for the administration of large tidal volumes, as well as the use of low fresh gas flows.

The volume reflector volume is approx. 1.2 liters.

6.7 Anesthetic gas scavenging (AGS)

When the AGS is connected to the rear of the system, the flowmeter on the front indicates the rate of evacuation.



The minimum level of evacuation flow is indicated by the floater being above the

6.8 Patient tubing

Maquet recommends using patient tubings measuring from 1.2 to 3 meters in length.

It is recommended only to use original tubes from MAQUET. Soft tubing or tubing with high

CAUTION: If the central scavenging system is inactive, the gases and agents used for anesthesia will be emitted into the operation room.

dashed area, corresponding to a flow of approx. 25 l/min.

The AGS evacuation flow must at least be 25 l/min (STPD), or 10 l/min (STPD) above the set minute volume, whichever is greatest.

FLOW-i can be connected to both low and high-flow EVAC systems.

If the hospital scavenging flow exceeds the recommended flow range (25-35 l/min), a flow restrictor unit can be used with the system. This ameliorates any undesired side effects due to high hospital evacuation flows. See Chapter 3, page 43 for more information on the EVAC restrictor.

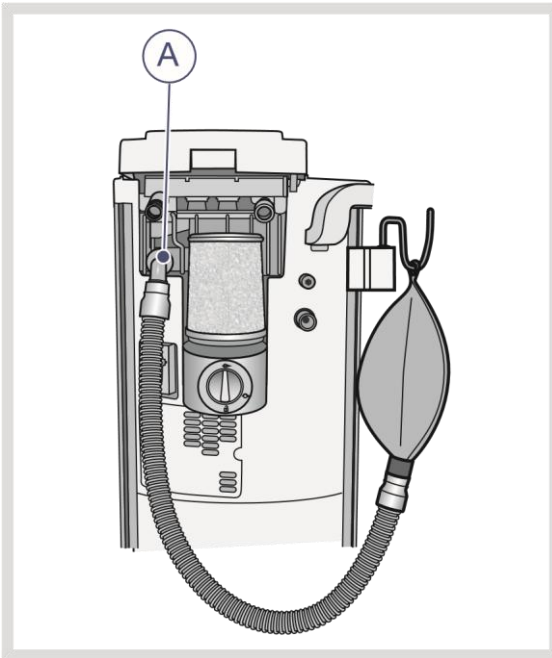
internal resistance may negatively effect the performance of the system.

A new Leakage check must be performed after every change of tubings, bags or filters, e.g. changing from adult to infant tubings. This

to prevent leakage and update the circuit compliance compensation.

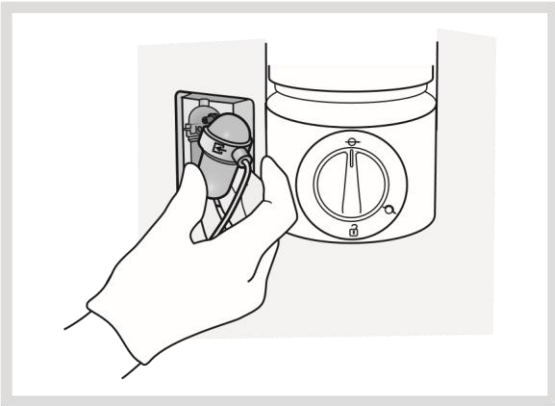
The manual breathing bag should comply with ISO 5362:2006.

6.9 Manual breathing bag



The manual breathing bag is connected to the marked port (A) on the patient cassette.

6.10 Water trap and sampling line



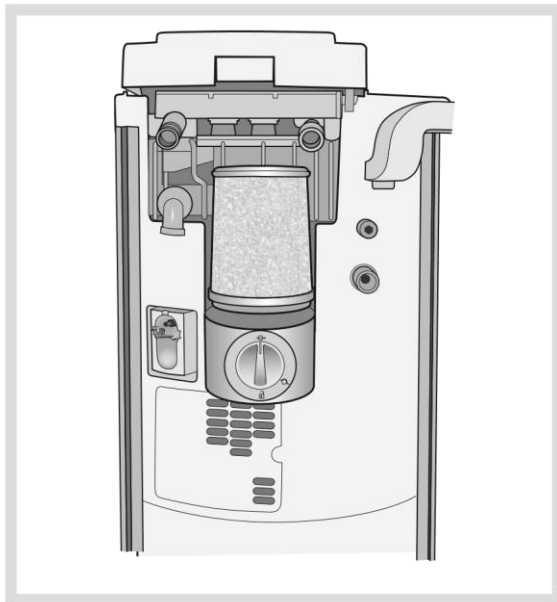
The water trap protects the internal gas analyzer from humidity, bacterial contamination, etc. It consists of a filter housing and a water container.

For a detailed description, see page 144.

24 fl oz (700 ml) of absorbent (soda lime).

The CO₂ absorber switch can be in two or three separate positions depending on configuration:

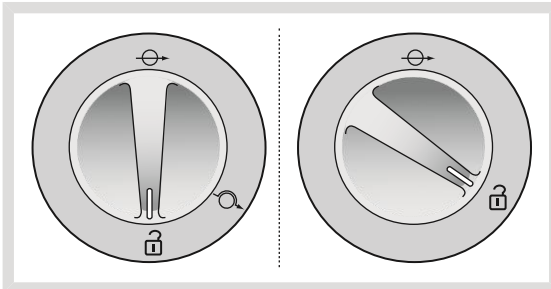
6.11 CO₂ absorber



The pre-packed CO₂ absorber contains

- Unlocked
- Bypassed (not applicable to all systems)
- Locked

Unlocked

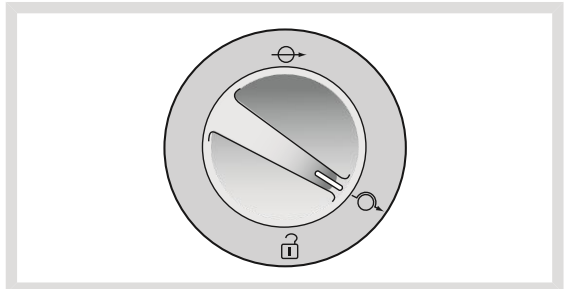


In this position, the patient cassette seals the gas flow and bypasses the absorber.

Turn the locking switch counter-clockwise to return the absorber into the breathing system.

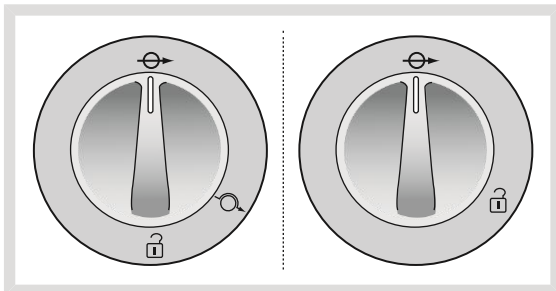
The system compensates for the change in compliance when the CO₂ absorber is disconnected.

Turn the locking switch counter-clockwise to



return the absorber into the breathing system.

The system compensates for the change in compliance when the CO₂ absorber is bypassed.



In this position, the expired gas flows through the CO₂ absorber, i.e. normal position.

The absorber can be mounted or dismantled from the system.

Locked

Bypassed (not applicable to all systems)

In this position, the patient cassette seals the gas flow and bypasses the absorber.

Replacement of absorber

During use, a pink (white-to-pink grade) or violet (white-to-violet grade) color will develop and successively deepen in intensity, indicating exhaustion of the soda lime. A slowly increasing inspiratory carbon dioxide concentration will usually begin to occur when the color change has penetrated to around half the depth of the absorber canister.

Color changes of the soda lime can however be misleading, and suitable monitoring of inspired carbon dioxide is required to assess when to replace absorber.

The change in CO₂ soda lime color may be reversed, i.e. loss of color may indicate exhaustion.

If the CO₂ absorber is not used for prolonged periods of time, the color indicating soda lime exhaustion may fade due to drying out.

The lifetime will depend not only on the original soda lime performance, but also on the amount of CO₂ absorbed.

Spent soda lime may still contain some absorbed or entrained anesthesia agent which needs to be assessed and managed according to hospital routines.

As a rough guide, the absorber will become saturated with CO₂ after about 8 hours of active ventilatory usage (low flow anesthesia).

If the CO₂ absorber is dropped and absorbent released, avoid inhaling the dust and avoid skin and eye contact. Sweep or vacuum up the absorbent.

Being alkaline in nature, soda lime should not be allowed to come into contact with sensitive skin, particularly with the eyes or mucous membranes. If contact is made, wash thoroughly with clean water and, where necessary, seek medical attention.

Replace the absorber immediately:

- If you suspect the soda lime has dried out
- If there is a sharp temperature increase during the washing-in phase
- If there is an unusual delay in the increase in inspired agent concentration.

CAUTION: If damage is suspected, do not use the absorber.

Storage

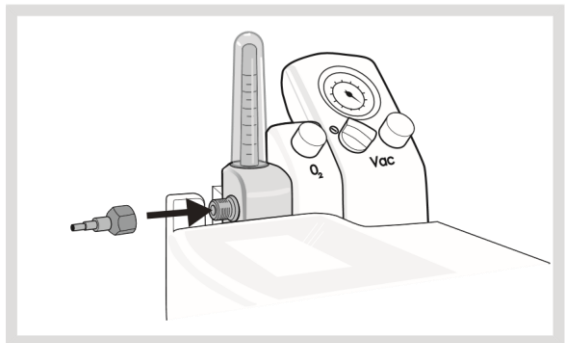
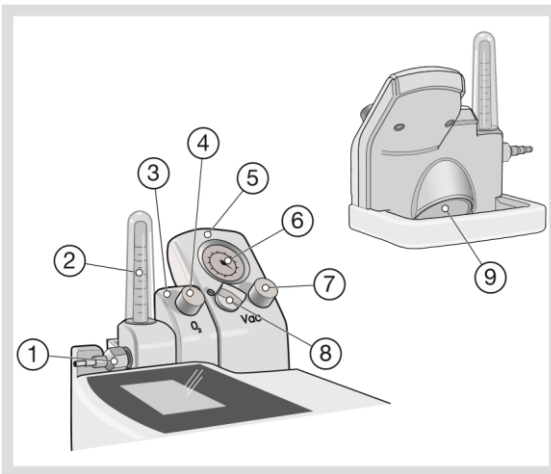
Containers holding unused absorbers should be stored in a clean, dry environment, and at temperatures between 0°C (32°F) and 35°C (95°F). Storage at higher temperatures can result in reduced efficiency and service life due to moisture loss. When correctly stored, canisters will maintain absorption capacity for a period of two years.

Canisters must NOT be stored where they can become subject to the following:

- Direct strong sunlight.
- Close proximity to incompatible chemicals or acids.
- Partial or total water immersion.
- Environment with abnormal concentrations of carbon dioxide, hydrogen sulphide or other acidic gases.
- Excessive stacking loads - maximum two pallets high.

6.12 Auxiliary O₂ and suction module (option)

With the tube adapter attached, the flowmeter outlet can be connected to 4, 6 and 8 mm tubings (0.16 in., 0.24 in. and 32 in. resp.)



Testing the Flowmeter

Test the Flowmeter unit before use, e.g. during

1. Flowmeter outlet (with tube adapter the System checkout. Turn on the O₂ flow attached) using the regulatory valve and ensure that
2. Flowmeter tube maximum flow can be reached, i.e. the
3. Flowmeter unit flowmeter ball reaches the top of the
4. Flowmeter unit regulatory valve flowmeter tube.
5. Suction unit
6. Vacuum gauge for the suction unit
7. Suction unit regulatory valve
8. On/Off switch for the suction unit

9. Mount for hydrophobic bacterial and viral filter

Flowmeter unit

The intended use for the Flowmeter unit is to provide oxygen for patient therapy.

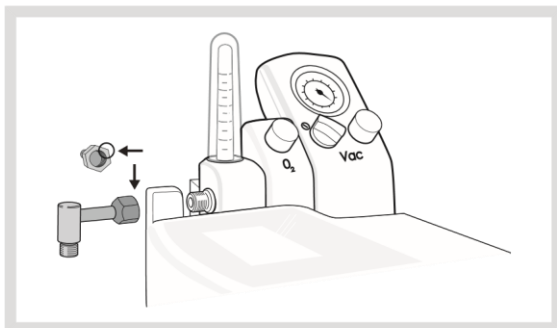
Outlet flow is regulated by the flowmeter regulating valve (4). Turning the knob clockwise decreases the flow, and turning it counter-clockwise increases the flow up to its maximum value. The current flow is indicated on the flowmeter tube when the center of the ball is aligned with the printed scale.

Bubble humidifier (option)

To protect patient lung tissue and airways from drying out, e.g. when bypassing the upper airways during treatment, the Flowmeter unit can be fitted with a bubble

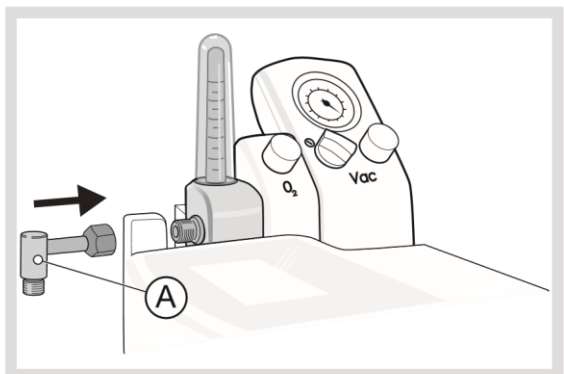
the angular adapter. Proceed according to the

Before assembly, ensure that the flat seals on the angular adapter and humidifier adapter are intact.



humidifier. This requires the tube adapter to be replaced with

1. Attach and secure the angular adapter (A) to the flowmeter unit by means of the cap nut and tighten by hand. The angular



CAUTION: The Flowmeter unit is connected to the FLOW-i O₂ outlet and is therefore dependant on the central gas supply pressure for its operation. The central gas supply pressure should be above 300 kPa (3.0 bar, 44 PSI) when the Auxiliary O₂ and suction module is connected to the system.

adapter shall point vertically downwards
when finished.

Suction unit

The intended use for the Suction unit is to extract body fluids from the stomach and airways.

Suction pressure is regulated by means of the On/Off switch and the suction unit regulatory valve. Turning the regulatory valve counter-clockwise decreases the pressure, i.e. increases the suction flow. The vacuum gauge shows the current suction pressure.

The Suction unit is connected to the FLOW-i Air outlet and is therefore dependant on the central gas supply pressure for its operation. The central gas supply pressure should be above 300 kPa (3.0 bar, 44 PSI) when the Auxiliary O₂ and suction module is connected to the system.

MAQUET recommends using single use

WARNING! The Suction unit must only be used by personnel who have received adequate instructions of use.

septic fluid jars and suction tubings. These shall comply with ISO 10079-3:2009.

The septic fluid jar used with the unit must have a usable volume of at least 500 ml and an integral overflow protection device. The inner diameter of the inlet to the fluid jar shall be at least 6 mm.

The length of tubings connecting the septic fluid jar with the unit shall not exceed 1 m. The tube material should be rigid enough to

prevent occlusion. The inner diameter of the tubing connecting the fluid jar with the suction unit shall be at least 6 mm.

MAQUET recommends using the following suction unit equipment items (where available):

- SERRES Suction Bag 1000 Art. nr 57157 ml
- SERRES Suction Canister Art. nr 57308 1000 ml
- SERRES Suction Tubing Art. nr 5823212

A hydrophobic bacterial/viral filter is located on the back of the suction unit. It protects the system against contagious particles and acts as a second overflow protection device in addition to the device located in the septic fluid jar. The suction unit cannot be operated without a correctly placed bacterial/viral filter.

The hydrophobic bacterial and viral filter needs to be replaced at regular intervals, see Chapter 13, page 245, for more details.

Testing the Suction unit

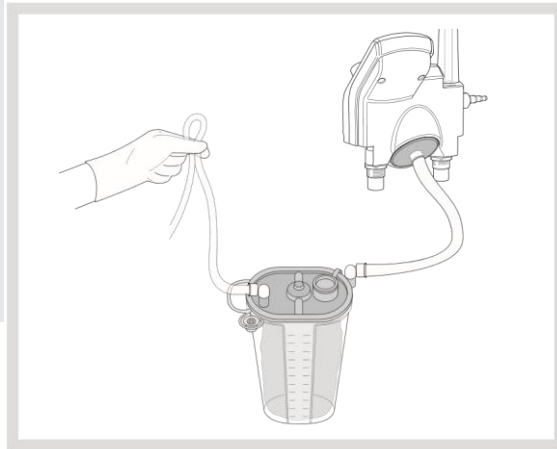
The Suction unit shall be tested before use, i.e. during the System checkout. Follow the steps outlined below:

CAUTIONS:

- Negative pressure during tracheobronchial suction with a closed suction system (CSS) must not be lower than - 200 cmH₂O.
- The septic fluid jar used with the Suction unit must have an overflow protection device.
- If treating newborn, or prematurely born infants, carefully monitor the suction pressure to prevent inadvertent oversuctioning.

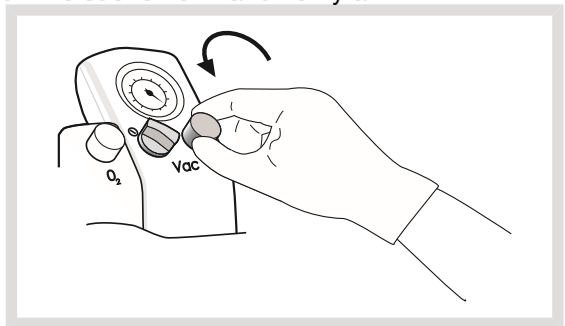
Close the suction system by occluding the end of the suction hose.

1.



negative pressure build-up as the Suction unit regulatory valve is turned counter-clockwise. The current pressure is displayed on the Vacuum gauge.

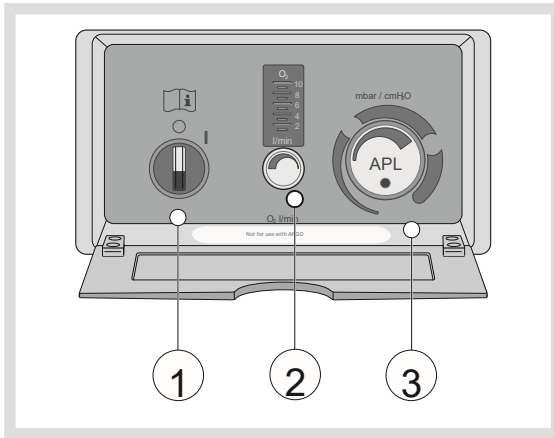
2. Turn on the suction unit and verify a



6.13 Emergency ventilation system

In case of a total power (mains power and battery) or other electronic failures, the built-in emergency ventilation system can be used.

Open the cover to access the emergency ventilation:



1. On/Off switch for emergency ventilation
2. O₂ gas supply and flowmeter, graded up to 10 l/min.
3. Mechanical APL

To operate the Emergency Ventilation, perform the following:

- a. Activate emergency ventilation
- b. Adjust oxygen flow - the current flow is indicated on the flowmeter when the center of the ball is aligned with the printed scale.
- c. Adjust pressure level

When settings have been made, ventilate the patient using the manual breathing bag.

If the emergency ventilation is activated, either during a patient case or when testing the system, the emergency ventilation APL valve must be set to minimum before resuming normal ventilation (AUTO or MAN) or starting a new patient case.

WARNING! If the emergency ventilation system is activated while the anesthesia

system is in operation, the anesthesia system will be shutdown.

CAUTION: The emergency flow is always delivered via the patient cassette outlet and thus the patient tubings must always be connected to patient cassette outlets when emergency ventilation is used.

6.14 Vaporizers



The vaporizer unit can hold two vaporizers, where the following agents can be used:

- Isoflurane
- Sevoflurane
- Desflurane

Do not use Flammable Anesthetic Agents close to the FLOW-i system.

6.14.1 Connecting a vaporizer

Vaporizers can be connected during Standby and Run mode.

Before inserting the vaporizer in the vaporizer slot, a visual check of the exterior and the mesh on the vaporizer inlet and outlet shall be performed.

Just above the vaporizer, an LED lamp shows the status of the vaporizer or the vaporizer slot (see table).

LED color	Vaporizer status
Green	The selected vaporizer has passed the System checkout or Vaporizer check, and is ready to deliver agent.
Yellow	• The vaporizer slot is selected but no vaporizer is present. • The selected vaporizer slot holds an unchecked vaporizer. • The selected slot's vaporizer is being filled. • The selected slot's vaporizer has triggered an alarm, i.e. Vaporizer nearly empty.
Red	The vaporizer is disabled. The power to the vaporizer is shut off (the condition is applicable to both slots, i.e. not only the selected slot).
Not lit/Off	The vaporizer slot is not selected.

There is a short delay before a vaporizer can be used after it has been connected due to calibration.

A vaporizer check should always be performed after connecting a vaporizer to the system but, in an emergency, or if the system is in Run mode, the check can be bypassed. A 'Vaporizer check bypassed' system message will be displayed in the message area and an entry is made in the event log.

A leakage check is performed together with the vaporizer check. The logs and system are subsequently updated with leakage measurements and circuit compliance compensation data.

Connecting a vaporizer in Standby mode

Insert the vaporizer into the vaporizer slot and push into place until a click is heard.

A prompt window appears: *'A vaporizer check should be performed before the vaporizer is used'*. Press Start to initiate the check.

If a patient case is started and the active vaporizer hasn't been checked, the user is required to confirm this action.

See Chapter 4, page 60, for a description of the Vaporizer check.

Connecting a vaporizer during Run mode

Insert the vaporizer into the vaporizer slot and push into place until a click is heard.

The vaporizer check cannot be performed during Run mode. A 'Vaporizer check bypassed' message will be displayed on the screen, and an entry is made in the event log.

When connecting a vaporizer during Run mode, only connect vaporizers that previously have passed System checkout/Vaporizer check.

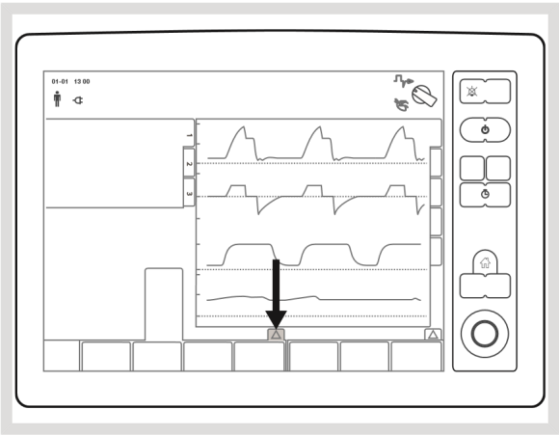
WARNING! Bypassing the Vaporizer check removes the possibility to detect leakage between the vaporizer and main system. Agent may thus leak out into the room undetected.

6.14.2 Selecting active vaporizer

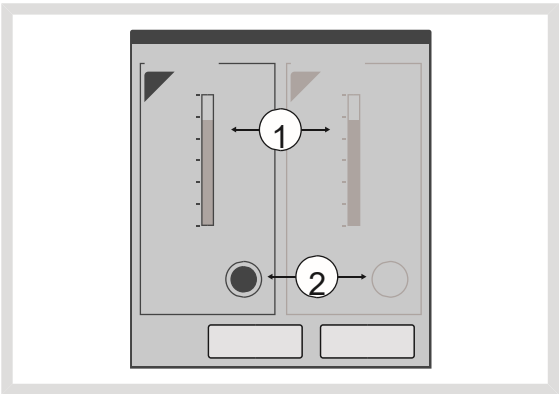
Selecting an active vaporizer is necessary if:

- The vaporizer is connected into the non-selected slot.
- There is a need to change the active vaporizer when there are two vaporizers connected to the system.

1. Press the vaporizer touch button on the screen.



2. The Select vaporizer window appears. It displays the connected vaporizer's current liquid level (1) and vaporizer type (2).



3. To select a vaporizer, press the required vaporizer type and confirm by pressing Accept. This activates the vaporizer. The

agent name is shown on the Vaporizer setting touch pad.

When a second vaporizer is activated, the other vaporizer is automatically de-activated; the agent concentration is set to OFF, the vaporizer is depressurized, flushed and unlocked.

It is possible to manually set the agent concentration to OFF, and the vaporizer to standby, using the Agent concentration touch pad, see page 141 for details.

6.14.3 Re-filling a vaporizer

The liquid level can be seen in the monitor tube on the front of the vaporizer, or in the Select vaporizer window.

An alarm message will be displayed when the liquid level of the vaporizer is low.

A vaporizer can be re-filled while connected to the system as well as when it is disconnected from the system.

The vaporizer agent level shall never be below the min level, or exceed the max level. The min and max levels are indicated by markings located on the vaporizer monitor tube.

To ensure that the correct agent is used when refilling the vaporizers, agent specific filling adapters are used. These are as follows:

Agent	Adapter
Isoflurane	MAQUET filling adapter
Sevoflurane	MAQUET filling adapter Quik-fil [®]
Desflurane	SAFE-FIL ^{TM3}

CAUTION: Agent-specific filling cannot be assured when bottles without collars are used.

3. Baxter International Inc.

Filling adapters are either fastened onto the opening of the bottle by the manufacturer (SAFE-FILTM, Quik-fil[®]), or are attached onto the bottle by the user before refilling (MAQUET filling adapter). Adapters will only fit designated vaporizers due to the keying of the filling adapter.

The sevoflurane vaporizer is approved for the following brands:

- AbbVie Sevoflurane[®]
- Sevoflurane Baxter
- Sevoflurane Piramal

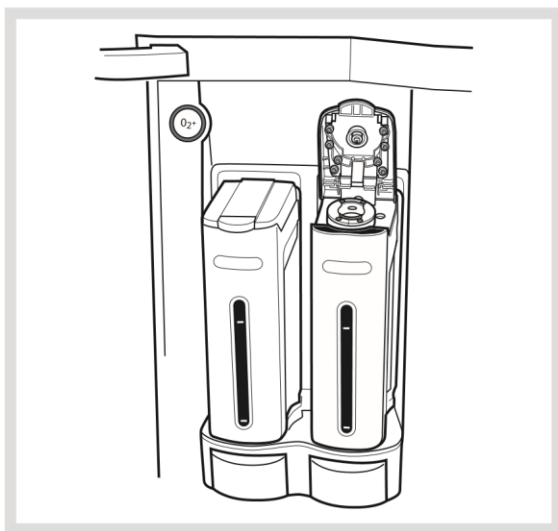
Do not use worn or damaged filling adapters.

Re-filling while connected

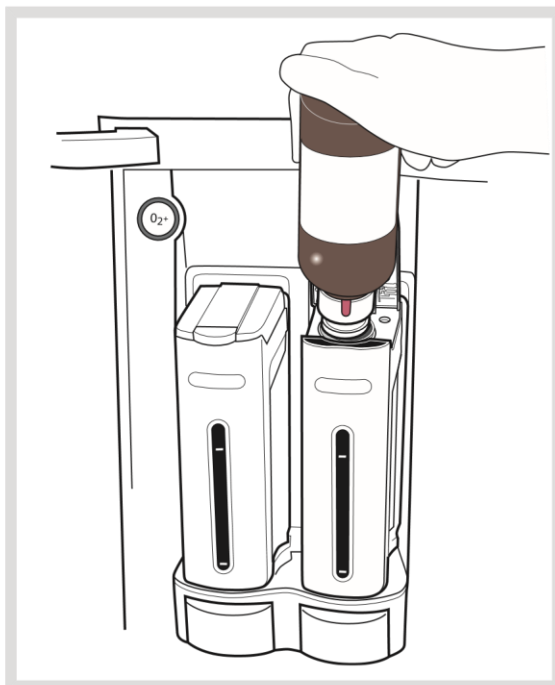
There is no need to turn off the vaporizer. the correct vaporizer. Press the bottle

Perform the following:

1. Lift the vaporizer lid. The vaporizer is depressurized and agent delivery is shut off. The LED above the vaporizer turns yellow. The Agent concentration touch pad is grayed out and cannot be selected when the lid is open. A system message is displayed in the system message area until the lid is closed.



2. Position the adapter onto the vaporizer. Again, the specific adapter will only match the correct vaporizer. Press the bottle downwards so that the agent liquid flows into the vaporizer. The liquid level can be monitored by looking at the monitor tube.



3. Remove the bottle, close the vaporizer lid. The LED above the vaporizer turns green and agent delivery resumes with the same concentration used prior to the refilling procedure.

Press down the outer ring on the Desflurane vaporizer connector to release the adapter.

Re-filling when disconnected

The procedure is identical to when the vaporizer is connected to the system. Refer to hospital routines regarding handling anesthetic agents and refilling processes.

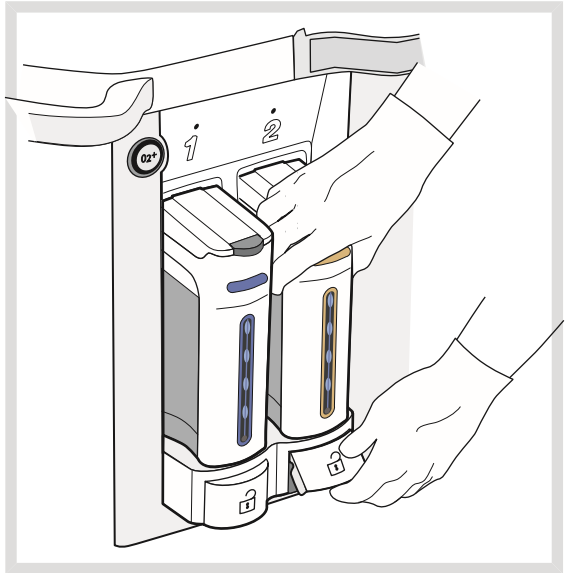
6.14.4 Disconnecting a vaporizer

Before disconnecting a vaporizer, it needs to be deactivated. This is done either by setting the vaporizer to non-active in the Select vaporizer window, or by using the rotary knob to deactivate the vaporizer. It is not possible to remove a vaporizer that is not deactivated.

To deactivate the vaporizer using the rotary knob, perform the following:

1. Turn the rotary knob counter-clockwise past '0% agent' until 'OFF' is displayed.
2. Press the knob to confirm the new setting. The vaporizer is depressurized, flushed and unlocked.
3. A system message appears when the process is completed: *'Vaporizer OFF, unlocked and ready to remove'*.

When the system message appears in the message area, lift the locking handle upwards and remove the vaporizer.



If a vaporizer is dropped, it should be checked by a service technician trained and authorized by MAQUET.

CAUTION: When a vaporizer is removed from the system, the gas connector on the rear of the vaporizer can be hot (>47°C).

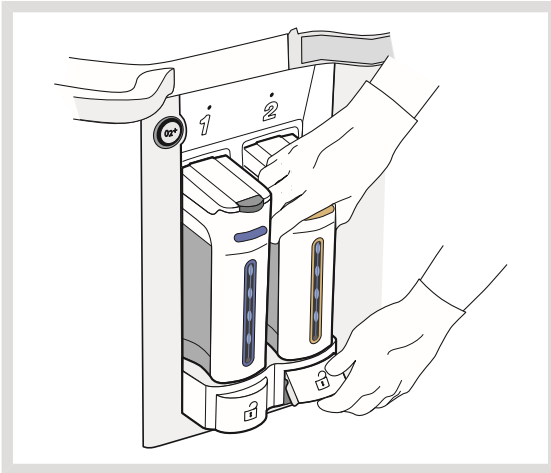
6.14.5 Emptying a vaporizer

MAQUET recommends that the vaporizers are emptied before long distance transportation or long term storage.

Connect the vaporizer to the corresponding filling adapter and turn it upside down. Press the bottle into the vaporizer to have the fluid flow back into the bottle.

6.14.6 Vacant vaporizer slots

When not currently occupied by a vaporizer, a protective cover can be used to cover the electronics and gas connections in the vaporizer slot.



6.15 Gas analyzer

Located inside the main unit, the gas analyzer provides a DIR (dispersive infrared) measurement of respiratory and anesthetic gases, and paramagnetic measurement of oxygen.

For sidestream monitoring of O₂, CO₂, N₂O and agents, connect the sampling line to the sampling port located between the Y-piece and the patient.

The gas analyzer continuously takes sidestream gas samples from the breathing circuit and directs these through a water trap into an internal measurement chamber.

The partial pressure and the percentage volume of CO₂, N₂O, O₂ and anesthetic agent depend on the amount of water vapor in the breathing gas. A partial H₂O pressure of up to 11 cmH₂O is automatically compensated for by the analyzer. Higher H₂O partial pressures will further dilute the gas sample; at 30 cmH₂O the general error of all measured gases is - 2%.

The respiratory gas monitoring system is equipped with automatic barometric pressure compensation.

The tidal volume in the system remains unaffected by the sampling flow since all sampled gas is continuously returned to the breathing circuit.

If an external gas analyzer is used which does not return its sample flow into the breathing circuit, the fresh gas flow and tidal volume should be set to compensate for the lost volume (Volume Control only).

All gas concentration readings are normally referenced to dry gas conditions, ambient room temperature and atmospheric pressure (ATPD).

Accuracy cannot be guaranteed with a respiratory rate > 60 bpm.

When operating the system at high altitudes, i.e. low barometric pressures, gas monitoring may not detect high Desflurane concentrations accurately. The gas analyzer may incorrectly identify secondary agents and present these on the screen. The following false alarms may be triggered:

- Agent mixture: MAC > 3

- Agent mixture

Agent delivery and internal safety systems are not affected and will prevent any hazardous overdose in case of system malfunction.

6.15.1 Change of anesthetic agent

During the anesthesia procedure, the change of anesthetic agent will result in a temporary mix of agents in the breathing system. Both agents are measured and displayed on the screen.

The agent with the highest concentration is automatically identified as the primary agent.

6.15.2 Water trap and sampling line

The water trap protects the internal gas analyzer from humidity, bacterial contamination, etc. It consists of a filter housing and a water container.

The sampling line is not reusable and must be replaced between patients or when the cleanliness of the tubing or connector is compromised.

To maintain measurement accuracy:

- perform a System checkout before connecting the system to the patient
- do not connect several sampling lines in order to extend the line's reach.

If the water trap is poorly fastened or not installed, the gas analyzer's internal pump is disabled and there is no flow through the measuring system.

The water trap, sampling line (marked 'gas sample') and connectors are all parts of a system optimized for the gas analyzer. The time delay of gas data and waveforms displayed on the touch screen is dependant on the length and inner diameter of the sampling line. The use of kinked or damaged sampling lines, non-approved brands, extra connectors or extension lines may all severely distort the displayed waveforms and possibly misguide the user.

WARNINGS!

- If a broken water trap has been used during operation, the gas analyzer might be contaminated and damaged. If this is the case, take the anesthesia system out of operation and contact a service technician trained and authorized by MAQUET.
- Do not connect tubing other than gas sampling lines to the water trap, even though other tubing (e.g. IV lines) might fit the connector on the trap.

CAUTION: Always use accessories approved by MAQUET. Other accessories may impair the measurements, affect accuracy, and result in slower response times.

6.15.3 MAC values

The presented MAC ($MAC^{rel,barcomp}$) is relative to MAC_{40} , and pressure compensated to account for fluctuations in ambient pressure. It is calculated as follows:

$MAC_{rel} = 10^{(-0.00269 * (\%Agent1_{age-40}))}$
 $* MAC_{Agent1} +$

$$\frac{(-0.00269 * (\%Agent2_{age-40})) * MAC_{Agent2} + 10 \%N_2O}{10^{(-0.00269 * (age-40))} * MAC_{N_2O}}$$

and,

$$MAC^{rel,barcomp} = MAC^{rel} * \frac{P_{amb}}{1013\text{ mbar}}$$

where,

- %Agent1 is the end-tidal concentration of primary agent.
- MAC_{Agent1} is the MAC value for the primary agent in percent of 1 atm.

- %Agent2 is the end-tidal concentration of the secondary agent.
- MAC_{Agent2} is the MAC value for the secondary agent in percent of 1 atm.
- %N₂O is the end-tidal concentration of N₂O.
- MAC_{N_2O} is the MAC value for N₂O in percent of 1 atm.

MAC_{Agent} is a constant related to the delivered anesthetic agent:

Agent and N ₂ O	MAC ₄₀
Isoflurane (Forene)	1.15%
Sevoflurane (Sevorane)	2.1%
Desflurane (Suprane)	6.6%
N ₂ O (Nitrous Oxide)	105 % (=MAC _{N₂O})

The above values are based on the anesthetic requirements of a test group comprised of healthy males, age 40, i.e. MAC_{40} .

The presented MAC is the sum of the individual $MAC^{rel,barcomp}$ values for all detected anesthetic agents and N₂O.

The equation compensates for the age difference, as defined by the current patient case settings, between the patient and the 'MAC₄₀' reference group. If the set age is below 1 or above 80, '1' or '80' will be used to calculate the MAC^{rel} respectively.

6.15.4 Warm-up

During startup, the gas analyzer passes through an initialization and warm-up period, where the etCO₂, O₂, and agent values are displayed as '***' for 60 seconds before ISO standard accuracy is achieved and the unit is operational. Delivery of anesthetics and N₂O can begin once the analyzer is operational. Normal/full accuracy is reached within 10 minutes after power on.

6.16 Lift (C30 model only)

With the mains power connected, the lift function is operated by means of the Up and Down buttons located to the right of the ON/OFF button:

The lift has a 10 % duty cycle; it may be used continuously for a 2 minute period, after which it must cool down for 18 minutes. If the lift is used before it has cooled down completely, the time of continuous use will be less than 2 minutes.

The maximum capacity is 150 kg (330 lbs), equivalent to approx. 30 kg (66 lbs) of auxiliary equipment.

For a detailed description of equipment weight restrictions refer to Chapter 15, page 268.

6.17 Ceiling pendant (C40 model only)

The FLOW-i C40 system can be attached onto a variety of ceiling pendant systems. Contact your local MAQUET supplier for more information.

Only personnel trained and authorized by MAQUET may install the C40 model.

The maximum lift capacity of the ceiling pendant may limit the number of accessory equipment that can be mounted onto the C40 system. For a detailed description equipment weight restrictions refer to Chapter 15, page 268.

WARNINGS!

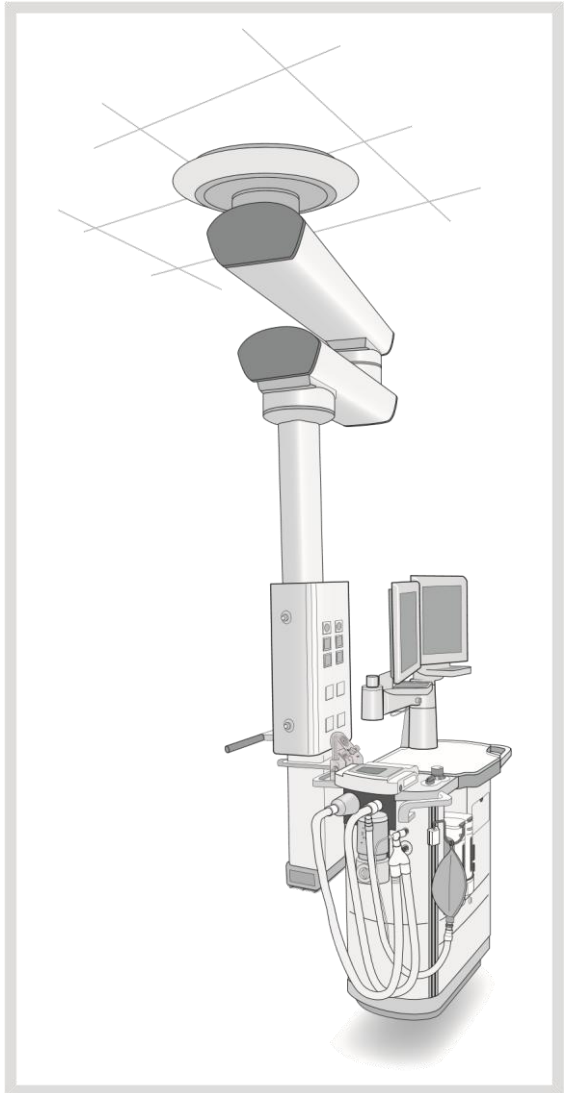
- The system must not be moved to other types or models of ceiling pendants, except by personnel trained and authorized by MAQUET.
- Ensure that the system is properly docked and secured before lifting the C40 using the ceiling pendant.
- Beware of crush injury when moving the C40 system using the ceiling pendant.

CAUTIONS:

- There shall be no objects or equipment underneath the system when it is attached to the ceiling pendant.

6.17.1 Docking and undocking

The C40 system must always stand on the floor when it is being docked or undocked onto or from the ceiling pendant.



The C40 system is intended for transport inside the operation room only when undocked from the ceiling pendant. Transportation to other locations requires the use of a suitable cart or trolley according to hospital routines.

All personnel working with the system shall be well acquainted with the docking/undocking procedure so that, in the event of an emergency, the system can be undocked and removed.

WARNING!

Ensure that the system is properly docked and secured before manoeuvring the C40 using the ceiling pendant.

This device, when removed from its wall or ceiling mount, does not meet the stability requirements of ISO 80601-2-13 and IEC 60601-1 respectively. Special caution has to be taken.

CAUTION: Disconnect all electrical cables and gas connections before undocking the system from the ceiling pendant.

7 AFGO (Additional Fresh Gas Outlet, Option) Table of contents

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7.2	Preparations	151
7.3	Manage AFGO settings	151
7.4	Further information about the use of AFGO	153

7.1 General

Additional Fresh Gas Outlet (AFGO) is an option to the anesthesia system that allows use of an external partial rebreathing system, such as Bains, Jackson Rees or Mapleson D.

When using AFGO the circle system is bypassed and the following can be set:

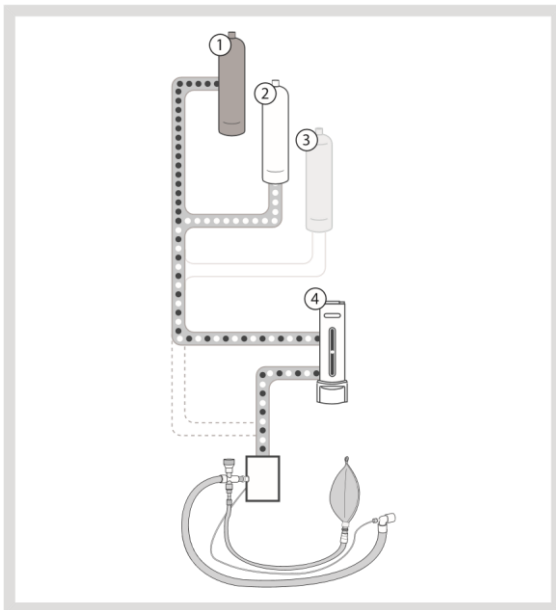
- Gas mix
- O₂ concentration
- Fresh gas flow
- Agent concentration

Fresh gas flow ranges are from 1.0 to 20 l/min when using the AFGO and the default settings are the same as in Manual ventilation.

Pre-selecting or activating AFGO in the ventilation dialog window requires user confirmation. On selecting the AFGO touchpad, a pop-up dialog appears asking the user to confirm the action.

When the MAN/AUTO switch is set to AUTO it is possible to preset MAN mode to AFGO in the ventilation dialog window. AFGO is activated when the MAN/AUTO switch is then turned to MAN.

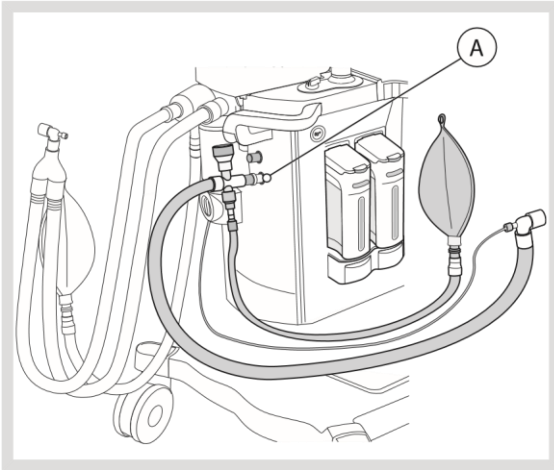
When the MAN/AUTO switch is set to MAN it is possible to activate AFGO in the ventilation dialog window.



1. Air gas module
2. O₂ gas module
3. N₂O gas module
4. Vaporizer

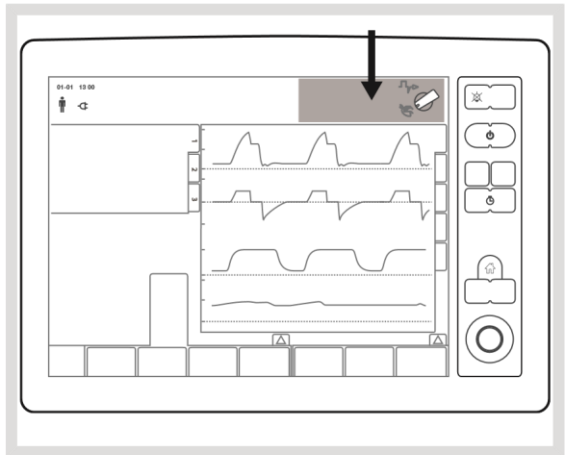
7.2 Preparations

Before using the AFGO outlet, make sure that normal system checkout procedures have been performed and the following requirements are fulfilled:



1. An external breathing circuit is connected to the AFGO outlet (A).
2. The external breathing circuit is connected to the patient gas monitor, or to an external patient gas monitor.
3. The function of the external breathing circuit has been tested according to hospital routines.

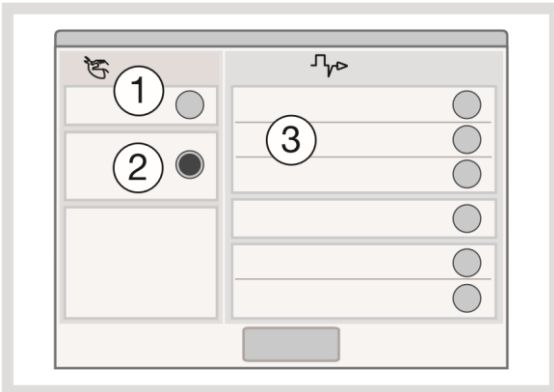
CAUTION: Only external breathing circuits equipped with a pressure relief valve or a manual bag with an opened end shall be connected to the AFGO outlet.



When the AFGO function is installed on the system, AFGO ventilation is available for selection in the dialog window produced by pressing the ventilation mode area.

7.3 Manage AFGO settings

Pressing the ventilation mode touch pad produces the ventilation mode selection dialog:



1. Manual
2. Additional Fresh Gas Outlet (AFGO)
3. Automatic ventilation modes

7.3.1 Preset and activate AFGO in Standby

- Select 'Additional Fresh Gas Outlet (AFGO)' and confirm selection in the dialog
- Set the AFGO ventilation parameters by selecting each parameter and adjusting them to the desired value using the rotary knob
- Adjust the external circuit pressure relief valve to an appropriate value.
- Turn the MAN/AUTO ventilation switch to MAN
- To activate AFGO when preset, press 'Start case' membrane button

- Verify there is flow at the Y-piece of the external breathing circuit connected to the AFGO outlet

Note that gas metrics and RR are displayed

when AFGO is active.

Ensure that surplus gas is evacuated by the hospital's gas evacuation system.

Depending on the type of external system and current fresh gas settings, expired gas may be diluted with fresh gas, causing the measured gas metrics displayed on the control panel to be lower than their actual values.

Ensure that the fresh gas flow is sufficient to avoid unwanted CO₂ rebreathing.

7.3.2 Activate AUTO from AFGO

Parameters for Automatic Ventilation can be preset during AFGO via the ventilation dialog window.

- Turn the MAN/AUTO ventilation switch to AUTO.

- Switch breathing circuits at the y-piece from the external breathing system to the circle system. Make sure that the sampling line is still connected.

The fresh gas flow is automatically temporarily increased to fill the circle system with the correct gas mix. This procedure lasts for a few seconds, during which time a dialog is visible.

Fresh gas mix is delivered according to the preset parameters for the selected automatic ventilation mode.

Waveforms and metrics according to selected automatic ventilation mode are displayed.

Clinical alarms are pre-muted for a duration of 30 seconds after the activation of Auto mode, indicated by the pre-mute symbol and count-down timer on the control panel screen. Check and adjust relevant settings and alarms.

WARNING! Check upper pressure limit.

7.3.3 Activate AFGO from AUTO

- Ensure that 'Additional Fresh Gas Outlet (AFGO)' is pre-set in the ventilation mode touch pad.
- Turn the MAN/AUTO ventilation switch to MAN.
- Verify there is flow at the Y-piece of the external breathing circuit connected to the AFGO outlet

- Switch breathing circuits at the endotracheal tube from the circle system to the external breathing system connected to AFGO. Transfer the sampling line if needed.

Curves and metrics associated with ventilation using AFGO are displayed.

Clinical alarms are pre-muted for a duration of 30 seconds after the activation of AFGO, indicated by the pre-mute symbol and count-down timer on the control panel screen.

Check and adjust relevant settings and alarms.

7.3.4 AFGO to End Case

Press 'End case' membrane button.

Confirm the dialog and press 'Accept' to return to Standby.

7.4 Further information about the use of AFGO

Gas scales and waveforms for gases are displayed according to user configuration. The sampling line must be connected to the breathing system to display valid data.

For more information on waveforms refer to Chapter 5, page 107.

The following functions are **not accessible** when using the AFGO:

- Loops
- Waveforms for flow and volume
- Ppeak, flow and volume metrics

Perform the following steps if a manual recruitment maneuver is required during automatic ventilation:

- Press the ventilation mode touch pad
- Select 'Manual' in the dialog and accept the confirmation dialog.
- Turn the MAN/AUTO switch to 'MAN'
- Perform the recruitment maneuver
- Adjust PEEP and switch back to AUTO using the MAN/AUTO switch

WARNING! The emergency flow is always delivered via the patient cassette outlet and thus the patient tubings must always be connected to patient cassette outlets when emergency ventilation is used.

8 Membrane buttons Table of contents

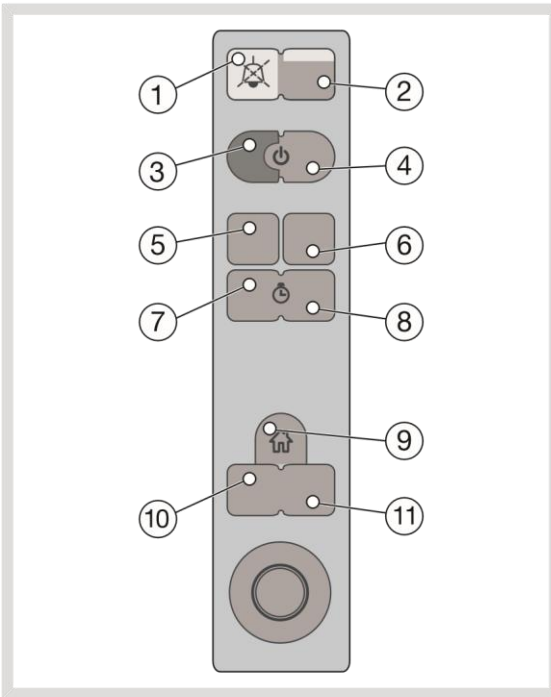
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8.1 Overview

8.2 Audio pause

The membrane buttons are used to quickly The Audio pause membrane button has the access information in the system and/or action following applications:

certain specific functions.

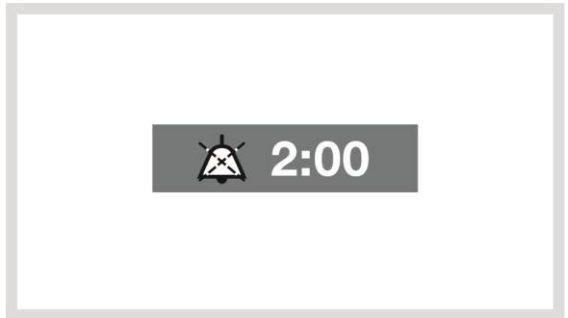


1. Audio pause
2. Alarm profile
3. Start case
4. End case
5. Save screen
6. Trends
7. Start/Stop timer
8. Reset timer
9. Home
10. Screen layout
11. Menu

page .

- Mutes active alarm sound signals for two minutes
- Pre-mutes alarms activated within a two minute period.
- Mutes some alarms until the alarm condition resolves. This action must be confirmed via a dialog window.

When pressed, a timer counting down from two minutes is displayed in the 'Audio pause and Countdown timer' area.

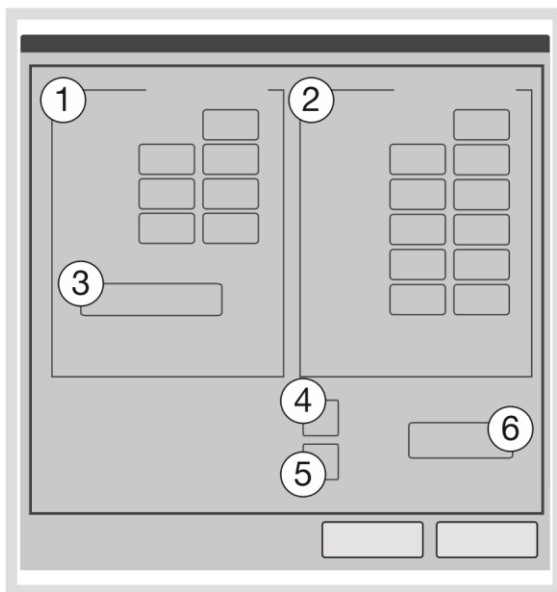


For a detailed description, see Chapter 10,

Refer to the following sections for full details on each membrane button.

8.3 Alarm profile

Shows all applicable alarms for the currently **case** selected ventilation mode (MAN or AUTO) along with their set upper and lower limits.



1. Ventilator alarms
2. Gas alarms
3. Alarm auto set
4. Mute Apnea alarms
5. HLM mode (CPB mode)
6. Alarm sound level

For a detailed description, see Chapter 10, page .

8.4 Start/End case

8.4.1 Start

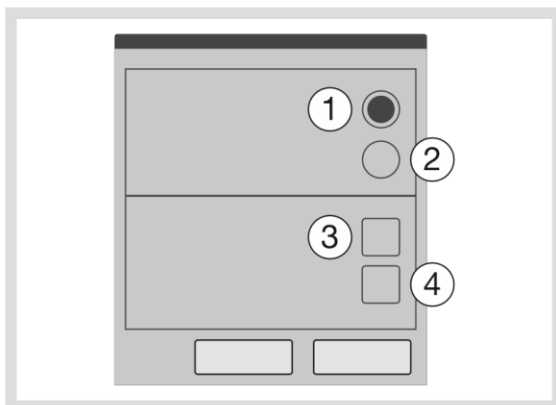
Starts a new patient case and activates the preset gas and ventilation settings depending on the setting of the AUTO/MAN ventilation

The agent concentration setting is only accessible during an ongoing patient case and must be manually set from its default position; OFF.

8.4.2 End case

Provides various options for completing a patient case.

The following is displayed:



1. Go to Standby switch.
2. Go to Standby and return to startup configuration
3. Save patient trends and clinical logs to USB memory
4. Delete patient log and trend data

Go to Standby

Return to Standby mode.

All settings (gas, ventilation and alarm limits) are unaltered, apart from agent concentration, which is set to OFF.

Go to Standby and return to startup configuration

Return to Standby and restore the system to the startup configuration.

The startup configuration is set using Menu/Service & Settings.

Save patient trends and clinical logs to USB memory

Saves all patient trends and clinical logs to an inserted USB memory device.

Delete patient trends and clinical logs

The data is deleted from working memory, but can still be accessed from Menu/Service and Settings.

When a patient case is completed and the system is switched 'Standby', the vaporizer is automatically set to 'OFF'.

8.5 Save screen

Saves a copy of the screen to a USB memory device.

A USB memory device must first be inserted and synchronized with the system. A system message will appear when the device is ready for use.

The user will be informed in the system message area that a picture has been taken. For the picture to be successfully saved, the system must be left alone until the message appears. The picture file will be given a unique ID number.

8.6 Trends

Pressing Trends opens a new window displaying the latest trend data in either numerical or graphical form.

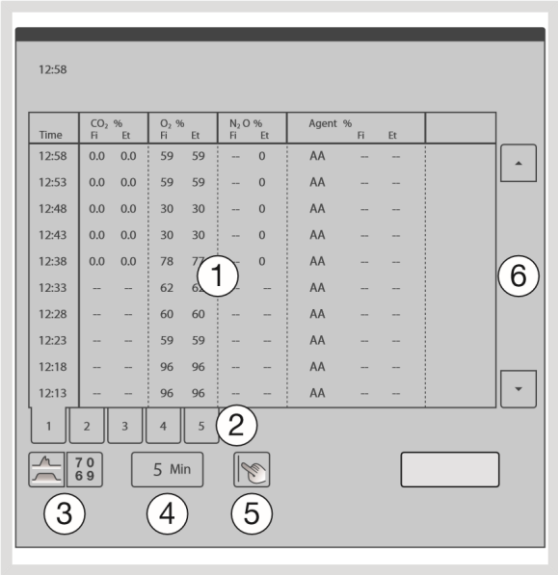
Switch between the graphic and numeric trends window by pressing the graphic (left) or numerical (right) touch pad button located at the lower left in the trends window.

The time scale is changed by pressing the time interval touch pad located under the trended data and using the rotary knob to increase or decrease the time.

The information displayed in the window is only updated by closing and re-opening the window.

Trend data is stored in the working memory for 24 hours or until erased by the End case function. Trend data can still be accessed from the Menu/Service and Settings.

8.6.1 Numerical display



1. Numerical display
2. Trend window page tab
3. Graphic/numerical touch pad
4. Time interval touch pad
5. Cursor
6. Scroll bar

The different trends are displayed as five pages of information. Each page is selected using the page tabs below the trend data.

In the numerical view, the trend data is displayed for the last 24 hours.

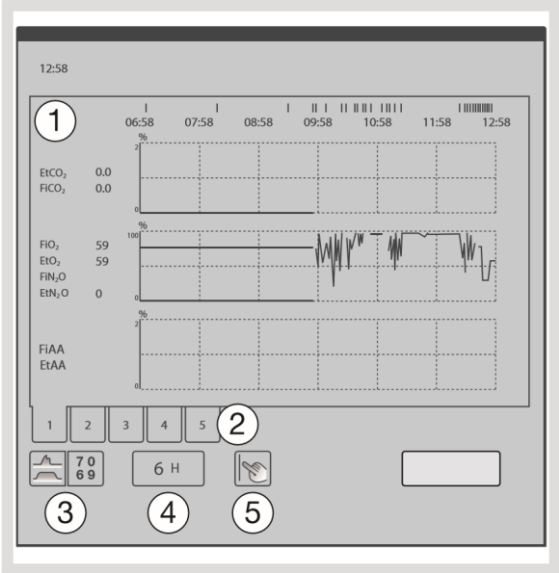
Different intervals can be selected from 1 minute to 60 minutes, with 10 values displayed at one time in the window.

A blue horizontal line marking the top row in the trends window can be activated and moved up and down to facilitate reading. It is activated either by pressing the trends window or by pressing the cursor touch pad. The line is moved using the rotary knob, or by pressing and moving a finger across the trends window.

8.6.2 Scroll bars

Scroll bars are used to navigate through listed information or available values. Select the scroll bar, positioned to the right of the window, by pressing it on the touch screen. Press the up and down arrows on the scroll bar, or use the rotary knob, to move up and down the list.

8.6.3 Graphical display



- 1. Graphical display
- 2. Trend window page tab
- 3. Graphic/numerical touch pad
- 4. Time interval touch pad
- 5. Cursor

The different trends are displayed as five pages of information. Each page is selected using the page tabs below the trend data.

The time scale for the graphs can be changed from 1 hour up to 24 hours.

A blue vertical line marking the latest sampling point in the graphical display window can be activated and moved left and right to facilitate reading the waveforms. It is activated either by pressing the display window or by pressing the cursor touch pad. The line is moved using the rotary knob, or by pressing and moving a finger across the display window. The exact

time and measured value corresponding to the current position of the line are displayed in the top left, and left part of the display window respectively.

At the top of the window, log events are displayed as small white vertical lines. These events are located at positions that correspond to the occurrence of the event. They can only be accessed and examined by moving the vertical line to match their position.

When a log event is accessed by the cursor, the related log information is displayed in the trend window.

8.6.4 Page information

Trended measurements are displayed on each of the five pages according to the table below:

Trend data - page information	
Page 1	Ppeak, Pmean, PEEP, Pplat, MVi, MVe
Page 2	RR, VTi, VT _e , VT/PBW
Page 3	V _{ee} , Ti, Ti/Tot, I:E ⁴ , Cdyn
Page 4	CO ₂ %, O ₂ %, N ₂ O %, AA % (Fi + Et)
Page 5	MAC, FGF

8.7 Timer

The Timer is controlled by two membrane buttons - Start/Stop and Reset.

Press Start/Stop once to start a timer visible at the top right of the screen. Press again to stop the timer.

Press Reset to reset the timer to zero.

The timer will count up to 24 hours and then stop.

4. Not shown when graphical display is selected

Once reset or stopped, the timer will remain displayed for 5 minutes.

Press Reset again to hide the timer.

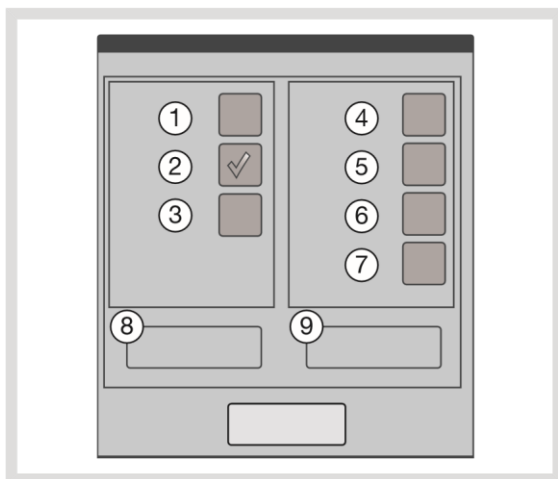
8.8 Home

Cancels current window settings, closes the window and returns to the main screen.

Pressing Home will unlock a control panel locked via the Lock panel option under Menu.

8.9 Screen layout

Allows the user to configure the screen layout and brightness. Layout options include display loops and rotameter, waveform combinations and the scaling of these.



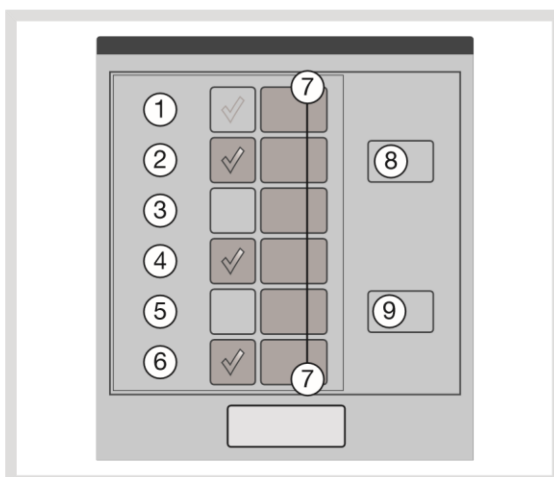
1. Display rotameter
2. Display numerical gas pressure
3. Display predicted body weight (PBW)
4. Display short trends tab
5. Display loops tab
6. Display volume reflector indicator (VRI) tab
7. Display graphical gas pressure tab
8. Waveforms and Scales settings
9. Screen brightness

The screen layout settings are not saved at system shutdown. However, the startup configurations for waveform and rotameter display can be selected using Menu/Service & Settings.

8.9.2 Screen brightness

The backlight on the control panel can be

8.9.1 Waveforms and scales



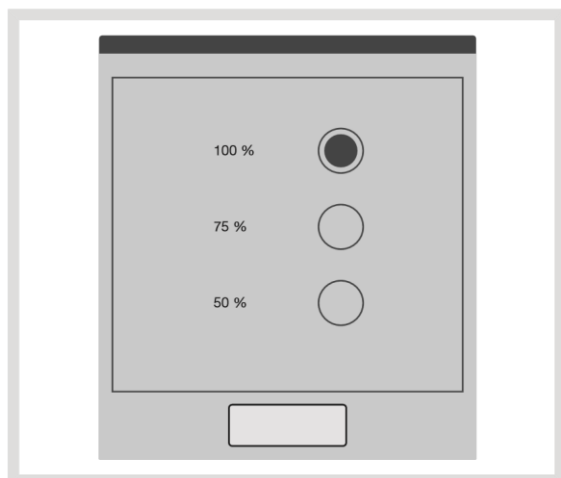
Waveform scaling can be set to manual or automatic. Waveform sweep speed can be set to either 5, 10 or 20 mm/s (0.2, 0.4 or 0.8 in./s)

The screen can be configured to display up to six different waveforms:

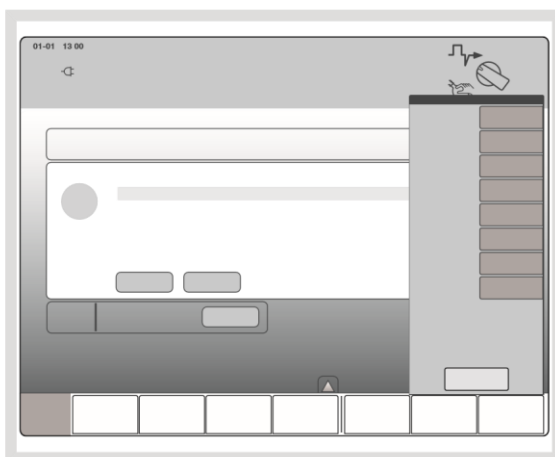
1. Pressure (mandatory)
2. Flow
3. Volume
4. CO₂
5. O₂
6. Anesthetic agent concentration
7. Waveform scaling
8. Sweep speed
9. Short trend range (10, 30, or 60 minutes)

8.10 Menu

Displays the main menu.



adjusted, depending on the surrounding lighting,
to provide the optimal screen readability.

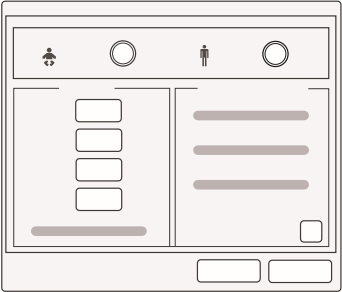


The menu gives access to the following
functions:

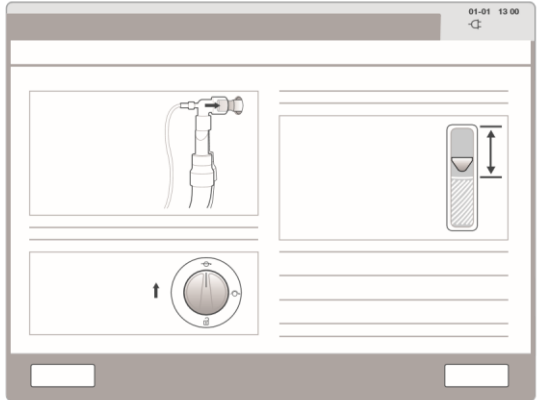
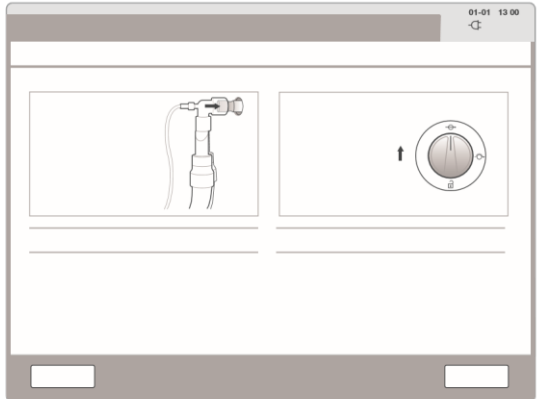
- Patient settings Press to adjust the brightness of the screen between 50% and 100%. P
- System checkout S
- System info

ogs	•	L
ave & delete data	•	S
anel lock	•	P
ervice & Settings	•	S

8.10.1 Patient settings

Menu - Patient settings	
Used to select patient settings. For a complete description, see section Patient settings on page 74.	

8.10.2 System checkout

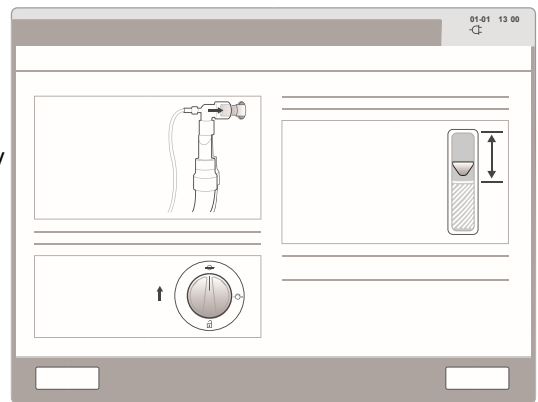
Menu - System checkout	
Result Brings up a detailed summary of the last System check, including leakage info and circuit compliance calculations, and the result of the System check before that.	
Full check Starts a full system checkout. Full check, Leakage check and Vaporizer check are only available in Standby mode See Chapter 4 for details for details on all system checks.	
Leakage check Performs a leakage check of the breathing system. The circuit compliance compensation is re-calculated during the Leakage check and the log is updated with the new measured values for leakage.	

Menu - System checkout

Vaporizer check

Performs a Leakage check and a Vaporizer check.

The additional Vaporizer check ensures functionality of the vaporizer and that no agent escapes into the operating room.



8.10.3 Status

Menu - Status

System info

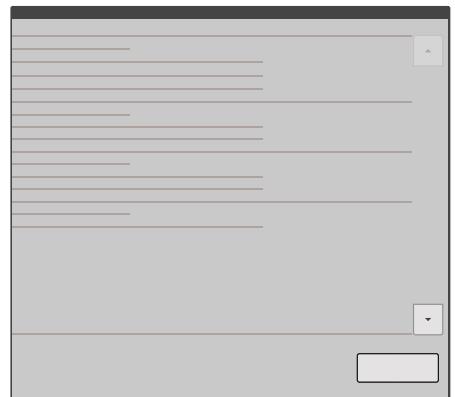
The following system status areas are covered:

Page 1:

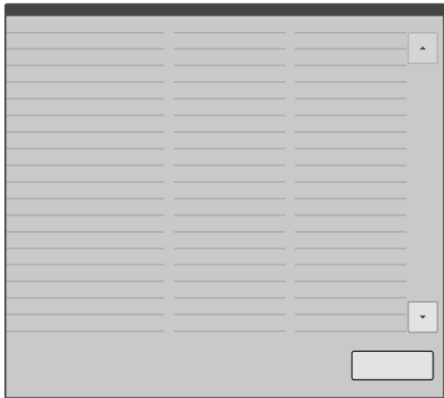
- Central gas pressure
- Vaporizer 1
- Vaporizer 2
- Battery status

Page 2:

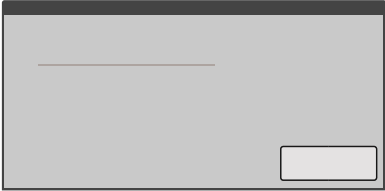
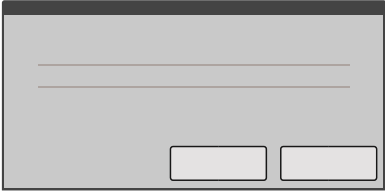
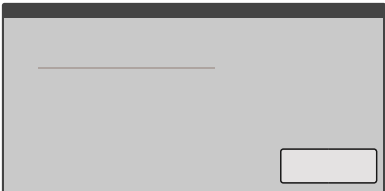
- System
- Vaporizer 1
- Vaporizer 2
- Patient cassette



8.10.4 Logs

Menu - Logs	
Alarms Displays the alarm log, with the most recent at the top of the list.	
All logs Displays a list of all logs - System checkout, Alarms, Ventilation settings and Events - in chronological order with the most recent at the top of the list.	
Agent Usage The list displays the amount of consumed agent for each of the twenty last performed patient cases. The oldest entry is automatically removed when the list is full and a new patient case is completed. The total agent usage since the last user reset is shown under the list. Pressing 'Reset total agent usage' will reset the total agent usage and the time/date information. This will not reset the list specifying agent consumption for each patient case.	

8.10.5 Save & delete data

Menu - Save & delete data	
Save trends & logs to USB Data can be stored onto a USB memory device for storage or further analysis.	
Delete trends & logs Delete trends and logs data from the memory. Trends and data in the working memory are either deleted by actively selecting that option in the 'End case' window, or automatically at system shut down.	
Save agent usage to USB Agent usage data is saved to the USB memory device.	

8.10.6 Service and settings

Only available in Standby mode.

Press the Service & Settings menu option. The following screen is displayed:



Press the required user category.

There are four user levels available for user category 'Technician'. This manual describes user level 'Biomed' only. The remaining three levels are intended only for personnel trained and authorized by MAQUET.

For Biomed, select Technician and thereafter enter the 4 digit code 1973 (default).

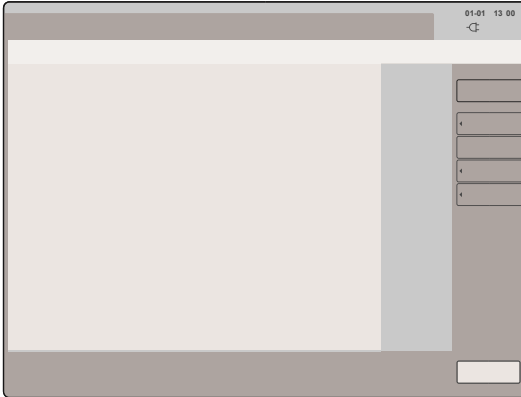
The system now displays the user category selected, along with the available menu options, i.e. the Overview screen.

The following sections regarding system overview are applicable to Biomed user

category. Less options are available for Clinician user category.

Overview

Confirms the user category selected and menu options available.



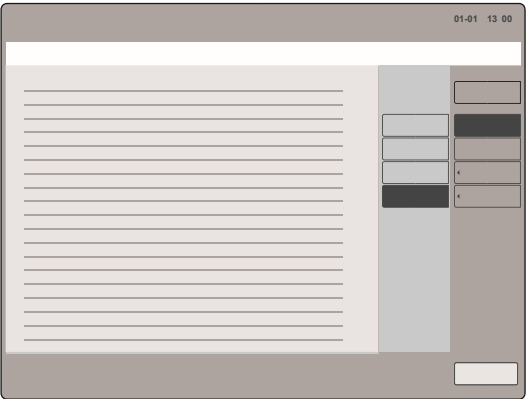
The following can be performed from the Overview menu by selecting from the list of options to the right:

- Status - View system information
- Logs - View and save logs to USB memory
- Startup configuration - Set system configuration
- Settings - Set date and time

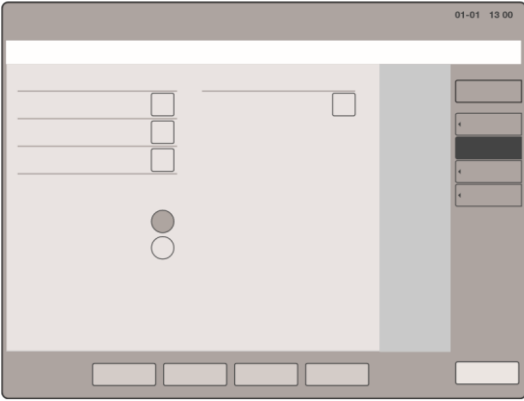
The menu options available will depend on the user category selected.

Status

Service and settings - Status	
System status A detailed view of the system split over three pages: • Gas pressure and vaporizer status • System version, operating time and patient cassette status • Panel screen status	
Hardware Information about the physical equipment comprising the anesthesia system, e.g. patient cassette.	
Software Information about system software version.	

Service and settings - Status	
Options Displays currently installed optional equipment.	

Logs

Service and settings - Logs • Different logs can be viewed or saved to a USB memory device using the touch pad options at the bottom of the screen. • Logs can be individually selected by checking the grey box to the right of each log option. Pressing the Select all touch pad will select all logs. • The latest log is automatically displayed or stored unless a specific time is specified by means of the Time selection touch pad.	
The following logs can be accessed from the logs window: • Alarms (shown to the right) • Ventilation settings • Events • Configuration (service log)	

Startup configuration

Service and settings - Configuration

General

(Read only)
Used to specify the date and time format, type of decimal separator, language and location used to specify the medical gas color code.

(User configurable)
Used to specify when to initiate remote services. The following options exist:

1. Manually from Standby

2. Prompted after SCO

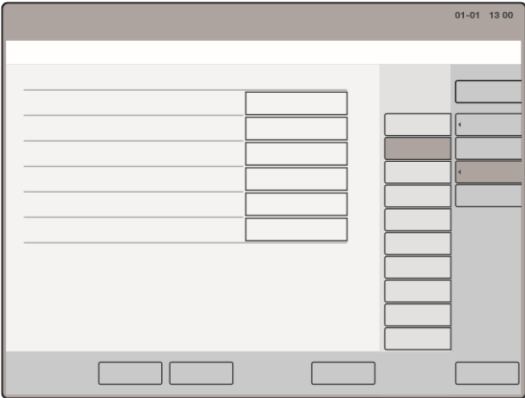
3. Always after SCO

4. Always after End case



Units

Used to specify the units to be used for displayed values.



Screen layout

Used to configure the layout of optional information presented on the control panel. The following tabs/information can be toggled ON and OFF:

1. Loops

2. Gas mix rotameter

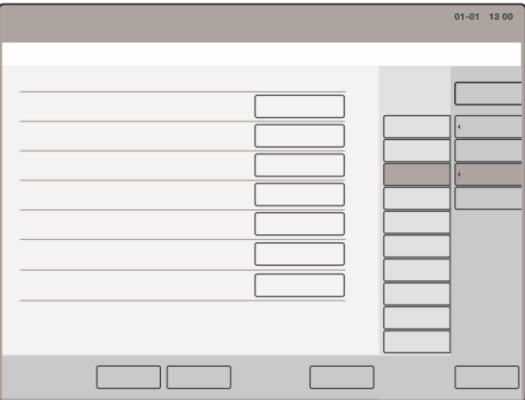
3. Numerical gas pressure

4. Graphical gas pressure

5. Short trends

6. Volume reflector indicator (VRI)

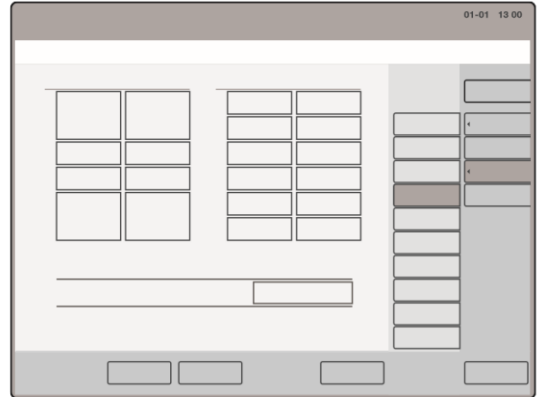
7. Predicted body weight (PBW)



Service and settings - Configuration

Displayed measurements

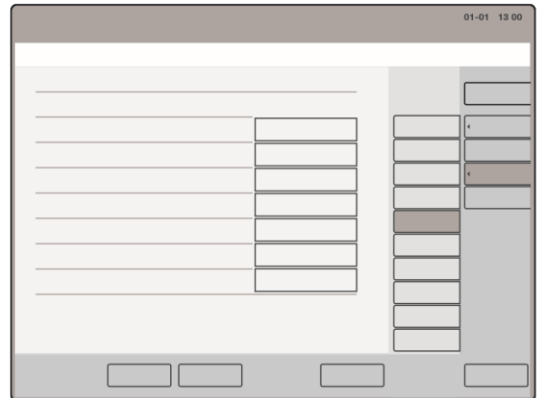
Used to select what measurements to display on the screen, and where to display them. Note that some positions are reserved for obligatory measurements, e.g. MVe and RR.



Waveforms

Used to manage settings for waveforms, loops and the rotameter.

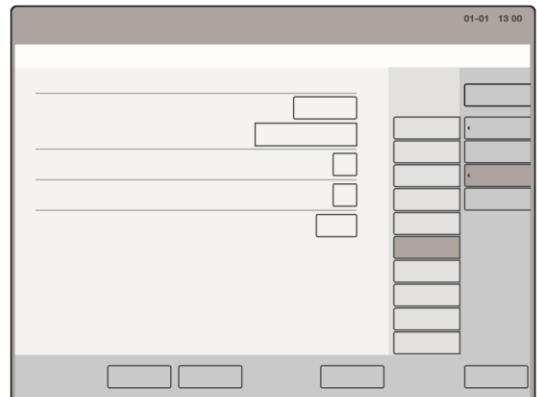
Waveform 1 is always set to Pressure.



Alarm general

Used to specify the sound level between 20% and 100% of maximum volume (approx. 85 dB)

Checking 'APNEA mute' and 'HLM/CPB mode' will make these available for activation in the alarm profile window.



Service and settings - Configuration

Adult AUTO alarm limits

Used to specify the required alarm limits:

- Page 1(2). Ventilation alarm settings (Adult patient category selected)
- Page 2(2). Gas alarm settings (Adult patient category selected)

01-01 13 00

The screen displays a grid of input fields for ventilation parameters. On the right side, there is a vertical column of buttons, including a '+' button at the top and a '-' button below it. The bottom of the screen features a row of four buttons.

01-01 13 00

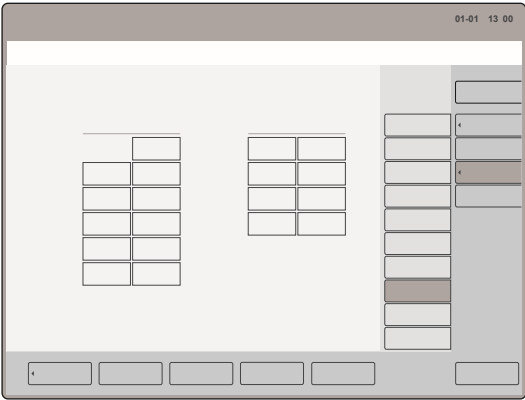
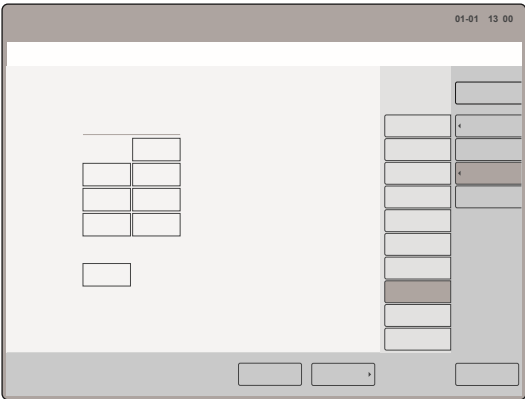
The screen displays a grid of input fields for gas parameters. On the right side, there is a vertical column of buttons, including a '+' button at the top and a '-' button below it. The bottom of the screen features a row of five buttons.

Service and settings - Configuration

Infant AUTO alarm limits

Used to specify the required alarm limits:

- Page 1(2). Ventilation alarm settings (Infant patient category selected)
- Page 2(2). Gas alarm settings (Infant patient category selected)



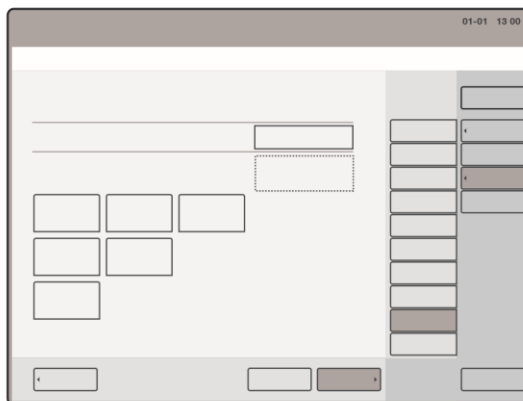
Service and settings - Configuration

Ventilation and gas

Used to specify the startup ventilation and gas configuration.

1. Begin by pressing the Start touch pad in the Config: Startup window.
2. Select any of the available parameters and toggle through available options using the rotary knob. Continue by pressing Next.
 - Patient range
 - Infant age
 - Infant, VT/PBW
 - Adult age
 - Adult, VT/PBW
 - Volume setting
 - Breath cycle setting
 - Activate Insp./Exp. hold (option)
 - PBW-based values used automatically
3. Press the ventilation mode touch pad and use the rotary knob to select the desired mode. Associated ventilation parameters become available and can be changed. Continue by pressing Next.

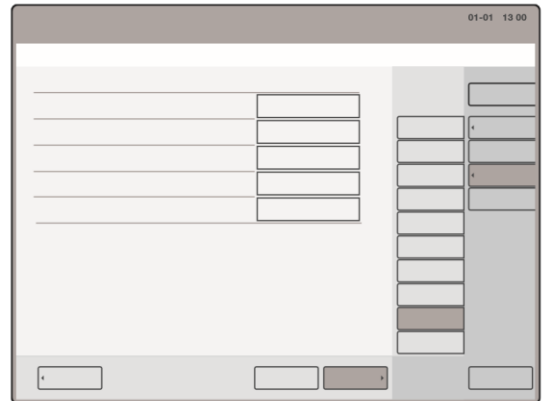
Pressing the Factory default touch pad will reset the parameters to default values, including default ventilation mode for each patient category.



Service and settings - Configuration

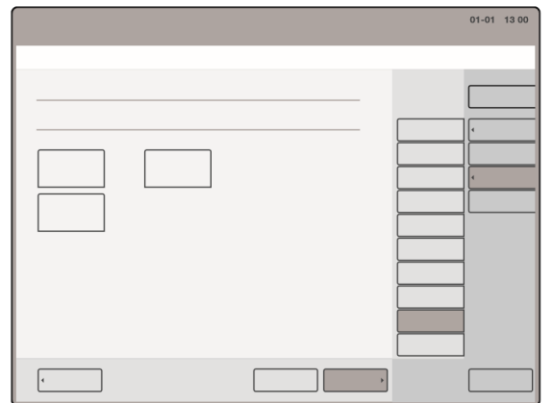
4. Select startup conditions for gas and gas delivery. The following can be activated or preset:
- Active vaporizer slot at startup
 - N₂O delivery
 - Active N₂O and Air cylinders during SCO
 - O₂Guard

N₂O and Air backup cylinder checks can each be individually disabled for the system checkout procedure. Continue by pressing Next.



5. Select fresh gas delivery startup values:
- O₂ concentration
 - Gas mix
 - Fresh gas flow

Continue by pressing Next.



6. The two last windows summarizes options and parameter settings made during the Startup configuration. Press Accept to confirm the new configuration or press Previous to go back and modify. Configuration setups can also be stored to, or copied from, a USB memory by means of the Save all to USB memory device and Copy from USB memory touch pads.

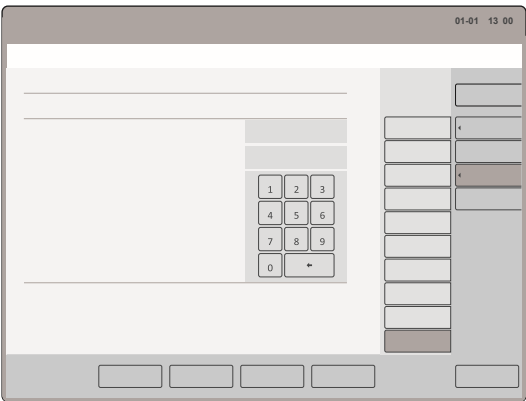
If the Startup configuration is modified, the system must be restarted for the changes to take effect. Any changes from the default parameter settings are only applicable to the currently selected ventilation mode and patient category.



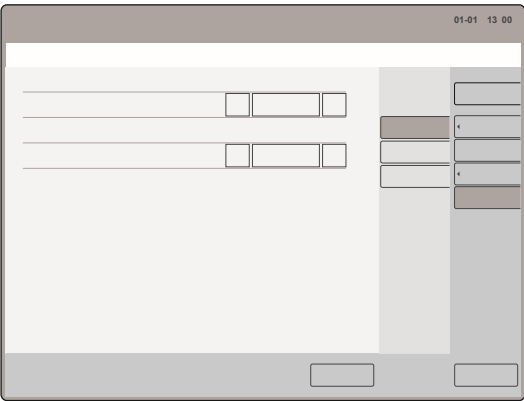
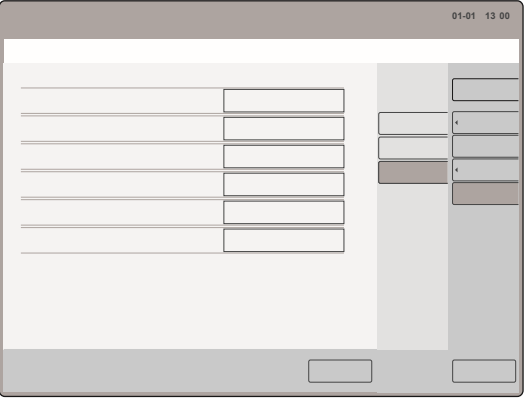
Service and settings - Configuration

Biomed access code

Used to change the access code.



Settings

Service and settings - Settings	
Sets the date and time for the system.	
Total agent usage since last Service&Settings reset	
Network settings for remote services	

8.10.7 Panel lock

Used to deactivate the touch pad functionality of the touch screen and membrane buttons. This is primarily used when cleaning the screen.

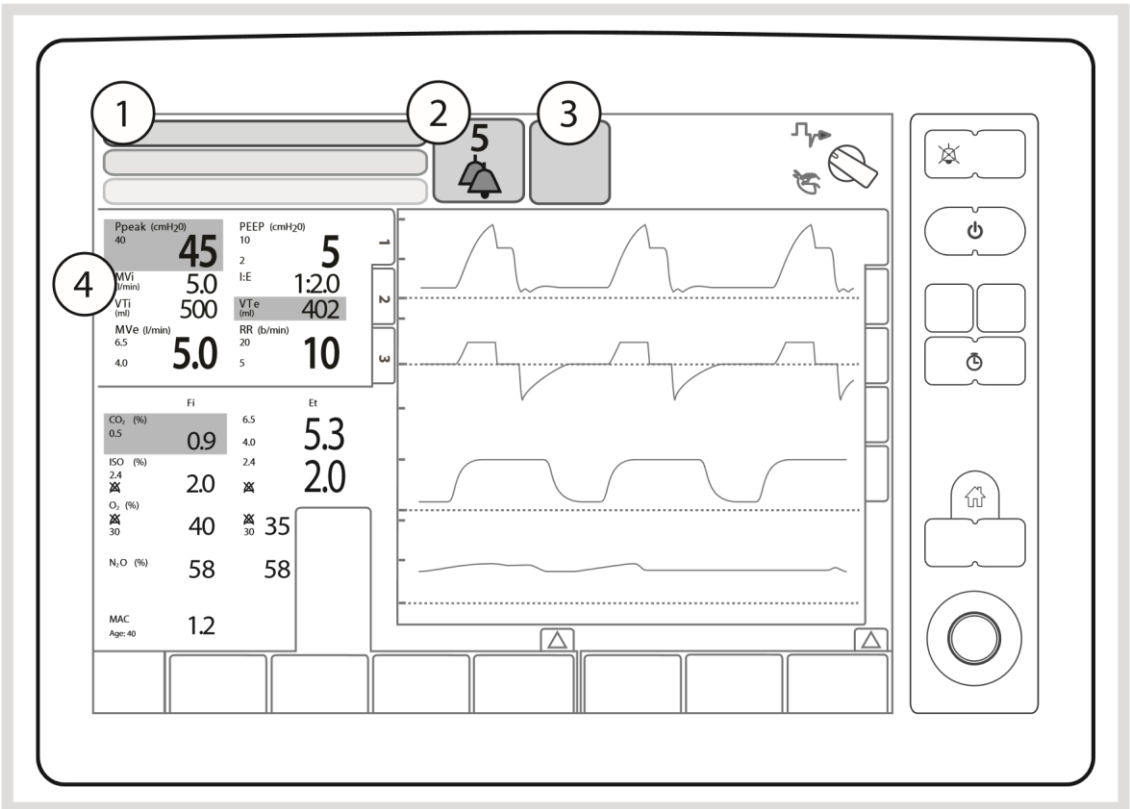
To unlock, press the Home membrane button.

9 Alarms and patient safety Table of contents

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9.2	Setting alarm limits	187
9.3	Audio pause	192
9.4	Alarm messages	196
9.5	Safety valves	201
9.6	AFGO alarms and alarm limits	202
9.7	Power supply	203
9.8	Gas supply	204

9.1 System alarms

FLOW-i has all alarms displayed on the control panel:



1. Alarm message area
2. Current alarm touch pad
3. Audio pause and countdown timer
4. Ventilation and gas measurement area with parameters affected by active alarms highlighted.

9.1.1 System alarms overview

The system has a number of alarms to alert the staff of changes in patient condition or possible system malfunctions.

Alarms are visually displayed on the screen. With each alarm, a sound is also emitted from

WARNINGS!

- Ensure that all alarm limits and ventilator settings are appropriately set before connecting the system to the patient.
- If several similar or identical systems are used within a ward, make sure to note if different alarm presets are used.

the system, indicating the severity of the alarm.

The alarm sound will cease once the alarm condition is no longer valid.

Each alarm is divided into groups of priority (see below). Each group has a characteristic sound associated with it, and is displayed in the alarm message area using color to reflect the severity of the alarm.

Current alarm touch pad	
	Single alarm condition
	Multiple alarm condition

Similarly, parameters affected by active alarms are highlighted in the ventilation and gas measurement area using the same color scheme.

Two classes of alarms exist:

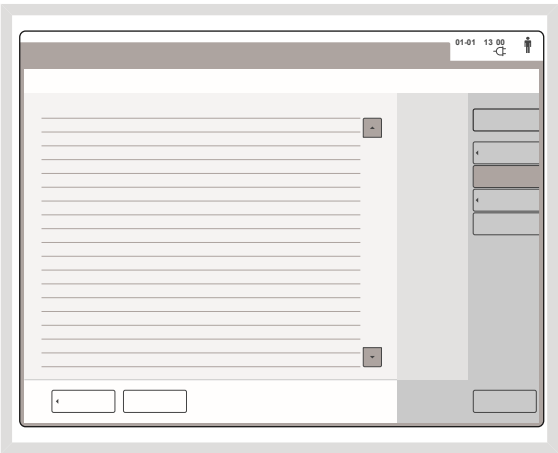
1. Clinical alarms
2. Technical alarms

Clinical and technical alarms are divided into groups of priority:

- High priority alarm (black text on red background, flashing)
- Medium priority alarm (black text on yellow background, flashing)
- Low priority alarm (black text on blue background)

Clinical alarms are related to the status and near surroundings of the patient. These can be corrected by the operator.

Technical alarms are activated when a technical error requiring the operators



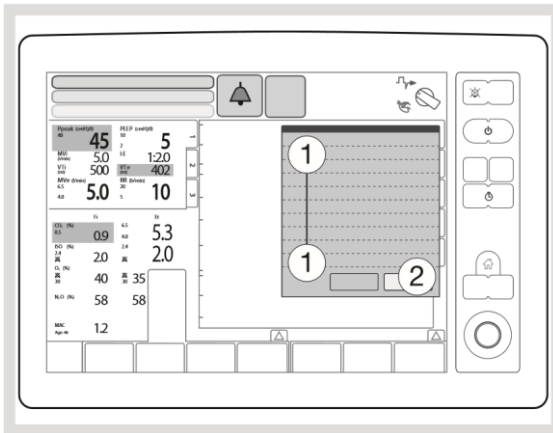
attention occurs.

All technical alarms are displayed using the prefix 'TE' along with a reference number linked to the specific problem, e.g. 'TE12: Ventilation error'.

9.1.2 Current alarms

Current alarms are displayed in the alarm message area.

The Current alarm touch pad indicates either a single alarm condition or multiple alarms. Press the Current alarm touch pad to show the current alarms window:



1. Active alarms
2. Alarm log touch pad

Ten alarms can be displayed in the Current alarms window, with the highest priority ones at the top of the list.

9.1.3 Alarm log

Press the Alarm log touch pad in the Current alarms window. The alarm log is displayed with the most recent alarms at the top of the list.

The Alarm log can also be displayed via the Menu membrane button.

Alarm log entries are maintained until the list is full, after which the oldest entries are automatically removed as new alarm entries are added.

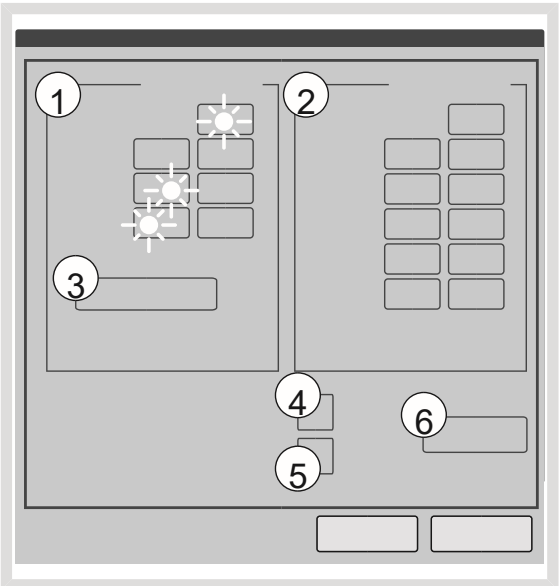
The alarm log is not affected by system shutdown or a temporary loss of power (supply mains and/or battery power).

The time and duration of system power down is recorded in the alarm log.

9.2 Setting alarm limits

The alarm limits for many of the parameters displayed on the touch screen can be user-specified. The alarm profile window is accessed via the Alarm profile membrane button or by pressing anywhere on the ventilation and gas measurements area.

The Alarm profile window is displayed.



1. Ventilator alarms
2. Gas alarms
3. Alarm auto set
4. Mute Apnea alarms
5. HLM mode (CPB mode)
6. Alarm sound level

Any active alarms will highlight associated parameter values in the Alarm profile window using red, yellow or blue to indicate the alarm priority.

The following can be performed from the Alarm profile window:

- Set upper and lower alarm limits
- Alarm sound level
- Enable/Disable certain alarms
- Alarm autoset

The contents of the Alarm profile window will differ depending on whether Manual or Automatic ventilation is selected, and the selected patient category. The default set-up is specified via Menu/Service and Settings/Configuration.

Any changes made to the settings in the Alarm profile window can be saved or deleted by use of the prompt that appears when the patient case is ended (End case), and when returning to Standby mode. When the system is shutdown, all settings return to their original status in Startup configuration.

The Alarm profile window can be accessed at any time by pressing anywhere in the measurement area of the touch screen.

In case of a total loss of power, i.e. mains power and battery power, alarm settings and stored patient data remain in effect if the duration of the power loss is less than 30 seconds.

Upper and lower alarm limits

To change a parameter setting, press its touch pad and use the rotary knob to increase or decrease the setting. A parameter can also be selected by moving through the values using the rotary knob, and pressing the rotary knob

when the blue box highlights the desired parameter touch pad.

New parameter settings are confirmed and applied to the system when the Accept touch pad is pressed.

Alarm sound level

The alarm sound level can be set from 20% (approx. 55 dB) to 100% of the maximum (approx. 85 dB).

WARNING! Ambient sounds levels may impede the recognition of auditory alarm signals. Make sure the alarm sound level is set sufficiently high.

Enabling/Disabling certain alarms

Some alarms can be disabled by selecting the associated parameter and turning the rotary knob counter-clockwise (lower limit settings) or clockwise (upper limit settings) until the Alarm OFF symbol appears. These include:

- FiO₂: High
- EtO₂: High
- EtO₂: Low
- EtCO₂: Low
- FiAA: Low
- EtAA: Low

In manual (MAN) mode, the following additional alarms can be disabled:

- Expiratory Minute Volume: Low
- Expiratory Minute Volume: High
- Respiratory Rate: Low

- Respiratory Rate: High
- Apnea

5. Available units include cmH₂O and mbar

Alarm autose

In controlled ventilation modes, the upper and lower limits of the ventilation metric alarms can be automatically set by the system during Run mode. Gas metric alarms cannot be automatically set this way.

The autose limits are derived from the measured patient values displayed on the screen. A percentage of the displayed value, or a fixed number, is either added to, or subtracted from the displayed value to give the upper and lower alarm limits. See the following tables for specific values.

9.2.1 Ventilation alarm limit settings

Alarm type

Alarm	Alarm type	Setting unit	Lower limit	Upper limit	Default	Step	Autose
Ppeak	Upper	cmH ₂ O ⁵	10 cmH ₂ O	120 cmH ₂ O	40 cmH ₂ O	1 cmH ₂ O	+10 cmH ₂ O (min 35)
MVe	Lower	l/min	0.5 l/min OFF (MAN)	40 l/min	2 l/min OFF (MAN)	0.5 l/min	-50 %
MVe	Upper	l/min	0.5 l/min	60 l/min OFF (MAN)	20 l/min OFF (MAN)	0.5 l/min	+50 %
RR	Lower	B/min	1 b/min OFF (MAN)	140 b/min	4 b/min OFF (MAN)	1 b/min	-40 %
RR	Upper	B/min	1 b/min	140 b/min OFF (MAN)	20 b/min OFF (MAN)	1 b/min	+40 %
PEEP (AUTO)	Lower	cmH ₂ O ⁶	0 cmH ₂ O	47 cmH ₂ O	2 cmH ₂ O	1 cmH ₂ O	-3 cmH ₂ O
PEEP (AUTO)	Upper	cmH ₂ O ⁶	0 cmH ₂ O	55 cmH ₂ O	10 cmH ₂ O	1 cmH ₂ O	+5 cmH ₂ O

(Adult)

9.2.2 Ventilation alarm limit settings (Infant)

Alarm	Alarm type	Setting unit	Lower limit	Upper limit	Default	Step	Autoset
Ppeak	Upper	cmH ₂ O ⁶	10 cmH ₂ O	90 cmH ₂ O	25 cmH ₂ O	1 cmH ₂ O	+10 cmH ₂ O (min 35)
MVe	Lower	l/min	0.01 l/min OFF (MAN)	20 l/min	0.8 l/min OFF (MAN)	0.01-0.1 l/min	-50 %
MVe	Upper	l/min	0.01 l/min	30 l/min OFF (MAN)	3 l/min OFF (MAN)	0.01-0.1 l/min	+50 %
RR	Lower	B/min	1 b/min OFF (MAN)	140 b/min OFF (MAN)	15 b/min OFF (MAN)	1 b/min	-40 %
RR	Upper	B/min	1 b/min	140 b/min	50 b/min OFF (MAN)	1 b/min	+40 %
PEEP (AUTO)	Lower	cmH ₂ O ⁶	0 cmH ₂ O	47 cmH ₂ O	1 cmH ₂ O	1 cmH ₂ O	-3 cmH ₂ O
PEEP (AUTO)	Upper	cmH ₂ O ⁶	0 cmH ₂ O	55 cmH ₂ O	7 cmH ₂ O	1 cmH ₂ O	+5 cmH ₂ O

6. Available units include cmH₂O and mbar

9.2.3 Gas alarm limit settings

Alarm	Alarm type	Setting unit ⁶	Lower limit	Upper limit	Default	Step	Autoset
FiCO ₂	Upper	[gas]%	0.1%	10%	1%	0.1%	---
EtCO ₂	Lower	[gas]%	OFF, 0.1%	9.9%	4% OFF (MAN)	0.1%	---
EtCO ₂	Upper	[gas]%	0.1%	10%	6.5%	0.1%	---
FiO ₂	Lower	[gas]%	18%	99%	25 %	1%	---
FiO ₂	Upper	[gas]%	23%	99%, OFF	OFF	1%	---
EtO ₂	Lower	[gas]%	OFF, 10%	99%	OFF	1%	---
EtO ₂	Upper	[gas]%	13%	99%, OFF	OFF	1%	---
FiAA	Lower	[gas]%	OFF, 0.1%	5.0% (ISO) 8.0% (SEV) 18.0%(DES) or OFF	OFF	0.1%	---
FiAA	Upper ⁷	[gas]%	0.1%	5.0% (ISO) 8.0% (SEV) 18.0%(DES) or OFF	4.0% (ISO) 8.0% (SEV) 16.0%(DES)	0.1%	---
EtAA	Lower	[gas]%	OFF, 0.1%	4.0% (ISO) 6.0% (SEV) 12.0%(DES)	OFF	0.1%	---
EtAA	Upper	[gas]%	0.1%	5.0% (ISO) 8.0% (SEV) 18.0%(DES) or OFF	2.3% (ISO) 4.0% (SEV) 12.0%(DES)	0.1%	---

6. Available units include [gas]%, kPa (partial pressure) and mmHg (partial pressure)

7. Alarm is also activated if the MAC is above a certain value for a defined period of time

9.3 Audio pause



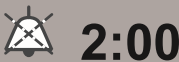
The Audio pause membrane button has the following applications:

- Mutes active alarm sound signals for two minutes
- Pre-mutes alarms activated within a two minute period.
- Mutes some alarms until the alarm condition resolves. This action must be confirmed via a dialog window.

9.3.2 Pre-muting alarms

Alarms can be pre-muted by depressing the membrane button for more than two seconds.

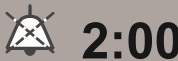
When pressed, a timer counting down from two minutes is displayed in the 'Audio pause and Countdown timer' area.



9.3.1 Muting alarms

Press once to silence all active alarms for a duration of two minutes.

When pressed, a timer counting down from two minutes is displayed in the 'Audio pause and Countdown timer' area.



Press again to reset the counter and start a new two minute countdown.

Silencing an alarm does not prevent other alarms indicating different conditions from being triggered.

Press again for less than two seconds and a new two minute pre-mute period is started. Pressing for more than two seconds during the countdown period will deactivate the pre-mute function.

Different symbols are used to differentiate between the mute alarm timer and the pre-mute alarm timer.

9.3.3 Mute until further notice (Audio OFF)

Some alarms, e.g. 'FiO2: High' and 'Apnea 60/120 s', can be muted until further notice, or until the alarm condition resolves.

Pressing the audio off button to silence any of these alarms will produce a dialog window:

'Do you want to turn <applicatory alarm> audio OFF?'. Selecting 'Yes' will mute the alarm until further notice. Selecting 'No' will mute the alarm for two minutes.

9.3.4 APNEA mute

Induction and certain emergency situations might warrant the possibility to turn off the audio signal of alarms associated with breathing and breathing parameters.

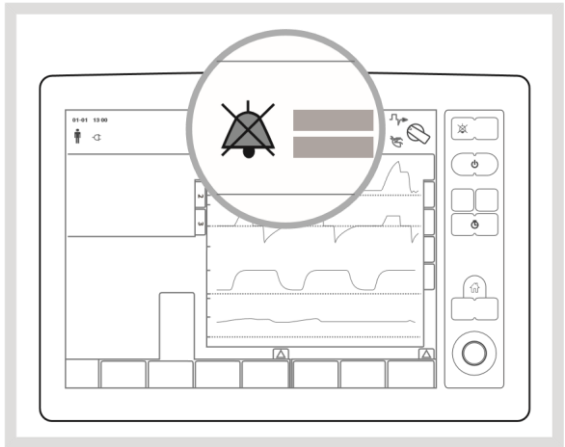
This is achieved by the 'APNEA mute' function. The function is by default enabled in the startup configuration settings, see Chapter 9, page 175. An activation button is thus visible in the alarm profile window. From there, APNEA mute can be activated and deactivated at any time.

The following alarms are affected:

- FiCO₂: High
- EtCO₂: Low
- EtCO₂: High
- Respiratory Rate: Low
- Apnea
- Apnea > 60 s / Apnea > 120 s
- Expiratory Minute Volume: Low
- Leakage

The APNEA mute function can be used in all ventilation modes. It is automatically deactivated by switching from MAN to AUTO, AUTO to MAN, or when going to standby mode.

'APNEA mute' and 'HLM mode' (CPB mode) cannot be active simultaneously. Selecting either in the alarm profile window, prevents the other from being activated.



When active, the APNEA mute symbol is visible next to the alarm area. Pressing this symbol (or the measurement area, or the membrane button 'Alarm profile'), brings up the alarm profile window.

APNEA mute can be preset in standby mode. In this case, the APNEA mute symbol is visible in the alarm function area in the standby window.

To inactivate, uncheck the APNEA mute button in the alarm profile window, press accept.

Log entries are made at 'activation' and 'deactivation'.

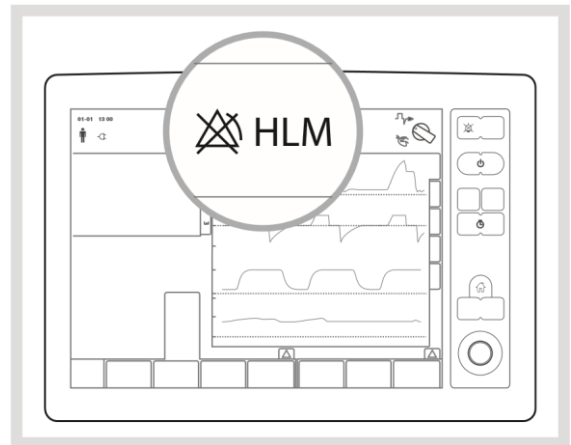
9.3.5 HLM mode (CPB mode)

The 'Heart Lung Machine' mode (CardioPulmonary Bypass mode) disables alarms that may otherwise be pointlessly triggered if the system is used to ventilate a patient undergoing open-heart surgery.

HLM/CPB mode will henceforth be referred to as HLM mode only.

The following alarms are disabled:

- FiCO₂: High
- EtCO₂: Low
- EtCO₂: High
- Respiratory Rate: Low
- Respiratory Rate: High
- Expiratory Minute Volume: Low
- Expiratory Minute Volume: High
- Apnea
- Apnea > 60 s / Apnea > 120 s



The HLM mode can be used in all ventilation modes except AFGO. It remains active when switching between available automatic modes, but is automatically deactivated when switching from MAN to AUTO, AUTO to MAN, or when ending the patient case. It is possible to preset HLM mode in standby in preparation of a patient case.

HLM mode is activated by checking the designated button in the alarm profile window. This requires that HLM is enabled in the startup configuration, see Chapter 9, page 175. HLM mode is by default set to inactive in the startup configuration.

When the check button is selected in the alarm profile window, affected alarms are disabled, as shown by the 'alarm off' symbol replacing the alarm limit settings. Effects take place when the accept button is pressed.

'APNEA mute' and 'HLM mode' (CPB mode) cannot be active simultaneously. Selecting either in the alarm profile window, prevents the other from being activated.

When active, the HLM symbol is visible next to the alarm area. Pressing this symbol (or the measurement area, or the membrane button 'Alarm profile'), brings up the alarm profile window.

HLM mode can be preset in standby mode. In this case, the HLM mode symbol is visible in the alarm function area in the standby window.

To inactivate, uncheck the HLM check button in the alarm profile window, press accept. Deactivated alarm are again active, alarm limit settings are identical to those before activation.

Log entries are made at 'activation' and 'deactivation'.

9.4 Alarm messages

9.4.1 High priority clinical alarms

Color = black text, red background, flashing.

Sound = A three tone pitch followed by a short pause and then a two tone pitch followed by a pause, repeated indefinitely.

The following high priority alarms exist on the system:

12. AFGO ventilation only

Alarm	Description
O ₂ supply pressure: Low	O ₂ central gas supply pressure <2.5 bar
Apnea ⁸	Alarm activated when time between breaths exceeds the adjustable time limit.
Apnea >60 s ⁹	No breath detection for >60 s
Apnea >120 s ¹⁰	No breath detection for >120 s
Airway pressure: High	Peak airway pressure exceeds the set upper pressure limit. The system shifts to 'exhale'. At peak airway pressure + 5cmH ₂ O, a safety valve opens.
Airway pressure: High	Backup version of the 'Airway pressure: High' alarm. Peak airway pressure exceeds set upper pressure limit by 2 cmH ₂ O. When peak airway pressure exceeds set upper pressure limit by 7 cmH ₂ O, a safety valve opens.
Airway pressure: High ¹¹	Peak airway pressure exceeds the set upper pressure limit by a value dependent on the current APL setting.
Airway pressure: High ¹²	Airway pressure exceeds 120 cmH ₂ O. A safety valve opens.
High continuous pressure ¹³	Airway pressure is constant above set PEEP level +15 cmH ₂ O more than 15 seconds.
PEEP: High	The measured end expiratory pressure is above the preset or default alarm limit for three consecutive breaths.
Negative airway pressure	Measured airway pressure is below -10 cmH ₂ O for more than one second
Expiratory Minute Volume: Low	Expiratory minute volume is below the lower alarm limit.

8. Pressure support, SIMV modes

9. Infant, manual mode only

10. Adult, manual mode only

11. Manual mode only

13. Automatic mode only

Alarm	Description
Check breathing circuit	Activated when inspiratory and expiratory pressures fail to meet preset requirements
Excessive leakage	Inspiration end pressure - Inspiration start pressure < 1 cmH ₂ O for three consecutive breaths
FiO ₂ : Low	Measured inspiratory O ₂ concentration is below the set Lower alarm limit.
FiN ₂ O: High	Inspiratory N ₂ O gas supply >80%
FiAA: High	Inspiratory agent concentration exceeds set upper limit.
EtAA: High	Expiratory agent concentration exceeds set upper limit.
Limited battery capacity	Less than 18 minutes left of battery operation.
No battery capacity	Less than 3 minutes left of battery operation.
Battery operation - battery capacity unknown	Connection error; unable to estimate battery time.
Patient cassette disconnected	No cassette is connected

9.4.2 Medium priority clinical alarms

Color = black text, yellow background, flashing.

Sound = A three tone pitch followed by a pause, repeated indefinitely.

The following medium priority alarms exist on the system:

Alarm	Description
Leakage ¹⁴	The system detects a shortage of fresh gas in the patient circuit due to possible leakage
Sampling tube leakage	Leakage detected on the Y-piece sampling tube.
Air/N ₂ O supply pressure: Low	Gas supply pressure <2.5 bar
Air/O ₂ supply pressure: High	Gas supply pressure >6.5 bar.
N ₂ O supply pressure: High	Gas supply pressure >4.5 bar.
PEEP: Low	The measured end expiratory pressure is below the preset or default alarm limit for three consecutive breaths.
Continuous APL pressure	Activated when the measured airway pressure exceeds predefined values for more than 15 seconds. Predefined values depend on current APL setting.
Expiratory Minute Volume: High	Expiratory minute volume exceeds upper alarm limit.
Regulated Pressure Limited ¹⁵	Permissible pressure limits pre-set tidal volume
Respiratory Rate: Low	RR is below the alarm limit
Respiratory Rate: High	RR exceeds the alarm limit.
Apnea ¹⁶	Alarm activated when time between breaths exceeds the adjustable time limit.
Occlusion in sampling line	Detected occlusion reported from Y-piece gas analyzer
FiO ₂ : High	Measured FiO ₂ concentration is above limit
EtO ₂ : Low	Measured expiratory O ₂ concentration is below the set Lower alarm limit.
FiCO ₂ : High	The measured inspiratory CO ₂ exceeds the alarm limit.

14. Automatic mode only

15. PRVC mode only

16. Manual mode, AFGO

Alarm	Description
EtCO ₂ : High	The measured expiratory CO ₂ exceeds the alarm limit.
EtCO ₂ : Low	The measured expiratory CO ₂ is below the alarm limit.
FiAA: Low	Measured AA concentration during insp. is below alarm limit
EtAA: Low	Measured AA concentration during expiration is below alarm limit
Agent mixture: MAC > 3	The MAC of the secondary agent is ≥ 0.6 and the total MAC value is ≥ 3
High continuous MAC ¹⁷	Measured MAC ¹⁸ exceeds alarm limit. ¹⁹
Vaporizer nearly empty	Vaporizer liquid level is below 5%
Patient cassette replaced	The last performed System checkout was not performed with the current patient cassette. Re-perform System checkout.
Water trap missing	The gas analyzer has detected that a water trap is missing
Replace water trap	The gas analyzer has detected that a water trap replacement is needed

17. Manual gas control only

18. In order to reflect time lag due to pharmacokinetics between alveolar concentration and the target organ brainconcentration, the high continuous MAC alarm activation and deactivation are delayed.

19. The high MAC alarm limit is 2.2 MAC, from starting a New Case, until 15 minutes after initial Vaporizer activation. Otherwise, it will be 1.8 MAC

9.4.3 Low priority clinical alarms

Color = black text, blue background.

Sound = A two tone pitch followed by a long pause, repeated indefinitely.

The following low priority alarms exist on the system:

Alarm	Description
EtO ₂ : High	Measured etO ₂ concentration is above alarm limit
Air/N ₂ O/O ₂ backup gas supply active	Air/N ₂ O or O ₂ central gas supply pressure <2.5 bar
Agent mixture	A mixture of anesthetic agents is detected. The MAC of the secondary agent is ≥ 0.6 , and the total MAC is <3.
Battery operation	Mains power failure, battery in operation.
Vaporizer liquid level low	Liquid level <10%.

9.4.4 Technical alarms

Color = black text, red, yellow or blue background (depending on technical alarm priority).

Sound = According to clinical alarm priority: high, medium and low priority alarm sound.

Technical alarms are characterized by each alarm having a Technical Alarm index, i.e. the letters 'TE' followed by a unique number. Examples include TE 12 Ventilation control error and TE 502 Communication error. For a comprehensive list of technical alarms and the appropriate action, see Chapter 15, page 287.

WARNING! When a technical alarm is activated, take the system out of use as soon as possible and contact a service technician trained and authorized by MAQUET unless otherwise stated.

9.4.5 System messages

Color = white text, black background.

Sounds = single pulse or no sound, depending on system message type.

Two types of system messages exist on the system:

- Short-lived system messages. These are displayed in the message areas for ten seconds and are accompanied by an initial single pulse sound, e.g. USB memory device connected.
- Lasting system messages. These are displayed in the message area until the message condition has been resolved, e.g.

Perform system checkout.

9.5 Safety valves

The 'Airway pressure: High' alarm will automatically force the system into its expiratory state, avoiding dangerous over-pressure situations.

In addition, the system has a built-in safety valve which opens at 117 cmH₂O.

9.6 AFGO alarms and alarm limits

If the system has the AFGO function (option) installed, the alarm profile changes accordingly when AFGO is activated:

- Unique alarms

- Inactivated alarms
- Alarm limit settings

9.6.1 Unique alarms

- Airway pressure: High (AFGO trigger condition)

'Airway pressure: High' in AFGO is triggered when the measured fresh gas pressure is above 120 cmH₂O. Check external circuit for blockage.

9.6.2 Inactivated alarms

- Expiratory Minute Volume: High
- Expiratory Minute Volume: Low
- Apnea > 60 s / Apnea > 120 s

- Continuous APL pressure

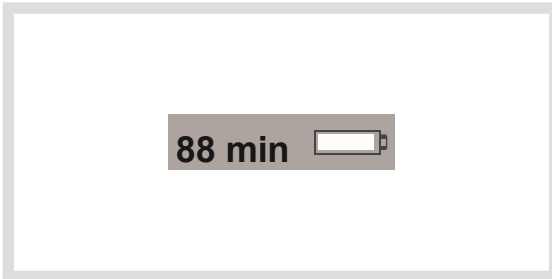
9.6.3 Alarm limits

Manual (MAN) mode alarm limits and default values also apply to AFGO mode except when a particular alarm is inactivated.

9.7 Power supply

In addition to the mains power supply, the system is equipped with a battery backup.

When the system is powered by battery backup, the estimated remaining battery time in minutes is displayed in the upper right corner of the screen.



The status of the power supply is continuously monitored by the system. If the status is unsatisfactory, an alarm is activated.

It takes six hours to recharge a depleted battery.

The batteries are charged while the system is connected to the mains power supply.

9.7.1 Mains power failure

In the event of a mains power failure or disconnection, the system switches to battery operation and activates an alarm.

During mains power failure, the backup battery in the anesthesia system will not power the patient monitor, equipment connected to the IEC outlets or the lift (C30 model).

Mains power failure will not affect the performance of other system functions, e.g. flow rate or composition of the fresh gas.

WARNING! The power cord shall remain connected to the power outlet when the internal battery back-up is used during a power failure. This maintains the protective earth connection.

9.7.2 Emergency ventilation system

In the event of mains power and battery power failure, the emergency ventilation system can be used to manually ventilate the patient.

If the emergency ventilation is activated, either during a patient case or when testing the system, the emergency ventilation APL valve must be set to minimum before resuming normal ventilation (AUTO or MAN) or starting a new patient case.

See Chapter 7, page 133 for more information.

9.8 Gas supply

9.8.1 Loss of central gas pressure

9.8.2 Backup gas supply (option)

The backup gas supply acts as a reserve system should the central gas supply become

WARNINGS!

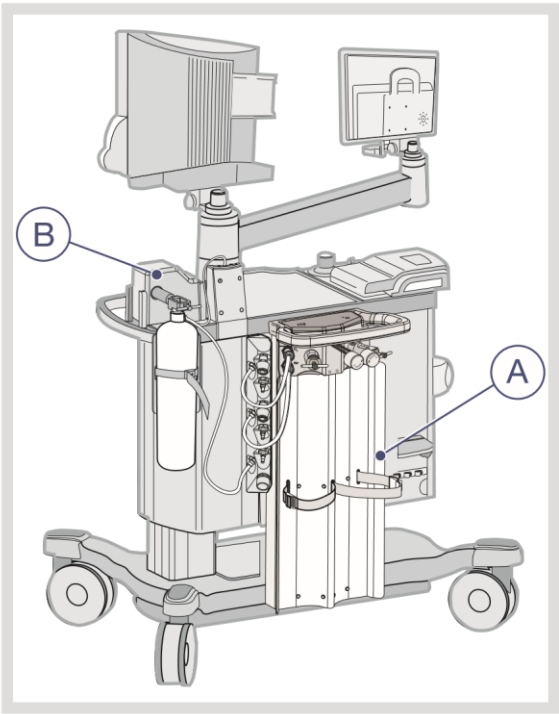
- If the O₂ supply is lost, delivery of anesthetic agent will be stopped.
- If the Air supply is lost, the patient suction unit (option) will not function.

If the O₂, Air or N₂O gas supply pressure falls below approx. 250 kPa (2.5 bar, 26 Psi), that particular gas supply is switched off and the 'Gas supply: Low' alarm is triggered. Gas delivery subsequently switches according to the table below. Supply of the missing gas automatically resumes when the supply pressure of that gas rises above approx. 280 kPa (2.8 bar, 40 Psi). Following this, the system reverts to delivering gas according to the configurations prior to the loss of gas supply.

Gas mix	Missing gas	Gas delivery
O ₂ /Air	N ₂ O	O ₂ /Air
O ₂ /N ₂ O	N ₂ O	O ₂
O ₂ /Air	Air	O ₂
O ₂ /N ₂ O	Air	O ₂ /N ₂ O
O ₂ /Air	O ₂	Air
O ₂ /N ₂ O	O ₂	Air
O ₂ /Air	AIR and O ₂	---

gas holder (B) delivering N₂O.

unavailable.



FLOW-i 4.2, User's Manual

The C20 and C30 systems can be equipped with a backup gas rack (A) configured to deliver either O₂/Air or O₂/N₂O and a backup

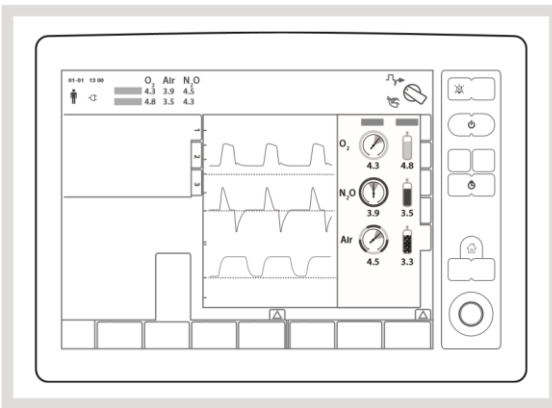
If the system has backup gas installed, the

2 2

- Central O₂/N₂O/Air pressure too low. Open O₂/N₂O/Air backup cylinder and press 'OK'.

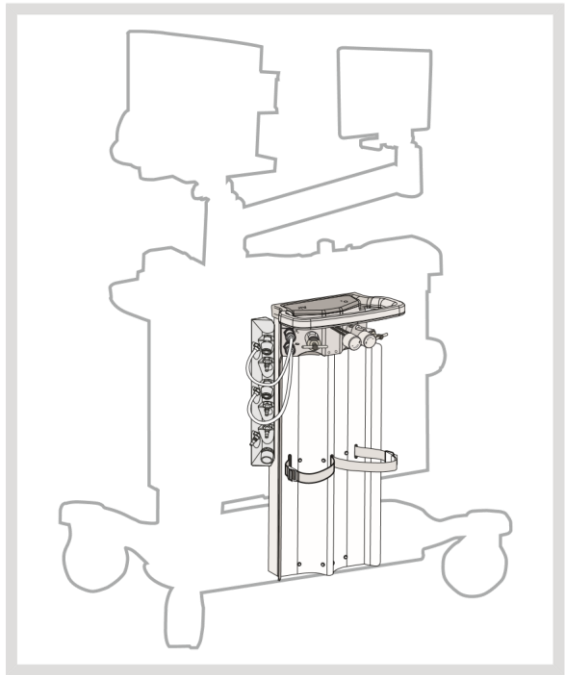
Open the appropriate cylinder valve and press 'OK'.

When the backup gas cylinder valves are opened, each cylinder pressure is displayed in the gas supply area:



alarm 'O₂/N₂O/Air central gas supply pressure: Low' will trigger a dialog window:

If the backup gas cylinder pressure decreases



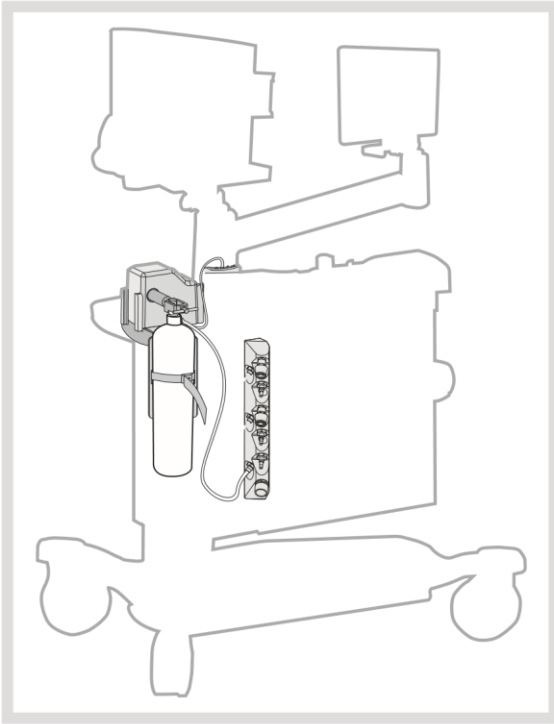
The backup gas rack, gas hoses and cable are permanently attached to the system, and shall only be removed by a service technician trained and authorized by MAQUET.

If the backup gas cylinder pressure decreases to 5.0 bar or lower, the gas supply pressure alarm re-appears.

If the central gas supply for the missing gas is restored, ventilation using the central gas supply is resumed. N₂O/Air/O₂ cylinder pressure information is removed from the message area. Close the backup gas cylinder valve.

The backup gas supply cylinders shall be turned off when not in use to avoid risk of emptying the bottles.

9.8.4 Backup gas holder (option)



An extra gas backup cylinder can be

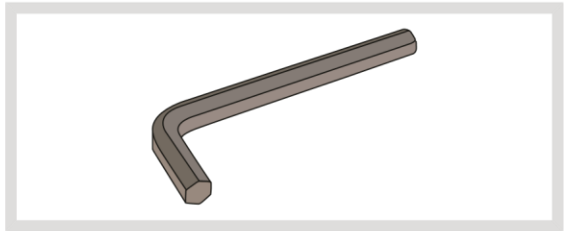
9.8.3 Backup gas rack (option)

The gas backup cylinders should be turned off at all times except during use or during the System checkout procedure.

The system may also be equipped with a backup gas trolley of similar capacity to that of the backup gas rack. The detachable backup gas trolley has wheels for transport to- and from storage.

Mounting the extra gas backup cylinder

1. Connect the cylinder as described in 'Backup gas cylinder replacement', page 207.
2. Use an allen key to adjust the support plate so that it is completely adjacent to the cylinder. Secure the cylinder using a restrainer belt.

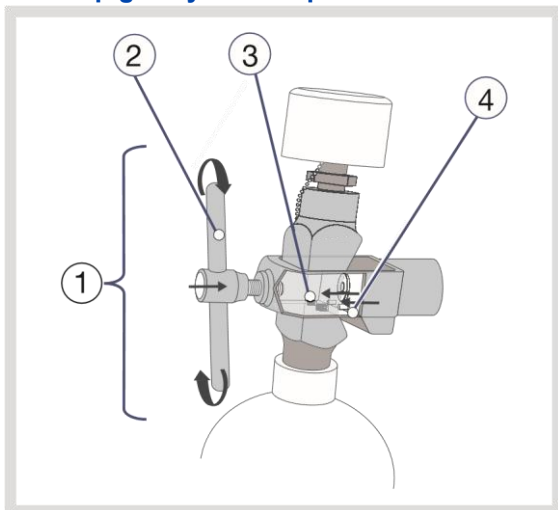


connected to the system with the addition of a backup gas holder.

If the installed backup gas holder is configured to deliver O₂, the backup gas system cannot be connected to the system.

The backup gas holder, gas hose and cable are permanently attached to the system, and shall only be removed by a service technician trained and authorized by MAQUET.

Backup gas cylinder replacement



1. Gas yoke assembly
2. Clamp screw
3. Cylinder pinholes
4. Pin-index safety system pins

The pin-index safety system (PISS) ensures that the correct gas cylinder (O_2 , Air or N_2O) is connected to each intended gas yoke assembly:

1. Fit the gas yoke pins into the cylinder pinholes and turn the clamp screw clockwise until it meets resistance.
2. Inspect the connection to verify that the seal is tight and the pins are fitted correctly.

CAUTION: Only gas cylinders which can be secured by a retaining belt may be installed in the backup gas system or onto the extra gas supply arm. Ensure that the cylinders are properly secured before use.

Storage

CAUTION: The backup gas trolley and backup gas cylinders shall be secured by a chain or wall mount when not used or in storage. If the cylinders are not equipped with a safety bolt, they must be secured only by a wall mount.

10 Ventilation modes Table of contents

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10.1 Introduction

The anesthesia system can operate in several different ventilation modes. This chapter describes these modes, their settings and associated safety information.

See Chapter 10, 'Alarms and patient safety', for a detailed description of setting alarm limits.

WARNING! To protect the patient's lungs from excessive pressure, it is important to always set the upper pressure limit to a suitable value.

10.2 Manual Ventilation

Manual Ventilation is carried out using a traditional breathing bag. The use of a manual breathing bag tube is required. The size of the breathing bag and tube may differ based on the size of the patient. The manual breathing bag model should comply with ISO 5362:2006.

The APL valve (Adjustable Pressure Limiting valve) relieves the patient circuit of excess pressure at user defined values and ventilates the gas into the AGS.

WARNING! When air only is used during manual ventilation, make sure the fresh gas flow is set sufficiently high to prevent re-breathing.

CAUTION: During spontaneous ventilation, always use CO₂ monitoring and maintain visual control of the manual breathing bag.

10.2.1 Important considerations

The following alarms are disabled during manual ventilation:

- Airway pressure: High
- High continuous pressure
- Excessive leakage.
- Leakage

10.3 AFGO

AFGO ventilation is carried out using an external breathing system connected to the system's AFGO (Additional Fresh Gas Outlet) connection.

This mode bypasses the circle system, creating an open system with gas and agents primarily moving in one direction. Re-use of gas and CO₂ clearance depends on the current fresh gas flow.

Refer to manufacturers description for recommendations on minimal flow required to carry out ventilation using any specific external breathing system.

For a complete description on AFGO, refer to Chapter 8 - AFGO.

CAUTION: Only external breathing circuits equipped with a pressure relief valve or a manual bag with an opened end shall be connected to the AFGO outlet.

10.3.1 Important considerations

The following alarms are disabled during AFGO ventilation:

- Expiratory Minute Volume: High
- Expiratory Minute Volume: Low

- Apnea > 60 s / Apnea > 120 s
- Continuous APL pressure

Refer to Chapter 10, page 202, for more information on alarm functions in AFGO mode.

10.4 Automatic ventilation modes

10.4.1 Controlled Ventilation

Pressure Control (PC)

Pressure controlled ventilation.

Volume Control (VC)

Volume controlled ventilation.

Pressure Regulated Volume Control (PRVC) Pressure regulated volume controlled ventilation.

10.4.2 Supported ventilation

Pressure Support (PS)

Pressure supported ventilation.

10.4.3 Combined ventilation

PS backup ventilation

Pressure supported ventilation combined with pressure controlled backup ventilation.

SIMV (PC) + PS

Synchronized intermittent mandatory ventilation based on pressure controlled ventilation with pressure support.

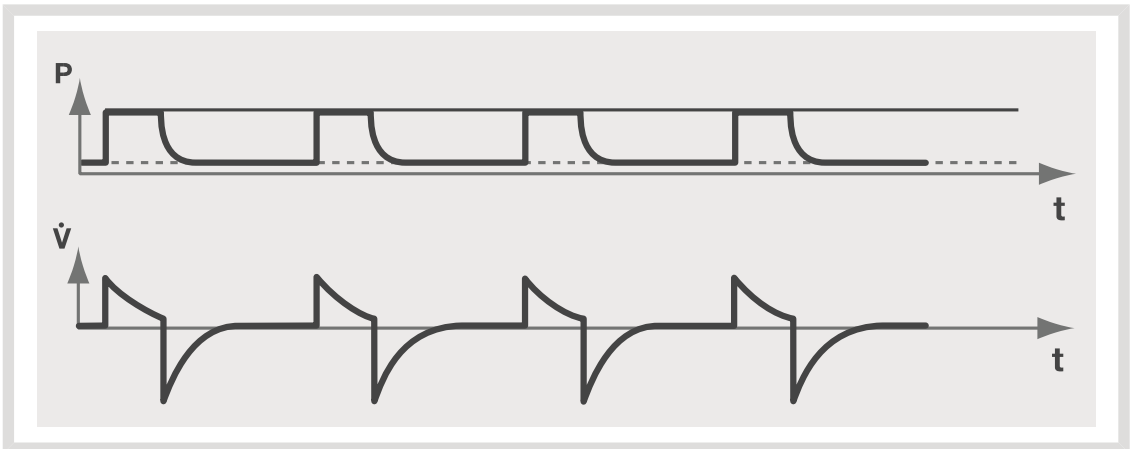
SIMV (VC) + PS

Synchronized intermittent mandatory ventilation based on volume controlled ventilation with pressure support.

The following pages describe each automatic ventilation mode in more detail.

10.5 Pressure Control (PC)

due to uncuffed endotracheal tube or in



In Pressure Control the system delivers breaths with a constant preset pressure, with a decelerating flow during a preset inspiratory time, and at a preset frequency.

The following parameters are set:

1. PC above PEEP (cmH₂O, solid line in illustration above)
2. Respiratory Rate (b/min)
3. PEEP (cmH₂O, dotted line in illustration above)
4. I:E ratio / Insp. time
5. Inspiratory rise time (%/s)
6. Trigg. Flow / Trigg. Pressure
7. Gas mix (O₂/Air or O₂/N₂O)
8. O₂ concentration (%)
9. Fresh gas flow (l/min)
10. Agent concentration (%)

The delivered volume is dependent upon the pressure above PEEP, lung compliance and resistance in the patient tube system and airways. This means that the Tidal Volume can vary. Pressure Control is preferred when there is leakage in the breathing system e.g.

situations when the maximum airway pressure must be controlled.

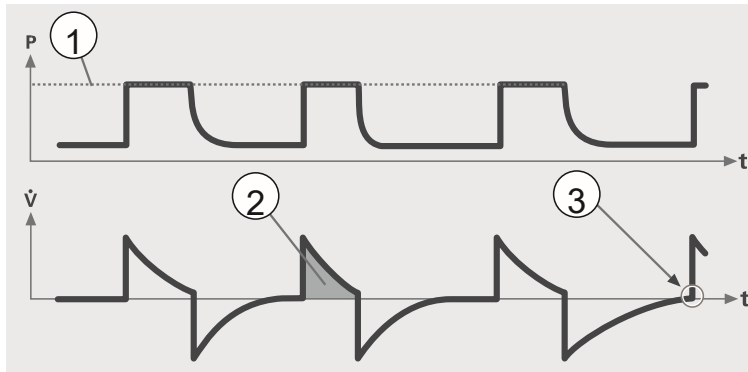
The patient can trigger extra breaths. Additionally, inspiration can be interrupted if the patient attempts to exhale, provided the generated pressure is more than 3 cmH₂O above the set pressure level. As the delivered tidal volume can vary it is very important to set alarm limits for Minute Volume to adequate levels.

The flow during inspiration is decelerating, i.e. the flow decreases as the pressure in the lungs approaches the set PC above PEEP.

Pressure control properties

1. Pressure Control assures that the preset inspiratory pressure level is maintained constantly during the entire inspiration. Breaths are delivered according to the preset frequency, inspiration time and inspiratory pressure level resulting in a decelerating flow.
2. The preset pressure level is controlled by the system. The resulting volume (shaded area) depends on the set

10.5.1 Pressure Control in detail



pressure level, I:E or T_i , and the patient's lung mechanical properties.

3. Inspiration starts according to the preset respiratory rate or when the patient triggers.

Numbering refers to illustration above.

Expiration criteria

Expiration begins:

- After the termination of preset inspiration time.
- If the upper pressure limit is exceeded.
- If the delivered tidal volume is twice that of the set upper limit for tidal volume for the current patient category.

Active expiratory valve

If a patient tries to exhale during the inspiration, pressure increases. When it increases 3 cmH₂O above the set inspiratory pressure level, the expiratory valve opens and regulates the pressure down to the set inspiratory pressure level.

If the pressure increases to the set upper pressure limit e.g. the patient is coughing, the

expiratory valve opens and the system switches to expiration.

Inspiration criteria

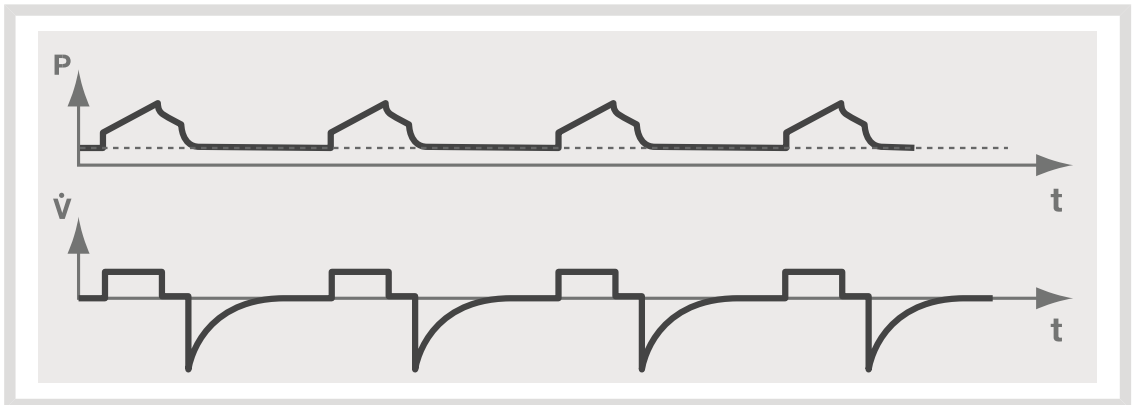
Inspiration will begin in accordance with the preset breathing frequency or when a breath is triggered by the patient.

10.5.2 Important considerations

- Set alarm limits for minute volume.
- Carefully monitor tidal volumes.
- The inspiratory flow is decelerating.
- Discrepancies in gas concentrations between the patient's inspiration and expiration depend on the patient's uptake.
- The patient can trigger extra breaths resulting in a higher delivered minute volume than the set one.

10.6 Volume Control (VC)

11. Agent concentration (%)



Volume Controlled ventilation ensures that the patient receives a certain pre-set Minute/Tidal Volume.

In this control mode, the system delivers the preset tidal volume with a constant flow during the preset inspiratory time and at the preset respiratory rate.

The time for inspiration and expiration is configured by setting the I:E ratio or T_i (inspiration time).

The following parameters are set:

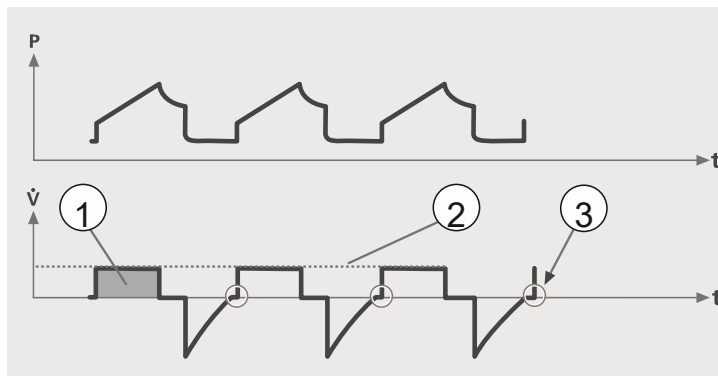
1. Tidal Volume (ml) or the Minute Volume (l/min)
2. Respiratory Rate (b/min)
3. PEEP (cmH₂O, dotted line in illustration above)
4. I:E ratio / Insp. time
5. Inspiratory rise time (%/s)
6. Pause time (%/s)
7. Trigg. Flow / Trigg. Pressure
8. Gas mix (O₂/Air or O₂/N₂O)
9. O₂ concentration (%)
10. Fresh gas flow (l/min)

The airway pressure is dependent on the tidal volume, PEEP setting, inspiration time and the resistance and compliance of the respiratory system. The set tidal volume will always be delivered. An increase in the resistance and decrease in compliance will lead to an increased airway pressure.

To protect the patient's lungs from excessive pressure, it is very important to set the upper pressure limit to a suitable value.

It is possible for patients to trigger extra breaths if they can overcome the pre-set trigger sensitivity. The flow during inspiration is constant.

10.6.1 Volume Control in detail



Volume control properties

1. Volume Control assures a preset tidal volume with constant flow during the inspiratory time at a preset respiratory rate.
2. The inspiratory flow is constant and depends on the user setting.
3. Inspiration starts according to the preset respiratory rate or when the patient triggers.

Numbering refers to illustration above.

Expiration criteria

Expiration begins:

- When the preset tidal volume is delivered and after the preset pause time. The

patient is however always guaranteed an expiration time corresponding to at least 20% of the total breath.

- If the upper pressure limit is exceeded.

Inspiration criteria

The initiation of inspiration is based on the preset breathing frequency, or when the patient triggers a breath. The duration of Inspiration is based on the breathing frequency and the I:E ratio, or the breathing frequency and T_i .

Pause time

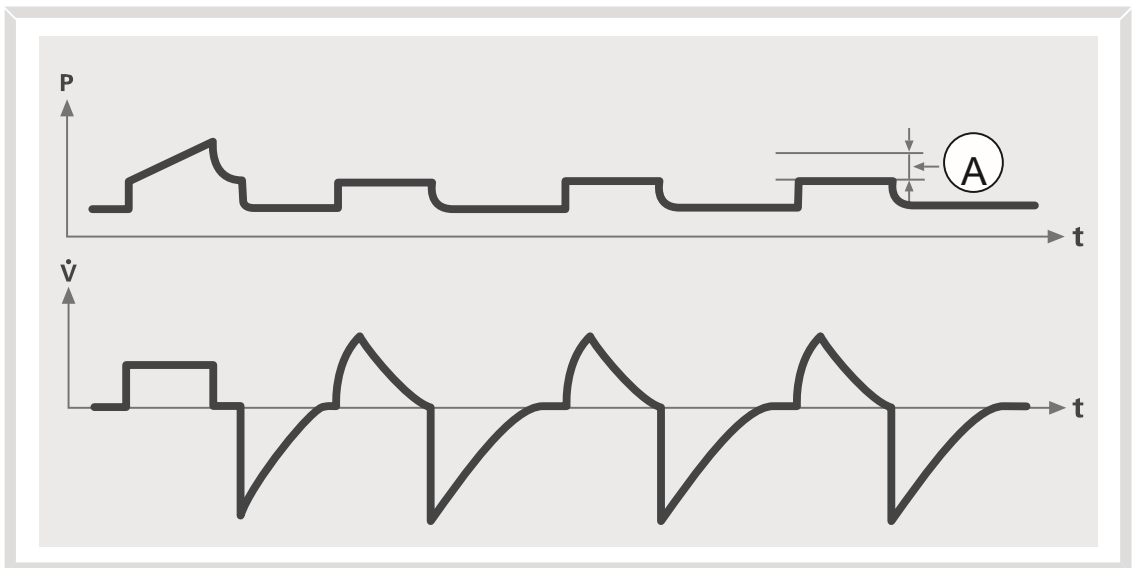
The pause time between inspiration and expiration can be set from 0% to 30% of the breath cycle time. Default setting is 10%.

10.6.2 Important considerations

WARNING! Volume control is not intended for use on neonates because the Tidal volume (V_T) cannot be set < 20 ml.

- Set upper pressure alarm limit. If this limit is reached, inspiration will cease. Expiration will begin and an alarm for high airway pressure will be triggered.
- Carefully monitor patient airway pressures. If the patient has poor lung characteristics there is a risk of high airway pressures.
- Discrepancies in gas concentrations between the patient's inspiration and expiration depend on the patient's uptake.
- The patient can trigger extra breaths resulting in a higher delivered minute volume than the set one.

10.7 PRVC



PRVC is a controlled mode of ventilation which combines the advantages of volume controlled and pressure controlled ventilation. The system delivers the preset tidal volume with the lowest possible pressure.

The flow during inspiration is decelerating. The patient can trigger extra breaths.

The system can be configured to set Tidal Volume or Minute Volume.

The time for inspiration and expiration is configured by setting the I:E ratio or T_i (inspiration time).

The following parameters are set:

1. Tidal Volume (ml) or Minute Volume (l/min)
2. Respiratory Rate (b/min)
3. PEEP (cmH₂O)
4. I:E ratio / Insp. time
5. Inspiratory rise time (%/s)
6. Trigg. Flow / Trigg. Pressure
7. Gas mix (O₂/Air or O₂/N₂O)
8. O₂ concentration (%)
9. Fresh gas flow (l/min)
10. Agent concentration (%)

The first breath of a start sequence is a volume-controlled test breath with Pause time set to 10%. The measured pause pressure of this breath is then used as the pressure level for the following breath. The pressure is automatically regulated to deliver the pre-set volume, but limited to 5 cmH₂O below the set upper pressure limit (A).

If the measured tidal volume deviates from the preset tidal volume, the system

PRVC properties

1. PRVC assures a set target minute volume to the patient. The target volume is based upon settings for Tidal Volume (shaded area), respiration rate and inspiration time.
2. The inspiratory pressure level is constant during each breath, but automatically adapts in small increments breath-by-breath to match the patient's lung mechanical properties for target volume delivery.
3. Inspiration starts according to the preset respiratory rate or when the patient triggers.

Expiration criteria

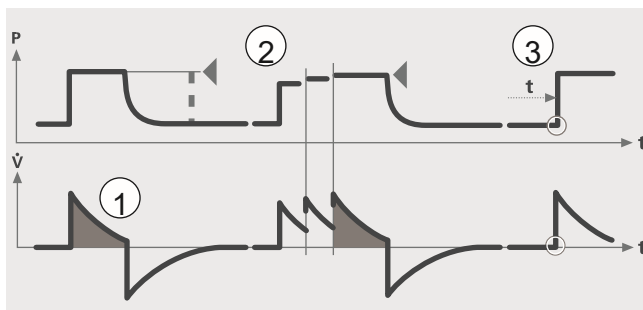
Expiration begins:

automatically changes the pressure level between consecutive breaths (in steps of maximum 3 cmH₂O) until the the preset tidal volume is delivered.

The maximum available pressure level is 5 cm H₂O below the preset upper pressure limit. If the pressure reaches this limit, the system will deliver as much volume as possible with this pressure. This will also trigger the alarm 'Regulation pressure limited'.

- After the termination of preset inspiration time
- If the upper pressure limit is exceeded.
- If the measured inspiratory tidal volume exceeds the pre-set tidal volume by 130%.

10.7.1 PRVC in detail



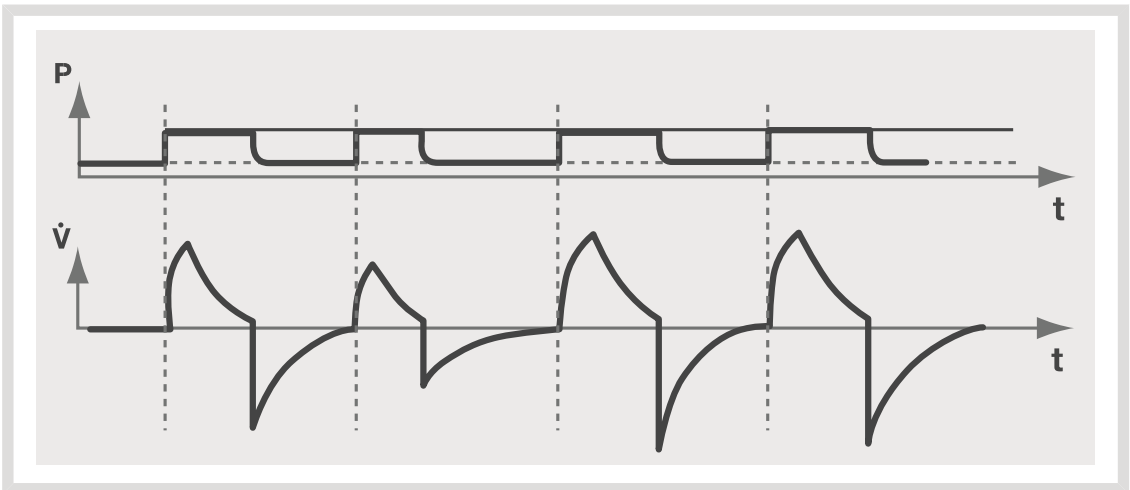
10.7.2 Important considerations

Fluctuations in compliance during a patient case, e.g. moving the patient, will cause the system to make adjustments to the compliance compensation. Consider the appropriate mode before starting the patient case. Monitor the patients dynamic compliance, 'C dyn', during run mode.

PRVC is not recommended when there is a leakage in the patient's breathing circuit

10.8 Pressure Support (PS, option)

9. Agent concentration (%)



Pressure Support (option) is a patient initiated breathing mode in which the system supports the patient with a set constant pressure. The following parameters are set:

1. PS above PEEP (cmH₂O, solid line in illustration above)
2. PEEP (cmH₂O, dotted line in illustration above)
3. Inspiratory rise time (s)
4. Inspiratory Cycle-off (%)
5. Trigg. Flow / Trigg. Pressure (trigg events indicated by vertical dotted lines in illustration above)
6. Gas mix (O₂/Air or O₂/N₂O)
7. O₂ concentration (%)
8. Fresh gas flow (l/min)

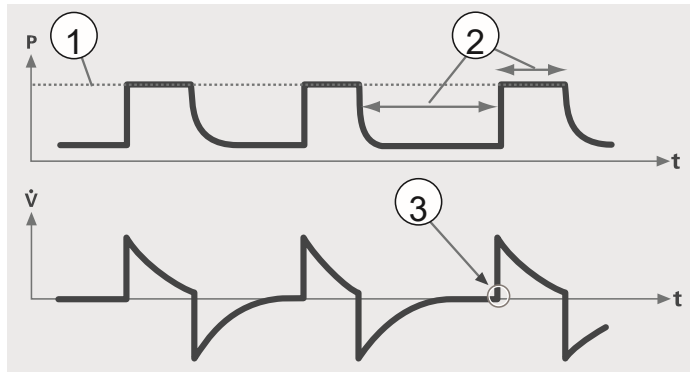
During Pressure Supported ventilation, the patient regulates the respiratory rate and the Tidal Volume with support from the system. The higher the pre-set inspiratory pressure level from the system, the more gas flows into the patient. As the patient becomes more active, the pressure support level may be gradually reduced.

It is important to set the Inspiratory rise time to a comfortable value for the patient. In Pressure Support, the Inspiratory rise time should normally be increased.

Pressure support properties

1. Pressure Support ensures that a preset inspiratory pressure level is constantly maintained upon patient effort.

10.8.1 Pressure Support in detail



2. The preset pressure level is controlled by the system, while the patient determines frequency and inspiration time.
3. Inspiration starts when the patient triggers.

Numbering refers to illustration above.

Expiration criteria

Expiration begins:

- When the inspiratory flow falls below a preset fraction of the inspiratory peak flow (Inspiratory cycle-off)
- If the upper pressure limit is exceeded.
- Maximum time for inspiration is exceeded.

Inspiration criteria

When the patient triggers a breath according to the set trigger sensitivity.

Triggered breaths are highlighted in the waveform display area, see Chapter 5, page 103 for more information.

10.8.2 Important considerations

- Pressure Support is not intended for use in conjunction with administration of muscle relaxants or high doses of opiates.
- It is important to set a lower and upper alarm limit for expired Minute Volume.
- Should the patient's inspiratory time exceed 2.5 s (Adult) or 1.5 s (Infant), inspiration will be terminated (valid also for PS Backup).

- If the Pressure Support Level above PEEP is set to zero, the system maintains the PEEP level during the breaths.
- Discrepancies in gas concentrations between the patient's inspiration and expiration depend on the patient's uptake.
- If the PEEP is set to a high level, the patient will have difficulty triggering the breaths.
- Infant patient category: 5-45 seconds
- Adult patient category: 15-45 seconds

10.8.3 Backup Ventilation

Pressure support backup ventilation is controlled by the backup respiration rate (Backup RR) and PC above PEEP parameters. If no breath is detected within the specified backup frequency, a pressure controlled breath is administered and the alarm 'Backup ventilation' (low priority) is activated.

Refer to Chapter 5, page 95, for a description on ventilation parameters.

The minimum backup pressure level is 5 cmH₂O when setting 'PC above PEEP'.

If no breath is detected (identified by changes in CO₂ concentration) within the set apnea time, the apnea alarm (high priority) is activated. The apnea alarm deactivates and replaces the backup ventilation alarm.

The apnea time is set in the alarm profile window, see Chapter 10 page 187. Apnea time depends on the patient category as follows:

10.8.4 Disabling Backup ventilation

Backup ventilation can be disabled in standby mode and during run mode. This is done by setting the backup RR parameter to 'OFF', see Chapter 5, page 97. The backup RR parameter can also be set to 'OFF' in 'Startup configuration', see Chapter 9 page 178.

10.9 SIMV

10.9.1 Functional description SIMV

SIMV is a combination mode where the patient receives mandatory breaths synchronized with his breathing efforts and according to the selected SIMV mode. The patient can breathe spontaneously with Pressure Support in between the mandatory breaths.

There are two different SIMV modes, depending on the modes installed:

- SIMV (Pressure Control) + Pressure Support • SIMV (Volume Control) + Pressure Support

During the SIMV period, the first triggered breath will be a mandatory breath. If the patient has not triggered a breath within the first 90% of the Breath Cycle time a mandatory breath will be delivered.

If the system is configured for setting of Inspiration time, an I:E ratio of 1:3 (Adult) or 1:2 (Infant) will be used to estimate the Breath cycle time.

The spontaneous/pressure supported breaths are defined by setting the Pressure support level above PEEP.

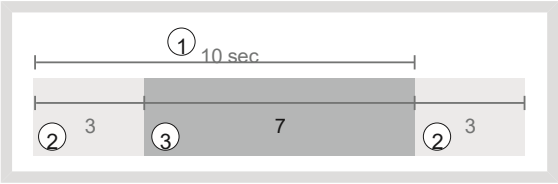
10.9.2 The mandatory breath

	SIMV (PC) + PS	SIMV (VC)+ PS
PC above PEEP	X	
Tidal volume /Minute volume		X
SIMV rate	X	X
Breath cycle time	X ²⁰	X ²¹
I:E ratio / Inspiration time	X	X
Insp. rise time	X	X
Pause time		X

The Mandatory breath is defined by the basic settings (as shown in the table above): Minute Volume/Tidal Volume (depending on configuration), PC above PEEP, I:E and the expiration time (depending on remaining for spontaneous breathing is 7 configuration), Pause time, Inspiratory rise time seconds. and Breath cycle time.

For example: A SIMV rate of 6, a breath cycle time of 3 seconds with an I:E ratio of 1:2 means that the inspiration will take 1 second and the expiration 2 seconds. The time ratio/Inspiration time (depending on configuration), Pause time, Inspiratory rise time seconds. and Breath cycle time.

In the Minute Volume configuration the Tidal Volume is determined by Minute Volume divided by SIMV rate.



The time for one SIMV cycle is 60 s divided by the SIMV rate.

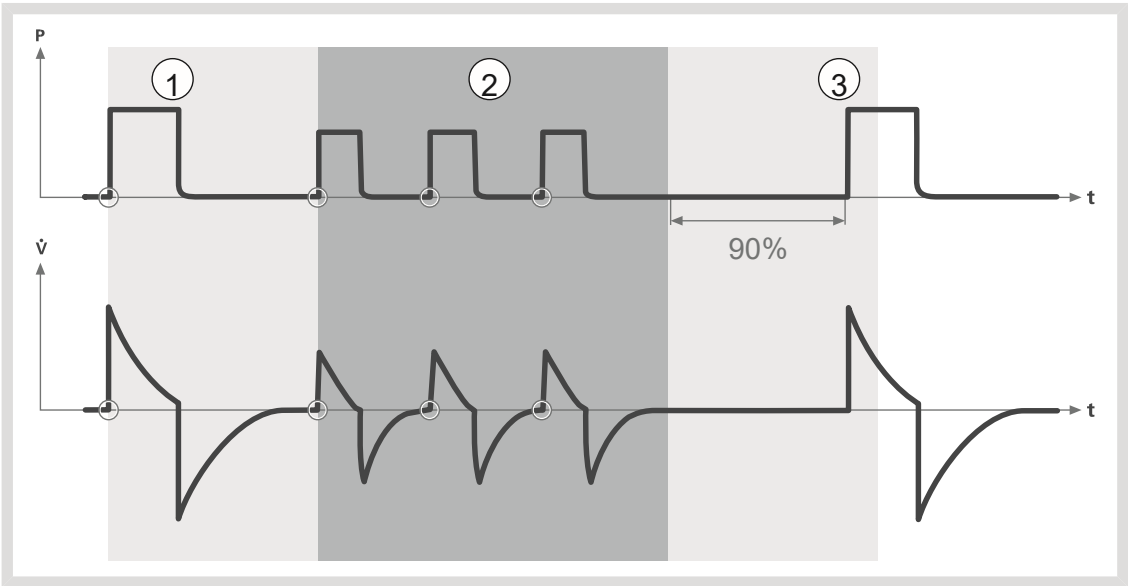
The Breath cycle time is the length of the mandatory breath in seconds.

The time for the spontaneous period is the SIMV cycle time subtracted by the breath cycle time.

1. SIMV cycle time
2. Breath cycle time
3. Spontaneous period

20. Only when the system is configured for I:E ratio setting.

10.9.3 SIMV (Pressure Control) + Pressure Support



The following parameters are set:

SIMV (PC) properties

1. PC (Pressure Control level) above PEEP (cmH₂O) This combined control and pressure
2. SIMV rate (b/min) support/spontaneous function allows for 3. PEEP (cmH₂O) preset mandatory breaths synchronized with
4. I:E ratio / Insp. time the patient's breathing.
5. Inspiratory rise time (%/s) 1. The mandatory breath is defined by the basic settings (mode of ventilation, breath cycle time, respiratory pattern and volumes/pressures).
6. Breath cycle time
7. Inspiratory Cycle-off (%)
8. Trigg. Flow / Trigg. Pressure
9. PS (Pressure Support level) above PEEP 2. The spontaneous/pressure supported breaths are defined by the setting for Pressure Support. (cmH₂O)
10. Gas mix (O₂/Air or O₂/N₂O)
11. O₂ concentration (%) 3. If there is no trigger attempt within a time window equal to 90% of the set Breath cycle time, a mandatory breath is delivered. (The Breath cycle time is the total time for one mandatory breath.)
12. Fresh gas flow (l/min)
13. Agent concentration (%)

Expiration criteria Expiration

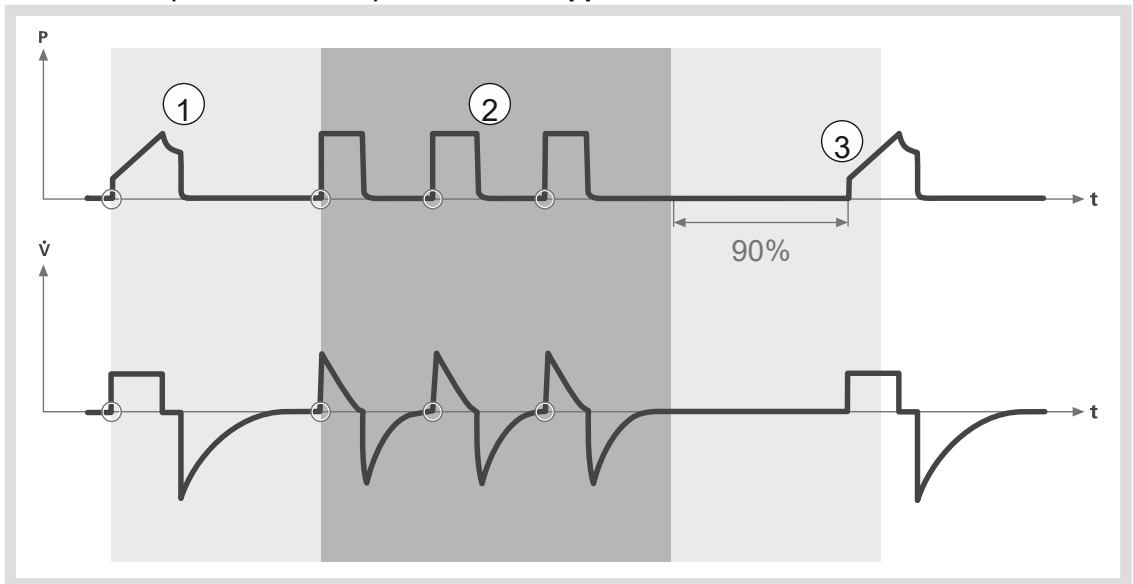
begins:

- When the inspiratory flow falls below a preset fraction of the inspiratory peak flow (Inspiratory cycle-off)
- If the upper pressure limit is exceeded.
- Maximum time for inspiration is exceeded.

Inspiration criteria

- When the patient triggers a breath according to the set trigger sensitivity.
- Mandatory breath requirement is met during the breath cycle time. Triggered breaths are highlighted in the waveform display area, see Chapter 5, page 103 for more information.

10.9.4 SIMV (Volume Control) + Pressure Support



The following parameters are set:

1. Tidal Volume (ml) or the Minute Volume

- (l/min)
2. SIMV rate (b/min)
 3. PEEP (cmH₂O)
 4. I:E ratio / Insp. time
 5. Breath cycle time
 6. Inspiratory rise time (%/s)
 7. Pause time
 8. Inspiratory Cycle-off (%)
 9. Trigg. Flow / Trigg. Pressure
 10. PS (Pressure Support level) above PEEP (cmH₂O)
 11. Gas mix (O₂/Air or O₂/N₂O)
 12. O₂ concentration (%)
 13. Fresh gas flow (l/min)
 14. Agent concentration (%)

SIMV (VC) properties

This combined control and pressure support/spontaneous function allows for preset mandatory breaths synchronized with the patient's breathing.

1. The mandatory breath is defined by the basic settings (mode of ventilation, breath cycle time, respiratory pattern and volumes/pressures).
2. The spontaneous/pressure supported breaths are defined by the setting for Pressure Support.
3. If there is no trigger attempt within a time window equal to 90% of the set Breath cycle time, a mandatory breath is delivered. (The Breath cycle time is the

total time for one mandatory breath.)

Expiration criteria Expiration begins:

- When the inspiratory flow falls below a preset fraction of the inspiratory peak flow (Inspiratory cycle-off)
- If the upper pressure limit is exceeded.
- Maximum time for inspiration is exceeded.

Inspiration criteria

- When the patient triggers a breath according to the set trigger sensitivity.
- Mandatory breath requirement is met during the breath cycle time. Triggered breaths are highlighted in the waveform display area, see Chapter 5, page 103 for more information.

Pause time

The pause time between inspiration and expiration can be set from 0% to 30% of the breath cycle time. Default setting is 10%.

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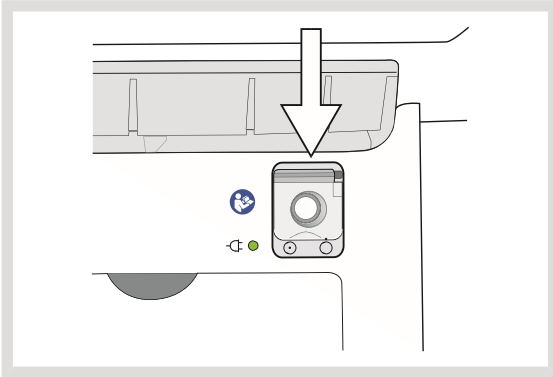
11 System shutdown

11.1 System shutdown

When the system is switched from On to Off, the system is turned off immediately. Patient monitoring ceases and the control panel shuts down.

To switch off the system, lift the protective glass cover and press the Power button.

A 'beep' sound is emitted and the system shuts down.



As long as the system is connected to the mains power outlet, the battery is charged.

If battery backup has been used to power the system at any time before, during, or after a patient case, it is important to immediately connect the system to a mains power after use. Leave it connected until the battery is fully charged. This will ensure battery performance and maximize battery life span.

12 Routine cleaning and maintenance Table of contents

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12.1 General information

MAQUET recommends methods that have been validated using the specified equipment and procedures outlined in this manual. Other methods may work but are not covered by the warranty unless MAQUET has given written permission.

As cleaning practices vary widely among healthcare institutions, it is not possible for MAQUET to specify particular practices that will meet all needs, or to be responsible for the effectiveness of cleaning procedures carried out in the patient care setting.

When the accessories and/or auxiliary equipment are not covered by the MAQUET CE mark on the anesthesia system, refer to cleaning instructions found in the equipment's user documentation.

CAUTION: The system must not be disassembled further than described in this chapter 'Routine cleaning and maintenance'.

21. Murphy *et al.* 1991

Follow the recommendations regarding softening of water provided by the manufacturer of the washer disinfectant.

All personnel should be aware of the risk of parts being contaminated when disassembling and cleaning the system.

All disposable parts must be discarded according to hospital routines and in an environmentally safe way.

12.2 Hygiene

The carbon dioxide absorbent has a potent cidal effect on microorganisms and only a low number of resistant spores pass through the absorber. Studies strongly suggest that patients rarely contaminate the absorber with

21

significant levels of bacteria/viruses. The bactericidal effect will however not be in effect if the absorber is bypassed.

MAQUET recommends that a bacterial/viral filter is always connected to the expiratory connection on the patient cassette. This will minimize the risk of cross-contamination.

If possible, cleaning should be performed immediately after use and always before disinfection/autoclaving. Blood or other residues should not be allowed to dry onto the devices.

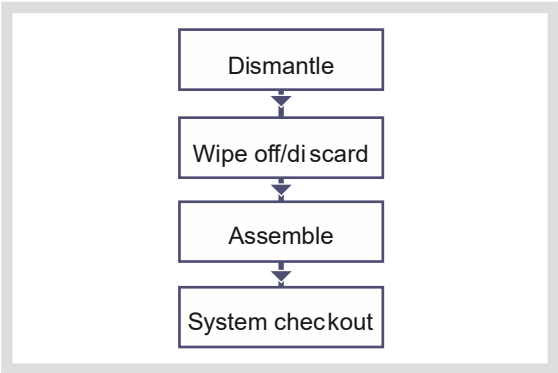
Water quality affects cleaning/disinfection. MAQUET recommends using deionized water for this purpose. The water quality must never be less than drinking water quality.

12.3 Cleaning/disinfection procedures Disinfection/steam autoclaving when no

- a summary

A summary of the procedures for cleaning, disinfection and steam autoclaving is given here. Details for each step are given in the following sections.

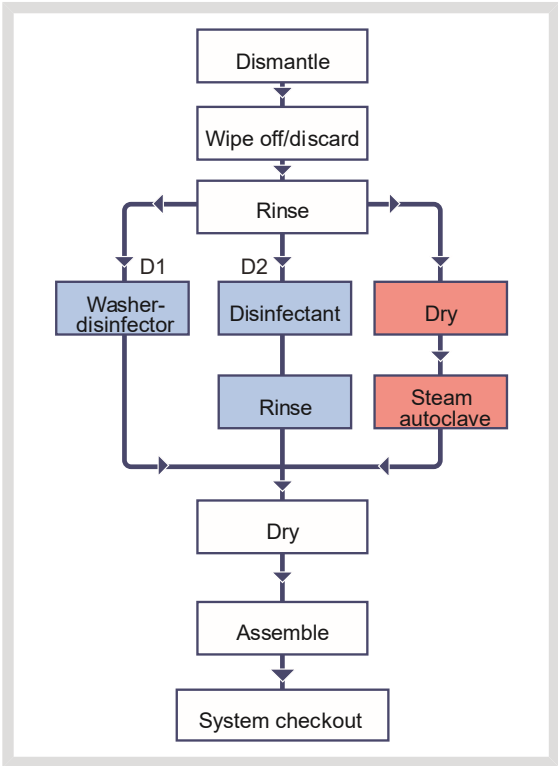
Cleaning when bacterial/viral filter is used



If a SERVO DuoGuard filter (or a filter with equal performance) is used, disinfection or steam autoclaving of the patient cassette, volume reflector and associated parts is usually not necessary.

A complete cleaning procedure should still be performed at regular intervals according to hospital routines, or after treating a known carrier of pulmonary contagious agents. disinfectant is used.

bacterial/viral filter is used



There are two disinfection procedures for the patient cassette, volume reflector and associated parts: D1 and D2. In D1 (recommended), a Washer-Disinfector (ISO-15883-1) is used, and in D2 only a

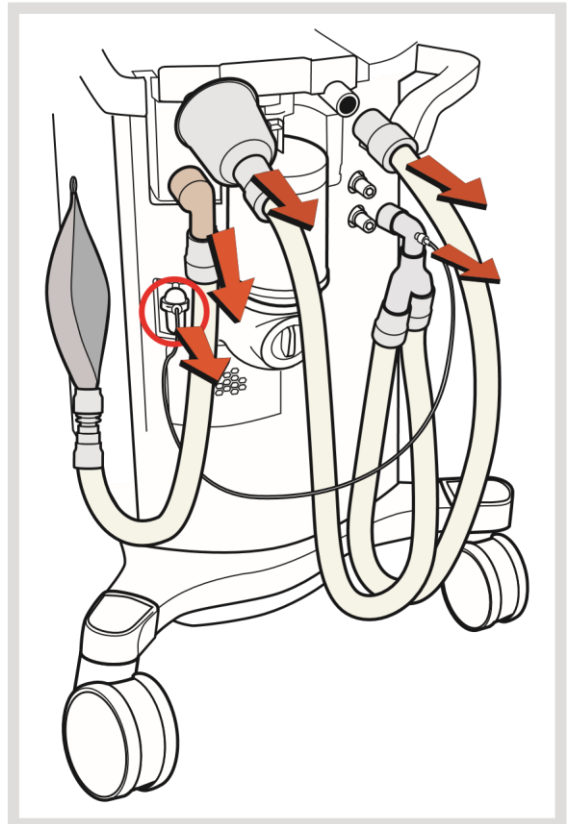
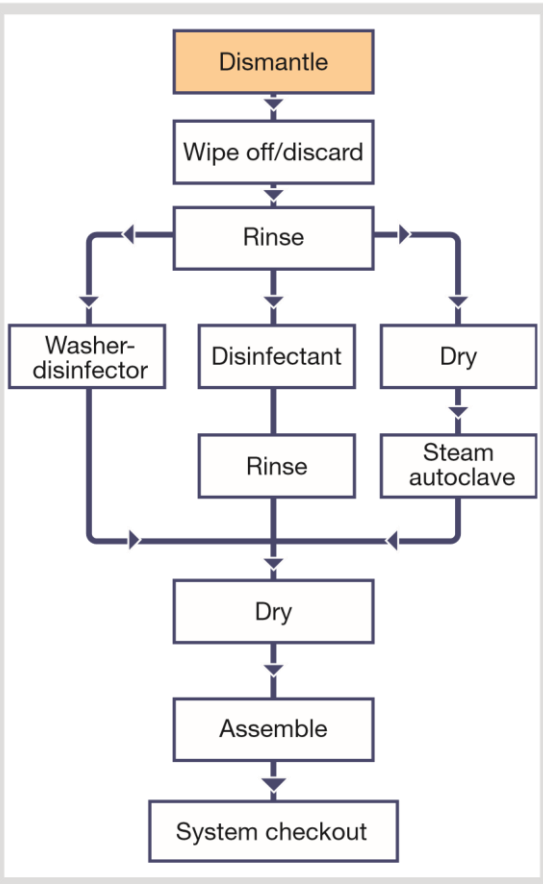
Use only validated processes when steam autoclaving the patient cassette, volume reflector and associated parts. See further instructions in section 11.8.

12.4 Preparations and dismantling

CAUTION: Steam autoclaving will reduce the lifetime of the components.

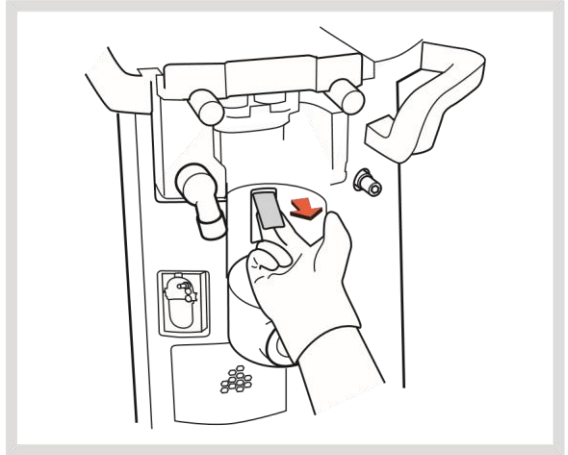
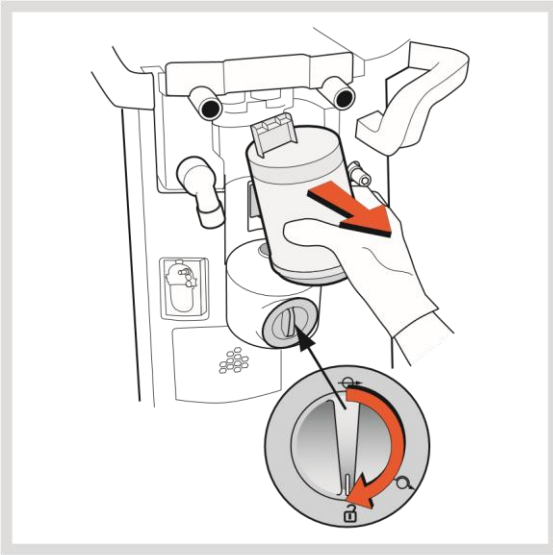
12.4.1 Remove external appliances

1. Shutdown the system.
2. Disconnect the system from the power and gas supply.
3. Disconnect any optional equipment from the anesthesia system.
4. Remove the water trap and gas sampling line. Discard/empty any water present in the water trap.
5. Remove patient tubing, Y-piece, and manual breathing bag.

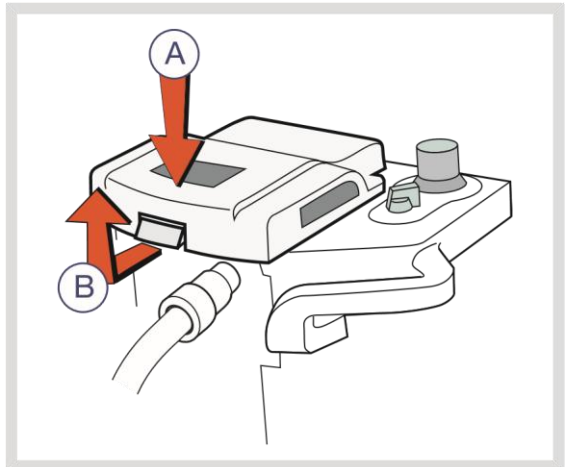


6. Remove the suction bottle/single use **12.4.2 Remove patient cassette** collection bag and the associated suction hose.
 1. Release the patient cassette locking device.

7. Release the CO₂ absorber using the



2. Open the patient cassette cover by:
 - A. Pressing the lid downwards
 - B. If necessary pressing the button on the right
 - C. Pulling the release catch

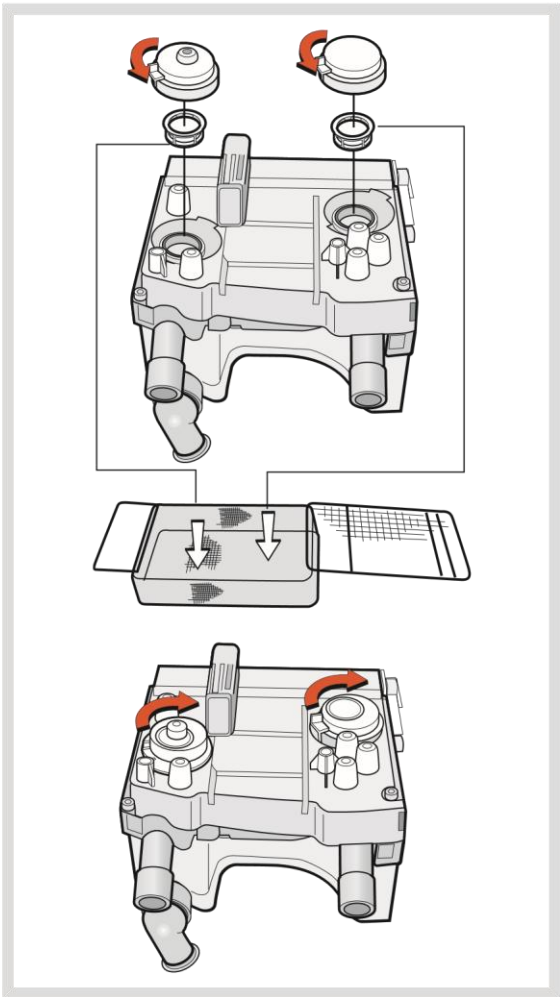
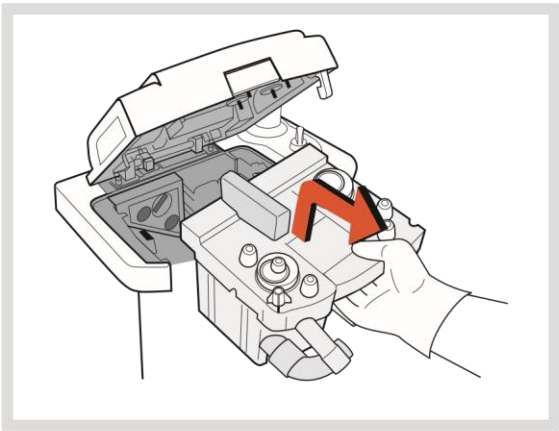


locking switch and remove it.

3.

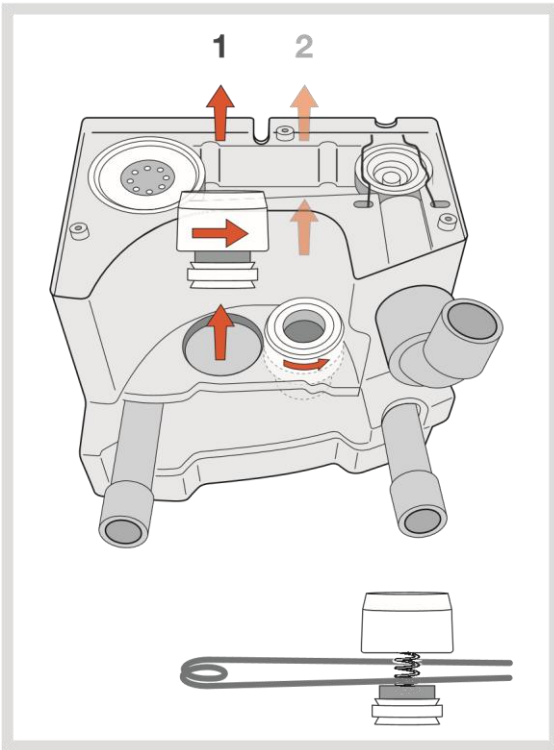
Remove the patient cassette by lifting the and valve cages from cassette upwards and out using the inspiratory/expiratory valves (turn intended grip located between the Place the valve cages inspiratory and expiratory connections. ceramic disks in a fine
4.

Remove the caps the counter-clockwise). containing the mesh box and re-attach the caps.



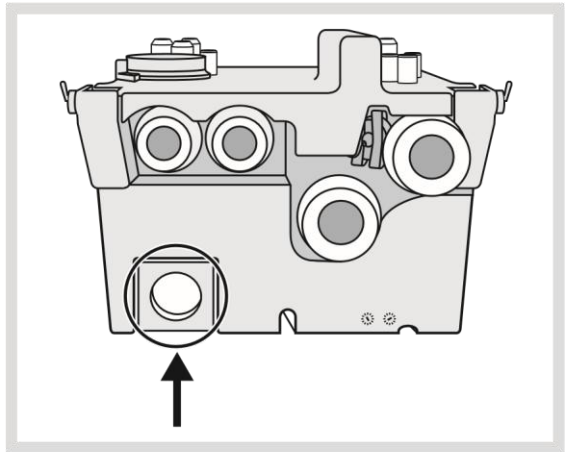
5.

Remove the two absorber valves (turn 6. expiratory counter-clockwise) in the order shown in the back the picture. Attach these to a metal
- Be careful not to damage the outlet one way valve located at of the cassette. framework.

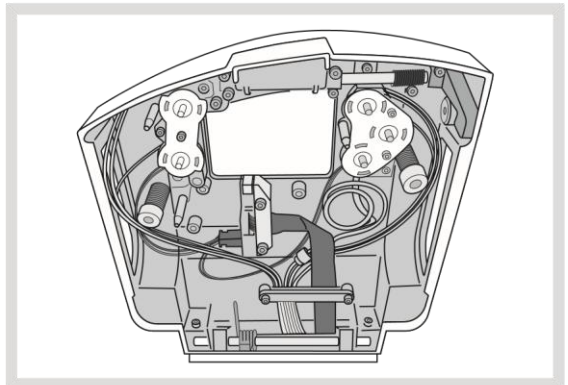


12.4.3 Remove volume reflector

1. Remove the volume reflector adaptor (A).
2. If considered necessary, remove the safety valve (B) using an Allen key and



7. Remove and replace the pressure sensor filters array (left) and gas sampling filters array (right) according to hospital routines.



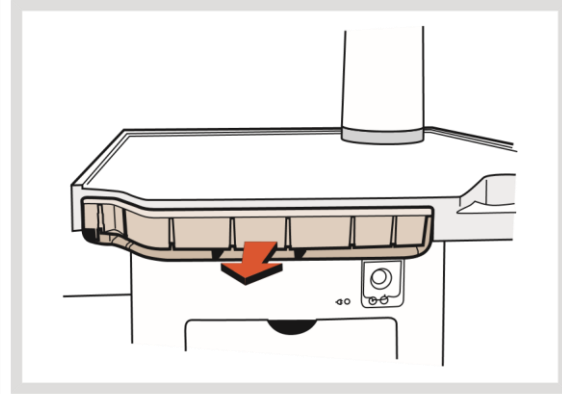
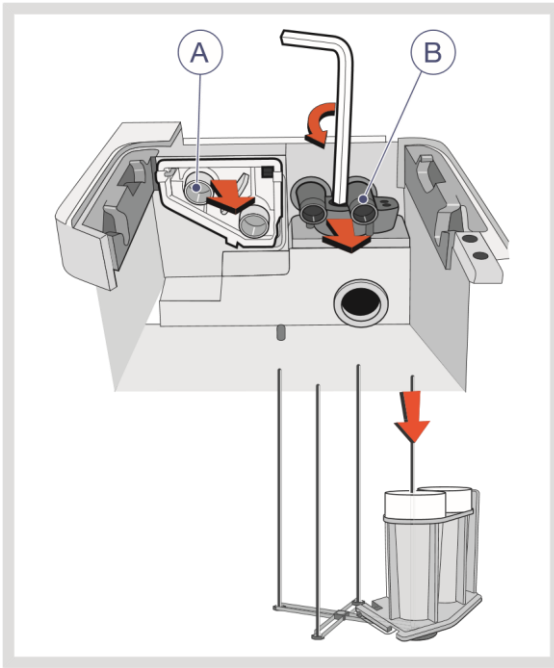
Newer systems are not equipped with the gas sampling filters array. Do not remove the protective covers replacing the array.

thread it onto the metal stand together with the volume reflector adaptor.

Cleaning the safety valve is usually not necessary unless special conditions apply;

the patient is a carrier of pulmonary contagious agents etc.

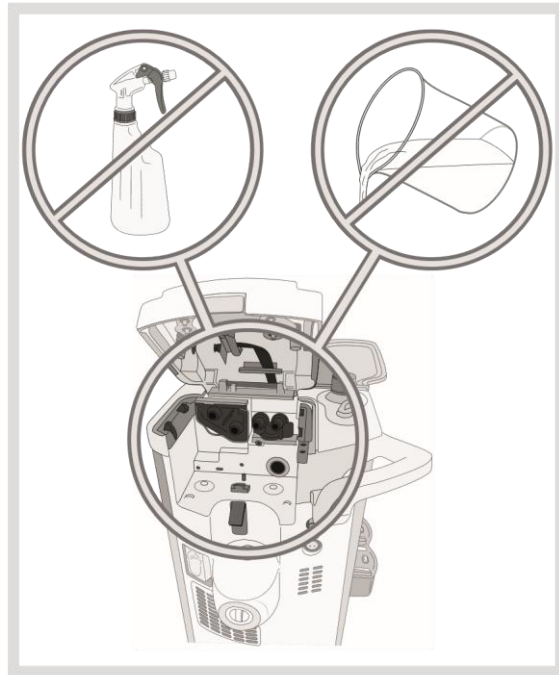
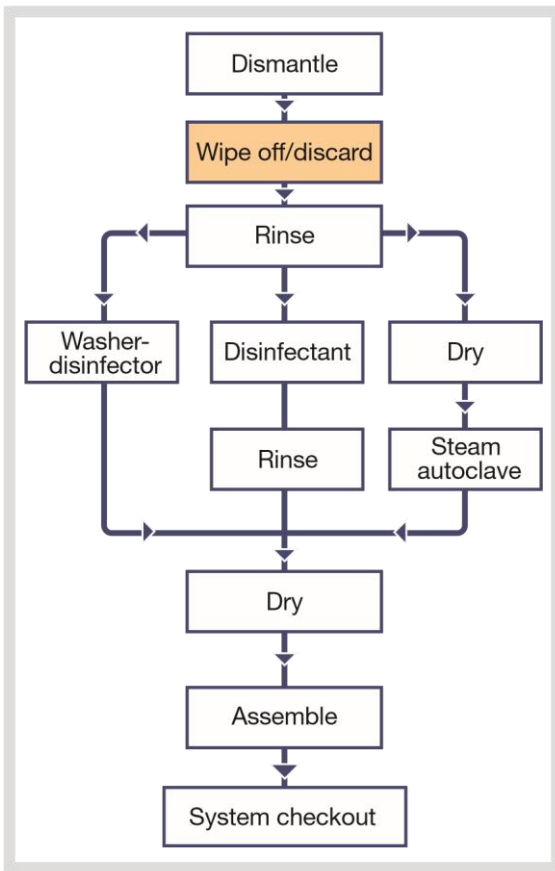
3. Firmly pull out the volume reflector from the system.



The volume reflector must also be cleaned when the patient cassette is cleaned.

For devices with serial numbers below 5001, the volume reflector adaptor has a steel wire retaining bracket. For devices with serial numbers above 5001, the volume reflector adaptor is delivered without this retaining bracket. The adaptor without the retaining bracket can be used on all FLOW-i devices, while the adaptor with the retaining bracket may NOT be used for devices with serial numbers above 5001. To avoid any risk of confusion, it is a good idea to remove the retaining bracket from the volume reflector adaptor.

12.5 Wipe off/discard



Do not pour or spray fluid into the patient cassette compartment.

Wipe the exterior of the system, the patient cassette compartment, the volume reflector compartment and all removable parts with a soft lint-free cloth moistened in soap (pH <10) and water or a mild detergent-based disinfectant.

In the case of more contaminated surfaces, the following products have been approved for use:

- Alcohol (ethyl or isopropyl alcohol 70%)

Wipe the system free of soap and disinfectant residues using a lint-free cloth moistened in water only.

Discard disposable items according to local and environmental regulations.

Regularly check that the fan filter looks clean (i.e. black). If dusty, remove (snap off/snap

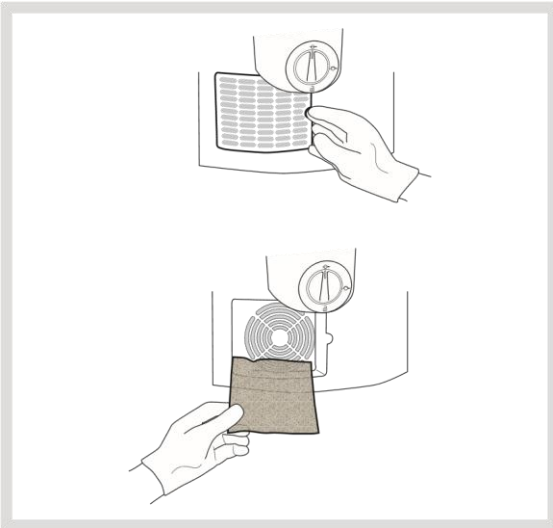
on) and rinse in water. Shake out all excess water from the filter.

After removing the patient cassette and volume reflector, avoid pouring or spraying any liquid into the open compartments. Avoid contact with electronics and electrical connectors when wiping.

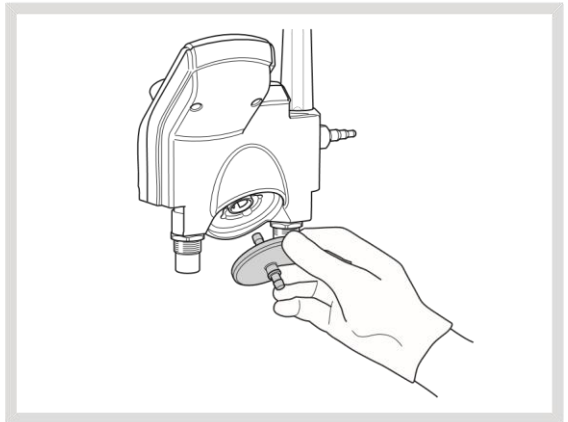
Check the Suction unit bacterial/viral filter after each patient case for signs of moisture or fluid and replace if necessary. Fluid in the filter will compromise the functionality of the filter and Suction unit.

Firmly pull the filter outwards and dispose it

Attach the new filter by fitting the unmarked end into the back of the suction module and press until a click is heard. Ensure the filter is correctly secured in the locked position.



according to hospital routines for contaminated waste products.



12.5.1 Discarding consumables

The following items need to be replaced at regular intervals or according to hospital routines:

Item	Interval	Notes
Water trap	Replace the entire water trap once a month	Empty the water trap after 17 hours of active use, or when half full. The water collector can be cleaned separately.
Patient suction collection bag	After each patient	Applicable only if used during treatment
Vacuum- and serial tubing used with the suction module	After each patient	Applicable only if used during treatment
Gas sampling line (single use)	After each patient if no other means has been used at the Y-piece	---
Exp. connection bacterial filter	After each patient if no other means has been used at the Y-piece	---
Heat and moisture exchanger (HME)	After each patient	---
Patient tubing	According to manufacturers instructions or hospital routines	---
Sensor and sampling filters	After each patient	---
Suction unit bacterial/viral filter	Once every three months, or according to hospital routines	Replace if filled with fluid
CO ₂ absorber	When exhausted	---

The CO₂ absorber is considered hazardous waste and should be disposed of in accordance with hospital regulations and appropriate industrial and environmental standards.

Discard the complete CO₂ absorber when used up. Never open the CO₂ absorber. Always use MAQUET approved absorbers.

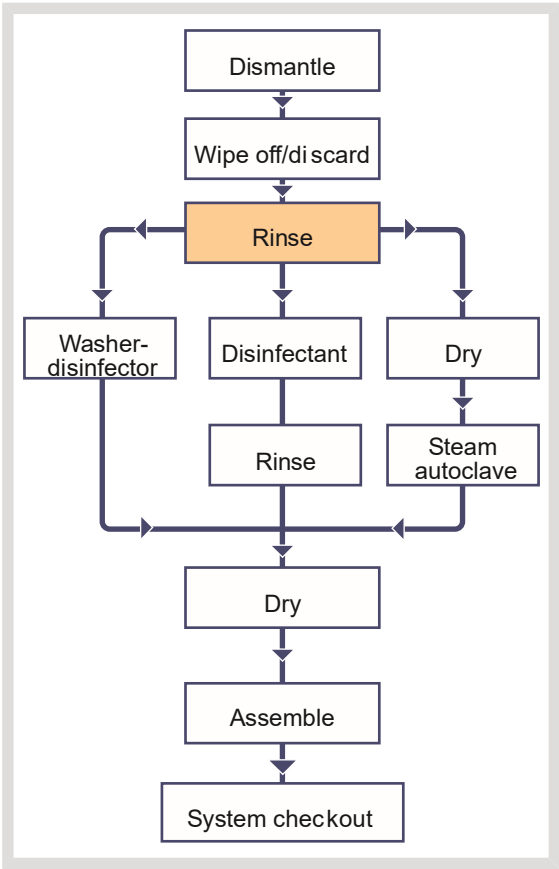
Do not clean or disinfect the bacterial/viral filter part of the water trap as this might cause irreversible damage to it.

The following disinfectants may be used to clean the water container part of the water trap:

- Ethanol 70%, Methanol 70% or Isopropanol 70%
- Glutaraldehyde (e.g. Cidex™)

- Chlorhexidine/ethanol (e.g. Hibitane™)
- Hypochlorite solution (e.g. Clorox™)

12.6 Rinsing before disinfection



and other residues. Rinsing with water warmer than 35°C (95°F) may cause organic material to coagulate.

If the step of rinsing before disinfecting is not included in the washer disinfection program, or if the washer disinfector will not be used immediately, then rinse the equipment manually.

Visually check the cleaning adapters for cracks and fissures. Discard broken adapters.

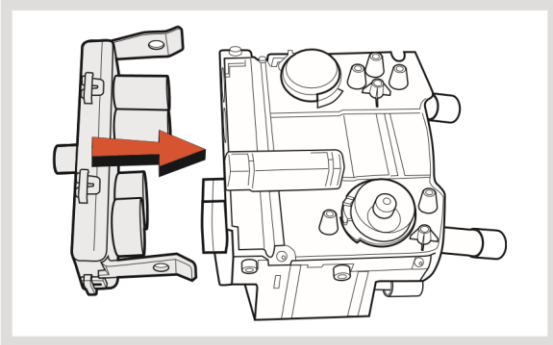
Rinse before disinfection as soon as possible after use to improve the disinfection and reduce the risk for cross-contamination between patients. Begin by rinsing the exterior surfaces under running water. Use the cleaning adaptors as described below to rinse the interior compartments of the equipment.

Rinse the parts thoroughly in water (<35°C/95°F). Let the water flow through the parts to remove organic matter, e.g. blood,

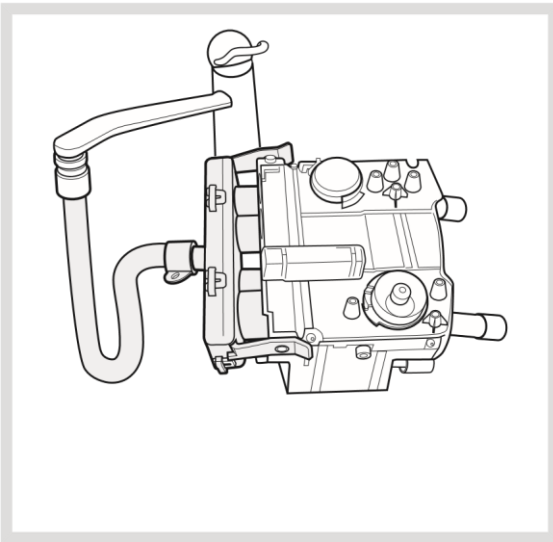
12.6.1 Patient cassette

the required cleaning adaptor when rinsing the patient cassette:

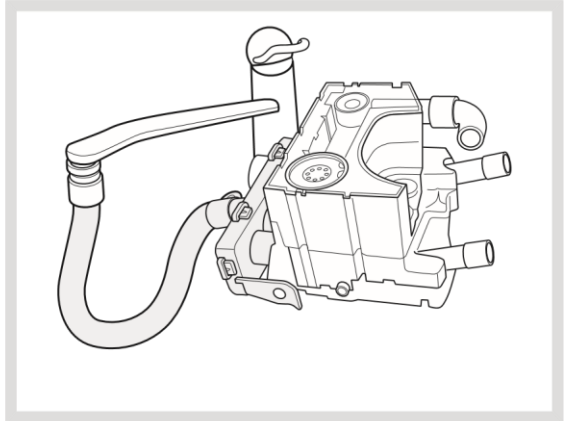
1. Connect the cleaning adaptor to fit into the back of the patient cassette.



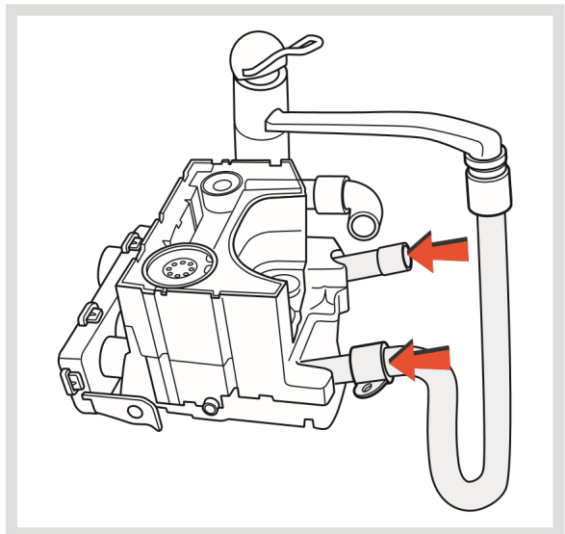
2. Connect the cleaning adaptor to a source of running water (flow > 10 l/min) using minute. Repeat for the inspiratory connective tubing. Let water flow through the cassette for at least one minute.



3. Place the patient cassette upside down and again let water flow through the Use cassette for at least one minute.



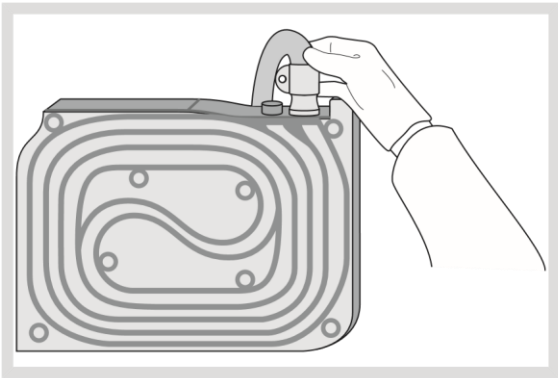
4. Attach the connective tubing to the expiratory connection and let water flow through the cassette for at least one minute. Repeat for the inspiratory connection.



5. Remove the cleaning adaptor, carefully

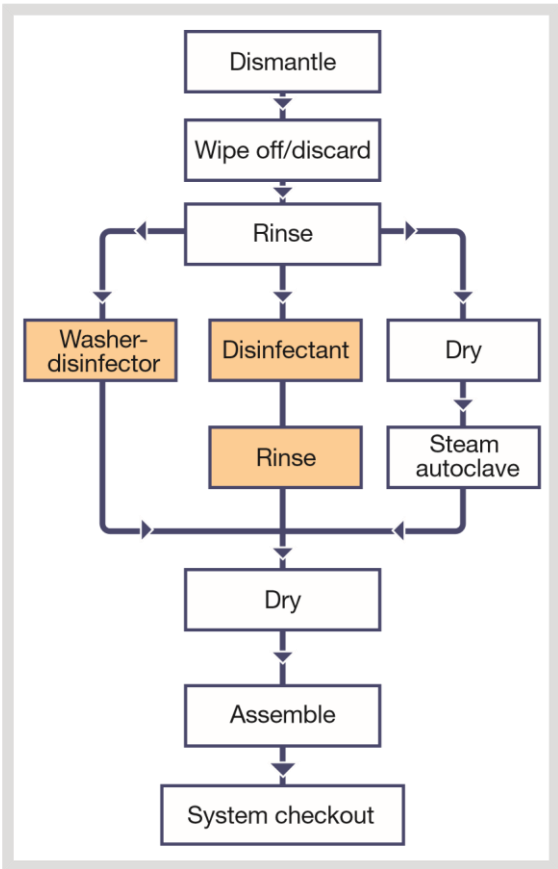
12.6.2 Volume reflector

Connect running water to one inlet of the volume reflector and let water flow through for at least 1 minute. Make sure no air pockets remain in the internal compartment. Place the volume reflector standing on its long end according to the illustration.



shake and tilt the cassette, turn upside down and repeat.

12.6.3 Volume reflector adaptor and safety



valve

Rinse the volume reflector adaptor and safety valve (if removed) under running water for at least one minute.

12.7 Disinfection procedures

This section describes two independent methods of disinfecting the patient cassette, volume reflector and associated parts:

- Washer-disinfector
- Treating the parts with disinfectant fluid

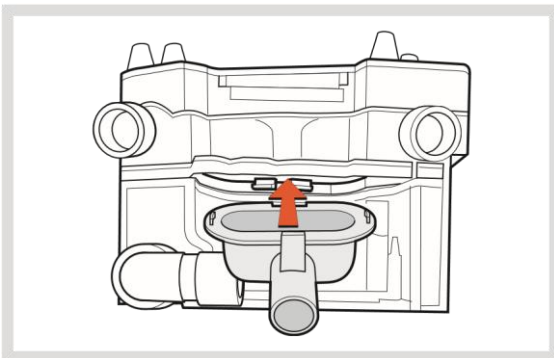
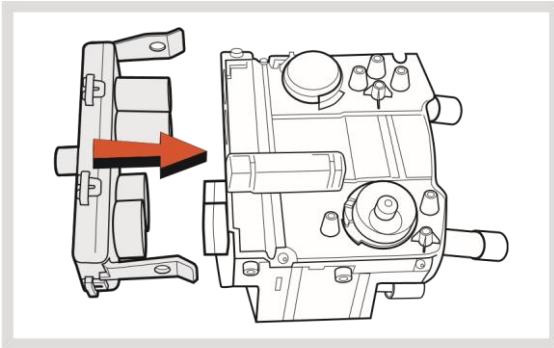
The patient cassette is a precision instrument and must be handled carefully.

12.7.1 Washer-Disinfector (D1)

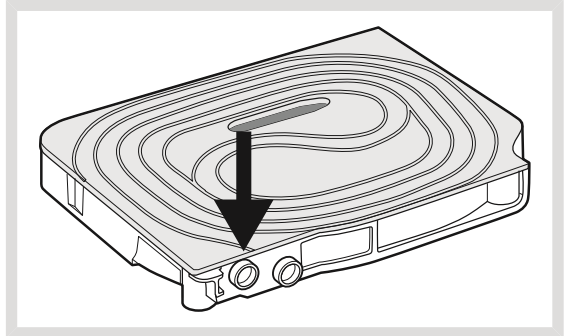
To minimize the effect on the environment, MAQUET recommends only using water when disinfecting the equipment. This is best performed in a washer-disinfector (with A_0 value 600), typically at a maximum temperature ranging from 90°C to 95°C (194°F to 203°F).

If detergents are used with the washer-disinfector, MAQUET recommends detergents with a pH <10. The following detergents have been validated: • neodisher MediClean forte

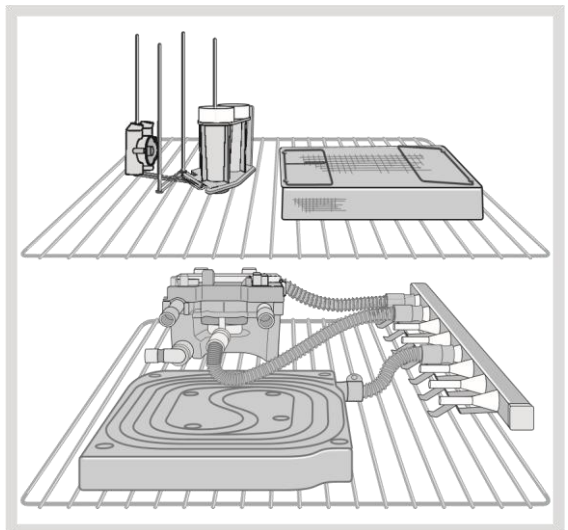
1. Connect the special cleaning adapters to the patient cassette.



2. Connect the special cleaning adapters to the washer-disinfector water outlet using connective tubings.
3. Connect a washer-disinfector water outlet to the patient end of the volume reflector (upper inlet).



4. For optimal cleaning results, fit the disassembled parts into the washer-disinfector according to the suggestion in the illustration below:



To further increase the washing efficiency, plug the remaining water outlets to increase the water flow through the patient cassette and volume reflector.

- The water pressure in the washer-disinfector should not exceed 1.5 bar.
- The water flow should never be less than 10 l/min.
- The water must have free passage through the cassette.

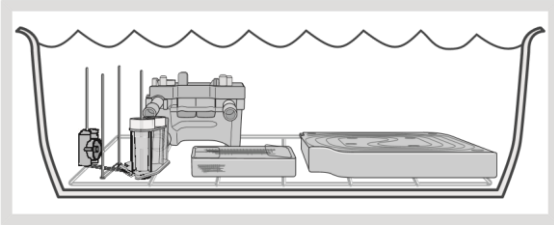
The patient cassette and volume reflector have been tested to withstand 200 disinfection cycles in a washer disinfector.

Proceed to section 11.9, page 253, for drying instructions.

CAUTION: Water remaining in the patient cassette and volume reflector after a completed washing program may be very hot.

12.7.2 Disinfectant (D2)

Remove the inspiratory and expiratory valve lids and place them in a fine mesh box together with the valve cages. Lower the parts into the disinfectant fluid and make sure they are completely submerged and no air pockets remain inside. Let the disinfectant act according to the manufacturers recommendations and instructions.



To ensure maintained functionality of the system, use only tested and validated disinfectants:

- Cidex OPA

12.7.3 Rinse after disinfection

Cleaning adapters used when rinsing after disinfection must have been disinfected using one of the methods outlined in this chapter.

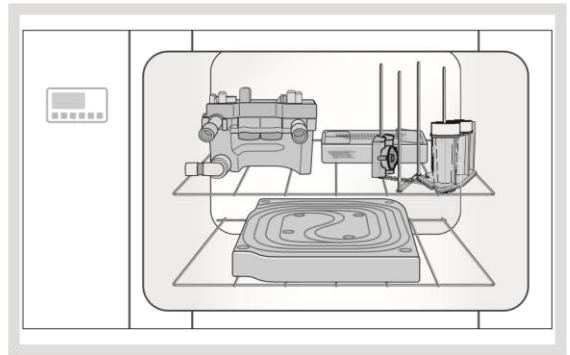
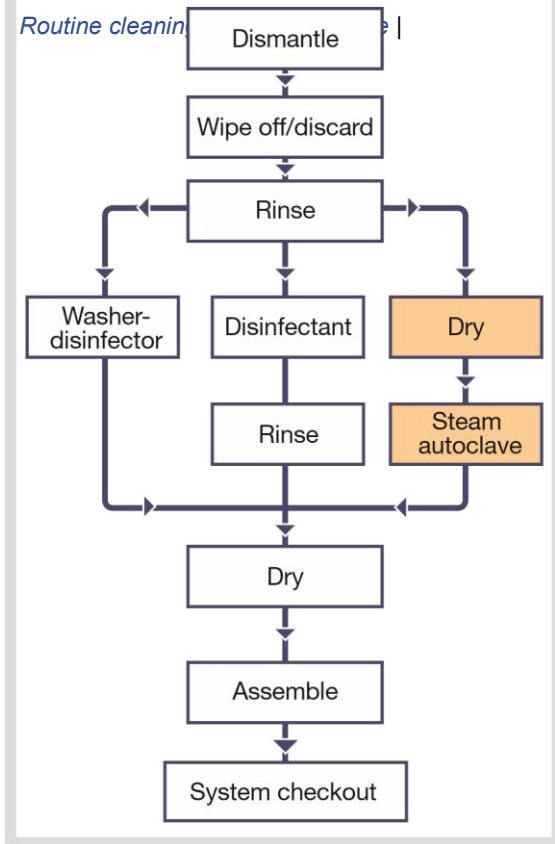
Use deionized water when rinsing after disinfection to avoid mineral scale build-up.

1. Rinse the parts thoroughly in water according to instructions in section 11.6, page 246, to remove all traces of disinfectant. Take care to rinse the inspiratory and expiratory valve screwthreads and lids thoroughly before attaching the lids back onto the patient cassette.

2. Make sure as little water as possible remains in the patient cassette and volume reflector before drying.
3. Proceed to section 11.9, page 253, for drying instructions.

CAUTION: Check for residuals on the equipment before proceeding to the next section.

| 12 Routine cleaning



- System parts should be autoclaved in a validated process, typically at a temperature of 121°C (250°F) for 15 minutes or 134°C (275°F) for 4 minutes.
- The system parts can also be autoclaved at a temperature of 134°C (275°F) for 18 minutes, a.k.a the Prion autoclave program. Using this method will further reduce the lifetime of the parts.
- The breathing bag and suction bottles should be autoclaved according to the manufacturer's instructions.

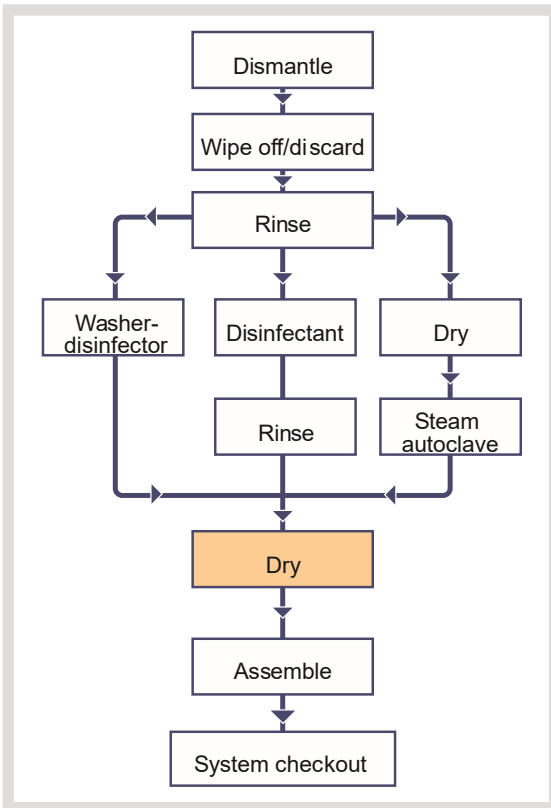
Position the parts inside the autoclave:

Rinse and dry the equipment prior to this procedure. Water remaining in the patient cassette can prevent complete energy transfer from the hot steam, resulting in an incomplete process.

For drying alternatives, see page 253.

CAUTION: Steam autoclaving will reduce the lifetime of the components.

12.9 Drying alternatives



- 100 autoclaving cycles with a 4 minute holding time at 134°C (275°F).
- 50 autoclaving cycles with an 18 minute holding time at 134°C (275°F).

The patient cassette, volume reflector and associated parts, must be dried before use.

If the assembled system parts are not dry, the System checkout may fail.

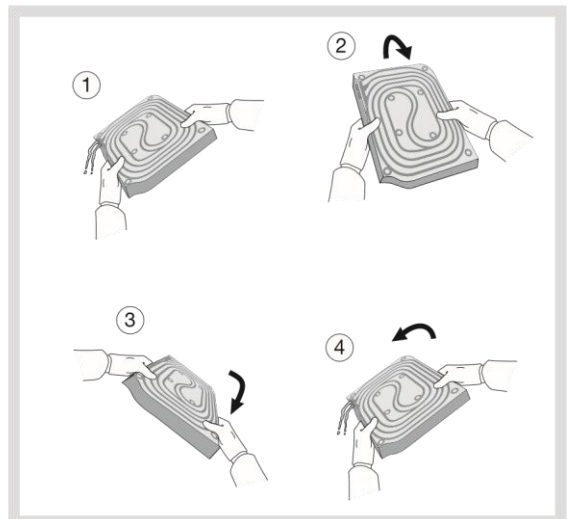
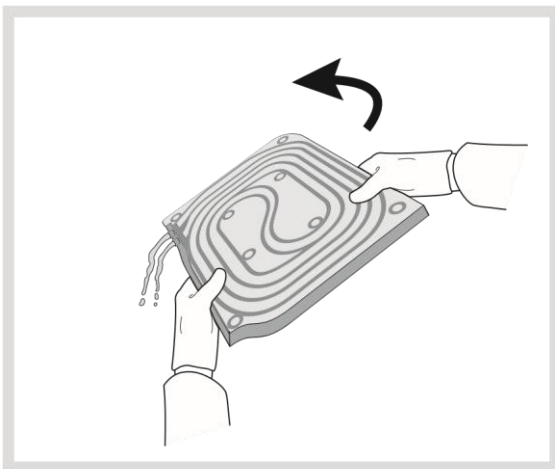
Drying may not be necessary, or the drying time may be reduced, if the washer-disinfector has a drying phase.

Prior to any drying procedure, manually remove as much water as possible by shaking and tilting the parts.

To remove water from the volume reflector the 2. following steps are recommended:

1. Hold the volume reflector horizontally and tilt the device so that water flows from the two openings into the sink.

Hold the volume reflector vertically, facing the top side (smooth side). Slowly rotate clockwise at least four times until the flow of water is minimal. Beware of residual hot water if the equipment has been cleaned using a washer-disinfector.

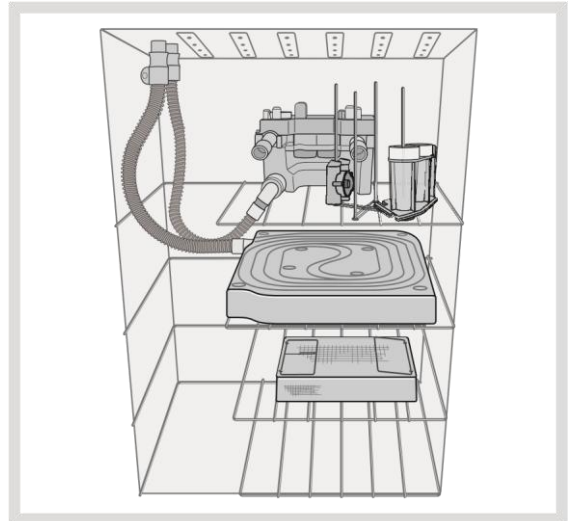


3. Visually check for residual water in the channels by holding the volume reflector in front of a light source.

Using a drying cabinet

To dry the patient cassette, volume reflector and associated parts, perform the following:

1. Disconnect the cleaning adapter.
2. Make sure as little water as possible remains in the patient cassette and volume reflector before drying.
3. Place the patient cassette, volume reflector and associated parts in a drying cabinet for a minimum of 1.5 hours at a maximum temperature of 90°C (194°F) maximum 0.3 bar pressure. If the volume reflector was manually disinfected and rinsed, the minimum required time under these conditions is 2.5 hours.

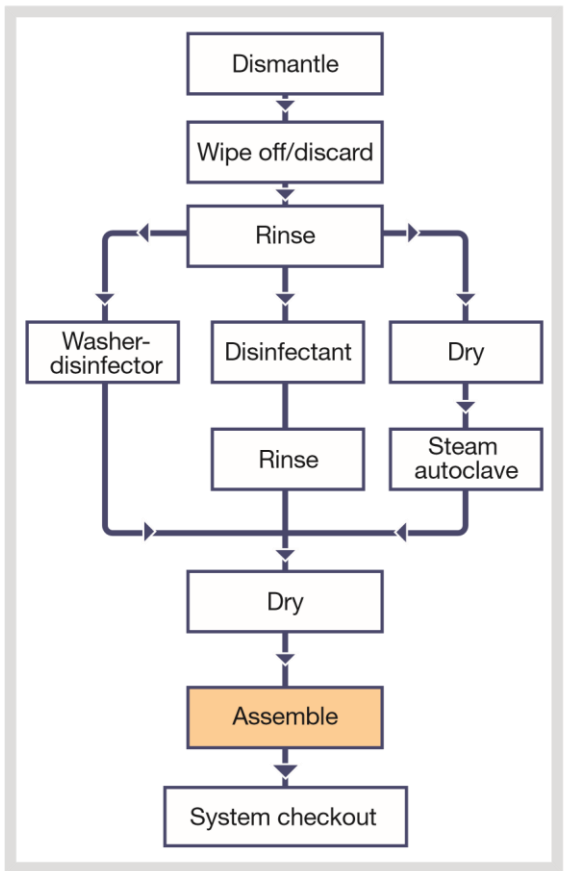


For extra drying capacity, connect a 22mm air **Forced ventilation**

To dry the patient cassette, volume reflector and associated parts using forced ventilation, assemble the system according to section 10.10 and ventilate using a MAQUET test lung for 15 minutes with a flow rate of at least 10 l/min.

Forced ventilation should not be considered a stand alone procedure for drying the system equipment. It acts as a complement to the drying cabinet only.

hose to the expiratory connection of the patient cassette and to the volume reflector.



Ensure that all parts are dry, free from dust

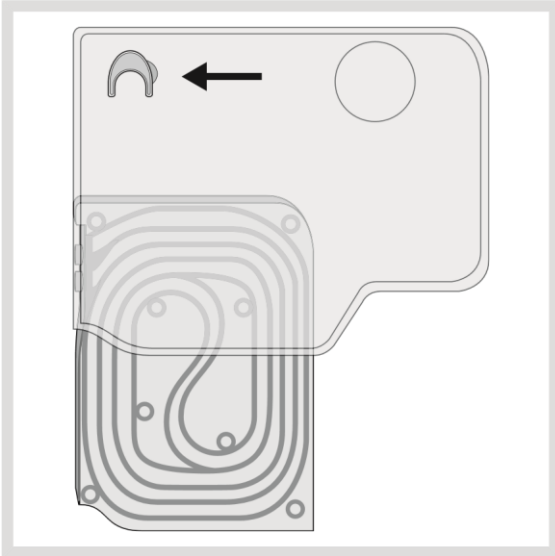
12.10 Assembling

and other particles and are at room temperature prior to assembly.

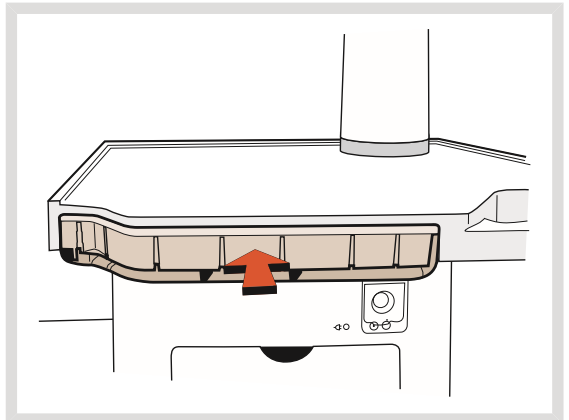
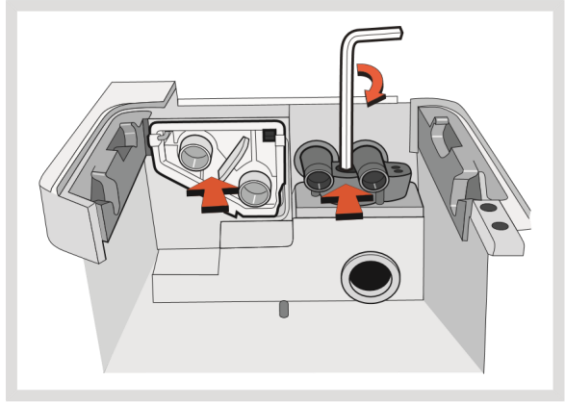
Thoroughly inspect all removed parts for cracks, deformities, fissures and other material wear. Replace if necessary. Ensure that all parts are correctly assembled.

12.10.1 Assembling volume reflector

1. Make sure the lever in the cassette
2. Re-insert the volume reflector.



compartment securing the volume reflector is correctly positioned.



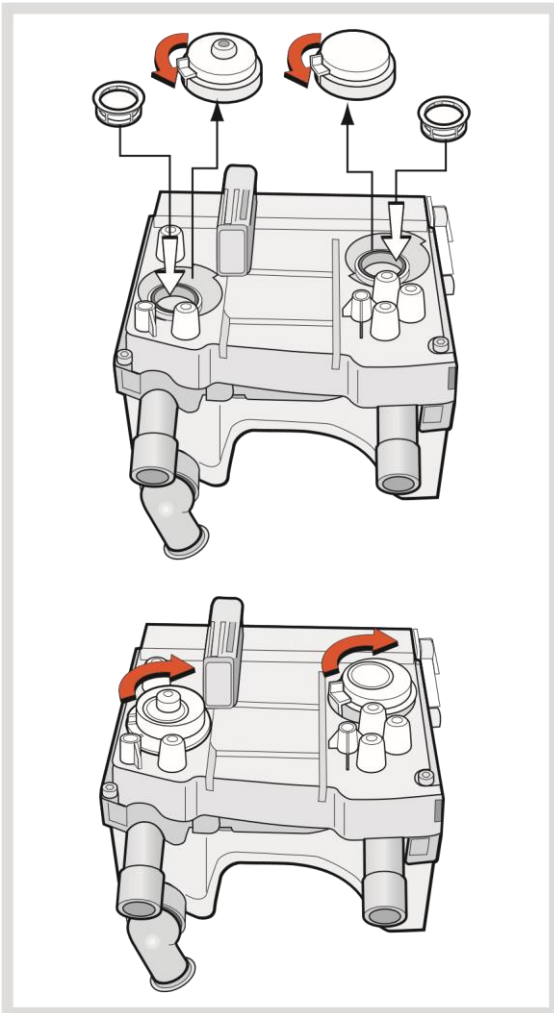
3. Plug in the volume reflector adaptor and re-attach the safety valve if removed.

For devices with serial numbers below 5001, the volume reflector adaptor has a steel wire retaining bracket. For devices with serial numbers above 5001, the volume reflector adaptor is delivered without this retaining bracket. The adaptor without the retaining

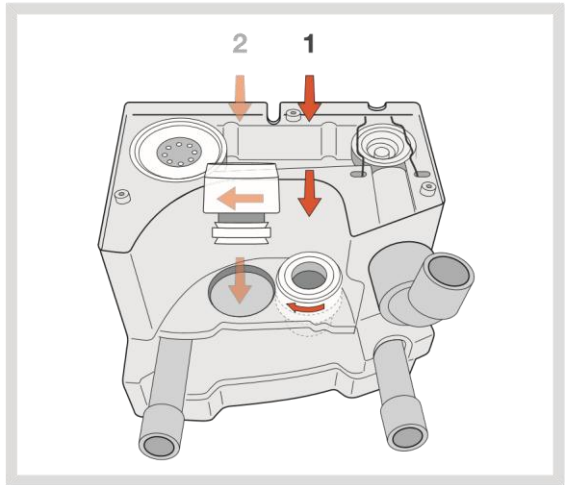
bracket can be used on all FLOW-i devices, while the adaptor with the retaining bracket may NOT be used for devices with serial numbers above 5001. To avoid any risk of confusion, it is a good idea to remove the retaining bracket from the volume reflector adaptor.

12.10.2 Assembling patient cassette 2. Re-attach the absorber valves in the order shown in the picture.

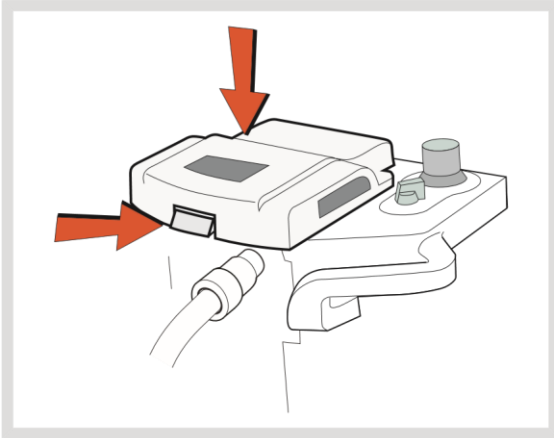
1. Re-position the valve cages holding the ceramic plates into the inspiratory and expiratory valves. Check for damages and



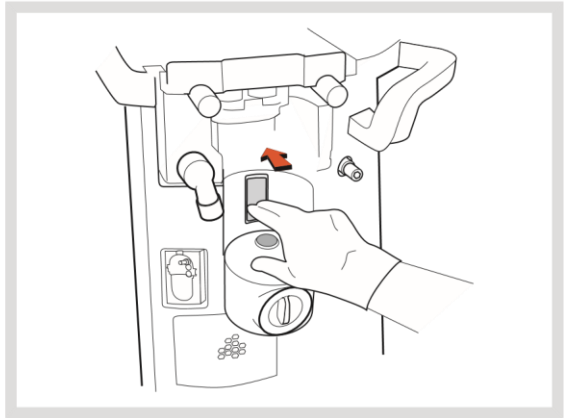
material wear.



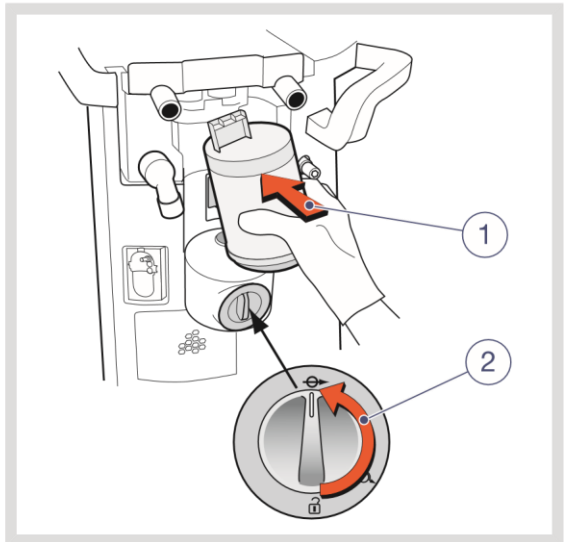
3. Re-position the patient cassette into the 5. locking system by fitting the back part of the cassette into the mounting rails and firmly press it down and forward.
4. Close the lid by pulling the release catch while pressing the lid downwards.



Tighten the patient cassette device.



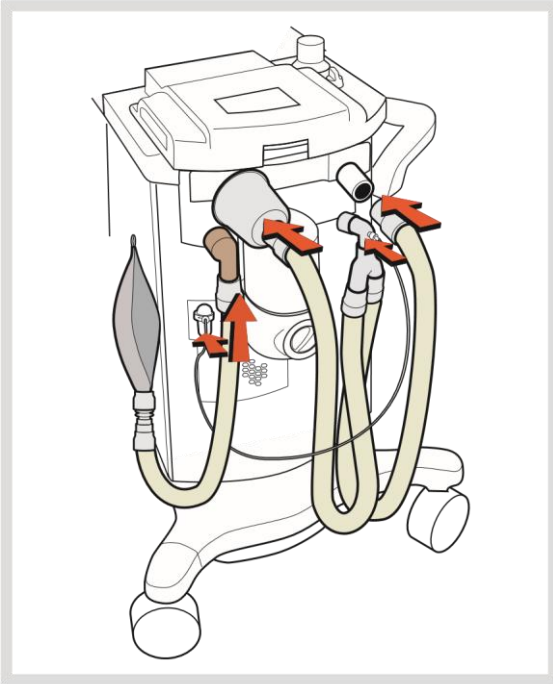
6. Re-dock the CO₂ absorber.



12.10.3 Connect patient tubing and manual breathing bag.

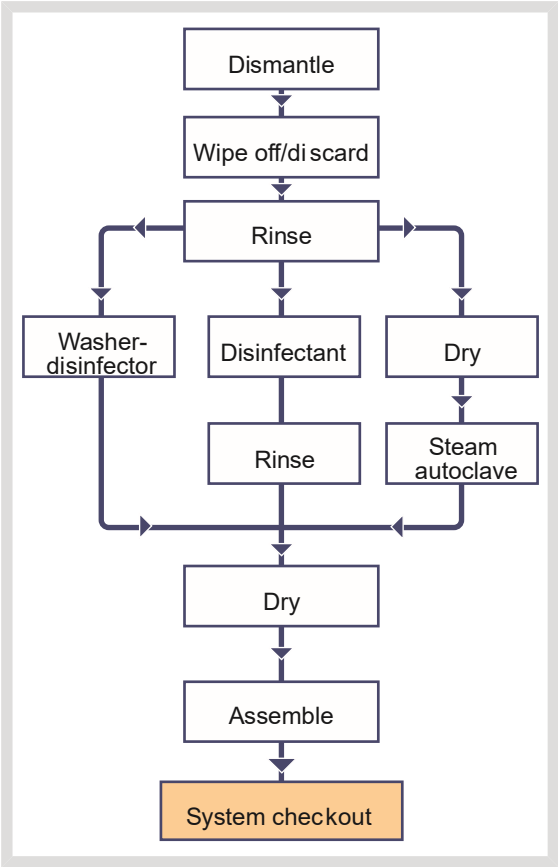
1. Attach a new bacterial/viral filter to the expiratory connection.
2. Attach patient tubing, Y-piece, and manual breathing bag.
3. Attach the water trap and the gas sampling line.

CAUTION: When replacing the gas sampling line, make sure that the dimensions and material of the new consumables comply with that specified by MAQUET, see Chapter 15, page 280. Deviating from these recommendations may affect the gas analysis process.



Note on a log sheet that a routine cleaning has been performed. Refer to hospital guidelines.

12.11 System checkout



12.12 Cleaning accessories and auxiliary equipment

When the accessories and/or auxiliary equipment are not covered by the MAQUET CE mark on the anesthesia system, refer to cleaning instructions found in the equipment's user documentation.

After cleaning, always perform a System checkout. See Chapter 4.

If the System checkout fails due to unknown reasons, repeat the forced ventilation drying process described in 11.9, page 253. Repeat the System checkout.

12.13 Preventive maintenance

MAQUET recommends that the system, associated parts and auxiliary equipment is inspected regularly for wear and damages.

Preventive maintenance and calibration of the system must be performed by personnel trained and authorized by MAQUET at least once a year, or every 5000 hours of operation, whichever comes first. The Status menu on the control panel shows the current operating time.

Service repair must be performed by personnel trained and authorized by MAQUET.

All technical documentation is available for use by MAQUET authorized personnel.

Original parts from MAQUET must be used.

12.14 MCare Remote Services

12.14.1 General information

When initiated, MCare remote services transfers information stored in the system to a remote MCare server located at a central facility.

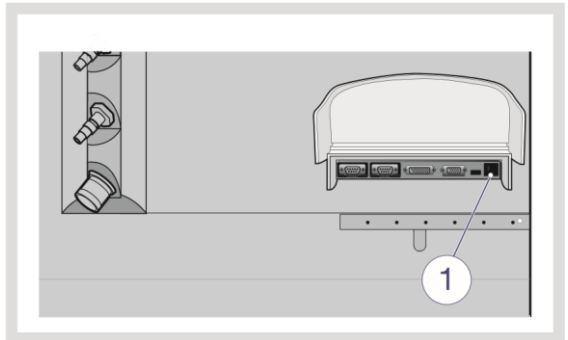
This enables service personnel to evaluate the system off-site and perform troubleshooting/rudimentary maintenance. It also prepares visiting service personnel for system service.

The transferred information consists of the following:

- All logs
- Startup configuration

12.14.2 Preparations

Make sure the network cable is connected to the network connection port on the system (1).



12.15 Starting MCare remote services

There are several methods to start the remote service process:

- A remote service touch pad is available on the log-in screen when selecting Menu/Service & Settings.
- The startup configuration can be set so that the system connects to the MCare portal at specified occasions:
 - OFF

- Prompts start of remote services after a completed SCO.
- Always performs remote services after a completed SCO.
- Always performs remote services after a completed patient case.

12.15.1 Startup configuration

The startup configuration settings for remote service is available in Service & Settings, see Chapter 9, page 174.

12.15.2 Running MCare remote services

When remote services is active and running, a status window is displayed on the control panel. It shows the progress of transfer, as well as any issues with the process. The window closes automatically upon completion.

To start a patient case when a remote services session is active, press 'Cancel' in the remote services status window. A patient case cannot be started when remote services is active and running.

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Gas volumes, flows and leakages associated with the breathing system are stated in the technical specifications and adhere to BTPS reference conditions. (Body temperature, ambient pressure, Saturated).

All gas concentration readings are normally referenced to dry gas conditions, ambient room temperature and atmospheric pressure (ATPD).

The condition for measured inlet gas pressures and flows is STPD (Standard Temperature and Pressure Dry); 20° C, standard pressure at 101.3 kPa and 0 % relative humidity (dry).

13.1 System properties

Weight and dimensions	C20	C30	C40
Base system weight (out of the box weight)	138 kg (2 drawers) 143 kg (3 drawers)	140 kg	140 kg
²² System nominal weight	164 kg (2 drawers) 169 kg (3 drawers)	166 kg	166 kg
System max weight including maximum load	353 kg (2 drawers) 363 kg (3 drawers)	243 kg	273 kg
Dimensions of base plate	27.2" x 38.8"	27.2" x 38.8"	16.0" x 35.3" (measured over table-top and grips)
Height (1 drawer)	N/A	32.9" - 40.4" (adjustable)	24.9"
Height (2 drawers)	33.9"	N/A	31.6"
Height (3 drawers)	40.6"	N/A	N/A
Drawers	2-3 (1 lockable)	1 (lockable)	1-2 (1 lockable)
Vertical shafts for optional horizontal rails	4	4	4
Wheels	Four wheels (diameter 5.9") with separate brakes	Four wheels (diameter 5.9") with separate brakes	Three wheels (diameter 2.0")
Working surface/writing table	Approx. 16.5" x 23.6"		
Additional writing table	15.6" x 9.8"		
Reading lamp	Adjustable LED light integrated into the display arm		

Display	
Type	LED touch screen, complete with 11 membrane switches and one rotary knob
Size	17.0" x 11.6"
Placement	Attached to display arm
Viewing area	15"
Waveforms	Up to 6 waveforms, user configurable
Trends	• Graphic display, 1 to 24 hour resolution • Numeric display, 1 to 60 minute resolution

22. Equipped with two full vaporizers, two gas outlets, one auxiliary O2 and suction module, two rails, two rail clamps, one absorber, breathing tubes, and one patient monitor.

13.2 Essential performance

ESSENTIAL PERFORMANCE (term defined in IEC 60601-1)	
Essential performance	• Oxygen flow under all conditions except the failure of the oxygen supply or generation of a clinical and/or technical alarm • Delivery of a non-hypoxic gas mixture to the patient or generation of a clinical and/or technical alarm • Non-delivery of excessive concentrations of a volatile anesthetic agent or generation of a clinical alarm • Airway pressure monitoring and associated clinical alarms (Ppeak, PEEP)

- Gas measurement accuracy (for Isoflurane, Desflurane, Sevoflurane, CO₂, N₂O, O₂) and generation of gas measurement associated clinical alarms or generation of a technical alarm
- Delivery of ventilation at the patient connection port within the alarm limits set by the operator or generation of a clinical or technical alarm

13.3 Weight and maximum load - accessories

Accessory/Option	Equipment weight	C20, max load (lbs)	C30, max ²³ load (lbs)	C40, max ²⁴ load (lbs)
Additional writing table	4.5	11	11	11
Drawer	N/A	11	11	11
Patient monitor mounting position	N/A	30	30	30
Auxiliary O ₂ & Suction module mounting position	N/A	5.5	5.5	5.5
Parameter box support arm	6.6	11	11	11
Working surface/writing table	N/A	44	11	11
4 vertical shafts	2	11 lbs per shaft	11 lbs per shaft	11 lbs per shaft
Horizontal rails for hanging accessories	N/A	6.6	6.6	6.6
Additional arm	5.8	26	²⁵ 26	26
Backup Gas Rack	20	14	14	N/A
Extra backup gas holder (incl. full gas cylinder)	5.5	7.5	7.5 ¹²	7.5
²⁶ Top shelf	1	44	44	44
Vaporizer holder	N/A	4	4	4
Universal bracket	2	66	66	N/A
Universal bracket (C20)	2	66	N/A	N/A

23. To maintain the lift function, the combined load on the extra equipment and accessories should not exceed 66 lbs.

24. Maximum configuration is dependant on the lift capacity (minimum requirement 440 lbs) of ceiling pendants approved for C40 and the maximum capacity of the C40-pendant interface (550 lbs). This gives a range of approx.

110-220 lbs for accessories and extra equipment, depending on the ceiling pendant.

25. C30 can only be equipped with the additional arm or the extra backup gas holder.

26. The system can be equipped with either Top shelf or Patient monitor.

13.4 Environment

Environment	Operating conditions	Non-operating conditions
Ambient temperature	+15°C to +35°C (+60°F to +95°F) (Desflurane: +15°C to +30°C, +60°F to +85°F)	-25°C to +60°C (-15°F to +140°F)
Battery	+15°C to +35°C (+60°F to +95°F)	+5°C to 40°C (40°F to 104°F)
Relative humidity (non-condensing)	15% to 95%	< 95%
Atmospheric pressure	700 hPa to 1060 hPa	470 hPa to 1060 hPa

13.5 Standards

Standards - Safety and Functionality

Safety	IEC 60601-1 (Class 1, Type B), IEC 60601-1-2, IEC 60601-1-8, ISO 80601-2-13, IEC 62304, ISO 5360
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Electromagnetic compatibility:	IEC 60601-1-2
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Respiratory gas monitoring:	ISO 80601-2-55
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Anesthetic gas delivery	ISO 80601-2-13
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Usability	IEC 62366
-----------	-----------

Cleaning:	IEC 60601-1, ISO 80601-2-13
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Classification according to IEC 60601-1:

Class I equipment:	according to the type of protection against electrical shock
--------------------	--

Type B equipment:	according to the degree of protection against electrical shock
-------------------	--

Continuous operation:	according to the mode of operation (not applicable to the lift, 10% duty cycle)
-----------------------	---

Classification according to EU Medical Directive 93/42/EEC:

The anesthesia system is classified as IIb.

Classification according to IEC 60529

Ingress Protection Rating (IP)	²⁷ IPX1
--------------------------------	-----------------------

27. Valid when the patient cassette is in place and the patient cassette lid is closed. Make sure any fluid has been wiped from the connections in the vaporizer slots before connecting a vaporizer.

13.6 Power supply

Power supply

Mains power

Mains power	100-120V, 220-240V, AC 50-60Hz
-------------	--------------------------------

Power consumption (C30)	500 VA (auxiliary power outlet not included)
-------------------------	--

Power consumption (patient monitor and maximum auxiliary configuration)	1220 VA
---	---------

Battery

Type	Sealed acid-lead rechargeable
Capacity	38 Ah
Operating time	Approx. 90 minutes
Charging time	Approx. 6 hours

Auxiliary power outlets (option)

Power outlet All auxiliary outlets and the patient monitor outlet are connected to an isolation transformer. Voltage depends on mains power supply

Type of electrical outlet	Max load from each outlet	Max load total
3 x IEC outlets	2A (230V)/4A (110V)	2A (230V)/4A (110V)
1 x IEC patient monitor power cable	1A (230V)/2A (110V)	1A (230V)/2A (110V)
3 x 15A NEMA 5-15R (Option)	2A-1A-1A (110V)	2A-1A-1A (110V)

13.7 Gas supply

Gas supply inlets and gas outlets

Central gas

Supply pressure:

Air	250-650 kPa/2.5-6.5 bar/ 36-94 PSI
O ₂	250-650 kPa/2.5-6.5 bar/ 36-94 PSI
N ₂ O	250-650 kPa/2.5-6.5 bar/ 36-94 PSI

Hospital central gas supply must be able to deliver a flow of at least 60 l/min at a supply pressure of 2.8 bar (280 kPa, 41 PSI)

Connection standards	• AGA • DISS • NIST • French standard
----------------------	---

Maximum levels:

Air	• H₂O < 7 g/m³ • Oil < 0.5 mg/m³ • Chlorine must not be detectable
O ₂	• H₂O < 20 mg/m³

If the compressed air is generated by a liquid ring compressor there is a potential risk of chlorine in the supplied air.

Maximum inlet gas temperature	<35°C (<95°F)
-------------------------------	---------------

Gas supply outlets

O ₂ outlet	Dependent central gas supply pressure or cylinder inlet pressure
Air outlet	Dependent central gas supply pressure or cylinder inlet pressure

Gas backup unit (Option)

Pin Index cylinders (5l)	Electronically measured cylinder pressure, Quick-release system for docking and release
Weight and size	Approx. 20 kg / 44 lbs excluding gas cylinders, 320 mm x 730 mm x 220 mm (12.6" x 28.8" x 8.7") (WxHxD)
Cylinder configuration	Max 2 cylinders: O ₂ /N ₂ O or O ₂ /Air
Cylinder pressure, O ₂ /Air	Max. 20,000 kPa / 200 bar / 2900 PSI
Cylinder pressure, N ₂ O	Max. 8,000 kPa / 80 bar / 1160 PSI
Cylinder safety valve opening pressure	Air/N ₂ O and O ₂ 650 kPa/6.5 bar/94 PSI

All gases and anesthetic agents must conform to the European and American Pharmacopeia.

13.8 Fresh gas flow

Fresh gas flow

Gas mix:

Air/O ₂	Electronic Servo controlled
O ₂ /N ₂ O	Electronic Servo controlled

Fresh gas flow range	- MAN = 0.3 - 20 l/min - AUTO = 0.3 - 20 l/min (FGF delivery depending on set MV) • AFGO = 1.0 - 20 l/min
----------------------	---

Fresh gas O ₂ /air Flow (numerical/bargraph)	Selectable
---	------------

Fresh gas O ₂ /N ₂ O Flow (numerical/bargraph)	Selectable
--	------------

Preset O ₂ concentration	Air/O ₂	21% - 100% ±5%
	O ₂ /N ₂ O	28% – 100% ±5%

Setting resolution	1%
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O ₂ Flush	• Approximately 55 l/min
	• 2 cmH ₂ O expiratory resistance when APL is set to SP

13.9 Anesthesia gas scavenging (AGS)

Anesthesia gas scavenging (AGS)

Type	Passive system (including a flow indicator) integrated into the system
Scavenging flow	Minimum 25 l/min (STPD), or 10 l/min (STPD) over the set minute volume, whichever is greater.
Outlet connection type options	• 30 mm ISO taper • DISS EVAC • 12.7 mm / ½" in hose Barb • 25 mm / 1" Barb AGA EVAC WAGD-to-Vacuum connector 22 mm out. diam. connector and 22 mm int. diam. connection tube

13.10 Breathing system

Breathing system

Type	Circle system with Volume Reflector		
System volume (incl. absorber, without patient tubings and manual breathing bag)	Approx. 2.9 l		
Maximum volume allowed for patient tubings and optional equipment forming part of the circle system	3000 ml		
The patient circuit configurations are intended to provide the following range of inspired tidal volumes	• Adult: 22 mm tubing, Tidal Volumes 100 - 2000 ml. • Infant: 15 mm tubing, Tidal Volumes 25 - 350 ml. • Infant: 10-12 mm tubing, Tidal Volumes 5-100 ml .	28	
Drive gas	O ₂		
CO ₂ absorber	• Volume • Absorbent material	Approx. 0.7 l Sofnolime™	
Patient tube connections	22/15 mm ISO cone		
Type of material (breathing circuit system)	PPSU (Polyphenylsulphone).		
System compliance (volume of gas lost due to internal compliance - manual mode only)	Approx. 3 ml/cmH ₂ O, i.e. 90 ml at a pressure of 30 cmH ₂ O		
Inspiratory/expiratory flow resistance of the system (the figures here apply to the breathing tubes recommended by Maquet)	• 10 mm breathing circuits (including Y-piece): - <2.8 cmH₂O at a flow of 2.5 l/min - <5.5 cmH₂O at a flow of 15 l/min • 15 mm breathing circuits (including Y-piece): - <1.3 cmH₂O at a flow of 2.5 l/min - <2.1 cmH₂O at a flow of 15 l/min • 22 mm breathing circuits (including Y-piece): - <4.4 cmH₂O at a flow of 30 l/min - <1.1 cmH₂O at a flow of 2.5 l/min - <2.0 cmH₂O at a flow of 15 l/min		

- <3.7 cmH₂O at a flow of 30 l/min
- <6.0 cmH₂O at a flow of 60 l/min

Manual ventilation

Electronic APL valve	Spontaneous breathing (SP) and adjustable pressure up to 80 cmH ₂ O
----------------------	--

AFGO - Additional Fresh Gas Outlet (option)

Type	• 22 mm coaxial/15 mm conical outlet connections • Pneumatic powered SW controlled (from control panel)
------	--

Emergency backup ventilation

Emergency APL valve	SP - 80 cmH ₂ O
O ₂ emergency flow	0 - 10 l/min

28. VC: 20-100 ml, PC: 5-100 ml

13.11 Breathing circuits and accessories

Note that the table applies to the breathing circuits recommended by Maquet.

Breathing circuits and accessories

Compliance	• 10 mm breathing circuits: <0.4 ml/cmH₂O • 15 mm breathing circuits: <0.7 ml/cmH₂O • 22 mm breathing circuits: <1.4 ml/cmH₂O
Internal volume	• 10 mm breathing circuits: 0.4 l • 15 mm breathing circuits: 0.7 l • 22 mm breathing circuits: 1.4 l
Flow resistance in each limb including Y-piece	• 10 mm breathing circuits: • At 2.5 l/min: <3.0 cmH₂O/l/s • At 15 l/min: <6.0 cmH₂O/l/s • At 30 l/min: <10 cmH₂O/l/s (30 l/min not applicable for intended patient tidal volume range) • 15 mm breathing circuits: • At 2.5 l/min: <1.5 cmH₂O/l/s • At 15 l/min: <1.5 cmH₂O/l/s • At 30 l/min: <2 cmH₂O/l/s • 22 mm breathing circuits: • At 2.5 l/min: <0.5 cmH₂O/l/s

	• At 15 l/min: <0.7 cmH₂O/l/s • At 30 l/min: <1.0 cmH₂O/l/s
Flow resistance for angled adapter	• At 2.5 l/min: <0.2 cmH₂O/l/s • At 15 l/min: <0.6 cmH₂O/l/s • At 30 l/min: <1.0 cmH₂O/l/s
Flow resistance for 22 mm joint adapter	• At 2.5 l/min: <0.2 cmH₂O/l/s • At 15 l/min: <0.2 cmH₂O/l/s • At 30 l/min: <0.2 cmH₂O/l/s

13.12 Ventilator

Ventilator

Type	Pneumatic powered Servo controlled	
Patient range	Neonatal to Adult	
Ventilation modes	• Manual/Bag • AFGO (option) • Volume Control (VC) • Pressure Control (PC) • Pressure Support (PS, option)	
Tidal volume range (volume controlled modes)	20 – 2000 ml, ±10% or 10 ml, whichever is greater	
Tidal volume setting range	Infant range:	20 - 350 ml, resolution 1 ml
	Adult range:	100 - 2000 ml, resolution 10 ml
Minute volume setting range	Infant range:	0.3 - 20 l/min
	Adult range:	0.5 - 60 l/min
Inspiratory pressure (Pressure controlled modes)	0 - 120 cmH ₂ O	± 15% or ± 2 cmH ₂ O, whichever is greater
Inspiratory pressure setting range	Infant range:	0 - 80 cmH ₂ O, resolution 1 cmH ₂ O
	Adult range:	0 - 120 cmH ₂ O, resolution 1 cmH ₂ O
Compressible volume compensation	Yes	
Max. inspiratory flow	200 l/min (3.3 l/s)	
Breathing frequency	4 – 100 ±1 breaths/minute	
I:E (VC, PC)	1:10 - 4:1	
PEEP	0 – 50 cmH ₂ O	
Trigger	Flow / Pressure	
Inspiratory pause (VC)	0 to 30% or 0 - 1.5 s	

13.13 Respiratory monitoring

Respiratory monitoring

Administered breaths	1-100 ±1 breath/minute
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Loops	- Flow – Volume - Volume – Pressure
Lung characteristics	- Expiratory resistance (Re) - Compliance (Cdyn)
Inspiratory Minute Volume	0.3 - 60 l/min
Accuracy Insp. Minute Volume	±15% or ±15 ml multiplied by the breathing frequency, whichever is greater
Expiratory Minute Volume	0.3 - 60 l/min
Accuracy Exp. Minute Volume	±15% or ±10 ml multiplied by the breathing frequency, whichever is greater
Inspiratory Tidal Volume	5 - 2000 ml
Accuracy Insp. Tidal Volume	±4 ml (5-20 ml range) ²⁹ ±15% or 15 ml, whichever is greater (20-2000 ml range)
Expiratory Tidal Volume	5 - 2000 ml
Accuracy Exp. Tidal Volume	±4 ml (5-20 ml range) ³⁵ ±15% or 10 ml, whichever is greater (20-2000 ml range)
Mean Airway Pressure	0 – 100 cmH ₂ O
Peak Airway Pressure	0 - 140 cmH ₂ O
End Expiratory Airway Pressure	-40 – 100 cmH ₂ O
Airway Pressure	-30 to 140 cmH ₂ O
Airway pressure accuracy (applicable to all pressure measurements)	±5% or ±2 cmH ₂ O, whichever is greater

29. Accuracy valid for O₂/Air gas mix, O₂ concentration at 60%, RR at 30 and I:E ≥ 1:2

13.14 Auxiliary O₂ and suction module (option)

Auxiliary O₂ and suction module

Weight (lbs)	Approx. 4.9
Patient suction supply pressure (Air)	30 300-650 kPa/3-6.5 bar/44-94 PSI
Auxiliary O ₂ supply pressure	300-650 kPa/3-6.5 bar/44-94 PSI ³⁶
Auxiliary O ₂ flow range	0 to 10 l/min. Apparent gas flow will increase with altitude as the ambient pressure decreases.
Compressed air consumption (Suction unit)	50 - 90 NI/min ³¹ at a supply pressure ranging from 3.0 to 6.5 bar
Max free flow (suction flow)	28 NI/min
Max. vacuum (suction) ³²	-60 kPa down to -90 kPa at a supply pressure ranging from 3.0 to 6.5 bar

13.15 Misc. optional equipment

Misc. optional equipment

Vaporizer holder

Weight	Approx. 0.8 kg
Weight including full vaporizer	Approx. 4.0 kg
Dimensions	W 90 x H 220 x L 215 mm

EVAC Restrictor

Weight	Approx. 60 g
Dimensions	W 32 x H 48 x L 48 mm

Manual breathing bag support arm

Weight	Approx. 2.2 lbs
Maximum load	2.2 lbs

30. The unit will not meet specified performance if supply pressure is below this range
31. Normal liter (NI) - volume of gas given ambient conditions, e.g. current atmospheric pressure
32. Max. vacuum varies as a function of atmospheric pressure and supply pressure. Highest performance is obtained at sea level when the supply pressure is approx. 4 bar. Performance decreases with increased altitude.

13.16 Vaporizer

Vaporizer

Agents	Isoflurane, Sevoflurane and Desflurane		
Type	Electronic Injector		
Weight (full)	ca. 7.1 lbs		
Dimensions	2.8" x 8.5" x 7.0"		
Agent capacity	300 ml		
Residual capacity	30 ml (triggering the low level alarm)		
Setting range	Isoflurane	0, 0.3-5%, OFF	
	Sevoflurane	0, 0.3-8%, OFF	
	Desflurane	0, 1.0-18%, OFF	
Accuracy	±15% of set value or ±5% of maximum possible user setting (whichever is greater)		
Filling system	Isoflurane	Maquet filling adapter	
	Sevoflurane	Quik Fil® and Maquet filling adapter	

Desflurane

SAFE-FIL™

Vaporizer filling speed	Approx. 4 ml/s
Overfill protection	Overfilling prevention systems built into the vaporizer
Tank liquid level	Optical and electronic

13.17 Gas analyzer

Gas analyzer		
Measuring technology	O ₂	Paramagnetic sensor
	Agents, CO ₂ , N ₂ O	IR sensor
Warm-up time	ISO standard accuracy	Within 60 s
	Full accuracy	Within 10 minutes
Sampling flow and tolerance	225 ml/min +/- 10% (Return to circuit), BTPS condition	
Sampling line	- Length: 2.0 m - Inner diameter: 0.05"	

Measured parameters

Resp. rate	2 - 100 breaths/minute			
Respiration rate measurement accuracy	<60 breaths/minute	± 1 breath/minute		
	>60 breaths/minute	Unspecified		
Gas measurement accuracy				
O ₂	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 25%	± 1	N ₂ O	0.1
	25 - 80%	± 2	CO ₂	0
	80 - 100%	± 3	Any agent 0.3	
N ₂ O	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 20%	± 2	CO ₂	0
	20 - 80%	± 3	O ₂	0
			Any agent 0	
CO ₂	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 1%	± 0.1	N ₂ O	0.1
	1 - 5%	± 0.2	O ₂	0.1
	5 - 7%	± 0.3	Any agent 0.3	
	7 - 10%	± 0.5		
	> 10%	Unspecified		

Isoflurane	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 1%	± 0.15	CO ₂	0
	1 - 5%	± 0.2	N ₂ O	0.1
	> 5%	Unspecified	O ₂	0.1
Sevoflurane			2nd agent 0.2	
	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 1%	± 0.15	CO ₂	0
	1 - 5%	± 0.2	N ₂ O	0.1
	5 - 8%	± 0.4	O ₂	0.1
	> 8%	Unspecified	2nd agent 0.2	

Gas analyzer				
Desflurane	Gas conc.	Accuracy [% _{ABS}]:	Interference [% _{ABS}]	
	0 - 1%	± 0.15	CO ₂	0
	1 - 5%	± 0.2	N ₂ O	0.1
	5 - 10%	± 0.4	O ₂	0.1
	10 - 15%	± 0.6	2nd agent 0.2	
	15 - 18%	± 1.0		
	> 18%	Unspecified		

Drift of measurement accuracy The accuracy includes stability and effects of device drift during operation between calibrations.

- The respiration rate limit for accurately measured end-tidal values is < 60 breaths/minute for I:E = 1:1, < 40 breaths/minute for I:E = 1:2 and < 30 breaths/minute for I:E = 1:3.
- The accuracy of the gas measurements may be affected if the Ethanol concentration is higher than 0.1%, the Methane concentration is higher than 1% or the Acetone concentration is higher than 1%.
- The partial pressure and the percentage volume of CO₂, N₂O, O₂ and anesthetic agent depend on the amount of water vapor in the breathing gas. A partial H₂O pressure of up to 11 cmH₂O is automatically compensated for by the analyzer. Higher H₂O partial pressures will further dilute the gas sample; at 30 cmH₂O the general error of all measured gases is -2%.

Inspiratory and End-Tidal O₂ Concentration YES

Inspiratory and End-Tidal CO₂ Concentration YES

Inspiratory and End-Tidal N₂O Concentration YES

Inspiratory and End-Tidal Agent Concentration YES

Gas measurement typical response times 10-90%

O₂ 500 ms at a sampling rate of 225 ml/min

N₂O 250 ms at a sampling rate of 225 ml/min

CO₂ 250 ms at a sampling rate of 225 ml/min

AA	300 ms at a sampling rate of 225 ml/min
----	---

Gas measurement typical response times 0-90%

O ₂	3800 ms at a sampling rate of 225 ml/min
----------------	--

N ₂ O	3600 ms at a sampling rate of 225 ml/min
------------------	--

CO ₂	3600 ms at a sampling rate of 225 ml/min
-----------------	--

AA	3700 ms at a sampling rate of 225 ml/min
----	--

Primary anesthetic agent identification

A primary anesthetic agent will be identified by the gas analyzer if the concentration is higher than 0.4% (ISO accuracy) or 0.15% (Full accuracy)

Gas analyzer

Secondary anesthetic agent identification

A secondary anesthetic agent will be identified by the gas analyzer if the concentration is higher than:

- 0.5% or 5% (10% for Isoflurane) of primary agent if primary agent concentration is >10% (ISO accuracy)
- 0.3% or 5% (10% for Isoflurane) of primary agent if primary agent >10% (full accuracy)

Primary anesthetic agent level

For a primary anesthetic agent in a gas mixture, the highest gas level that is concealed when anesthetic concentration falls is 0.3% (ISO accuracy) and 0.15% (Full accuracy).

13.18 Gas analyzer validation

Requirements according to ISO 80601-2-55	Description
--	-------------

Test method used to determine The test method consists of sampling gas from two different sources the rated respiration rate range connected to an electrically controlled pneumatic valve to permit and the corresponding effects of rapid switching between the two sources. During the test, the valve end-tidal gas reading accuracy as is set to switch gas source at a number of frequencies (simulating a function of respiratory rate the range of specified breath rates) and for each frequency the

end-tidal value presented by the gas analyzer is noted. From a diagram of end-tidal value over frequency, the frequency at which the gas analyzer is no longer able to resolve end-tidal values according to specification is identified.

Data sampling rate	25 Hz
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Method used to calculate End-tidal CO₂ concentration readings are identified by the gas end-tidal gas readings analyzer using the highest values of the CO₂-curve. Corresponding readings of N₂O and anesthetic agents are taken at the same point in time. End-tidal O₂ concentration readings are identified by the O₂ mean value during the expiratory phase as identified by the CO₂ curve. Once correctly identified, the lowest O₂ concentration readings during the expiratory phase will be presented as end-tidal O₂.

13.19 Alarms

Alarms	
Expiratory Minute Volume: High	0.5 - 60 l/min
Expiratory Minute Volume: Low	0.01 - 40 l/min
Excessive leakage	The difference between the maximum and minimum pressures during inspiration is too low
Airway pressure: High	10 to 120 cmH ₂ O
Continuous APL pressure (manual mode only)	Activated when the measured airway pressure exceeds predefined values for more than 15 seconds. Predefined values depend on current APL setting.
High continuous pressure (automatic mode only)	Airway pressure is constant above set PEEP level +15 cmH ₂ O more than 15 seconds.
Negative airway pressure	Measured airway pressure is below -10 cmH ₂ O for more than one second
Regulated Pressure Limited (PRVC mode only)	Permissible pressure limits pre-set tidal volume
PEEP: High	0 - 55 cmH ₂ O
PEEP: Low	0 - 47 cmH ₂ O
Respiratory rate: High	1 - 140 B/min and OFF
Respiratory rate: Low	1 - 140 B/min and OFF
Apnea	5 - 45 s and OFF
Long apnea (manual mode only)	No breath detection for up to 120 s
Backup ventilation	Pressure controlled administered breath detected in pressure support backup mode due to time out of the backup respiratory rate parameter.
Check breathing circuit	Activated when inspiratory and expiratory pressures fail to meet preset requirements
Limited battery capacity	Less than 18 minutes left of battery operation.
No battery capacity	Less than 3 minutes left of battery operation.
Water trap missing/Replace water trap	The gas analyzer has detected that a water trap replacement is needed

Alarms

Gas alarms

FiO ₂ : High	23 - 99% and OFF
FiO ₂ : Low	18 - 99%
EtO ₂ : High	13 - 99% and OFF
EtO ₂ : Low	10 - 99% and OFF
FiCO ₂ : High	0.1 - 10%
EtCO ₂ : High	0.1 - 10%
EtCO ₂ : Low	0.1 - 9.9% and OFF
FiAA: High	• 0.1 - 5.0% and OFF (ISO) • 0.1 - 8.0% and OFF (SEV) • 0.1 - 18% and OFF (DES)
FiAA: Low	• 0.1 - 5.0% and OFF (ISO) • 0.1 - 8.0% and OFF (SEV) • 0.1 - 18% and OFF (DES)
EtAA: High	• 0.1 - 5.0% and OFF (ISO) • 0.1 - 8.0% and OFF (SEV) • 0.1 - 18% and OFF (DES)
EtAA: Low	• 0.1 - 4.0% and OFF (ISO) • 0.1 - 6.0% and OFF (SEV) • 0.1 - 12% and OFF (DES)
Agent mixture: MAC > 3	The MAC ₄₀ of the secondary agent is ≥0.6 and the total MAC ₄₀ value is ≥3
Agent mixture	The second agent is MAC ≥0.6 and the total MAC value is <3
High continuous MAC	Measured MAC exceeds time limit: MAC > 2.2; from starting a New Case, until 15 minutes after the first vaporizer activation. MAC > 1.8 otherwise
FiN ₂ O: High	Inspiratory N ₂ O gas supply >80%
Occlusion in sampling line	Detected occlusion reported from Y-piece gas analyzer

Measurements of volume of auditory alarm signals and information signals

Microphone of the sound level meter complying with type 1 instruments specified in IEC 61672-1

Alarm priority	Volume level setting (highest/lowest)	Positions as specified in Table F.1 of ISO 3744	Measured sound pressure level (dB)	A-weighted sound pressure level averaged over the measurement surface (dB)
High	100	1, 3, 5, 7	90.9, 82.8, 71.2, 75.6	85
High	20	1, 3, 5, 7	64.6, 55.8, 44.2, 51.0	59
Medium	100	1, 3, 5, 7	84.4, 82.7, 70.6, 75.6	81
Medium	20	1, 3, 5, 7	57.5, 55.5, 43.5, 50.1	54
Low	100	1, 3, 5, 7	84.6, 82.3, 67.9, 75.4	81
Low	20	1, 3, 5, 7	57.9, 55.6, 43.6, 50.0	54
Background	-	1, 3, 5, 7	30.1, 33.0, 36.9, 29.4	33

Technical alarms

Error code number	Alarm text	Solution
-------------------	------------	----------

Power and gas supply

TE 1-6, 32-51, 65-68, 608	Power failure.	Abort operation. Contact Maquet service.
TE 61	Power button error	Contact Maquet service
TE 77	Battery error	Contact Maquet service
TE 83, 88	Airway pressure sensor error	Contact Maquet service
TE 20, 501, 614	Memory backup battery depleted	Contact Maquet service to replace the battery

Breathing circuit

TE 7, 8, 10, 12, 24, 503, 801	Ventilation control error	Contact Maquet service
TE 9	Safety valve open	Contact Maquet service
TE 21, 23	Ventilation error	Contact Maquet service
TE 25, 26	Barometric sensor error	Contact Maquet service
TE 27	APL pressure exceeded	Perform system checkout
TE 29	Monitoring error	Contact Maquet service
TE 52	Gas analyzer error	Contact Maquet service
TE 58	Gas analyzer error	Restart system. Contact Maquet service if condition persists.
TE 53-56	Gas analyzer error	Contact Maquet service
TE 70	Patient cassette error	Contact Maquet service

Vaporizer

TE 63, 64	Anesthesia delivery error	Contact Maquet service
TE 59, 60, 900, 902, 904-916, 918, 920-923, 926, 950, 952, 954-966, 968, 970-973, 976	Vaporizer error	Change the vaporizer
TE 924, 974	The vaporizer software is incompatible with the system software.	Contact Maquet service
TE 89	Very high AA konc.	Contact Maquet service
TE 917, 967	Vaporizer error	Be observant of vaporizer liquid level
TE 919, 969	Vaporizer error	Perform service of Vaporizer and core system as soon as possible.
TE 925, 975	Vaporizer lid leakage	Check the filling lid or change vaporizer

Control panel

TE 14, 15, 18	Audible backup	Contact Maquet service
TE 16, 31, 502, 601	Communication error	Contact Maquet service
TE 22, 605	Control panel error	Contact Maquet service
TE 28	System date and time error	Contact Maquet service
Technical alarms		
TE 81	Control panel disconnected	Contact Maquet service
TE 606	Control panel high temperature	Contact Maquet service
TE 604	Audible alarm error	System is still usable. Alarm sound signals will be different, non-alarm sound signals will not be generated.
TE 607, 611	Communication error	Check video cable connection. Contact Maquet service.
TE 603	Key press timeout	Check Control panel. Contact Maquet service.

Alarms and patient safety

TE 11	Re-specify alarm limits	Contact Maquet service
TE 62	Emergency ventilation On/Off error	Contact Maquet service
TE 30	Alarm limit error, check alarm profile	Check alarm profile
TE 57	APL valve malfunction	Switch to emergency ventilation
TE 85-87	Backup cylinder pressure unknown: Air/O ₂ /N ₂ O	Replace cylinder

Miscellaneous

TE 17	Fan malfunction	Contact Maquet service
TE 78	System temperature: High	Contact Maquet service
---	Settings lost. Restart the system!	Contact Maquet service

13.20 Health and Environment

13.20.1 Pollution control

This product complies with environmental protection use period as defined in People's Republic of China Electronic Industry Standard SJ/T11364-2006.



Toxic or hazardous substances will not leak or mutate under normal operating conditions for 50 years.

13.20.2 Hazardous substances

The following table shows the names and contents of toxic or hazardous substances in this product as defined in People`s Republic of China Electronic Industry Standard SJ/T11364-2006.

Hazardous substances						
Parts	Pb	Hg	Cd	Cr6+	PBB	PBDE
Metal parts	0	0	0	X	0	0
Plastic and polymeric parts	0	0	0	0	0	0
Electrical components	X	0	0	0	0	0
LCD display	0	0	0	0	0	0

0: Indicates that this toxic or hazardous substance contained in all of the homogeneous materials for this part is below the limit and meets the requirement in SJ/T11363-2006.

X: Indicates that this toxic or hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement in SJ/T11363-2006.

Some internal electronic cables contain the phthalate DEHP classified as SVHC (Substance of Very High Concern) according to REACH (European Union Regulation EC/2006/1907). When the ventilator system is used as intended, no phthalates will be transferred to the environment.

13.21 External communication

External communication	
Serial ports	2 with FCI (FLOW-i Communication Interface) protocol
USB	1
Video out	1 VGA
Ethernet	1 Network connection for service use

13.22 UDI Label

UDI Label	
Unique Device Identification number	Global standard for identifying Medical Equipment, example: (01)07325710000007(11)140625(21)01311141

Application Identifier (AI)	Each UDI number can be divided into several parts, each referred to by their AI number '(#)'.
(01)	GTIN - Global Trade Item Number
(241)	Part number
(10)	Batch no.
(11)	Manufacturing date (YYMMDD)
(17)	Exp. date (YYMMDD)
(20)	Revision
(21)	Serial number
(30)	Count of items

The GTIN consists of four parts:

0 732571 000021 2

a. Package level



a b c d

b. GS-1 company prefix

c. Item reference

d. Check digit

14 EMC Declaration Table of contents

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14.1 Introduction

The EMC declaration serves as information for the safe use of the device.

(EMC = Electromagnetic Compatibility)

14.1.1 FLOW-i Anesthesia System

- FLOW-i
- Vaporizers
(Sevoflurane/Desflurane/Isoflurane)

14.1.2 Cables

- Mains cable (length 3.5 m)
- RS232 cables (shielded, length 2.5 m)
- Ethernet cable (shielded)
- Potential equalization conductor
- VGA cable (shielded, length 2.0 m)

14.1.3 Information to the responsible organization

EMC declaration according to IEC 60601-1-1-2:2007, "Electromagnetic compatibility" for ME equipment.

Electromagnetic immunity levels correspond to test levels for Life Supporting devices.

14.2 Electromagnetic emissions

Guidance and manufacturer's declaration - electromagnetic emissions		
The FLOW-i is intended for use in the electromagnetic environment specified below. The customer or the user of the FLOW-i should assure that it is used in such an environment.		
Emissions test	Compliance	Electromagnetic environment - guidance
RF-emissions CISPR11	Group 1	The FLOW-i uses RF energy only for its internal function. Therefore its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF-emissions CISPR11	Class A	
Harmonic emissions IEC 61000-3-2	Not applicable	
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not applicable	

14.3 Electromagnetic immunity

Guidance and Manufacturer's Declaration - electromagnetic immunity			
The FLOW-i is intended for use in the electromagnetic environment specified below. The customer or the user of the FLOW-i should assure that it is used in such an environment.			
Immunity test	IEC 60601 test level	FLOW-i Compliance level	Electromagnetic environment - guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV Contact discharge ± 8 kV Air discharge	± 6 kV Contact discharge ± 8 kV Air discharge	Floors should be made of concrete, wood or ceramic tile. If floors are covered with synthetic material, then the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV power supply lines ± 1 kV input/output lines	± 2 kV power supply lines ± 1 kV input/output lines	Mains power quality should be that of a typical hospital environment.
Surge IEC 61000-4-5	± 1 kV line to line ± 2 kV line(s) to earth	± 1 kV line to line ± 2 kV line(s) to earth	Mains power quality should be that of a typical hospital environment.
Power frequency (50/60 Hz) magnetic fields. IEC 61000-4-8	3 A/m	30 A/m	The power frequency of magnetic fields should be at levels characteristic of a typical location in a typical hospital environment.

Voltage dips, short interruptions and voltage variations on power supply input lines. IEC 61000-4-11	>95% reduction for 0.5 cycle, 60% reduction for 5 cycles, 30% reduction for 25 cycles and >95% reduction for 5 s.	>95% reduction for 0.5 cycle, 60% reduction for 5 cycles, 30% reduction for 25 cycles and >95% reduction for 5 s.	Mains power quality should be that of a typical hospital environment. The anesthesia system requires continuous operation during power mains interruptions. FLOW-i is equipped with internal batteries. The responsible organisation/user shall make sure that batteries are replaced before their lifetime has expired, according to the instructions of use.
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Electromagnetic immunity

The FLOW-i is intended for use in the electromagnetic environment specified below. The customer or the user of the FLOW-i should assure that it is used in such an environment.

Electromagnetic environment - guidance^a

Portable and mobile RF communications equipment should be used no closer to any part of the FLOW-i, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter.

P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation in distances in metres (m).^b

Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey^c, should be less than the compliance level in each frequency range.

Interference may occur in the vicinity of equipment marked with the following symbol: 

Immunity test	IEC 60601 test level	FLOW-i Compliance level	Recommended separation distance (d):
Conducted RF IEC 61000-4-6	3 Vrms, 150 kHz to 80 MHz outside ISM bands ^{d,e}	10 Vrms, 150 kHz to 80 MHz	$d = 0.35 \sqrt{P}$, 150 kHz to 80 MHz outside ISM bands ^{d,e}
	10 Vrms, 150 kHz to 80 MHz in ISM bands ^{d,e}		$d = 1.2 \sqrt{P}$, 150 kHz to 80 MHz in ISM bands ^{d,e}
Radiated RF IEC 61000-4-3	10 V/m, 80 MHz to 2.5 GHz	10 V/m, 80 MHz to 2.5 GHz	$d = 1.2 \sqrt{P}$, 80 MHz to 800 MHz ^e $d = 2.3 \sqrt{P}$, 800 MHz to 2.5 GHz ^e

- a. These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structure, objects and people.
- b. The compliance levels in ISM bands (parts of the tested bandwidth) are intended to decrease the likelihood that mobile/portable communications equipment could cause interference if it is inadvertently brought into patient areas. For this reason a safety factor of 10/3 is used in calculating the recommended separation distance for transmitters in these frequency ranges.
- c. Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the FLOW-i is used exceeds the applicable RF compliance level above, the FLOW-i should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as relocating the FLOW-i.
- d. The ISM (industrial, scientific and medical) bands between 150 kHz and 80 MHz are 6,765 MHz to 6,795 MHz; 13,553 MHz to 13,567 MHz; 26,957 to 27,283 MHz; and 40,66 MHz to 40,70 MHz.
- e. At 80 MHz and 800 MHz, the higher frequency range applies.

WARNING! The FLOW-i anesthesia system should not be used adjacent to or stacked with other equipment. If such a configuration is necessary, the FLOW-i anesthesia system should be observed to verify normal operation during use.

14.4 Recommended separation distances

Recommended separation distances between portable and mobile RF communications equipment and the FLOW-i.

The FLOW-i is intended for use in an electromagnetic environment in which radiated RF-disturbances are controlled. The customer or the user of the FLOW-i can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the FLOW-i as recommended below, according to the maximum power of the communications equipment.

Rated maximum output power of transmitter (W)	Separation distance according to frequency of transmitter (m)			
	150 kHz to 80 MHz outside ISM bands. $d = 0.35 \sqrt{P}$	150 kHz to 80 MHz in ISM bands. $d = 1.2 \sqrt{P}$	80 MHz to 800 MHz $d = 1.2 \sqrt{P}$	800 MHz to 2.5 GHz $d = 2.3 \sqrt{P}$
0.01	0.04	0.12	0.12	0.23
0.1	0.11	0.38	0.38	0.73
1	0.35	1.20	1.20	2.30
10	1.11	3.79	3.79	7.27
100	3.50	12.00	12.00	23.00
See the notes in the previous tables.				

15 Definitions

AFGO - Additional Fresh Gas Outlet

AGS system - Anesthetic Gas Scavenging system

ATPD - Ambient Temperature and Pressure, Dry (no humidity) **b/min**—Breaths per minute

Breath cycle time—Total cycle time per mandatory breath (inspiratory + pause + expiratory). Set in seconds.

BTPS - Body Temperature and Pressure Saturated: 37° C, ambient pressure and 100 % relative humidity (saturated). Defined standard condition for gas flows in the breathing circuit.

Cdyn—Dynamic characteristics **CPAP**—Continuous Positive Airway Pressure

etCO₂—End tidal carbon dioxide concentration

FGF - Fresh Gas Flow

HME—Heat and moisture exchanger

I:E—Inspiration to Expiration ratio (only during controlled ventilation)

Inspiratory cycle-off—Fraction of maximum flow at which inspiration should switch to expiration (%)

Inspiratory rise time—Time to full inspiratory flow or pressure at the start of each breath, as a percentage or in seconds of the breath cycle time (% or s)

Leakage—Leakage during inspiration (%)

MAC - Minimum Alveolar Concentration is an

index of the anesthetic effect and potency of an inhalation agent in relation to alveolar concentration and is the concentration required for lack of reflex response to skin incision in 50% of patients.

ME equipment - Medical Electrical equipment

Minute Volume—Volume per minute or target volume (l)

MVe—expiratory Minute Volume

MVe sp—Spontaneous expiratory minute volume

MVe sp / MVe—The ratio of spontaneous expired minute volume to total expired minute volume (only applicable in Bi-Vent/APRV)

MVi—inspiratory Minute Volume

O₂—Oxygen concentration in vol. %

Option—Optional, add-on functionality or accessory

P—Pressure

Pause time—Time for no flow or pressure delivery (%)

PBW - Predicted Body Weight

PC—Pressure Control

PEEP—Positive end expiratory pressure

PEEP_{tot}— Set PEEP + Intrinsic PEEP

Phigh—High pressure level

Pmean—Mean airway pressure

Ppeak—Maximum inspiratory pressure

Pplat—Pressure during end-inspiratory pause

Pressure trigger level — The negative pressure that the patient has to create to trigger the ventilator to deliver a breath.

PRVC—Pressure-regulated volume control

PS—Pressure support

Re—expiratory resistance

Respiratory Rate—Rate of controlled mandatory breaths or used for calculating target volume (b/min)

RH—Relative humidity

RR—Respiratory rate

SIMV—Synchronized Intermittent Mandatory Ventilation

SIMV rate—Rate of controlled mandatory breaths (b/min)

Start breath—Manually triggered set breath

STPD - Standard Temperature and Pressure Dry: 20° C, standard pressure at 101.3 kPa and 0 % relative humidity (dry). Standard condition for inlet gas.

T—Time

Ti—Inspiration time

Ti/Ttot—Duty cycle or ratio of inspiration time to total breathing cycle time (only during spontaneous breathing)

Tidal Volume—Volume per breath or target volume (ml)

$\dot{V}$ —Flow

$\dot{V}_{ee}$ —End expiratory flow **leak**—Leakage

$\dot{V}$ flow (l/min)

VC—Volume Control

VRI - Volume Reflector Indicator

VT_e—Expiratory Tidal Volume

VT_i—Inspiratory Tidal Volume

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16.1 China

16.1.1 FLOW-i

CFDA registration no	CFDA(I)20131542673
Product standard no	YZB/SWE 2969-2013
Manufacturing date	For manufacturing date, see label on the device
Manufacturer	Maquet Critical Care AB
Manufacturer/Manufacturer site address	Rontgenvagen 2, SE-17154 Solna, Sweden
Agent for registration and after sales	Maquet (Shanghai) Medical Equipment Co., Ltd.
Agent address	Room 227, 2nd floor, No. 56, Meisheng Road, Pilot Free Trade Zone, Shanghai, China
Agent contacts	800 820 0207
IFU revise date	150904

CFDA registration no	SFDA(I)20123542189
Product standard no	YZB/SWE 2546-2012
Manufacturing date	For manufacturing date, see label on the device
Manufacturer	Maquet Critical Care AB
Manufacturer/Manufacturer site address	Rontgenvagen 2, SE-17154 Solna, Sweden
Agent for registration and after sales	Maquet (Shanghai) Medical Equipment Co., Ltd.
Agent address	Room 227, 2nd floor, No. 56, Meisheng Road, Pilot Free Trade Zone, Shanghai, China
Agent contacts	800 820 0207
IFU revise date	150904

16.1.2 CO₂ absorber

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This chapter provides a sample log sheet to keep a record of system checkouts, routine cleaning and maintenance performed on the system. These log sheets may be copied as they are, or used to provide ideas for designing individual log sheets.

18.1 System checkout

The System checkout log can be viewed via Menu/Logs/System checkout. See page 173 for details.

Date	Passed	Leakage (ml/min)	Other	Signature

18.2 Routine cleaning

Date	Steam autoclave	Disinfection	Signature

18.3 Maintenance

Yearly maintenance / 5000 hours

Date	Signature

MAQUET
GETINGE GROUP

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171 54 Solna, Sweden
Phone: +46 (0) 10 335 73 00
www.maquet.com

For local contact:
Please visit our website
www.maquet.com



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